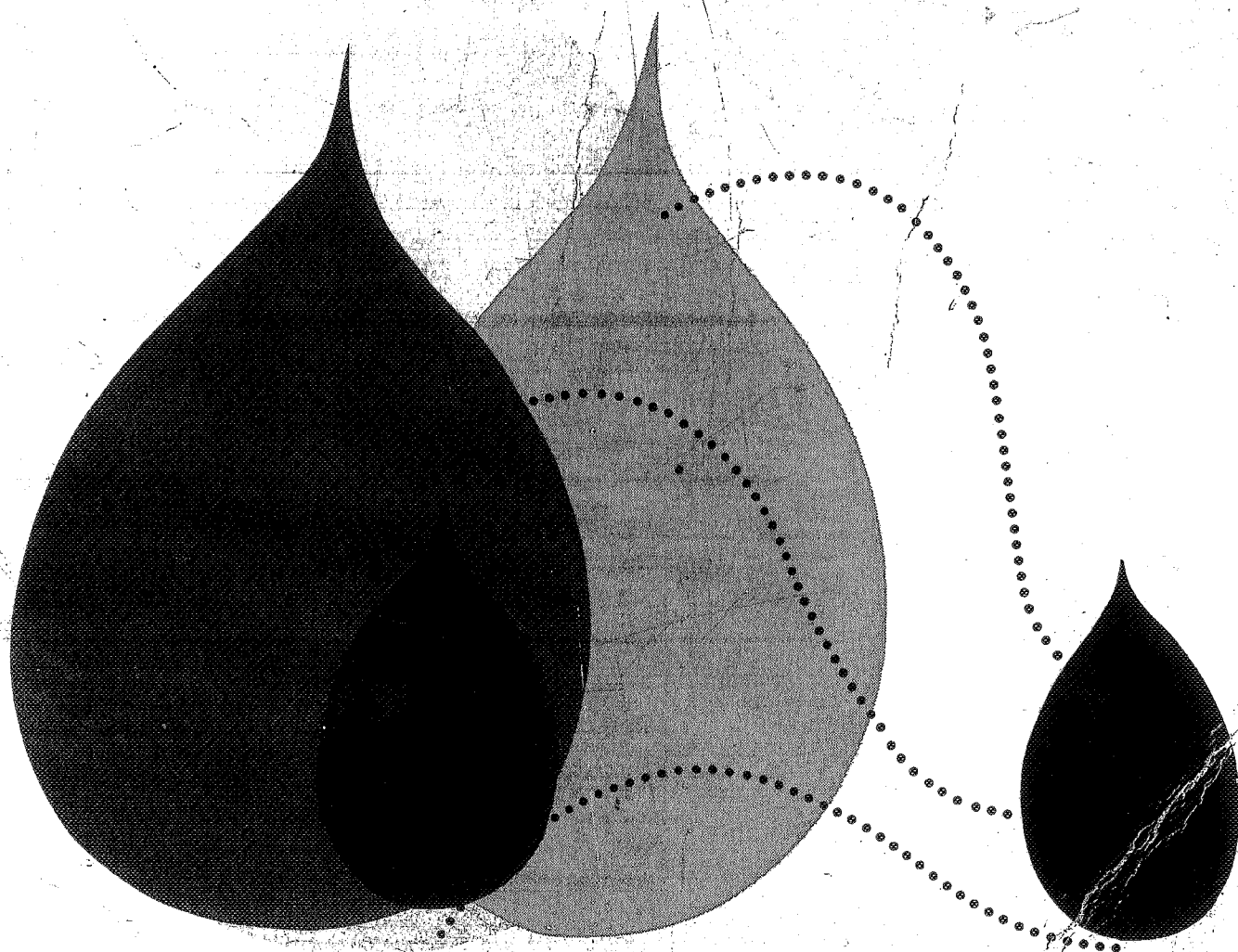


PROCESS HEAT TRANSFER

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CHAPTER 1

PROCESS HEAT TRANSFER

Heat Transfer. The science of thermodynamics deals with the quantitative transitions and rearrangements of energy as heat in bodies of matter. *Heat transfer* is the science which deals with the rates of exchange of heat between hot and cold bodies called the *source* and *receiver*. When a pound of water is vaporized or condensed, the energy change in either process is identical. The rates at which either process can be made to progress with an independent source or receiver, however, are inherently very different. Vaporization is generally a much more rapid phenomenon than condensation.

Heat Theories. The study of heat transfer would be greatly enhanced by a sound understanding of the nature of heat. Yet this is an advantage which is not readily available to students of heat transfer or thermodynamics because so many manifestations of heat have been discovered that no simple theory covers them all. Laws which may apply to mass transitions may be inapplicable to molecular or atomic transitions, and those which are applicable at low temperatures may not apply at high temperatures. For the purposes of engineering it is necessary to undertake the study with basic information on but a few of the many phenomena. The phases of a single substance, solid, liquid, and gaseous, are associated with its energy content. In the solid phase the molecules or atoms are close together, giving it rigidity. In the liquid phase sufficient thermal energy is present to extend the distance of adjacent molecules such that rigidity is lost. In the gas phase the presence of additional thermal energy has resulted in a relatively complete separation of the atoms or molecules so that they may wander anywhere in a confined space. It is also recognized that, whenever a change of phase occurs outside the critical region, a large amount of energy is involved in the transition.

For the same substance in its different phases the various thermal properties have different orders of magnitude. As an example, the specific heat per unit mass is very low for solids, high for liquids, and usually intermediate for gases. Similarly in any body absorbing or losing heat, special consideration must be given whether the change is one of sensible or latent heat or both. Still further, it is also known that a hot source is

capable of such great subatomic excitement that it emits energy without any direct contact with the receiver, and this is the underlying principle of radiation. Each type of change exhibits its own peculiarities.

Mechanisms of Heat Transfer. There are three distinct ways in which heat may pass from a source to a receiver, although most engineering applications are combinations of two or three. These are *conduction*, *convection*, and *radiation*.

Conduction. Conduction is the transfer of heat through fixed material such as the stationary wall shown in Fig. 1.1. The direction of heat

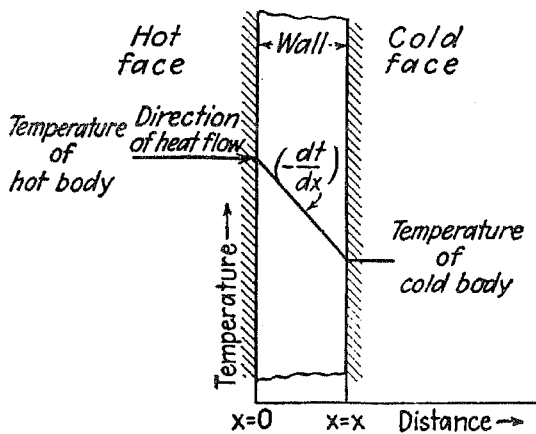


FIG. 1.1. Heat flow through a wall.

flow will be at right angles to the wall if the wall surfaces are isothermal and the body homogeneous and isotropic. Assume that a source of heat exists on the left face of the wall and a receiver of heat exists on the right face. It has been known and later it will be confirmed by derivation that the flow of heat per hour is proportional to the change in temperature through the wall and the area of the wall A . If t is the temperature at any point in the wall

and x is the thickness of the wall in the direction of heat flow, the quantity of heat flow dQ is given by

$$dQ = kA \left(- \frac{dt}{dx} \right) \quad \text{Btu/hr} \quad (1.1)$$

The term $-dt/dx$ is called the *temperature gradient* and has a negative sign if the temperature has been assumed higher at the face of the wall where $x = 0$ and lower at the face where $x = X$. In other words, the instantaneous quantity of heat transfer is proportional to the area and temperature difference dt , which drives the heat through the wall of thickness dx . The proportionality constant k is peculiar to conductive heat transfer and is known as the *thermal conductivity*. It is evaluated experimentally and is basically defined by Eq. (1.1). The thermal conductivities of solids have a wide range of numerical values depending upon whether the solid is a relatively good conductor of heat such as a metal or a poor conductor such as asbestos. The latter serve as *insulators*. Although heat conduction is usually associated with heat transfer through solids, it is also applicable with limitations to gases and liquids.

Convection. Convection is the transfer of heat between relatively hot and cold portions of a fluid by mixing. Suppose a can of liquid were

placed over a hot flame. The liquid at the bottom of the can becomes heated and less dense than before owing to its thermal expansion. The liquid adjacent to the bottom is also less dense than the cold upper portion and rises through it, transferring its heat by mixing as it rises. The transfer of heat from the hot liquid at the bottom of the can to the remainder is *natural* or *free convection*. If any other agitation occurs, such as that produced by a stirrer, it is *forced convection*. This type of heat transfer may be described in an equation which imitates the form of the conduction equation and is given by

$$dQ = hA \, dt \quad (1.2)$$

The proportionality constant h is a term which is influenced by the nature of the fluid and the nature of the agitation and must be evaluated experimentally. It is called the *heat-transfer coefficient*. When Eq. (1.2) is written in integrated form, $Q = hA \, \Delta t$, it is called Newton's law of cooling.

Radiation. Radiation involves the transfer of radiant energy from a source to a receiver. When radiation issues from a source to a receiver, part of the energy is absorbed by the receiver and part reflected by it. Based on the second law of thermodynamics Boltzmann established that the rate at which a source gives off heat is

$$dQ = \sigma \epsilon \, dA \, T^4 \quad (1.3)$$

This is known as the fourth-power law in which T is the absolute temperature. σ is a dimensional constant, but ϵ is a factor peculiar to radiation and is called the *emissivity*. The emissivity, like the thermal conductivity k or the heat-transfer coefficient h , must also be determined experimentally.

Process Heat Transfer. *Heat transfer* has been described as the study of the rates at which heat is exchanged between heat sources and receivers usually treated independently. *Process heat transfer* deals with the rates of heat exchange as they occur in the heat-transfer equipment of the engineering and chemical processes. This approach brings to better focus the importance of the temperature difference between the source and receiver, which is, after all, the driving force whereby the transfer of heat is accomplished. A typical problem of process heat transfer is concerned with the quantities of heats to be transferred, the rates at which they may be transferred because of the natures of the bodies, the driving potential, the extent and arrangement of the surface separating the source and receiver, and the amount of mechanical energy which may be expended to facilitate the transfer. Since heat transfer involves an *exchange* in a *system*, the loss of heat by the one body will equal the heat absorbed by another *within the confines of the same system*.

In the chapters which follow studies will first be made of the three individual heat-transfer phenomena and later of the way in which their combination with a simultaneous source and receiver influences an apparatus as a whole. A large number of the examples which follow have been selected from closely related processes to permit gradual comparisons. This should not be construed as limiting the broadness of the underlying principles.

Many of the illustrations and problems in the succeeding chapters refer to liquids derived from petroleum. This is quite reasonable, since petroleum refining is a major industry, petroleum products are an important fuel for the power industry, and petroleum derivatives are the starting point for many syntheses in the chemical industry.

Petroleum is a mixture of a great many chemical compounds. Some can be isolated rather readily, and the names of common hydrocarbons present in petroleum may be identified on Fig. 7 in the Appendix. But more frequently there is no need to obtain pure compounds, since the ultimate use of a mixture of related compounds will serve as well. Thus lubricating oil is a mixture of several compounds of high molecular weight, all of which are suitable lubricants. Similarly, gasoline which will ultimately be burned will be composed of a number of volatile combustible compounds. Both of these common petroleum products were present in the crude oil when it came from the ground or were formed by subsequent reaction and separated by distillation. When dealt with in a process or marketed as mixtures, these products are called *fractions* or *cuts*. They are given common names or denote the refinery operation by which they were produced, and their specific gravities are defined by a scale established by the American Petroleum Institute and termed either degrees API or °API. The °API is related to the specific gravity by

$$^{\circ}\text{API} = \frac{141.5}{\text{sp gr at } 60^{\circ}\text{F}/60^{\circ}\text{F}} - 131.5 \quad (1.4)$$

Being mixtures of compounds the petroleum fractions do not boil isothermally like pure liquids but have boiling ranges. At atmospheric pressure the lowest temperature at which a liquid starts to boil is identified as the *initial boiling point*, IBP, °F. A list of the common petroleum fractions derived from crude oil is given below:

Fractions from crude oil	Approx °API	Approx IBP, °F
Light ends and gases.....	114	
Gasoline.....	75	200
Naphtha.....	60	300
Kerosene.....	45	350
Absorption oil.....	40	450
Straw oil.....	40	500
Distillate.....	35	550
Gas oil.....	28	600
Lube oil.....	18-30	
Reduced crude.....		
Paraffin wax and jelly.....		
Fuel oil (residue).....	25-35	500
Asphalt.....		

A method of defining the chemical character of petroleum and correlating the properties of mixtures was introduced by Watson, Nelson, and Murphy.¹ They observed that, when a crude oil of uniform distilling behavior is distilled into narrow cuts, the ratio of the cube root of the absolute average boiling points to the specific gravities of the cuts is a constant or

$$K = \frac{T_B^{1/3}}{s} \quad (1.5)$$

where K = characterization factor

T_B = average boiling point, °R

s = specific gravity at 60°/60°

NOMENCLATURE FOR CHAPTER 1

A	Heat-transfer surface, ft ²
h	Individual heat-transfer coefficient, Btu/(hr)(ft ²)(°F)
K	Characterization factor
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
Q	Heat flow, Btu/hr
s	Specific gravity, dimensionless
T	Temperature, °R
T_B	Absolute average boiling temperature, °R
t	Temperature in general, °F
x, X	Distance, ft
σ	A constant, Btu/(hr)(ft ²)(°R ⁴)
ϵ	Emissivity, dimensionless

¹ Watson, K. M., E. F. Nelson, and G. B. Murphy, *Ind. Eng. Chem.*, **25**, 880 (1933) 27, 1460 (1935).

CHAPTER 2

CONDUCTION

The Thermal Conductivity. The fundamentals of heat conduction were established over a century ago and are generally attributed to Fourier. In numerous systems involving flow such as heat flow, fluid flow, or electricity flow, it has been observed that the flow quantity is directly proportional to a driving potential and inversely proportional to the resistances applying to the system, or

$$\text{Flow} \propto \frac{\text{potential}}{\text{resistance}} \quad (2.1)$$

In a simple hydraulic path the pressure along the path is the driving potential and the roughness of the pipe is the flow resistance. In an electric circuit the simplest applications are expressed by Ohm's law: The voltage on the circuit is the driving potential, and the difficulty with which electrons negotiate the wire is the resistance. In heat flow through a wall, flow is effected by a temperature difference between the hot and cold faces. Conversely, from Eq. (2.1) when the two faces of a wall are at different temperatures, a flow of heat and a resistance to heat flow are necessarily present. The *conductance* is the reciprocal of the resistance to heat flow and Eq. (2.1) may be expressed by

$$\text{Flow} \propto \text{conductance} \times \text{potential} \quad (2.2)$$

To make Eq. (2.2) an equality the conductance must be evaluated in such a way that both sides will be dimensionally and numerically correct. Suppose a measured quantity of heat Q' Btu has been transmitted by a wall of unknown size in a measured time interval θ hr with a measured temperature difference Δt °F. Rewriting Eq. (2.2)

$$Q = \frac{Q'}{\theta} = \text{conductance} \times \Delta t \quad \text{Btu/hr} \quad (2.3)$$

and the conductance has the dimensions of Btu/(hr)(°F). The conductance is a measured property of the entire wall, although it has also been found experimentally that the flow of heat is influenced independently by the thickness and the area of the wall. If it is desired to design a wall to have certain heat-flow characteristics, the conductance obtained above

is not useful, being applicable only to the experimental wall. To enable a broader use of experimental information, it has become conventional to report the conductance only when all the dimensions are referred to unit values. When the conductance is reported for a quantity of material 1 ft thick with heat-flow area 1 ft², time unit 1 hr, and temperature difference 1°F, it is called the *thermal conductivity* k . The relationship between the thermal conductivity and the conductance of an entire wall of thickness L and area A is then given by

$$\text{Conductance} = k \frac{A}{L}$$

and

$$Q = k \frac{A}{L} \Delta t \quad (2.4)$$

where k has the dimensions resulting from the expression $QL/A \Delta t$ or Btu/(hr)(ft² of flow area)(°F of temperature difference)/(ft of wall thickness).¹

Experimental Determination of k : * **Nonmetal Solids.** An apparatus for the determination of the thermal conductivity of nonmetal solids is shown in Fig. 2.1. It consists of an electrical heating plate, two identical test specimens through which heat

passes, and two water jackets which remove heat. The temperatures at both faces of the specimens and at their sides are measured by thermocouples. A guard ring is provided to assure that all the measured heat input to the plate passes through the specimens with a negligible loss from their sides. The guard ring surrounds the test assembly and consists of an auxiliary heater sandwiched between pieces of the material being tested. While current enters the heating plate, the input to the

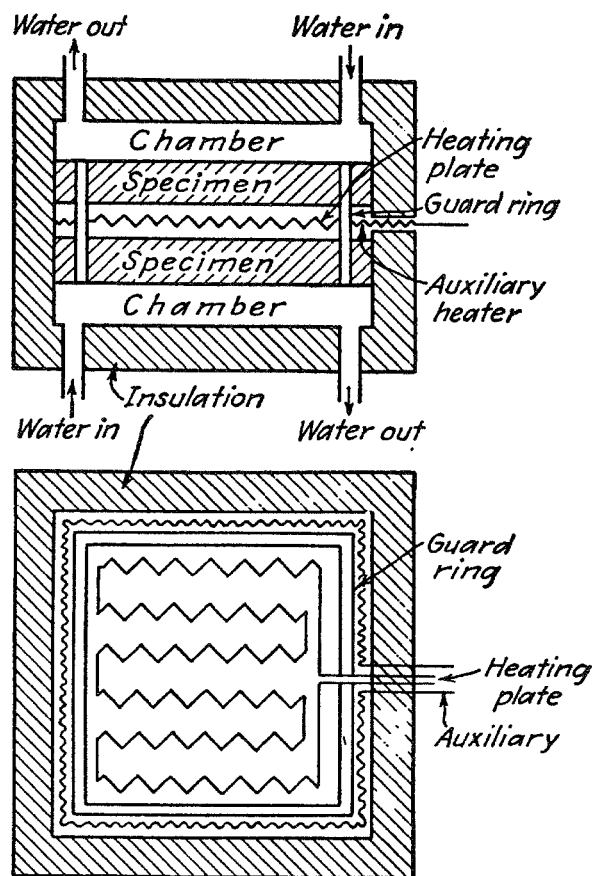


FIG. 2.1. Guarded conductivity apparatus.

¹ In the metric system it is usual to report the thermal conductivity as cal/(sec)-(cm²)(°C/cm).

* An excellent review of experimental methods will be found in Saha and Srivastava, "Treatise on Heat," The Indian Press, Calcutta, 1935. Later references are Bates, O. K., *Ind. Eng. Chem.*, **25**, 432 (1933); **28**, 494 (1936); **33**, 375 (1941); **37**, 195 (1945). Bolland, J. L. and H. W. Melville, *Trans. Faraday Soc.*, **33**, 1316 (1937). Hutchinson, E., *Trans. Faraday Soc.*, **41**, 87 (1945).

auxiliary heater is adjusted until no temperature differences exist between the specimens and adjacent points in the guard ring. Observations are made when the heat input and the temperatures on both faces of each specimen remain steady. Since half of the measured electrical heat input to the plate flows through each specimen and the temperature difference and dimensions of the specimen are known, k can be computed directly from Eq. (2.4).

Liquids and Gases. There is greater difficulty in determining the conductivities of liquids and gases. If the heat flows through a thick

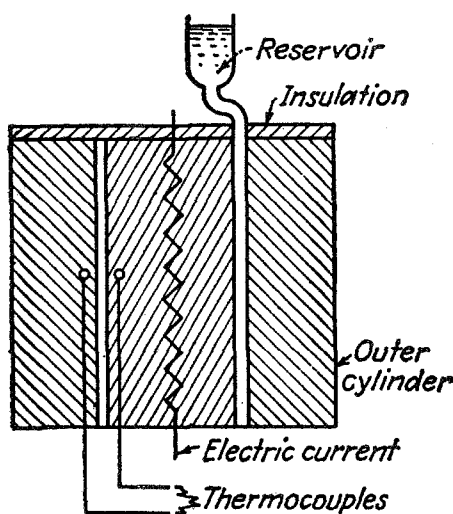


FIG. 2.2. Liquid conductivity apparatus. (After J. F. D. Smith.)

layer of liquid or gas, it causes free convection and the conductivity is deceptively high. To reduce convection it is necessary to use very thin films and small temperature differences with attendant errors of measurement. A method applicable to viscous fluids consists of a bare electric wire passing through a horizontal tube filled with test liquid. The tube is immersed in a constant-temperature bath. The resistance of the wire is calibrated against its temperature. For a given rate of heat input and for the temperature of the wire obtained from resistance measurements the conductivity can be calculated

by suitable equations. A more exact method, however, is that of Bridgman and Smith,¹ consisting of a very thin fluid annulus between two copper cylinders immersed in a constant-temperature bath as shown in Fig. 2.2. Heat supplied to the inner cylinder by a resistance wire flows through the film to the outer cylinder, where it is removed by the bath. This apparatus, through the use of a reservoir, assures that the annulus is full of liquid and is adaptable to gases. The film is $\frac{1}{64}$ in. thick, and the temperature difference is kept very small.

Influence of Temperature and Pressure on k . The thermal conductivities of solids are greater than those of liquids, which in turn are greater than those of gases. It is easier to transmit heat through a solid than a liquid and through a liquid than a gas. Some solids, such as metals, have high thermal conductivities and are called *conductors*. Others have low conductivities and are poor conductors of heat. These are *insulators*. In experimental determinations of the type described above the thermal conductivity has been assumed independent of the temperature at any point in the test material. The reported values of k are consequently

¹ Smith, J. F. D., *Ind. Eng. Chem.*, **22**, 1246 (1930); *Trans. ASME*, **58**, 719 (1936).

the averages for the entire specimen, and the error introduced by this assumption can be estimated by an examination of Tables 2 to 5 in the Appendix. The conductivities of solids may either increase or decrease with temperature and in some instances may even reverse their rate of change from a decrease to an increase. For the most practical problems there is no need to introduce a correction for the variation of the thermal conductivity with temperature. However, the variation can usually be expressed by the simple linear equation

$$k = k_0 + \gamma t$$

where k_0 is the conductivity at 0°F and γ is a constant denoting the change in the conductivity per degree change in temperature. The conductivities of most liquids decrease with increasing temperature, although water is a notable exception. For all the common gases and vapors there is an increase with increasing temperature. Sutherland¹ deduced an equation from the kinetic theory which is applicable to the variation of the conductivity of gases with temperature

$$k = k_{32} \frac{492 + C_k}{T + C_k} \left(\frac{T}{492} \right)^{3/2}$$

where C_k = Sutherland constant

T = absolute temperature of the gas, °R

k_{32} = conductivity of the gas at 32°F

The influence of pressure on the conductivities of solids and liquids appears to be negligible, and the reported data on gases are too inexact owing to the effects of free convection and radiation to permit generalization. From the kinetic theory of gases it can be concluded that the influence of pressure should be small except where a very low vacuum is encountered.

Contact Resistance. One of the factors which causes error in the determination of the thermal conductivity is the nature of the bond formed between the heat source and the fluid or solid specimen which contacts it and transmits heat. If a solid receives heat by contacting a solid, it is almost impossible to exclude the presence of air or other fluid from the contact. Even when a liquid contacts a metal, the presence of minute pits or surface roughness may permanently trap infinitesimal bubbles of air, and it will be seen presently that these may cause considerable error.

Derivation of a General Conduction Equation. In Eqs. (2.1) to (2.4) a picture of heat conduction was obtained from an unqualified observation of the relation between heat flow, potential, and resistance. It is now feasible to develop an equation which will have the broadest applicability

¹ Sutherland, W., *Phil. Mag.*, **36**, 507 (1893).

and from which other equations may be deduced for special applications. Equation (2.4) may be written in differential form

$$\frac{dQ'}{d\theta} = k dA \frac{dt}{dx} \quad (2.5)$$

In this statement k is the only property of the matter and it is assumed to be independent of the other variables. Referring to Fig. 2.3, an elemental cube of volume $dv = dx dy dz$ receives a differential quantity of heat dQ'_1 Btu through its left yz face in the time interval $d\theta$. Assume all

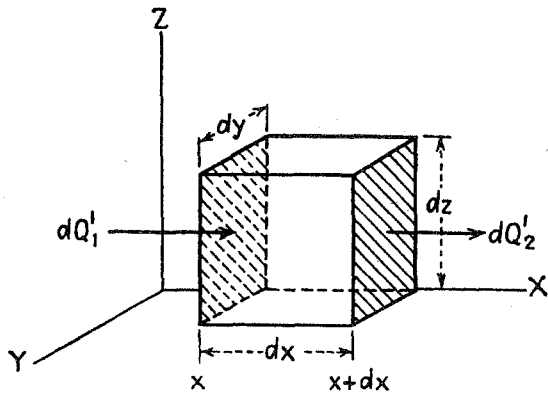


FIG. 2.3. Unidirectional heat flow.

but the left and right yz faces are insulated. In the same interval the quantity of heat dQ'_2 leaves at the right face. It is apparent that any of three effects may occur: dQ'_1 may be greater than dQ'_2 so that the elemental volume stores heat, increasing the *average* temperature of the cube; dQ'_2 may be greater than dQ'_1 so that the cube loses heat; and lastly, dQ'_1 and dQ'_2 may be equal so that the heat will simply

pass through the cube without affecting the storage of heat. Taking either of the first two cases as being more general, a storage or depletion term dQ' can be defined as the difference between the heat entering and the heat leaving or

$$dQ' = dQ'_1 - dQ'_2 \quad (2.6)$$

According to Eq. (2.5) the heat entering on the left face may be given by

$$\frac{dQ'_1}{d\theta} = k dy dz \left(- \frac{\partial t}{\partial x} \right) \quad (2.7)$$

The temperature gradient $-\frac{\partial t}{\partial x}$ may vary with both time and position in the cube. The variation of $-\frac{\partial t}{\partial x}$ as $f(x)$ only is $-\frac{\partial(\partial t/\partial x)}{\partial x}$. Over the distance dx from x to $x + dx$, if $dQ'_2 > dQ'_1$, the total change in the temperature gradient will be $-\frac{\partial(\partial t/\partial x)}{\partial x} dx$ or $-\frac{\partial^2 t}{\partial x^2} dx$. Then at x the gradient is $-\frac{\partial t}{\partial x}$, and at $x + dx$ the temperature gradient is

$$-\frac{\partial t}{\partial x} - \frac{\partial^2 t}{\partial x^2} dx$$

dQ'_2 out of the cube at the right face and in the same form as Eq. (2.7) is given by

$$\frac{dQ'_2}{d\theta} = k \, dy \, dz \left(-\frac{\partial t}{\partial x} - \frac{\partial^2 t}{\partial x^2} dx \right) \quad (2.8)$$

from which

$$\frac{dQ'}{d\theta} = \frac{dQ'_1}{d\theta} - \frac{dQ'_2}{d\theta} = k \, dy \, dz \left(\frac{\partial^2 t}{\partial x^2} \right) dx \quad (2.9)$$

The cube will have changed in temperature by $-dt$ deg. The change in temperature per unit time will be $dt/d\theta$ and over the time interval $d\theta$ it is given by $(dt/d\theta) d\theta$ deg. Since the analysis has been based on an elemental *volume*, it is now necessary to define a volumetric specific heat, c_v Btu/(ft³)(°F) obtained by multiplying the weight specific heat c Btu/(lb)(°F) by the density ρ . To raise the volume $dx \, dy \, dz$ by

$$\frac{dt}{d\theta} d\theta \text{ } ^\circ\text{F}$$

requires a heat change in the cube of

$$\frac{dQ'}{d\theta} = c\rho \, dx \, dy \, dz \frac{\partial t}{\partial \theta} \quad (2.10)$$

and combining Eqs. (2.9) and (2.10)

$$c\rho \, dx \, dy \, dz \frac{\partial t}{\partial \theta} = k \, dy \, dz \left(\frac{\partial^2 t}{\partial x^2} \right) dx \quad (2.11)$$

from which

$$\frac{\partial t}{\partial \theta} = \frac{k}{c\rho} \left(\frac{\partial^2 t}{\partial x^2} \right) \quad (2.12)$$

which is *Fourier's general equation*, and the term $k/c\rho$ is called the thermal diffusivity, since it contains all the properties involved in the conduction of heat and has the dimensions of ft²/hr. If the insulation is removed from the cube so that the heat travels along the X , Y , and Z axes, Eq. (2.12) becomes

$$\frac{\partial t}{\partial \theta} = \frac{k}{c\rho} \left(\frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2} \right) \quad (2.13)$$

When the flow of heat into and out of the cube is constant as in the *steady state*, t does not vary with time, and $dt/d\theta = 0$, in Eq. (2.12). $\partial t/\partial x$ is a constant and $\partial^2 t/\partial x^2 = 0$. $dQ'_1 = dQ'_2$, and Eq. (2.8) reduces to Eq. (2.5) where $dx \, dy = dA$. Substituting dQ for $dQ'/d\theta$, both terms having

dimensions of Btu/hr, the *steady-state* equation is

$$dQ = k dA \frac{dt}{dx} \quad (2.14)$$

Equation (2.14) applies to many of the common engineering problems.

Thermal Conductivity from Electrical-conductivity Measurements.

The relationship between the thermal and electrical conductivities of metals demonstrates an application of Fourier's derivation incorporated

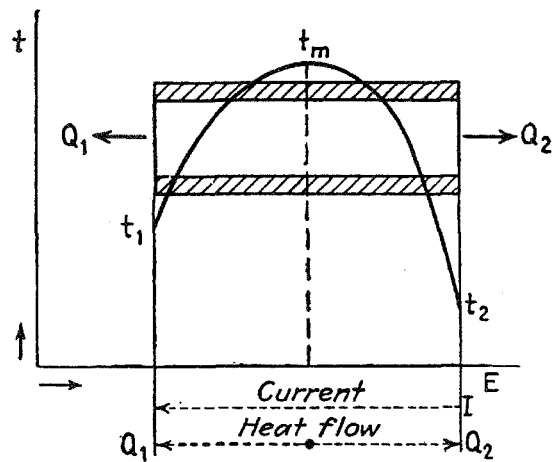


FIG. 2.4. Heat flow in a metal.

in Eq. (2.9) and is a useful method of determining the thermal conductivities of metals. An insulated bar of metal as shown in Fig. 2.4 has its left and right cross-sectional faces exposed to different constant-temperature baths at t_1 and t_2 , respectively. By fastening electric leads to the left and right faces, respectively, a current of I amp may be passed in the direction indicated, generating heat throughout the length of the bar. The quantities

of heat leaving both ends of the bar in the steady state must be equal to the amount of heat received as electrical energy, $I^2 R^\omega$, where R^ω is the resistance in ohms. From Ohm's law

$$I = \frac{E_1 - E_2}{\sigma(L/A)} = \frac{A}{\sigma} \frac{dE}{dx}$$

where $E_1 - E_2$ is the voltage difference, σ the resistivity of the wire in ohm-ft and K , the reciprocal of the resistivity, is the electrical conductivity.

$$I = KA \frac{dE}{dx} \quad (2.15)$$

$$R^\omega = \frac{\sigma dx}{A} = \frac{dx}{KA} \quad (2.16)$$

Substituting Eqs. (2.15) and (2.16) for $I^2 R^\omega$,

$$dQ = I^2 R^\omega = K^2 A^2 \left(\frac{dE}{dx} \right)^2 \frac{dx}{KA} = KA \left(\frac{dE}{dx} \right)^2 dx \quad (2.17)$$

But this is the same as the heat transferred by conduction and given by

Eq. (2.9). When $t_1 = t_2$ and equating (2.9) and (2.17),

$$k \frac{d^2 t}{dx^2} dx - K \left(\frac{dE}{dx} \right)^2 dx = 0 \quad (2.18)$$

But

$$\frac{dt}{dx} = \frac{dt}{dE} \frac{dE}{dx} \quad (2.19)$$

Differentiating,

$$\frac{d^2 t}{dx^2} = \left(\frac{dE}{dx} \right)^2 \frac{d^2 t}{dE^2} + \frac{dt}{dE} \left(\frac{d^2 E}{dx^2} \right) \quad (2.20)$$

If I and A are constant for the bar, then $K(dE/dx)$ is constant. Since K does not vary greatly with t or x , dE/dx is constant, $d^2 E/dx^2 = 0$, and from Eq. (2.18) substituting Eq. (2.20) for $d^2 t/dx^2$

$$k \frac{d^2 t}{dE^2} - K = 0 \quad (2.21)$$

$$\frac{d^2 t}{dE^2} = \frac{K}{k} \quad (2.22)$$

$$\frac{kt}{K} = \frac{1}{2} E^2 + C_1 E + C_2 \quad (2.23)$$

where C_1 and C_2 are integration constants. Since there are three constants in Eq. (2.23), C_1 , C_2 , and k/K , three voltages and three temperatures must be measured along the bar for their evaluation. C_1 and C_2 are determined from the end temperatures and k is obtained from k/K using the more simply determined value of the electrical conductivity K .

Flow of Heat through a Wall. Equation (2.14) was obtained from the general equation when the heat flow and temperatures into and out of the two opposite faces of the partially insulated elemental cube $dx dy dz$ were constant. Upon integration of Eq. (2.14) when all of the variables but Q are independent the steady-state equation is

$$Q = \frac{kA}{L} \Delta t \quad (2.24)$$

Given the temperatures existing on the hot and cold faces of a wall, respectively, the heat flow can be computed through the use of this equation. Since kA/L is the conductance, its reciprocal R is the resistance to heat flow, or $R = L/kA$ (hr)(°F)/Btu.

Example 2.1. Flow of Heat through a Wall. The faces of a 6-in. thick wall measuring 12 by 16 ft will be maintained at 1500 and 300°F, respectively. The wall is made of kaolin insulating brick. How much heat will escape through the wall?

Solution. The average temperature of the wall will be 900°F. From Table 2 in the

Appendix the thermal conductivity at 932°F is 0.15 Btu/(hr)(ft²)(°F/ft). Extrapolation to 900°F will not change this value appreciably.

$$Q = \frac{kA}{L} \Delta t$$

Where $\Delta t = 1500 - 300 = 1200^\circ\text{F}$

$A = 16 \times 12 = 192 \text{ ft}^2$

$L = \frac{1}{2} = 0.5 \text{ ft}$

$$Q = 0.15 \times \frac{192}{0.5} \times 1200 = 69,200 \text{ Btu/hr}$$

Flow of Heat through a Composite Wall: Resistances in Series. Equation (2.24) is of interest when a wall consists of several materials placed together in series such as in the construction of a furnace or boiler firebox. Several types of refractory brick are usually employed,

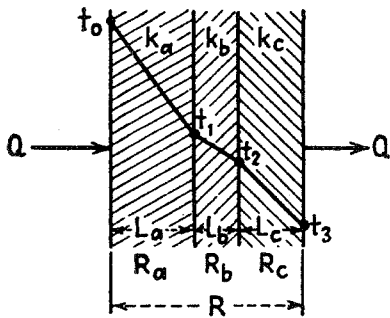


FIG. 2.5. Heat flow through a composite wall.

since those capable of withstanding the higher inside temperatures are more fragile and expensive than those required near the outer surface, where the temperatures are considerably lower. Referring to Fig. 2.5, three different refractory materials are placed together indicated by the subscripts *a*, *b*, and *c*. For the entire wall

$$Q = \frac{\Delta t}{R} \quad (2.25)$$

The heat flow in Btu per hour through material *a* must overcome the resistance R_a . But in passing through material *a* the heat must also pass through materials *b* and *c* in series. The heat flow entering at the left face must be equal to the heat flow leaving the right face, since the steady state precludes heat storage. If R_a , R_b , and R_c are unequal, as the result of differing conductivities and thicknesses, the ratio of the temperature difference across each layer to its resistance must be the same as the ratio of the total temperature difference is to the total resistance or

$$Q = \frac{\Delta t}{R} = \frac{\Delta t_a}{R_a} = \frac{\Delta t_b}{R_b} = \frac{\Delta t_c}{R_c} \quad (2.26)$$

For any composite system using actual temperatures

$$Q = \frac{\Delta t}{R} = \frac{t_0 - t_1}{R_a} = \frac{t_1 - t_2}{R_b} = \frac{t_2 - t_3}{R_c} \quad (2.27)$$

Rearranging and substituting,

$$Q = \frac{\Delta t}{R} = \frac{t_0 - t_3}{(L_a/k_a A) + (L_b/k_b A) + (L_c/k_c A)} \quad (2.28)$$

Example 2.2. Flow of Heat through a Composite Wall. The wall of an oven consists of three layers of brick. The inside is built of 8 in. of firebrick, $k = 0.68$ Btu/(hr)(ft²)(°F/ft), surrounded by 4 in. of insulating brick, $k = 0.15$, and an outside layer of 6 in. of building brick, $k = 0.40$. The oven operates at 1600°F and it is anticipated that the outer side of the wall can be maintained at 125°F by the circulation of air. How much heat will be lost per square foot of surface and what are the temperatures at the interfaces of the layers?

Solution:

For the firebrick, $R_a = L_a/k_a A = 8/12 \times 0.68 \times 1 = 0.98$ (hr)(°F)/(Btu)

Insulating brick, $R_b = L_b/k_b A = 4/12 \times 0.15 \times 1 = 2.22$

Building brick, $R_c = L_c/k_c A = 6/12 \times 0.40 \times 1 = 1.25$

$$R = 4.45$$

Heat loss/ft² of wall, $Q = \Delta t/R = (1600 - 125)/4.45 = 332$ Btu/hr

For the individual layers:

$$\Delta t = QR \quad \text{and} \quad \Delta t_a = QR_a, \text{ etc.}$$

$$\Delta t_a = 332 \times 0.98 = 325^\circ\text{F} \quad t_1 = 1600 - 325 = 1275^\circ\text{F}$$

$$\Delta t_b = 332 \times 2.22 = 738^\circ\text{F} \quad t_2 = 1275 - 738 = 537^\circ\text{F}$$

Example 2.3. Flow of Heat through a Composite Wall with an Air Gap. To illustrate the poor conductivity of a gas, suppose an air gap of $\frac{1}{4}$ in. were left between the insulating brick and the firebrick. How much heat would be lost through the wall if the inside and outside temperatures are kept constant?

Solution. From Table 5 in the Appendix at 572°F air has a conductivity of 0.0265 Btu/(hr)(ft²)(°F/ft), and this temperature is close to the range of the problem.

$$R_{\text{air}} = 0.25/12 \times 0.0265 = 0.79 \text{ (hr)(°F)/Btu}$$

$$R = 4.45 + 0.79 = 5.24$$

$$Q = \frac{1600 - 125}{5.24} = 281 \text{ Btu/hr}$$

It is seen that in a wall 18 in. thick a stagnant air gap only $\frac{1}{4}$ in. thick reduces the heat loss by 15 per cent.

Heat Flow through a Pipe Wall. In the passage of heat through a flat wall the area through which the heat flows is constant throughout the entire distance of the heat flow path. Referring to Fig. 2.6 showing a unit length of pipe, the area of the heat flow path through the pipe wall increases with the distance of the path from r_1 to r_2 . The area at any radius r is given by $2\pi r l$ and if the heat flows out of the cylinder the temperature gradient for the

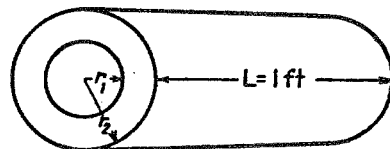


FIG. 2.6. Heat flow through a pipe wall.

incremental length dr is dt/dr . Equation (2.14) becomes

$$q = 2\pi rk \left(-\frac{dt}{dr} \right) \quad \text{Btu/(hr)(lin ft)} \quad (2.29)$$

Integrating,

$$t = -\frac{q}{2\pi k} \ln r + C_1 \quad (2.30)$$

When $r = r_i$, $t = t_i$; and when $r = r_o$, $t = t_o$; where i and o refer to the inside and outside surfaces, respectively. Then

$$q = \frac{2\pi k(t_i - t_o)}{2.3 \log r_o/r_i} \quad (2.31)$$

and if D is the diameter,

$$\frac{r_o}{r_i} = \frac{D_o}{D_i}$$

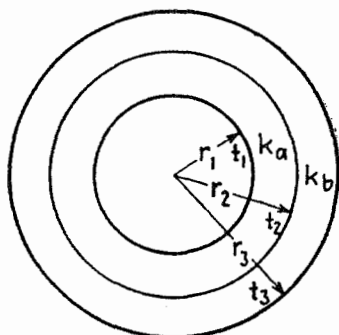


FIG. 2.7. Cylindrical resistances in series.

Referring to Fig. 2.7 where there is a composite cylindrical resistance,

$$t_1 = t_2 + \frac{2.3q}{2\pi k_a} \log \frac{D_2}{D_1} \quad (2.32)$$

$$t_2 = t_3 + \frac{2.3q}{2\pi k_b} \log \frac{D_3}{D_2} \quad (2.33)$$

Adding,

$$t_1 - t_3 = \frac{2.3q}{2\pi k_a} \log \frac{D_2}{D_1} + \frac{2.3q}{2\pi k_b} \log \frac{D_3}{D_2} \quad (2.34)$$

Example 2.4. Heat Flow through a Pipe Wall. A glass pipe has an outside diameter of 6.0 in. and an inside diameter of 5.0 in. It will be used to transport a fluid which maintains the inner surface at 200°F. It is expected that the outside of the pipe will be maintained at 175°F. What heat flow will occur?

Solution. $k = 0.63$ Btu/(hr)(ft²)(°F/ft) (see Appendix Table 2).

$$q = \frac{2\pi k(t_i - t_o)}{2.3 \log D_o/D_i} = \frac{2 \times 3.14 \times 0.63(200 - 175)}{2.3 \log 6.0/5.0} = 538 \text{ Btu/lin ft}$$

If the inside diameter of a cylinder is greater than 0.75 of the outside diameter, the mean of the two may be used. Then per foot of length

$$q = \frac{\Delta t}{R} = \frac{\Delta t}{L_a/k_a A_m} = \frac{t_1 - t_2}{\frac{(D_2 - D_1)/2}{\pi k_a (D_1 + D_2)/2}} \quad (2.35)$$

where $(D_2 - D_1)/2$ is the thickness of the pipe. Within the stated limitations of the ratio D_2/D_1 , Eq. (2.35) will differ from Eq. (2.34) by

about 1 per cent. Actually there are 1.57 ft² of external surface per linear foot and 1.31 ft² of internal surface. The heat loss per square foot is 343 Btu/hr based on the outside surface and 411 Btu/hr based on the inside surface.

Heat Loss from a Pipe. In the preceding examples it was assumed that the cold external surface could be maintained at a definite temperature. Without this assumption the examples would have been indeterminate, since both Q and Δt would be unknown and independent in a single equation. In reality the temperature assigned to the outer wall depends not only on the resistances between the hot and cold surfaces but also on the ability of the surrounding colder atmosphere to remove the heat arriving at the outer surface. Consider a

pipe as shown in Fig. 2.8 covered (lagged) with rock wool insulation and carrying steam at a temperature t_s , considerably above that of the atmosphere, t_a . The overall temperature difference driving heat out of the pipe is $t_s - t_a$. The resistances to heat flow taken

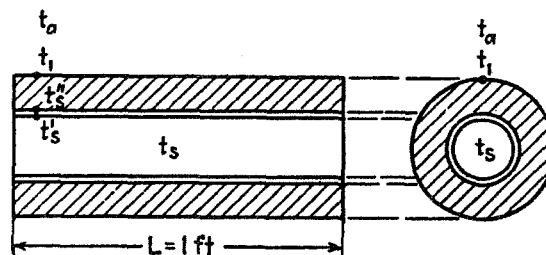


FIG. 2.8. Heat loss from an insulated pipe.

in order are (1) the resistance of the steam to condense upon and give up heat to the inner pipe surface, a resistance which has been found experimentally to be very small so that t_s and t'_s are nearly the same; (2) the resistance of the pipe metal, which is very small except for thick-walled conduits so that t'_s and t''_s are nearly the same; (3) the resistance of the rock wool insulation; and (4) the resistance of the surrounding air to remove heat from the outer surface. The last is appreciable, although the removal of heat is effected by the natural convection of ambient air in addition to the radiation caused by the temperature difference between the outer surface and colder air. The natural convection results from warming air adjacent to the pipe, thereby lowering its density. The warm air rises and is replaced continuously by cold air. The combined effects of natural convection and radiation cannot be represented by a conventional resistance term $R_a = L_a/K_a A$, since L_a is indefinite and the conductance of the air is simultaneously supplemented by the transfer of heat by radiation. Experimentally, a temperature difference may be created between a known outer surface and the air, and the heat passing from the outer surface to the air can be determined from measurements on the fluid flowing in the pipe. Having Q , A , and Δt , the combined resistance of both effects is obtained as the quotient of $\Delta t/Q$. The flow of heat from a pipe to ambient air is usually a heat loss, and it is therefore desirable to report the data as a *unit conductance* term k/L Btu/(hr)(ft² of

external surface)(°F of temperature difference). The unit conductance is the reciprocal of the *unit resistance* L/k instead of the reciprocal of the resistance for the entire surface L/kA . In other words, it is the conductance per square foot of heat-flow surface rather than the conductance of the total surface. The unit resistance has the dimensions

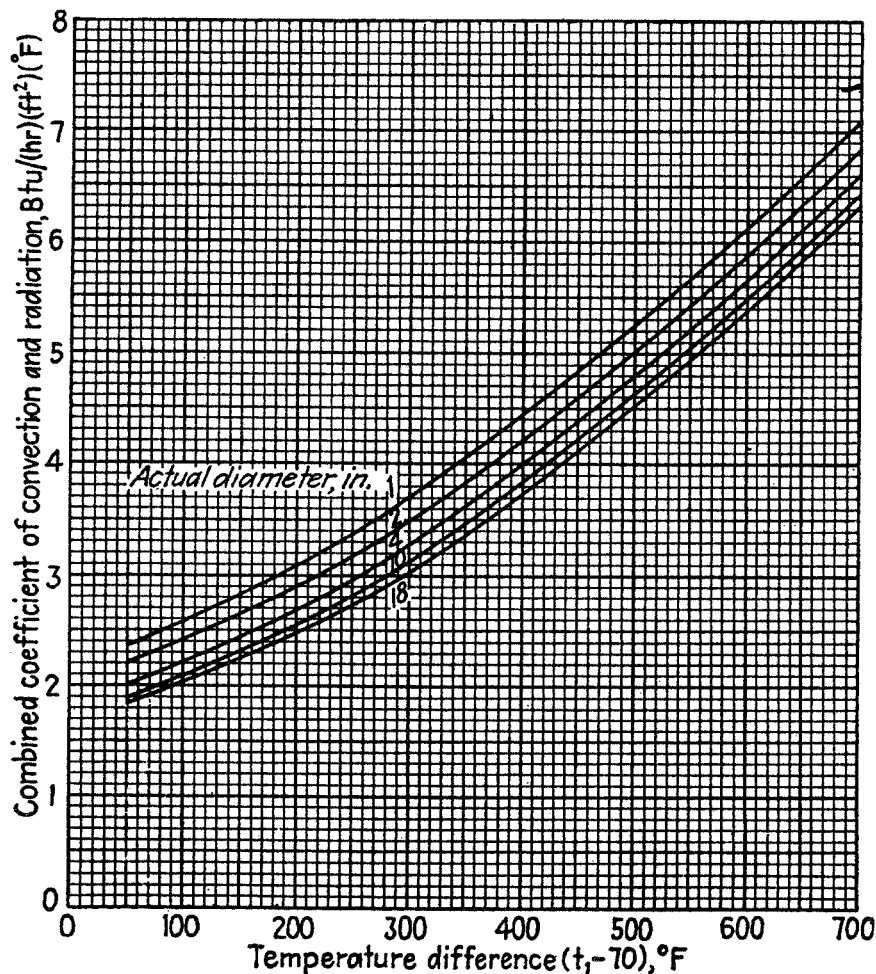


FIG. 2.9. Heat transfer by convection and radiation from horizontal pipes at temperature t_1 to air at 70°F.

(hr)(ft²)(°F)/Btu. The reciprocal of the unit resistance, h_a , has the dimensions of Btu/(hr)(ft²)(°F) and is sometimes designated the *surface coefficient* of heat transfer. Figure 2.9 shows the plot of the surface coefficient from pipes of different diameters and surface temperatures to ambient air at 70°F. It is based upon the data of Heilman,¹ which has been substantiated by the later experiments of Bailey and Lyell.² The four resistances in terms of the equations already discussed are

Condensation of steam:

$$q = h_s \pi D'_s (t_s - t'_s) \quad (1.2)$$

¹ Heilman, R. H., *Ind. Eng. Chem.*, **16**, 445-452 (1924).

² Bailey, A., and N. C. Lyell, *Engineering*, **147**, 60-62 (1939).

Pipe wall:

$$q = \frac{2\pi k_b}{2.3 \log D_s''/D_s} (t_s' - t_s'') \quad (2.31)$$

Insulation:

$$q = \frac{2\pi k_c}{2.3 \log D_1/D_s''} (t_s'' - t_1) \quad (2.31)$$

Radiation and convection to air:

$$q = h_a \pi D_1 (t_1 - t_a) \quad (1.2)$$

or combining

$$t_s - t_a = q \left(\frac{1}{h_s \pi D_s'} + \frac{2.3}{2\pi k_b} \log \frac{D_s''}{D_s'} + \frac{2.3}{2\pi k_c} \log \frac{D_1}{D_s''} + \frac{1}{h_a \pi D_1} \right)$$

The terms inside the parentheses are the four resistances, and of these the first two can usually be neglected. The equation then reduces to

$$q = \frac{\pi(t_s - t_a)}{\frac{2.3}{2k_c} \log \frac{D_1}{D_s''} + \frac{1}{h_a D_1}}$$

From the abscissa of Fig. 2.9 it is seen that h_a depends upon not only the temperature difference but the actual temperatures at the outside of the insulation and of the air. Its reciprocal is also one of the resistances necessary for the calculation of the total temperature difference, and therefore the surface coefficient h_a cannot be computed except by trial-and-error methods.

Example 2.5. Heat Loss from a Pipe to Air. A 2-in. steel pipe (dimensions in Table 11 in the Appendix) carries steam at 300°F. It is lagged with ½ in. of rock wool, $k = 0.033$, and the surrounding air is at 70°F. What will be the heat loss per linear foot?

Solution. Assume $t_1 = 150^\circ\text{F}$, $t_1 - 70 = 80^\circ\text{F}$, $h_a = 2.23 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$.

$$q = \frac{3.14(300 - 70)}{\frac{2.3}{2 \times 0.033} \log \frac{3.375}{2.375} + \frac{1}{2.23 \times 3.375/12}} = 104.8 \text{ Btu}/(\text{hr})(\text{lin ft})$$

Check between t_s and t_1 , since $\Delta t/R = \Delta t_c/R_c$.

$$q = 104.8 = \frac{2 \times 3.14 \times 0.033(300 - t_1)}{2.3 \log 3.375/2.375}$$

$$t_1 = 123.5^\circ\text{F} \quad \text{No check}$$

Assume $t_1 = 125^\circ\text{F}$, $t_1 - 70 = 55^\circ\text{F}$, $h_a = 2.10 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$.

$$q = \frac{3.14(300 - 70)}{\frac{2.3}{2 \times 0.033} \log \frac{3.375}{2.375} + \frac{1}{2.10 \times 3.375/12}} = 103.2 \text{ Btu}/(\text{hr})(\text{lin ft})$$

Check between t_s and t_1 .

$$q = 103.2 = \frac{2 \times 3.14 \times 0.033(300 - t_1)}{2.3 \log 3.375/2.375}$$

$$t_1 = 125.8^\circ\text{F} \quad \text{Check}$$

The total heat loss q does not appear to vary significantly for the different assumed values of t_1 . This is because the insulation and not the small surface coefficient affords the major resistance to heat flow. When the variation in q is considerable for different assumed temperatures of t_1 , it indicates insufficient insulation.

The Maximum Heat Loss through Pipe Insulation. It would seem at first that the thicker the insulation the less the total heat loss. This is always true for flat insulation but not for curved insulation. Consider a pipe with successive layers of cylindrical insulation. As the thickness of

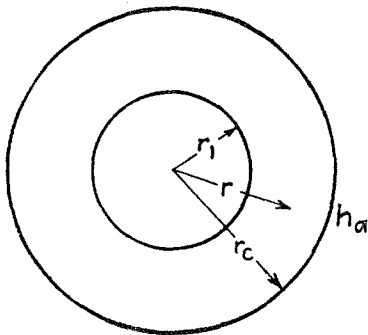


FIG. 2.10. The critical radius.

the insulation is increased, the surface area from which heat may be removed by air increases and the total heat loss may also increase if the area increases more rapidly than the resistance. Referring to Fig. 2.10, the resistance of the insulation per linear foot of pipe is

$$R_b = \frac{1}{2\pi k_b} \ln \frac{r}{r_1} \quad (2.36)$$

and the resistance of the air per linear foot of pipe, although a function of the surface and air temperatures, is given by

$$R_a = \frac{1}{h_a 2\pi r} \quad (2.37)$$

The resistance is a minimum and the heat loss a maximum when the derivative of the sum of the resistances R with respect to the radius r is set equal to zero or

$$\begin{aligned} \frac{dR}{dr} = 0 &= \frac{1}{2\pi k_b} d \ln \frac{r}{r_1} + \frac{1}{h_a 2\pi} d \frac{1}{r} \\ &= \frac{1}{2\pi k_b r} - \frac{1}{h_a 2\pi r^2} \end{aligned} \quad (2.38)$$

At the maximum heat loss $r = r_c$, the critical radius, or

$$r_c = \frac{k_b}{h_a} \quad (2.39)$$

In other words, the maximum heat loss from a pipe occurs when the critical radius equals the ratio of the thermal conductivity of the insulation to the surface coefficient of heat transfer. The ratio has the dimension of ft. It is desirable to keep the critical radius as small as possible

so that the application of insulation will result in a reduction and not an increase in the heat loss from a pipe. This is obviously accomplished by using an insulation of small conductivity so that the critical radius is less than the radius of the pipe, or $r_c < r_1$.

The Optimum Thickness of Insulation. The optimum thickness of insulation is arrived at by a purely economic approach. If a bare pipe were to carry a hot fluid, there would be a certain hourly loss of heat whose value could be determined from the cost of producing the Btu in the plant heat-generating station. The lower the heat loss the greater the thickness and initial cost of the insulation and the greater the annual fixed charges (maintenance and depreciation) which must be added to the annual heat loss. The fixed charges on pipe insulation will be about 15 to 20 per cent of the initial installed cost of the insulation. By assuming a number of thicknesses of insulation and adding the fixed charges to the value of the heat lost, a minimum cost will be obtained and the thickness corresponding to it will be the optimum economic thickness of the insulation. The form of such an analysis is shown in Fig. 2.11. The most difficult part is obtaining reliable initial-installation-cost data, since they vary greatly with plant to plant and with the amount of insulating to be done at a single time.

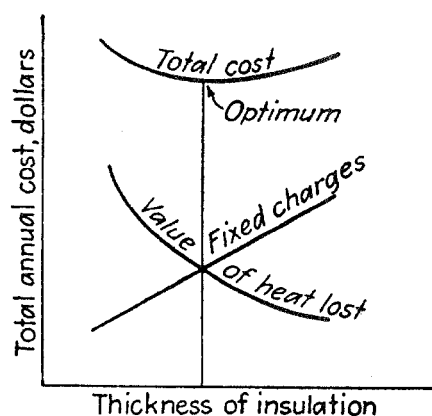


FIG. 2.11. Optimum thickness of insulation.

Graphical Solutions of Conduction Problems. Thus far in the treatment of conduction, only those cases have been considered in which the heat input per square foot of surface was uniform. It was also characteristic of these cases that the heat removal per square foot of surface was also uniform. This was likewise true for the cylinder, even though the internal and external surfaces were not identical. Some of the common problems of steady-state conduction in solids involve the removal or input of heat where it is not uniform over a surface, and although the solution of such problems by mathematical analysis is often complicated it is possible to obtain close approximations graphically. The method employed here is that of Awbery and Schofield¹ and earlier investigators.

Consider the section of a metal-sheathed wall, as shown in Fig. 2.12, with hot side ABC at the uniform temperature t_1 . At recurring intervals DF on the cold side DEF at the uniform temperature t_2 , metal bracing

¹ Awbery, J. and F. Schofield, *Proc. Intern. Congr. Refrig.*, 5th Congr., **3**, 591-610 (1929).

ribs are attached to the outer sheath and imbedded two-thirds into the thickness of the wall. Since the sheath and metal rib both have a high thermal conductivity compared with the wall material itself, the rib and sheath may both be considered to be at very nearly the same temperature. The predominantly horizontal lines indicated on the drawing represent isothermal planes perpendicular to the plane of the drawing. Consequently there is no heat flow to be considered in the direction perpendicular to the plane of the drawing.

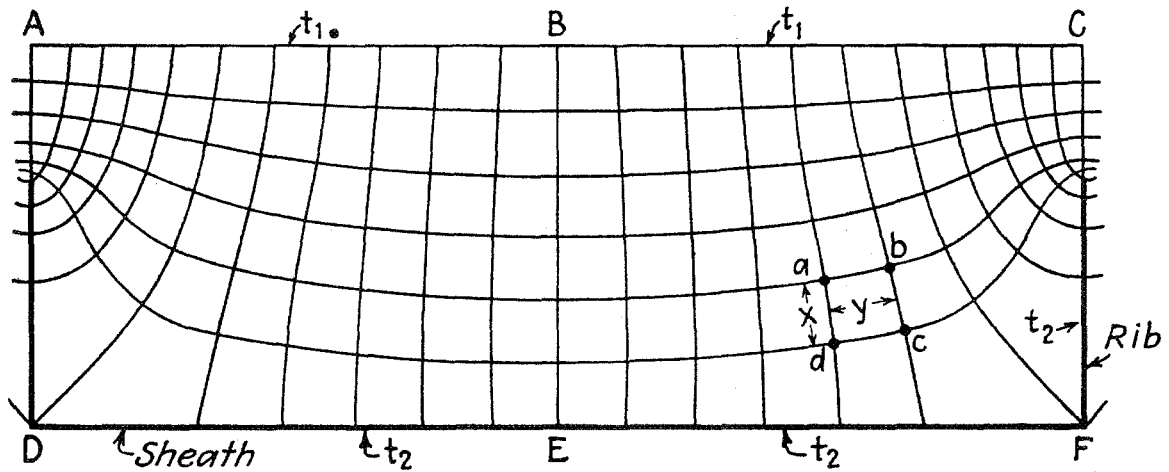


FIG. 2.12. Graphical representation of heat conduction.

Since the drawing is symmetrical about the vertical line BE , consider only the right half of the drawing bounded by $BCFE$. Assume an arbitrary number of isotherms n_x in the direction from B to E so that, if k is constant, $\Delta t = n_x \Delta t_x$. If k varies with t then $k \Delta t_x = \frac{1}{n_x} \int_1^2 k dt$. The greater the assumed number of isotherms the greater the precision of the solution. Next, consider the heat to flow from t_1 to metal at t_2 through n_l lanes emanating from BC and forming the network indicated. Now refer to any small portion of any lane, such as $abcd$ with length x , mean width y , where $y = (ab + cd)/2$, and unit depth $z = 1$ perpendicular to the drawing. The steady-heat flow into each lane is Q_l . The conduction equation is then $Q_l = k(yz) \Delta t_x / x$. The temperature difference from one isotherm to the next is naturally the same, and since Q_l is constant for the lane it is evident from the conduction equation that the ratio y/x must also be constant, although y and x may vary. The network of the drawing is constructed such that, for each quadrilateral, $y = x$. Where x is small it is because the isotherms are crowded together owing to the high heat removal by the rib. The heat flow per lane is then given by

$$k \frac{(t_1 - t_2)}{n_x}.$$

The total heat flow from BC thus requires $n_i = Qn_x/k(t_1 - t_2)$ lanes, where Q is the total heat flow. Figure 2.12 was constructed in this manner starting with six isotherms.

Although the individual portions of the network are neither squares nor rectangles their corners are at right angles in accordance with the steady-state principle that the flow of heat is always at right angles to the isotherms comprising the temperature difference. In Fig. 2.12 it is seen that 11 lanes were obtained for each half of a symmetrical section. If the isotherms were undisturbed by the rib, the portions $abcd$ would then be squares and the heat entering BC would flow normal to it and 8.3 lanes would be required. The rib is therefore equivalent to increasing the heat removal by 33 per cent. When the ribs are spaced more closely together, the fractional heat removal increases.

PROBLEMS

2.1. A furnace is enclosed by walls made (from inside out) of 8 in. of kaolin firebrick, 6 in. of kaolin insulating brick, and 7 in. of fireclay brick. What is the heat loss per square foot of wall when the inside of the furnace is maintained at 2200°F and the outside at 200°F?

2.2. A furnace wall is to consist in series of 7 in. of kaolin firebrick, 6 in. of kaolin insulating brick, and sufficient fireclay brick to reduce the heat loss to 100 Btu/(hr)(ft²) when the face temperatures are 1500 and 100°F, respectively. What thickness of fireclay brick should be used? If an effective air gap of $\frac{1}{8}$ in. can be incorporated between the fireclay and insulating brick when erecting the wall without impairing its structural support, what thickness of insulating brick will be required?

2.3. A furnace wall consists of three insulating materials in series. 32 per cent chrome brick, magnesite bricks, and low-grade refractory bricks ($k = 0.5$). The magnesite bricks cannot withstand a face temperature above 1500°F, and the low-grade bricks cannot exceed 600°F. What thickness of the wall will give a heat loss not in excess of 1500 Btu/(hr)(ft²) when the extreme face temperatures are 2500 and 200°F, respectively?

2.4. A 6-in. IPS pipe is covered with three resistances in series consisting from the inside outward of $\frac{1}{2}$ in. of kapok, 1 in. of rock wool, and $\frac{1}{2}$ in. of powdered magnesite applied as a plaster. If the inside surface is maintained at 500°F and the outside at 100°F, what is the heat loss per square foot of outside pipe surface?

2.5. A 2-in. IPS line to a refrigerated process covered with $\frac{1}{2}$ in. of kapok carries 25% NaCl brine at 0°F and at a flow rate of 30,000 lb/hr. The outer surface of the kapok will be maintained at 90°F. What is the equation for the flow of heat? Calculate the heat leakage into the pipe and the temperature rise of the fluid for a 60-ft length of pipe.

2.6. A vertical cylindrical kiln 22 ft in diameter is enclosed at the top by a hemispherical dome fabricated from an 8-in. layer of interlocking and self-supporting 32 per cent chrome bricks. Derive an expression for conduction through the dome. When the inside and outside of the hemispherical dome are maintained at 1600 and 300°F, respectively, what is the heat loss per square foot of internal dome surface? How does the total heat loss for the dome compare with the total heat loss for a flat structurally supported roof of the same material when exposed to the same difference in temperature?

2.7. A 4-in. steel pipe carrying 450°F steam is lagged with 1 in. of kapok surrounded by 1 in. of powdered magnesite applied as a plaster. The surrounding air is at 70°F. What is the loss of heat from the pipe per linear foot?

2.8. A 3-in. IPS main carries steam from the powerhouse to the process plants at a linear velocity of 8000 fpm. The steam is at 300 psi (gage), and the atmosphere is at 70°F. What percentage of the total heat flow is the bare-tube heat loss per 1000 ft of pipe? If the pipe is lagged with half a thickness of kapok and half a thickness of asbestos, what total thickness will reduce the insulated heat loss to 2 per cent of the bare-tube heat loss?

2.9. In a 6-in. steam line at 400°F the unit resistance for the condensation of steam at the inside pipe wall has been found experimentally to be 0.00033 (hr)(ft²)(°F)/Btu. The line is lagged with ½ in. of rock wool and ½ in. of asbestos. What is the effect of including the condensation and metal pipe-wall resistances in calculating the total heat loss per linear foot to atmospheric air at 70°F?

NOMENCLATURE FOR CHAPTER 2

A	Heat-flow area, ft ²
C_1, C_2	Constants of integration
C_k	Sutherland constant
c_v	Volumetric specific heat, Btu/(ft ³)(°F)
c	Specific heat at constant pressure, Btu/(lb)(°F)
D	Diameter, ft
E	Voltage or electromotive force
h_a	Surface coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
I	Current, amp
K	Electrical conductivity, 1/ohm-ft
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Thickness of wall or length of pipe, ft
n_l	Number of heat-flow lanes
n_x	Number of isotherms
Q	Heat flow, Btu/hr
Q_l	Heat flow per lane, Btu/hr
Q'	Heat, Btu
q	Heat flow, Btu/(hr)(lin ft)
R	Resistance to heat flow, (hr)(°F)/Btu
R^w	Resistance to electric flow, ohms
r	Radius, ft
t	Temperature at any point, °F
Δt	Temperature difference promoting heat flow, °F
T	Absolute temperature, °R
v	Volume, ft ³
x, y, z	Coordinates of distance, ft
γ	Change in thermal conductivity per degree
θ	Time, hr
ρ	Density, lb/ft ³
σ	Resistivity, ohm-ft

CHAPTER 3

CONVECTION

Introduction. Heat transfer by convection is due to fluid motion. Cold fluid adjacent to a hot surface receives heat which it imparts to the bulk of the cold fluid by mixing with it. Free or natural convection occurs when the fluid motion is not implemented by mechanical agitation. But when the fluid is mechanically agitated, the heat is transferred by forced convection. The mechanical agitation may be supplied by stirring, although in most process applications it is induced by circulating the hot and cold fluids at rapid rates on the opposite sides of pipes or tubes. Free- and forced-convection heat transfer occur at very different speeds, the latter being the more rapid and therefore the more common. Factors which promote high rates for forced convection do not necessarily have the same effect on free convection. It is the purpose of this chapter to establish a general method for obtaining the rates of heat transfer particularly in the presence of forced convection.

Film Coefficients. In the flow of heat through a pipe to air it was seen that the passage of heat into the air was not accomplished solely by conduction. Instead, it occurred partly by radiation and partly by free convection. A temperature difference existed between the pipe surface and the *average* temperature of the air. Since the distance from the pipe surface to the region of average air temperature is indefinite, the resistance cannot be computed from $R_a = L_a/k_a A$, using k for air. Instead the resistance must be determined experimentally by appropriately measuring the surface temperature of the pipe, the temperature of the air, and the heat transferred from the pipe as evidenced by the quantity of steam condensed in it. The resistance for the entire surface was then computed from

$$R_a = \frac{\Delta t_a}{Q} \quad (\text{hr})(^\circ\text{F})/\text{Btu}$$

If desired, L_a can also be calculated from this value of R_a and would be the length of a fictitious conduction film of air equivalent to the combined resistance of conduction, free convection, and radiation. The length of the film is of little significance, although the concept of the fictitious film finds numerous applications. Instead it is preferable to

deal directly with the reciprocal of the unit resistance h , which has an experimental origin. Because the use of the unit resistance L/k is so much more common than the use of the total surface resistance L/kA , the letter R will now be used to designate L/k (hr)(ft²)(°F)/Btu and it will simply be called the resistance.

Not all effects other than conduction are necessarily combinations of two effects. Particularly in the case of free or forced convection to liquids and, in fact, to most gases at moderate temperatures and temperature

differences the influence of radiation may be neglected and the experimental resistance corresponds to forced or free convection alone as the case may be.

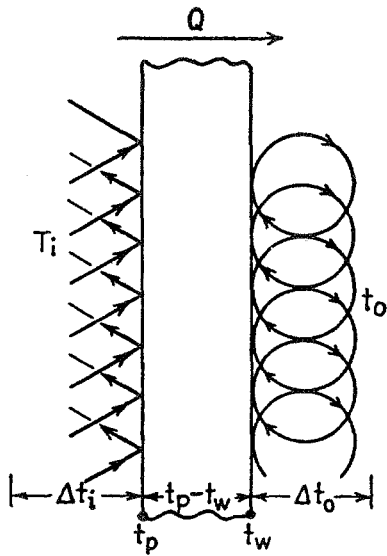


FIG. 3.1. Two convection coefficients.

Consider a pipe wall with forced convection of different magnitudes on both sides of the pipe as shown in Fig. 3.1. On the inside, heat is deposited by a hot flowing liquid, and on the outside, heat is received by a cold flowing liquid. Either resistance can be measured independently by obtaining the temperature difference between the pipe surface and the average temperature of the liquid. The heat transfer can be determined from the sensible-heat change in either fluid over the length of the pipe in which the heat

transfer occurs. Designating the resistance on the inside by R_i and on the outside by R_o , the inside and outside pipe-wall temperatures by t_p and t_w , and applying an expression for the steady state,

$$Q = \frac{A_i(T_i - t_p)}{R_i} = \frac{A_o(t_w - t_o)}{R_o} \quad (3.1)$$

where T_i is the temperature of the hot fluid on the inside and t_o the temperature of the cold fluid on the outside. Replacing the resistances by their reciprocals h_i and h_o , respectively,

$$Q = h_i A_i \Delta t_i = h_o A_o \Delta t_o \quad (3.2)$$

The reciprocals of the heat-transfer resistances have the dimensions of Btu/(hr)(ft²)(°F of temperature difference) and are called *individual film coefficients* or simply *film coefficients*.

Inasmuch as the film coefficient is a measure of the heat flow for unit surface and unit temperature difference, it indicates the rate or speed with which fluids having a variety of physical properties and under varying degrees of agitation transfer heat. Other factors influence the

film coefficient such as the size of the pipe and whether or not the fluid is considered to be on the inside or outside of the pipe. With so many variables, each having its own degree of influence on the rate of heat transfer (film coefficient), it is fairly understandable why a rational derivation is not available for the direct calculation of the film coefficient. On the other hand, it is impractical to run an experiment to determine the coefficient each time heat is to be added or removed from a fluid. Instead it is desirable to study some method of correlation whereby several basic experiments performed with a wide range of the variables can produce a relationship which will hold for any other combinations of the variables. The immediate problem is to establish a method of correlation and then apply it to some experimental data.

The Viscosity. It is not possible to proceed very far in the study of convection and fluid flow without defining a property which has an important bearing upon both, *viscosity*. In order to evaluate this property by fluid dynamics two assumptions are required: (1)

Where a solid-liquid interface exists, there is no slip between the solid and liquid, and (2) Newton's rule: Shear stress is proportional to the rate of shear in the direction perpendicular to motion.

An unstressed particle of liquid as shown in Fig. 3.2a will assume the form in Fig. 3.2b when a film of liquid is subjected to shear.

The *rate of shear* is proportional to the velocity gradient du/dy . Applying Newton's rule, if τ is the shear stress,

$$\tau = \mu \frac{du}{dy} \quad (3.3)$$

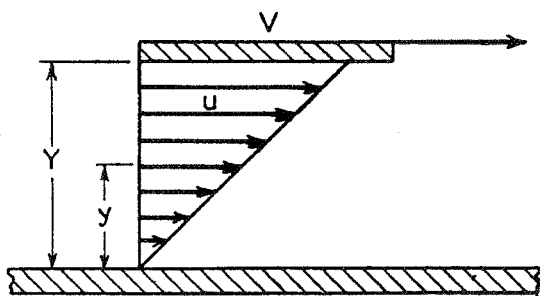


FIG. 3.3. Fluid shear.

where μ is the proportionality constant or

$$\mu = \frac{\tau}{du/dy} \quad \left\{ \begin{array}{l} \text{In solids it results in deformation and is equivalent} \\ \text{to the modulus of elasticity.} \\ \text{In liquids it results in deformation at a given rate.} \end{array} \right.$$

To evaluate μ refer to Fig. 3.3 where shear is produced by maintaining the liquid film between a stationary plate at distance Y and a moving plate with velocity V . At any point in the film the velocity u is given

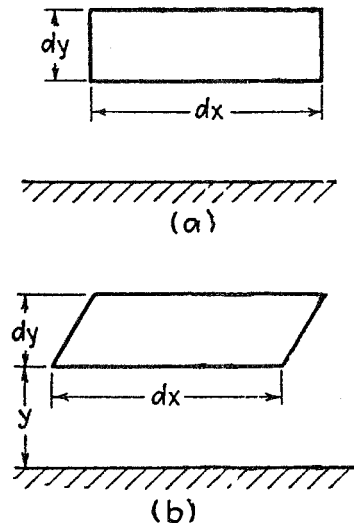


FIG. 3.2. Fluid strain.

by $u = Vy/Y$.

$$\text{Rate of shear} = \frac{du}{dy} = \frac{V}{Y} \quad (3.3)$$

$$\tau = \mu \frac{V}{Y} \quad (3.4)$$

where μ is called the viscosity when V and Y have unit values.

The actual force required to move the plate is τA . If F is the pound-force, L the length, and θ the time, the dimensions of the viscosity are

$$\mu = \tau \frac{Y}{V} = \frac{F}{L^2} \frac{L}{L/\theta} = \frac{F\theta}{L^2}$$

or using the pound-mass M , where $F = Mg$ and $g = L/\theta^2$, the acceleration of gravity,

$$\mu = \frac{ML}{L^2\theta^2} \frac{L}{L/\theta} = \frac{M}{L\theta}$$

When evaluated in cgs metric units μ is commonly called the *absolute viscosity*.

$$\mu = \frac{\text{gram-mass}}{\text{centimeter} \times \text{second}}$$

This unit has been named the poise after the French scientist Poiseuille. This is a large unit, and it is customary to use and speak of the centipoise, or one-hundredth poise. In engineering units its equivalent is defined by

$$\mu = \frac{\text{pound-mass}}{\text{foot} \times \text{hour}}$$

Viscosities in centipoises can be converted to engineering units on multiplying by 2.42. This unit has no name. Another unit, the kinematic viscosity, is also used because it occurs frequently in physical systems and produces straighter graphs of viscosity vs. temperature on logarithmic coordinates. The kinematic viscosity is the absolute viscosity in centipoises divided by the specific gravity.

$$\text{Kinematic viscosity} = \frac{\text{absolute viscosity}}{\text{specific gravity}}$$

The unit of kinematic viscosity is the stokes, after the English mathematician Stokes, and the hundredth of the stokes is the centistokes.

The viscosity can be determined indirectly by measuring the time of efflux from a calibrated flow device having an orifice and a controlled temperature. The commonest is the Saybolt viscometer, and the time of efflux from a standard cup into a standard receiver is measured in

seconds and recorded as Saybolt Seconds Universal, SSU. Conversion factors from the time of efflux to centistokes for the Saybolt and other viscometers are given in Fig. 13.¹

Heat Transfer between Solids and Fluids: Streamline and Turbulent Flow. The Reynolds Number. When a liquid flows in a horizontal pipe, it may flow with a random eddying motion known as *turbulent flow*, as shown in Fig. 3.4 by a plot of the local velocity in the pipe vs. the distance from its center. If the linear velocity of the liquid is decreased below some threshold value, the nature of the flow changes and the turbulence disappears. The fluid particles flow in lines along the axis of the pipe, and this is known as *streamline flow*. An experiment used for the visual determination of the type of flow consists of a glass tube

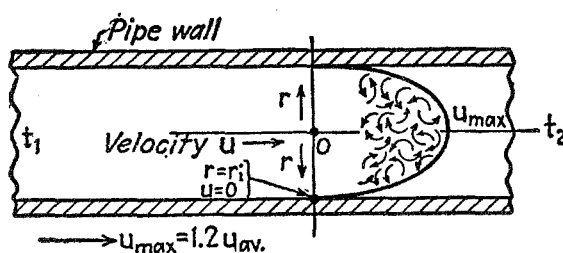


FIG. 3.4. Turbulent flow in pipes.

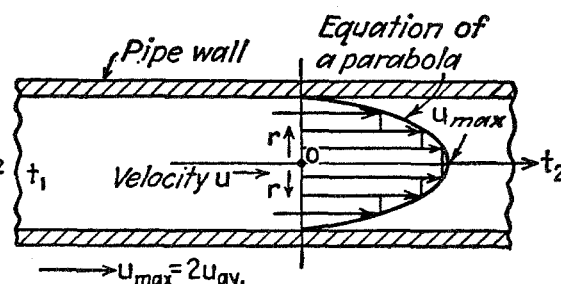


FIG. 3.5. Streamline flow in pipes.

through which water flows. A thin stream of ink is injected at the center of the tube, and if the ink remains at the center for a reasonable distance, it is indicative of streamline flow. Synonyms for streamline flow are viscous, laminar, and rodlike flow. Additional experimentation has indicated that streamline flow proceeds as if by the sliding of concentric thin cylinders of liquid one within the other as shown in Fig. 3.5. It appears that the distribution of the velocities of the cylinders is parabolic with the maximum at the center and approaching zero at the tube wall.

Reynolds² observed that the type of flow assumed by a liquid flowing in a tube was influenced by the velocity, density, and viscosity of the liquid and the diameter of the tube. When related as the quotient of $Du\rho/\mu$, called the Reynolds number, it was found that turbulent flow always existed when the numerical value of $Du\rho/\mu$ exceeded about 2300 and usually when it exceeded 2100. By definition, the transfer of heat by convection proceeds mainly as the result of mixing, and while this requirement appears to be satisfied by turbulent flow, it is not fulfilled by stream-

¹ Figure numbers which are not preceded by a chapter number will be found in the Appendix.

² Reynolds, O., "Scientific Papers of Osborne Reynolds," p. 81, Cambridge University Press, London, 1901.

line flow. Streamline flow is, in fact, a form of conduction whose study will be deferred to a later chapter. The speed with which heat is transferred to or from a given liquid to a tube is considerably less for streamline than for turbulent flow, and in industrial practice it is almost always desirable to avoid conditions such as low liquid velocity which promote streamline flow.

Dimensional Analysis. A method of correlating a number of variables into a single equation expressing an effect is known as *dimensional analysis*. Certain equations describing physical phenomena can be obtained rationally from basic laws derived from experiments. An example is the time of vibration of a pendulum from Newton's second law and the gravitational constant. Still other effects can be described by differential equations, and the course or extent of the phenomena deduced by means of the calculus. Numerous examples of this type are encountered in elementary physics. In still other types of phenomena there is insufficient information to permit the formulation of either differential equations or a clear picture by which fundamental laws may be applied. This last group must be studied experimentally, and the correlation of the observations is an empirical approach to the equation. Equations which can be obtained theoretically can also be obtained empirically, but the reverse is not true.

Bridgman¹ has presented by far the most extensive proof of the mathematical principles underlying dimensional analysis. Because it operates only upon the dimensions of the variables, it does not directly produce numerical results from the variables but instead yields a modulus by which the observed data can be combined and the relative influence of the variables established. As such, it is one of the important cornerstones of empirical study. It recognizes that any combination of a number and a dimension, such as 5 lb or 5 ft, possesses two identifying aspects, one of pure magnitude (numerical) and the other quantitative (dimensional). *Fundamental dimensions* are quantities such as length, time, and temperature which are directly measurable. Derived dimensions are those which are expressed in terms of fundamental dimensions such as velocity = length/time or density = mass/length³. The end results of a dimensional analysis may be stated as follows: If a dependent variable having given dimensions depends upon some relationship among a group of variables, the individual variables of the group must be related in such a way that the net dimensions of the group are identical with those of the dependent variable. The independent variables may also be related in such a way that the dependent variable is defined by the sum of several

¹ Bridgman, P. W., "Dimensional Analysis," Yale University Press, New Haven, 1931.

different groups of variables, each group having the net dimensions of the dependent variable. As a very simple illustration consider the *continuity equation* which is frequently written in elementary physics and thermodynamics texts in the form

$$w = \frac{ua}{v} \quad (3.5)$$

where w = fluid flow rate, weight/time

u = fluid velocity in conduit, length/time

a = cross section of the conduit, length $\times$ length = length²

v = specific volume, length $\times$ length $\times$ length/weight = length³/weight

Why does Eq. (3.5) have this particular form? u , a , and v must be related so that their net dimensions are the same as those of the dependent variable w , namely, weight/time. An equation involving both pure numbers and dimensions must be correct with respect to either and both. Checking the dimensions alone, writing for the variables in Eq. (3.5) their individual dimensions,

$$\frac{\text{Weight}}{\text{Time}} = \frac{\text{length}}{\text{time}} \times \text{length}^2 \times \frac{1}{\frac{\text{length}^3}{\text{weight}}} = \frac{\text{weight}}{\text{time}} \quad (3.6)$$

It is seen that the dimensions on the left side are identical with the net dimensions of the group only when the variables of the group are arranged in the particular manner indicated by the formula. The three independent variables above give an answer in weight/time only when arranged in a single way, ua/v . *Conversely it may be deduced that the form of an equation is determined only by its dimensions*; the form which produces dimensional equality represents the necessary relationship among the variables. Any physical equation may be written and evaluated in terms of a power series containing all the variables. If the form were not known in the illustration above and it was desired to find a relationship which must exist between the variables w , u , a , and v , it may be expressed by a power series such as

$$\phi(w, u, a, v) = \alpha w^b u^c a^d v^e + \alpha' w^{b'} u^{c'} a^{d'} v^{e'} + \dots = 0 \quad (3.7)$$

The factors α and α' are dimensionless proportionality constants. Since the dimensions of all the consecutive terms of the series are identical, it is not necessary to consider any of the terms beyond the first. Accordingly one can write

$$\phi'(w^b u^c a^d v^e) = 1 \quad (3.8)$$

where ϕ' indicates the function. Arbitrarily setting $b = -1$ so that w

will not appear in the final equation raised to a fractional exponent,

$$w = \alpha u^c a^d v^e \quad (3.9)$$

Substituting dimensions,

$$\frac{\text{Weight}}{\text{Time}} = \alpha \left(\frac{\text{length}}{\text{time}} \right)^c \times (\text{length}^2)^d \times \left(\frac{\text{length}^3}{\text{weight}} \right)^e \quad (3.10)$$

If a group of independent variables will establish numerical equality with a dependent variable, the same is true of their dimensions. Equation (3.6) imposes this condition. The exponents c , d , and e may then assume such values as are necessary to effect the dimensional equality between the left and right sides. The remainder of the solution is merely to evaluate c , d , and e by simple algebra. Summing the exponents of the dimensions on both sides and recalling that an exponent of zero reduces a number to unity,

$$\Sigma \text{ length, } 0 = c + 2d + 3e$$

$$\Sigma \text{ weight, } 1 = -e$$

$$\Sigma \text{ time, } -1 = -c$$

Solving for the unknown, d is found to be $+1$. The three exponents are then $c = +1$, $d = +1$, and $e = -1$. Substituting these in Eq. (3.9),

$$w = \alpha u^{+1} a^{+1} v^{-1} = \alpha \frac{ua}{v} \quad (3.11)$$

Inasmuch as this is an exact relationship, the proportionality constant α is equal to 1.0 and

$$w = \frac{ua}{v}$$

Thus by purely algebraic means the correct form of the equation has been established. This has been an extremely elementary illustration in which all of the exponents were integers and the dependent and independent variables were expressed in only three kinds of dimensions: weight, length, and time. In systems involving mechanics and heat it is often necessary to employ other dimensions such as temperature and a derived heat unit H , the Btu or calorie.

In mechanical and chemical engineering it is customary to use a set of six dimensions: force F , heat H , length L , mass M , temperature T , and time θ . One of the important alternatives, however, hinges about the unit of force and the unit of mass. In the preceding illustration the *weight* was employed. The relationship would hold whether the pound-mass or gram-mass or the pound-force (poundal) or gram-force (dyne) were used

TABLE 3.1. DIMENSIONS AND UNITS

Dimensions:* Force = F , heat = H , length = L , mass = M , temperature = T , time = θ . The force-pound is the poundal, the force-gram is the dyne

Sym- bol	Quantity: consistent engineering and metric units	Dimen- sion
g	Acceleration of gravity, ft/hr ² , cm/sec ²	L/θ^2
A	Area or surface, ft ² , cm ²	L^2
K_H	Conversion from kinetic energy to heat	$ML^2/H\theta^2$
K_M	Conversion from force to mass	$ML/F\theta^2$
ρ	Density, lb/ft ³ , g/cm ³	M/L^3
D	Diameter, ft, cm	L
k_d	Diffusivity (volumetric), ft ² /hr, cm ² /sec	L^2/θ
F	Force, force-pound (poundal), force-gram (dyne)	F
H	Heat, Btu, cal	H
L	Length, ft, cm	L
M	Mass, lb, g	M
W	Mass flow, lb/hr, g/sec	M/θ
G	Mass velocity, lb/(hr)(ft ²), g/(sec)(cm ²)	$M/\theta L^2$
J	Mechanical equivalent of heat, (force-lb)(ft)/Btu, (force-g)(cm)/cal	FL/H
P	Pressure, force-lb/ft ² , force-g/cm ²	F/L^2
P_0	Power, (force-lb)(ft)/hr, (force-g)(cm)/sec	FL/θ
r	Radius, ft, cm	L
c	Specific heat, Btu/(lb)(°F), cal/(g)(°C)	H/MT
v	Specific volume, ft ³ /lb, cm ³ /g	L^3/M
τ	Stress, force-lb/ft ² , force-g/cm ²	F/L^2
σ	Surface tension, force-lb/ft, force-g/cm	F/L
T	Temperature, °F, °C	T
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft), cal/(sec)(cm ²)(°C/cm)	$H/LT\theta$
α	Thermal diffusivity, ft ² /hr, cm ² /sec	L^2/θ
β	Thermal coefficient of expansion, 1/°F, 1/°C	$1/T$
R_t	Thermal resistivity, (°F)(ft)(hr)/Btu, (°C)(cm)(sec)/cal	$LT\theta/H$
θ	Time, hr, sec	θ
u	Velocity, ft/hr, cm/sec	L/θ
μ_g	Viscosity (force-lb)(hr)/ft ² , (force-g)(sec)/cm ²	$F\theta/L^2$
μ	Viscosity (abs), lb/(ft)(hr), g/(cm)(sec)	$M/L\theta$
M	Mass, lb, g	M
w_0	Work, (force-lb)(ft), (force-g)(cm)	FL

* For a system without heat changes these automatically reduce to $FLM\theta$.

as long as the weight was always treated in the same way. Consider a system in which the mass is a fundamental dimension as M , L , T , θ . From the acceleration equation, force = mass $\times$ acceleration,

$$F = ML\theta^{-2}.$$

In another set of dimensions it may be more convenient to consider force

the fundamental dimension, in which case mass is expressed by

$$M = FL^{-1}\theta^2.$$

When some of the variables are commonly expressed in units of force such as pressure FL^{-2} and other variables by units of mass such as the density ML^{-3} , it is necessary to introduce a dimensional constant into the series expression before solving for the exponents. The constant relating M and F naturally has the dimensions of the gravitational acceleration constant $L\theta^{-2}$. A similar situation also arises when describing a phenomenon by which a work or kinetic energy change occurs in a system. Some of the variables may ordinarily be expressed in terms of foot-pounds (force-pound $\times$ foot) and others in terms of heat energy such as the Btu. A conversion factor which is the heat equivalent of work must be introduced to convert FL to H , or vice versa. The constant is the kinetic energy equivalent of the heat $ML^2/H\theta^2$. A number of common variables and dimensional constants are given in Table 3.1 together with their net dimensions in a six-dimension system. Typical sets of engineering and metric units are included.

Analysis for the Form of a Fluid-flow Equation. When an incompressible fluid flows in a straight horizontal uniform pipe with a constant mass rate, the pressure of the fluid decreases along the length of the pipe owing to friction. This is commonly called the *pressure drop* of the system, ΔP . The pressure drop per unit length is referred to as the *pressure gradient*, dP/dL , which has been found experimentally to be influenced by the following pipe and fluid properties: diameter D , velocity u , fluid density ρ , and viscosity μ . What relationship exists between the pressure gradient and the variables?

Solution. The pressure has dimensions of force/area, whereas the density is expressed by mass/volume so that a dimensional constant relating M to F must be included, $K_M = ML/F\theta^2$. The same result may be accomplished by including the acceleration constant g along with the variables above. While the *viscosity* is determined experimentally as a force effect and has the dimensions $F\theta/L^2$, it is a very small unit, and it is more common in the engineering sciences to use the absolute viscosity $M/L\theta$ in which the conversion from force to mass has already been made.

Using the same method of notation as before,

$$\frac{dP}{dL} \propto D, u, \rho, \mu, K_M \quad (3.12)$$

$$\frac{dP}{dL} = \alpha D^a u^b \rho^c \mu^d K_M^e \quad (3.13)$$

Substituting dimensions and arbitrarily setting the exponent of dP/dL equal to 1,

$$\frac{F}{L^3} = \alpha(L)^a \left(\frac{L}{\theta}\right)^b \left(\frac{M}{L^3}\right)^c \left(\frac{M}{L\theta}\right)^d \left(\frac{ML}{F\theta^2}\right)^e \quad (3.14)$$

Summing exponents,

$$\begin{aligned} \Sigma F, \quad 1 &= -e \\ \Sigma L, \quad -3 &= a + b - 3c - d + e \\ \Sigma M, \quad 0 &= c + d + e \\ \Sigma \theta, \quad 0 &= -b - d - 2e \end{aligned}$$

Solving simultaneously,

$$\begin{aligned} a &= -1 - d \\ b &= 2 - d \\ c &= 1 - d \\ d &= d \\ e &= -1 \end{aligned}$$

Substituting back in Eq. (3.13),

$$\frac{dP}{dL} = \alpha D^{-1-d} u^{2-d} \rho^{1-d} \mu^d K_M^{-1} = \frac{\alpha u^2 \rho}{DK_M} \left(\frac{Du\rho}{\mu}\right)^{-d} \quad (3.15)$$

where α and $-d$ must be evaluated from experimental data. A convenient term of almost universal use in engineering is the *mass velocity* G , which is identical with $u\rho$ and corresponds to the weight flow per square foot of flow area. To obtain the pressure drop from Eq. (3.15), replace dP by ΔP , dL by the length of pipe L , or ΔL , and substituting for K_M its equivalent g ,

$$\Delta P = \frac{\alpha G^2 L}{D\rho g} \left(\frac{DG}{\mu}\right)^{-d} \quad (3.16)$$

where $Du\rho/\mu$ or DG/μ is the Reynolds number.

Analysis for the Form of a Forced-convection Equation. The rate of heat transfer by forced convection to an incompressible fluid traveling in turbulent flow in a pipe of uniform diameter at constant mass rate has been found to be influenced by the velocity u , density ρ , specific heat c , thermal conductivity k , viscosity μ , as well as the inside diameter of the pipe D . The velocity, viscosity, density, and diameter affect the thickness of the fluid film at the pipe wall through which the heat must first be conducted, and they also influence the extent of fluid mixing. k is the thermal conductivity of the fluid, and the specific heat reflects the variation of the average fluid temperature as a result of unit heat absorp-

tion. What relationship holds between the film coefficient or rate of heat transfer, $h_i = H/\theta L^2 T$ [such as Btu/(hr)(ft²)(°F)], and the other variables?

Solution. It is not known whether all energy terms will be expressed mechanically or thermally by the dimensions of the variables so that the dimensional constant $K_H = ML^2/H\theta^2$ must be included. If all the dimensions combine to give only thermal quantities such as the Btu, which appears in the dimensions of h_i , the exponent of K_H in the series expression should be zero and the constant will reduce dimensionally to 1.0, a pure number.

$$\begin{aligned} h_i &\propto u, \rho, c, D, k, \mu, K_H \\ h_i &= \alpha u^a \rho^b c^d D^e k^f \mu^g K_H^i \end{aligned} \quad (3.17)$$

$$\frac{H}{\theta L^2 T} = \alpha \left(\frac{L}{\theta}\right)^a \left(\frac{M}{L^3}\right)^b \left(\frac{H}{MT}\right)^d (L)^e \left(\frac{H}{\theta LT}\right)^f \left(\frac{M}{L\theta}\right)^g \left(\frac{ML^2}{H\theta^2}\right)^i \quad (3.18)$$

Summing exponents,

$$\begin{aligned} \Sigma H, \quad 1 &= d + f - i \\ \Sigma L, \quad -2 &= a - 3b + e - f - g + 2i \\ \Sigma M, \quad 0 &= b - d + g + i \\ \Sigma T, \quad -1 &= -d - f \\ \Sigma \theta, \quad -1 &= -a - f - g - 2i \end{aligned}$$

Solving simultaneously,

$$\begin{aligned} a &= a \\ b &= a \\ d &= 1 - f \\ e &= a - 1 \\ f &= f \\ g &= 1 - f - a \\ i &= 0 \end{aligned}$$

Substituting back,

$$h_i = \alpha u^a \rho^a c^{1-f} D^{a-1} k^f \mu^{1-f-a} K_H^0 \quad (3.19)$$

or collecting terms,

$$\frac{h_i D}{k} = \alpha \left(\frac{Du\rho}{\mu}\right)^a \left(\frac{c\mu}{k}\right)^{1-f} \quad (3.20)$$

where α , a , and $1 - f$ must be evaluated from a minimum of three sets of experimental data. Substituting the mass velocity for $u\rho$ in the above,

$$\frac{h_i D}{k} = \alpha \left(\frac{DG}{\mu}\right)^a \left(\frac{c\mu}{k}\right)^{1-f} \quad (3.21)$$

The dimensionless groups hD/k and $c\mu/k$, like the Reynolds number

$Du\rho/\mu$ or DG/μ , have been assigned names to honor earlier investigators in the field of fluid mechanics and heat transfer. A list of the common groups and the names assigned to them are included in Table 3.2.

TABLE 3.2. COMMON DIMENSIONLESS GROUPS

Symbol	Name	Group
Bi	Biot number	hr/k
Fo	Fourier number	$k\theta/\rho c r^2$
Gz	Graetz number	wc/kL
Gr	Grashof number	$D^3\rho^2g\beta\Delta t/\mu^2$
Nu	Nusselt number	hD/k
Pe	Peclet number	DGc/k
Pr	Prandtl number	$c\mu/k$
Re	Reynolds number	$DG/\mu, Du\rho/\mu$
Sc	Schmidt number	$\mu/\rho k_a$
St	Stanton number	h/cG

One of the useful aspects of dimensional analysis is its ability to provide a relationship among the variables when the information about a phenomenon is incomplete. One may have speculated that both fluid friction and forced convection are influenced by the surface tension of the fluid. The surface tension could have been included as a variable and new equations obtained, although the form of the equations would have been altered considerably. Nevertheless it would have been found that the exponents for any dimensionless groups involving the surface tension would be nearly zero when evaluated from experimental data. By the same token, the equations obtained above may be considered to be predicated on incomplete information. In either case a relationship is obtainable by dimensional analysis.

Consistent Units. In establishing the preceding formulas the dimensions were referred to in general terms such as length, time, temperature, etc., without specifying the units of the dimensions. The dimension is the basic measurable quantity, and convention has established a number of different basic units such as temperature, °F and °C; area, square foot, square inch, square meter, square centimeter; time, second or hour; etc. In order that the net dimensions of the variables may be obtained by cancellation among the fundamental and derived dimensions, all must employ the same basic measured units. Thus if several variables employ dimensions containing length such as velocity L/θ , density M/L^3 , and thermal conductivity $H/\theta LT$, each must employ the same basic unit of length such as the foot. Accordingly, when substituting values of the variables into a dimensionless group, it is not permissible to signify the

dimensions of some variables in feet, some in inches, and still others in centimeters. However, any unit of length is acceptable provided all the lengths involved in the variables are expressed in the same unit of length. The same rule applies to the other fundamental and derived dimensions. When a group of dimensions are expressed in this manner, they are called *consistent units*. Any group of consistent units will yield the same numerical result when the values of the variables are substituted into the dimensionless groups.

The Pi Theorem. One of the important mathematical proofs of dimensional analysis is attributed to Buckingham,¹ who deduced that the number of dimensionless groups is equal to the difference between the number of variables and the number of dimensions used to express them. Dimensional constants are also included as variables. The proof of this statement has been presented quite completely by Bridgman.² Designating dimensionless groups by the letters π_1, π_2, π_3 , the complete physical statement of a phenomenon can be expressed by

$$\phi(\pi_1, \pi_2, \pi_3 \dots) = 0 \quad (3.22)$$

where the total number of π terms or dimensionless groups equals the number of variables minus the number of dimensions. In the preceding example there were, including h_i , eight variables. These were expressed in five dimensions, and the number of dimensionless groups consequently was three. There is a notable exception which must be considered, however, or this method of obtaining the number of dimensionless groups by inspection can lead to an incorrect result. When two of the variables are expressed by the same dimension such as the length and diameter of a pipe, neither is a unique variable, since the dimensions of either are indistinguishable, and to preserve the identity of both they must be combined as a dimensionless constant ratio, L/D or D/L . When treated in this manner, the equation so obtained will apply only to a system which is geometrically similar to the experimental arrangement by which the coefficients and exponents were evaluated, namely, one having the same ratio of L/D or D/L . For this reason the form of a fluid-flow equation, Eq. (3.15), was solved for the pressure gradient rather than the pressure drop directly. Although the solution for a forced-convection equation has already been obtained algebraically, it will be solved again to demonstrate the Pi theorem and the extent to which it differs from the direct algebraic solution. In general, it is desirable to solve for the dimensionless groups appearing in Table 3.2.

¹ Buckingham, E., *Phys. Rev.*, **4**, 345-376 (1914).

² Bridgman, *op. cit.*

Analysis for a Forced-convection Equation by the Pi Theorem.

$$\begin{aligned}\phi(\pi_1, \pi_2, \pi_3 \dots) &= 0 \\ \pi &= \phi'(h_i^a u^b \rho^d c^e D^f k^g \mu^m K_H^i) = 1\end{aligned}\quad (3.23)$$

$$\pi = \alpha \left(\frac{H}{\theta L^2 T} \right)^a \left(\frac{L}{\theta} \right)^b \left(\frac{M}{L^3} \right)^d \left(\frac{H}{MT} \right)^e (L)^f \left(\frac{H}{\theta LT} \right)^g \left(\frac{M}{L\theta} \right)^m \left(\frac{ML^2}{H\theta^2} \right)^i \quad (3.24)$$

Summing exponents,

$$\begin{aligned}\Sigma H, 0 &= a + e + g - i \\ \Sigma L, 0 &= -2a + b - 3d + f - g - m + 2i \\ \Sigma M, 0 &= d - e + m + i \\ \Sigma T, 0 &= -a - e - g \\ \Sigma \theta, 0 &= -a - b - g - m - 2i\end{aligned}$$

π_1 , π_2 , and π_3 may be evaluated by simple algebra. All the exponents need not be summed up in one operation, since it has been seen in Eq. (3.20) that the dimensionless groups comprising it are only composed of three or four variables each. It is requisite in summing for the three groups individually only that all the exponents be included at some time and that three summations be made equal to the difference between the eight variables and five dimensions or three groups, π_1 , π_2 , and π_3 .

π_1 . Since it is desired to establish an expression for h_i as the dependent variable, it is preferable that it be expressed raised to the first power or $a = 1$. This will assure that in the final equation the dependent variable will not be present raised to some odd fractional power. Since all the exponents need not be included in evaluating π_1 , assume $b = 0$ and $e = 0$. Referring to Eq. (3.20) it will be seen that as a result of these assumptions neither the Reynolds number nor the Prandtl number will appear as solutions for π_1 . When $b = 0$, $u^b = 1$ and $Du\rho/\mu = 1$; and when $e = 0$, $c^e = 1$ and $c\mu/k = 1$.

Assume $a = 1$, $b = 0$, $e = 0$. Solving the simultaneous equations above, $d = 0$, $f = +1$, $g = -1$, $m = 0$, $i = 0$,

$$\pi_1 = \phi' \left(\frac{h_i D}{k} \right)$$

π_2 . Having already obtained h_i it is desirable not to have it appear again in either π_2 or π_3 . This can be accomplished in solving for the next group by assuming $a = 0$. The entire Nusselt number $h_i D/k$ will then reduce to 1. If a Reynolds number is desired, because it is a useful criterion of fluid flow, assume b or $f = 1$. Lastly, if the Prandtl number is to be eliminated, assume that the exponent of c or k is zero. If the exponent of the viscosity is assumed to be zero, it will not be possible to obtain either a Reynolds number or a Prandtl number.

Assume $f = 1$, $a = 0$, $e = 0$. Solving the simultaneous equations above, $b = 1$, $d = 1$, $g = 0$, $m = -1$, $i = 0$,

$$\pi_2 = \phi' \left(\frac{Du\rho}{\mu} \right)$$

π_3 . To prevent the h_i term and the velocity or density from appearing again, assume $a = 0$, $e = 1$, $f = 0$. All the exponents will have now appeared in one or more solutions.

Assume $a = 0$, $e = 1$, $f = 0$. Solving the simultaneous equations above, $b = 0$, $d = 0$, $g = -1$, $m = 1$, $i = 0$,

$$\pi_3 = \phi' \left(\frac{c\mu}{k} \right)$$

The final expression is

$$\phi \left(\frac{h_i D}{k}, \frac{Du\rho}{\mu}, \frac{c\mu}{k} \right) = 0 \quad (3.25)$$

or

$$\begin{aligned} \frac{h_i D}{k} &= \phi_1 \left(\frac{Du\rho}{\mu} \right) \phi_2 \left(\frac{c\mu}{k} \right) \\ &= \alpha \left(\frac{Du\rho}{\mu} \right)^p \left(\frac{c\mu}{k} \right)^q = \alpha \left(\frac{DG}{\mu} \right)^p \left(\frac{c\mu}{k} \right)^q \end{aligned} \quad (3.26)$$

where the proportionality constant and the exponents must be evaluated from experimental data.

Development of an Equation for Streamline Flow. Since streamline flow is a conduction phenomenon, it is subject to rational mathematical analysis. On the assumption that the distribution of velocities at any cross section is parabolic, the inside surface of the pipe is uniform, and the velocity at the wall is zero, Graetz¹ obtained for radial conduction to a fluid moving in a pipe in rodlike flow

$$\frac{t_2 - t_1}{t_p - t_1} = 1 - 8\phi \left(\frac{wc}{kL} \right) \quad (3.27)$$

where t_1 and t_2 are the inlet and outlet temperatures of the fluid, t_p is the uniform inside pipe surface temperature, $t_p - t_1$ the temperature difference at the inlet, and $\phi(wc/kL)$ is the numerical value of an infinite series having exponents which are multiples of wc/kL . Equation (3.27) may be replaced through dimensional analysis with an empirical expression which must be evaluated from experiments. If $t_2 - t_1$, the rise in the temperature of the fluid flowing in the pipe, is considered to be influenced

¹ Graetz, L., *Ann. Physik*, **25**, 337 (1885). For a review of the treatment of conduction in moving fluids see T. B. Drew, *Trans. AIChE*, **26**, 32 (1931).

in radial conduction by the length of path L , the rate of flow w , specific heat c , thermal conductivity k , and the temperature difference between the pipe inside surface and the fluid temperature so that $\Delta t_i = t_p - t_1$,

$$t_2 - t_1 = \alpha L^a w^b c^d k^e \Delta t_i^f \quad (3.28)$$

Solving by the method of dimensional analysis

$$t_2 - t_1 = \alpha \left(\frac{wc}{kL} \right)^d \Delta t_i \quad (3.29)$$

or

$$\frac{t_2 - t_1}{\Delta t_i} = \alpha \left(\frac{wc}{kL} \right)^d \quad (3.30)$$

which is similar to Eq. (3.27). Now note that neither Eq. (3.28) nor (3.29) contains h_i or the viscosity μ . But $Q = h_i A_i \Delta t_i$ or

$$wc(t_2 - t_1) = h_i \pi DL \Delta t_i$$

and substituting $G = \frac{w}{\pi D^2/4}$ in Eq. (3.30),

$$\frac{h_i L}{DG c} = \alpha \left(\frac{D^2 G c}{kL} \right)^d \quad (3.31)$$

Now *synthetically* introducing the viscosity by multiplying the right term of Eq. (3.31) by $(\mu/\mu)^d$, one obtains

$$\frac{h_i D}{k} = \alpha \left[\left(\frac{DG}{\mu} \right) \left(\frac{c\mu}{k} \right) \left(\frac{D}{L} \right) \right]^{d+1} = \alpha \left(\frac{wc}{kL} \right)^{d+1} \quad (3.32)$$

This is a convenient means of representing streamline flow using the dimensionless groups incidental to turbulent flow and including the dimensionless ratio D/L . It should not be inferred, however, that because of this method of representation the heat-transfer coefficient is influenced by the viscosity even though the Reynolds number, which is the criterion of streamline flow, is inversely proportional to the viscosity. The values of μ in Eq. (3.32) actually cancel out.

The Temperature Difference between a Fluid and a Pipe Wall. Before attempting to evaluate the constants of a forced-convection equation from experimental data, an additional factor must be taken into account. When a liquid flows along the axis of a tube and absorbs or transmits sensible heat, the temperature of the liquid varies over the entire length of the pipe. In the case of heat flow through a flat wall, the temperature over the entire area of each face of the flat wall was identical and the temperature difference was simply the difference between any

points on the two faces. If the temperature of the inner circumference of a pipe wall is nearly constant over its entire length, as it might be when the fluid inside the tube is heated by steam, there will be two distinct temperature differences at each end: one between the pipe wall and the inlet liquid and one at the other end between the pipe wall and the heated liquid. What is the appropriate temperature difference for use in the equation $Q = h_i A_i (t_p - t) = h_i A_i \Delta t_i$, where t_p is the constant temperature of the inside pipe wall and t is the varying temperature of the liquid inside the pipe?

Referring to Fig. 3.6, the constant temperature of the inside pipe wall is shown by the horizontal line t_p . If the specific heat is assumed con-

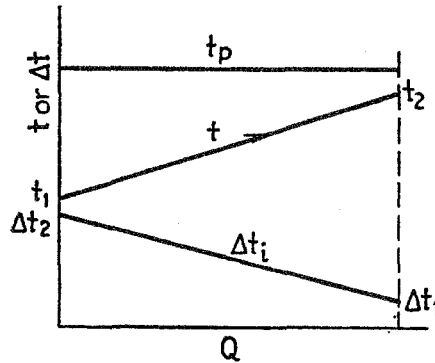


FIG. 3.6. Temperature difference between a fluid and a pipe wall.

stant for the liquid, the rise in temperature is proportional to the total heat received by the liquid in passing from the inlet temperature t_1 to the outlet temperature t_2 and if h_i is considered constant

$$dQ = h_i dA_i \Delta t_i \quad (3.33)$$

The slope of the lower line defining the temperature difference Δt_i as a function of Q is

$$\frac{d \Delta t_i}{dQ} = \frac{\Delta t_2 - \Delta t_1}{Q} \quad (3.34)$$

where $\Delta t_2 = t_p - t_1$ and $\Delta t_1 = t_p - t_2$. Eliminating dQ from Eqs. (3.33) and (3.34),

$$\frac{h_i dA_i}{Q} (\Delta t_2 - \Delta t_1) = \int_{\Delta t_1}^{\Delta t_2} \frac{d \Delta t_i}{\Delta t_i} \quad (3.35)$$

Integrating

$$Q = \frac{h_i A_i (\Delta t_2 - \Delta t_1)}{\ln \Delta t_2 / \Delta t_1} \quad (3.36)$$

The expression $\frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1}$ is the *logarithmic mean temperature difference*, abbreviated LMTD, and the value of h_i , which has been computed from $Q = h_i A_i \Delta t_i$ when Δt_i is the logarithmic mean, is a distinct value of

h_i . If the value of Δt_i were arbitrarily taken as the arithmetic mean of Δt_2 and Δt_1 , the value of h_i would have to be designated to show that it does not correspond to the logarithmic mean. This is usually accomplished by affixing the subscript a or m for arithmetic mean, as h_a or h_m . When Δt_2 and Δt_1 approach equality, the arithmetic and logarithmic means approach each other.

Experimentation and Correlation. Suppose experimental apparatus was available with known diameter and length and through which liquid could be circulated at various measurable flow rates. Suppose, furthermore, that it was equipped with suitable devices to permit measurement of the inlet and outlet liquid temperatures and the temperature of the pipe wall. The heat absorbed by the liquid in flow through the pipe would be identical with the heat passing into the pipe in directions at right angles to its longitudinal axis, or

$$Q = wc(t_2 - t_1) = h_i A_i \Delta t_i \quad (3.37)$$

From the observed values of the experiment and the calculation of Δt_i , as given in Eq. (3.36), h_i can be computed from

$$h_i = \frac{wc(t_2 - t_1)}{A_i \Delta t_i} \quad (3.38)$$

The problem encountered in industry, as compared with experiment, is not to determine h_i but to apply experimental values of h_i to obtain A_i , the heat-transfer surface. The process flow sheet ordinarily contains the heat and material balances about the various items of equipment which enter into a process. From these balances are obtained the conditions which must be fulfilled by each item if the process is to operate as a unit. Thus between two points in the process it may be required to raise the temperature of a given weight flow of liquid from t_1 to t_2 while another fluid is cooled from T_1 to T_2 . The question in industrial problems is to determine how much heat-transfer surface is required to fulfill these process conditions. The clue would appear to be present in Eq. (3.38), except that not only A_i but also h_i is unknown unless it has been established by former experiments for identical conditions.

In order to prepare for the solution of industrial problems it is not practical to run experiments on all liquids and under an infinite variety of experimental conditions so as to have numerical values of h_i available. For example, h_i will differ for a single weight flow of a liquid absorbing identical quantities of heat when the numerical values of t_1 and t_2 differ, since the liquid properties are related to these temperatures. Other factors which affect h_i are those encountered in the dimensional analysis

such as the velocity of the liquid and the diameter of the tube through which heat transfer occurs. It is here that the importance of the equations obtained through dimensional analysis becomes apparent. If the values of the exponents and coefficients of the dimensionless equation for extreme conditions of operation are established from experiments, the value of h_i can be calculated for any intermediate combination of velocity and pipe and liquid properties from the equation.

A typical apparatus for the determination of the heat-transfer coefficient to liquids flowing inside pipes or tubes is shown in Fig. 3.7. The principal part is the test exchanger, which consists of a test section of pipe enclosed by a concentric pipe. The annulus is usually connected

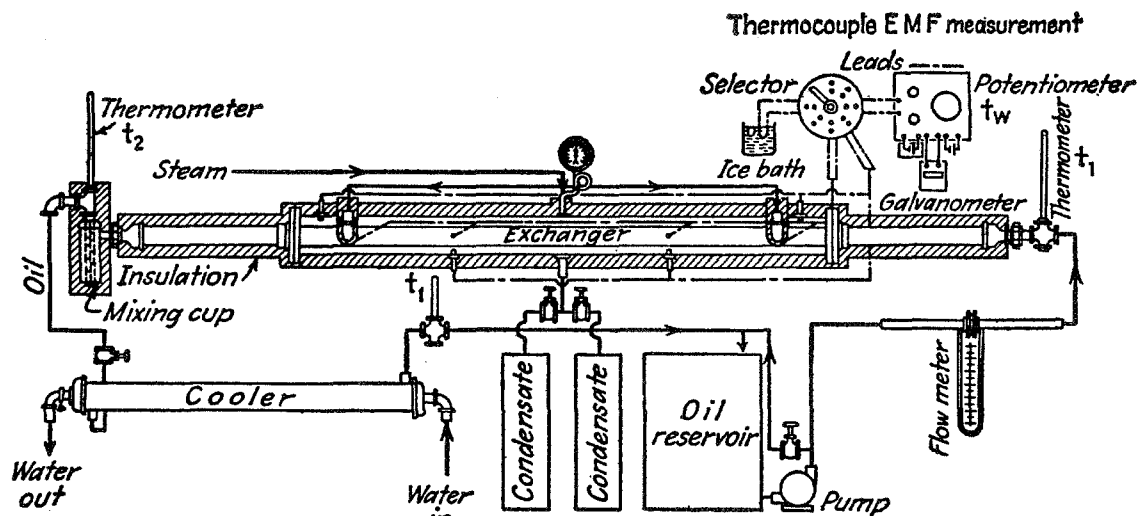


FIG. 3.7. Test apparatus for the tube transfer coefficient.

to permit the condensation of steam for liquid-heating experiments or the rapid circulation of water for liquid-cooling experiments. The auxiliary exchanger is connected to perform the opposing operation of the test section and cools when the test section is used for heating. For a heating experiment, cold liquid from the reservoir is circulated through the circuit by a centrifugal pump. A by-pass is included on the pump discharge to permit the regulation of flow through the flow meter. The liquid then passes through a temperature-measuring device, such as a calibrated thermometer or thermocouple, where t_1 is obtained. t_1 is taken at some distance in the pipe before the test section, so that the temperature-measuring device does not influence the convection eddies in the test section itself. The liquid next passes through the test section and another unheated length of pipe before it is mixed and its temperature t_2 is measured. The unheated extensions of the test pipe are referred to as *calming sections*. Next the liquid passes through a cooler where its temperature is returned to t_1 . The annulus of the test section is con-

nected to calibrated condensate drain tanks where the heat balance can be checked by measuring the steam, Dowtherm (a fluid which permits the attainment of high vapor temperatures at considerably lower pressures than are obtained with steam), or other vapor quantity. The pressure of steam can be adjusted by a pressure-reducing valve, and in the event that the heating vapor does not have a well-established temperature-pressure saturation curve, thermocouples or thermometers can also be inserted in the shell. The temperature of the *outside* surface of the heated length of the pipe is obtained by attaching a number of thermocouples about the pipe surface and obtaining their average temperature. The thermocouples may be calibrated by circulating preheated oil through the tube while the annulus outside the test pipe is kept under vacuum. The temperature at the outside of the test-pipe surface can then be calculated from the uniform temperature of the preheated oil after correcting for the temperature drop through the pipe wall and calibrating temperature against emf by means of a potentiometer. The leads from the pipe-surface thermocouples are brought out through the gaskets at the ends of the test section.

The performance of the experiment requires the selection of an initial reservoir temperature t_1 which may be attained by recirculating the liquid through the exchanger at a rapid rate until the liquid in the reservoir reaches the desired temperature. A flow rate is selected and cooling water is circulated through the cooler so that the temperature of the oil returning to the reservoir is also at t_1 . When steady conditions of t_1 and t_2 have persisted for perhaps 5 min or more, the temperatures t_1 and t_2 are recorded along with the flow rate, readings of the pipe-surface thermocouples, and gain in condensate level during the test interval. With versatile apparatus using good regulating valves an experiment can usually be carried out in less than an hour.

Several important points must be safeguarded in the design of experimental apparatus if consistent results are expected. The covers at the ends of the test section cannot be connected directly to the test pipe, since they act as gatherers of heat. Should they touch the test pipe in a metal-to-metal contact, they add large quantities of heat in local sections. To prevent inaccuracies from this source the covers and the test pipe should be separated by a nonconducting packing gland. Another type of error results from the accumulation of air in the annulus, which prevents the free condensation of vapor on the test pipe. This usually is apparent if there is a lack of uniformity in the readings of the pipe-surface thermocouples and annulus thermocouples when the latter are employed. This can be overcome by providing a bleed for the removal of any trapped air. Problems incidental to the installation and calibration of thermocouples

and thermometers appear elsewhere.¹ The same is true of equations to correct for fluid flow through standard orifices when the properties of the liquid vary.

Evaluation of a Forced Convection Equation from Experimental Data. As an example of correlation, data are given in Table 3.3 which were obtained by Morris and Whitman² on heating gas oil and straw oil with steam in a $\frac{1}{2}$ -in. IPS pipe with a heated length of 10.125 ft.

The viscosity data are given in Fig. 3.8 and are taken from the original publication. The thermal conductivities may be obtained from Fig. 1, and the specific heats from Fig. 4. Both are plotted in the Appendix

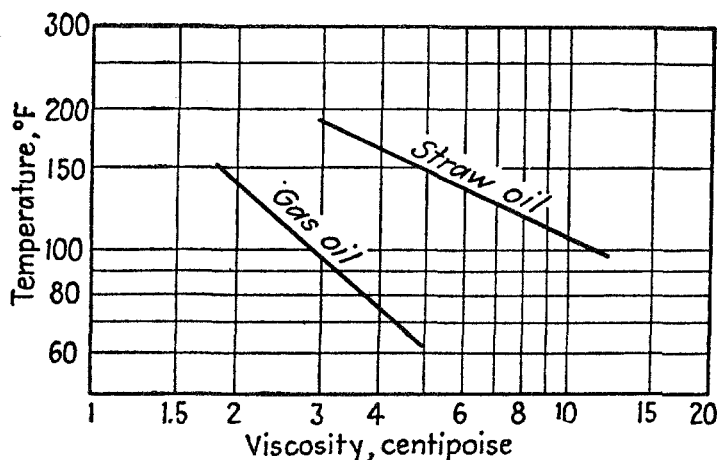


FIG. 3.8. Viscosities of test oils.

with °API as parameters. The thermal conductivity of the metal was taken by Morris and Whitman to be 35 Btu/(hr)(ft²)(°F/ft) and constant, although this is higher than the value given in Appendix Table 3. Only columns 2, 3, 4, and 5 in Table 3.3 were observed where

t_1 = oil inlet temperature, °F

t_2 = oil outlet temperature, °F

t_w = temperature of *outer* pipe surface averaged from thermocouples

w = weight flow, lb/hr

The first step in correlating a forced convection equation is to determine that the data are in turbulent flow, otherwise an attempted correlation by Eq. (3.26) would be incorrect. In column 11 the Reynolds numbers have been calculated using the diameter and flow area of a $\frac{1}{2}$ -in. IPS pipe which may be found in Table 11. Fluid properties have been obtained at the mean temperature $(t_1 + t_2)/2$. The Reynolds numbers all exceed

¹ American Institute of Physics, "Temperature: Its Measurement and Control in Science and Industry," Reinhold Publishing Corporation, New York, 1941.

² Morris, F. H., and W. G. Whitman, *Ind. Eng. Chem.*, **20**, 234 (1928).

TABLE 3.3. DATA OF MORRIS AND WHITMAN
Runs Heating 36.8°API Gas Oil with Steam

(1) Run No.	(2) w	(3) t_1	(4) t_2	(5) t_w	(6) Q	(7) Δt_i	(8) h_i	(9) $\frac{h_i D}{k}$	(10) G	(11) $\frac{DG}{\mu}$	(12) $\frac{c\mu}{k}$	(13) $j_H = \frac{Nu}{Pr}$	(14) $j_H = \frac{Nu}{Pr^{1/3}}$
B1	722	77.1	106.9	210.1	10,150	115.7	53.6	35.5	342,000	2,280	47.2	0.75	9.83
B2	890	77.9	109.3	209.0	13,150	112.6	71.4	46.3	421,000	2,825	46.7	0.99	12.85
B3	1056	85.6	117.6	208.9	16,100	104.0	94.5	62.3	501,000	3,710	43.3	1.44	17.75
B4	1260	89.8	121.9	208.0	19,350	98.5	120	79.5	597,000	4,620	41.4	1.92	23.0
B5	1497	91.6	123.3	207.5	22,700	96.0	144	95	709,000	5,780	40.7	2.33	27.7
B6	1802	99.1	129.2	207.2	26,200	88.5	181	120.5	854,000	7,140	38.7	3.12	35.5
B7	2164	102.3	131.7	206.9	30,900	84.7	223	147.5	1,026,000	8,840	37.7	3.91	44.0
B8	2575	106.5	134.3	206.4	35,000	80.3	266	176.5	1,220,000	10,850	36.5	4.83	53.2
B9	3265	111.5	137.1	205.0	41,100	74.2	338	223	1,548,000	14,250	35.3	6.32	68.0
B10	3902	113.9	138.2	203.8	46,600	70.6	403	266.5	1,850,000	17,350	35.1	7.60	81.4
B11	4585	116.8	139.7	203.0	51,900	66.9	474	313	2,176,000	20,950	34.1	9.18	96.5
B12	5360	122.2	142.9	202.9	54,900	62.3	538	356	2,538,000	25,550	32.9	10.82	111.2
B13	6210	124.8	144.1	202.2	59,500	59.1	615	407	2,938,000	30,000	32.7	12.43	127.5

Runs Heating 29.4°API Straw Oil with Steam

C8	2900	100.0	115.4	206.3	20,700	95.8	132	87.5	1,375,000	3,210	133	0.66	17.2
C9	2920	86.7	99.3	208.0	16,800	112.7	91.1	60.4	1,387,000	2,350	179	0.34	10.7
C10	3340	101.7	117.6	206.0	24,800	92.9	163	108	1,585,000	3,820	129.5	0.83	21.4
C11	3535	100.5	115.7	205.5	25,000	94.0	162	107.5	1,675,000	3,880	133.3	0.81	21.1
C12	3725	163.0	175.1	220.1	22,600	47.9	288	191	1,767,000	10,200	57.8	3.30	49.5
C13	3810	160.5	173.6	220.5	24,900	50.0	304	201.5	1,805,000	10,150	59.3	3.39	51.6
C14	3840	109.0	124.4	205.7	27,800	85.2	199	131.5	1,820,000	4,960	115	1.15	27.1
C15	4730	112.0	127.3	205.3	34,100	81.0	257	170	2,240,000	6,430	110	1.55	35.5
C16	5240	154.9	167.6	217.7	33,100	51.9	389	257.5	2,485,000	13,150	62.9	4.08	64.7
C17	5270	150.9	164.3	217.0	34,900	54.7	389	257.5	2,500,000	12,520	65.6	3.92	63.7
C18	5280	142.3	156.8	216.7	37,500	62.0	369	244	2,510,000	11,250	72.6	3.36	58.5
C19	5320	132.3	148.1	215.9	40,600	70.2	353	234	2,520,000	9,960	81.5	2.87	54.1
C20	5620	118.8	133.1	204.7	38,200	73.5	317	210	2,660,000	8,420	100.4	2.09	45.2
C21	6720	122.2	135.9	204.3	43,900	69.2	387	256	3,185,000	10,620	95.6	2.68	56.0
C22	8240	124.1	137.0	204.4	47,900	67.3	434	287	3,905,000	12,650	93.3	3.08	63.3

2100 in substantiation of turbulent flow. Equation (3.26) is given as

$$\frac{h_i D}{k} = \alpha \left(\frac{DG}{\mu} \right)^p \left(\frac{c\mu}{k} \right)^q$$

and α , p , and q can be found algebraically by taking the data for three test points.

Algebraic Solution. This method of correlation is demonstrated by using the three points B4, B12, and C12 in Table 3.3 which include a large range of $h_i D/k$, DG/μ , and $c\mu/k$ as calculated presently from flow and fluid properties and tabulated in columns 9, 11, and 12.

$$\begin{aligned} \frac{h_i D}{k} &= \alpha \left(\frac{DG}{\mu} \right)^p \left(\frac{c\mu}{k} \right)^q \\ \text{C12: } 191 &= \alpha (10,200)^p (57.8)^q \\ \text{B12: } 356 &= \alpha (25,550)^p (32.9)^q \\ \text{B4: } 79.5 &= \alpha (4,620)^p (41.4)^q \end{aligned}$$

Taking the logarithms of both sides,

$$\begin{aligned} \text{C12: } 2.2810 &= \log \alpha + 4.0086p + 1.7619q \\ \text{B12: } 2.5514 &= \log \alpha + 4.4065p + 1.5172q \\ \text{B4: } 1.9004 &= \log \alpha + 3.6646p + 1.6170q \end{aligned}$$

Eliminating the unknowns one by one gives $\alpha = 0.00682$, $p = 0.93$, and $q = 0.407$, and the final equation is

$$\frac{h_i D}{k} = 0.00892 \left(\frac{DG}{\mu} \right)^{0.93} \left(\frac{c\mu}{k} \right)^{0.407}$$

When the equation is to be used frequently, it can be simplified by fixing q as the cube root of the Prandtl number and solving for new values of α and p . The simplified equation would be

$$\frac{h_i D}{k} = 0.0089 \left(\frac{DG}{\mu} \right)^{0.965} \left(\frac{c\mu}{k} \right)^{1/3}$$

Graphical Solution. For the correlation of a large number of points the graphical method is preferable. Rewriting Eq. (3.26),

$$\frac{h_i D}{k} \left(\frac{c\mu}{k} \right)^{-q} = \alpha \left(\frac{DG}{\mu} \right)^p \quad (3.39)$$

which is an equation of the form

$$y = \alpha x^p \quad (3.40)$$

Taking the logarithms of both sides,

$$\log y = \log \alpha + p \log x$$

which reduces on logarithmic coordinates to an equation of the form

$$y = \alpha + px \quad (3.41)$$

On logarithmic coordinates the entire group $(h_i D/k)(c\mu/k)^{-q}$ is the ordinate y in Eq. (3.41), the Reynolds number is x , p is the slope of the data when plotted as y vs. x , and α is the value of the intercept when

$$p \log x = 0$$

which occurs when the Reynolds number is 1.0. To plot values of $j_H = (h_i D/k)(c\mu/k)^{-q}$, the exponent q must be assumed. The most satisfactory assumed value of the exponent will be the one which permits the data to be plotted with the smallest deviation from a straight line. A value of q must be assumed for the entire series of the experiments, and j_H computed accordingly. This is a more satisfactory method than an algebraic solution, particularly when a large number of tests on different oils and pipes are being correlated. If q is assumed too large, the data will scatter when plotted as y vs. x . If q is assumed too small, the data will not scatter but will give a large deviation by producing a curve. Preparatory to a graphical solution, run B1 in Table 3.3 will be computed completely from observed data alone. Run B1 consists of a test employing a 36.8°API gas oil in a ½-in. IPS pipe.

Observed test data:

Weight flow of gas oil, $w = 722$ lb/hr

Temperature of oil at pipe inlet, $t_1 = 77.1^\circ\text{F}$

Temperature of oil at pipe outlet, $t_2 = 106.9^\circ\text{F}$

Average temperature of outside pipe surface, $t_w = 210.1^\circ\text{F}$

Physical data and calculated results:

Heat load, Btu/hr:

$$\text{Average oil temperature} = \frac{77.1 + 106.9}{2} = 92.0^\circ\text{F}$$

Average specific heat, $c = 0.472$ Btu/(lb)(°F)

$$Q = wc(t_2 - t_1) = 722 \times 0.472(106.9 - 77.1) = 10,150 \text{ Btu/hr}$$

Temperature of pipe at inside surface, t_p :

I.D. of ½-in. IPS = 0.62 in.; O.D. = 0.84 in.

Length, 10.125 ft; surface, 1.65 ft²

Thermal conductivity of steel, 35 Btu/(hr)(ft²)(°F/ft)

$$Q \text{ per lin ft, } q = \frac{10,150}{10.125} = 1007 \text{ Btu}$$

$$t_p = t_w - \frac{2.3q}{2\pi k} \log \frac{D_2}{D_1} = 210.1 - \frac{2.3 \times 1007}{2 \times 3.14 \times 35} \log \frac{0.84}{0.62} = 208.7^\circ\text{F}$$

Δt_i in expression $Q = h_i A_i \Delta t_i$:

$$\text{At inlet, } \Delta t_2 = 208.7 - 77.1 = 131.6^\circ\text{F}$$

$$\text{At outlet, } \Delta t_1 = 208.7 - 106.9 = 101.8^\circ\text{F}$$

$$\Delta t_i = \text{LMTD} = \frac{131.6 - 101.8}{2.3 \log (131.6/101.8)} = 115.7^\circ\text{F} \quad (3.36)$$

$$h_i = \frac{Q}{A_i \Delta t_i} = \frac{10,150}{1.65 \times 115.7} = 53.6 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

The thermal conductivity of the oil will be considered constant at 0.078 Btu/(hr)(ft²)(°F/ft).

$$\text{Nusselt number, } Nu = \frac{h_i D}{k} = \frac{53.6 \times 0.62}{0.078 \times 12} = 35.5, \text{ dimensionless}$$

$$\text{Mass velocity, } G = \frac{w}{\pi D^2/4} = \frac{722}{(3.14 \times 0.62^2)/(4 \times 12^2)} = 342,000 \text{ lb/(hr)(ft}^2\text{)}$$

The viscosity from Fig. 3.8 at 92°F is 3.22 centipoises [gram-mass/100 (cm)(sec)] or $3.22 \times 2.42 = 7.80 \text{ lb/(ft)(hr)}$.

$$\text{Reynolds number, } Re = \frac{DG}{\mu} = \frac{0.62}{12} \times 342,000 \times \frac{1}{7.80} = 2280, \text{ dimensionless}$$

$$\text{Prandtl number, } Pr = \frac{c\mu}{k} = \frac{0.472 \times 7.80}{0.078} = 47.2, \text{ dimensionless}$$

Assume values of q of 1.0 and $1/3$ respectively.

First trial: $j_H = Nu/Pr = 0.75$ plotted in Fig. 3.9

Second trial: $j_H = Nu/Pr^{1/3} = 9.83$ plotted in Fig. 3.10

The values of the first trial in which the ordinate is $j_H = \frac{h_i D}{k} / \left(\frac{c\mu}{k} \right)$ for an assumed value of $q = 1$ are plotted in Fig. 3.9 where two distinct lines result, one for each of the oils. It is the object of a good correlation to provide but one equation for a large number of liquids and this can be accomplished by adjusting the exponent of the Prandtl number. By assuming a value of $q = 1/3$ and plotting the ordinate $j_H = \frac{h_i D}{k} / \left(\frac{c\mu}{k} \right)^{1/3}$, it is possible to obtain the single line as shown in Fig. 3.10. By drawing

the best straight line through the points of Fig. 3.10, the slope can be measured in the same way as on rectangular coordinates, which in this particular case is found to be 0.90. By extrapolating the straight line

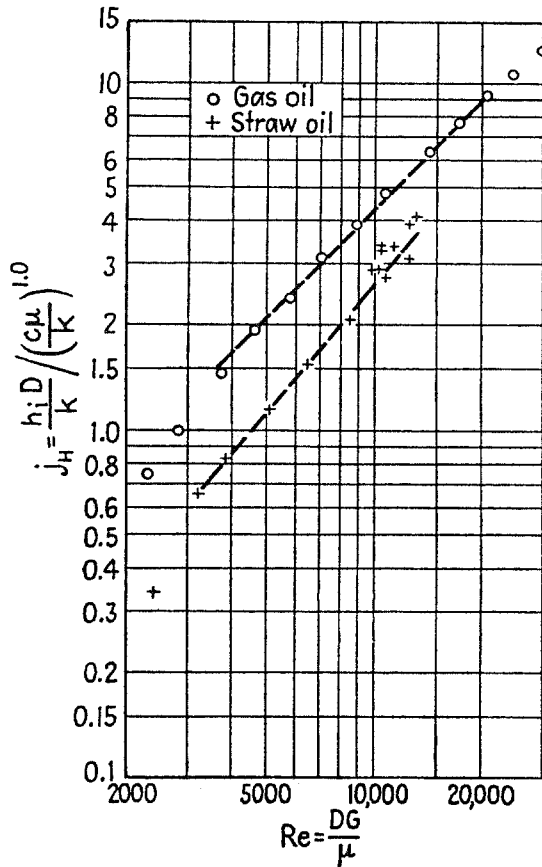


FIG. 3.9. Plot of Re vs. j_H with Prandtl exponent of 1.0.

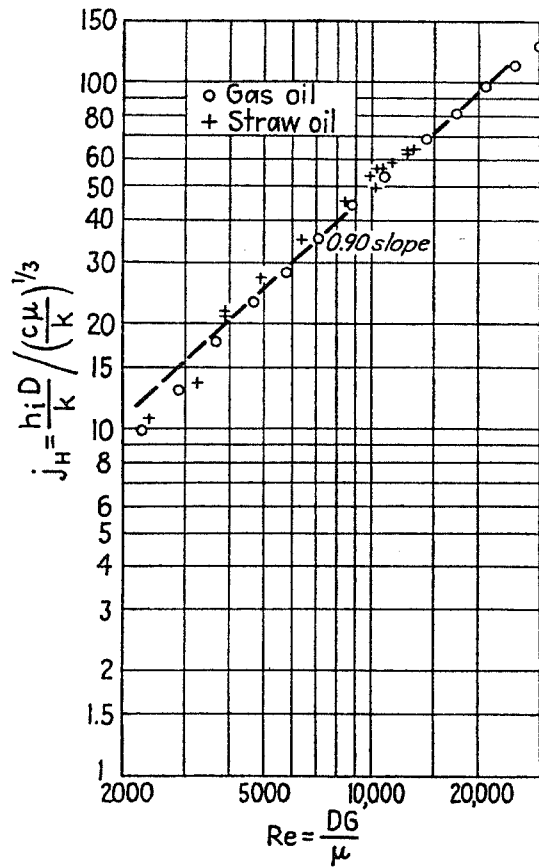


FIG. 3.10. Plot of Re vs. j_H with Prandtl exponent of $\frac{1}{3}$.

until the Reynolds number is 1.0, a value of $\alpha = 0.0115$ is obtained as the intercept. The equation for all the data is thus

$$\frac{h_i D}{k} = 0.0115 \left(\frac{DG}{\mu} \right)^{0.90} \left(\frac{c \mu}{k} \right)^{\frac{1}{3}} \quad (3.42)$$

A value of $q = 0.40$ would cause less bowing and a smaller deviation. The correlation of data need not be confined to heating or cooling runs on liquids separately. It is entirely possible to combine both types of data into a single correlation, called the isothermal heat-transfer equation, but the procedure involves an additional consideration which is deferred until Chap. 5.

Correlation of Fluid Friction in Pipes. When a fluid flows in a pipe isothermally, it undergoes a decrease in pressure. From Eq. (3.16) it is seen that for isothermal turbulent flow this pressure drop is a function of the Reynolds number and, in addition, to the roughness of the pipe

Rewriting Eq. (3.16) in dimensionless form,

$$f' = \frac{\Delta P g \rho D}{G^2 L} = \alpha \left(\frac{DG}{\mu} \right)^{-a}$$

where f' is one of the dimensionless factors found in the literature to designate the *friction factor* and ΔP is the pressure drop in pounds per square foot. For combination with other hydrodynamic equations it is more convenient to use a friction factor f so that

$$f = \frac{\Delta P 2g\rho D}{4G^2 L} = \frac{\alpha}{Re^a} \quad (3.43)$$

When experimental data are available, it is thus convenient to obtain a correlation by plotting f as a function of the Reynolds number and the conventional Fanning equation as shown in Fig. 3.11. The Fanning equation comprises the first and second terms of Eq. (3.43) and is usually written as $\Delta F = \Delta P/\rho$ where ΔF is the pressure drop expressed in *feet of liquid*, or

$$\Delta F = \frac{4fG^2 L}{2g\rho^2 D} \quad (3.44)$$

For the portion of the graph corresponding to streamline flow

$$(Re < 2100 \text{ to } 2300),$$

the equation for the pressure drop may be deduced from theoretical considerations alone and has been verified by experiment. The equation is

$$\Delta F = \frac{32\mu G}{g\rho^2 D^2} \quad (3.45)$$

By equating (3.44) and (3.45), since each applies at the transition point from streamline to turbulent flow, the equation of this line, known as the Hagen-Poiseuille equation, where f is used with Eq. (3.44), is

$$f = \frac{16}{DG/\mu} \quad (3.46)$$

To the right of the transition region in turbulent flow there are two lines, one for commercial pipes and the other for tubes. Tubes have smoother finishes than pipes and therefore give lower pressure drops when all other factors are the same. This is not so of streamline flow where the fluid at the pipe or tube wall is assumed to be stationary or nearly stationary and the pressure drop is not influenced by roughness. The equation of f in Eq. (3.44) for fluids in tubes in turbulent flow is given by

Drew, Koo, and McAdams¹ within ± 5 per cent by

$$f = 0.00140 + \frac{0.125}{(DG/\mu)^{0.32}} \quad (3.47a)$$

For clean commercial iron and steel pipes an equation given by Wilson,

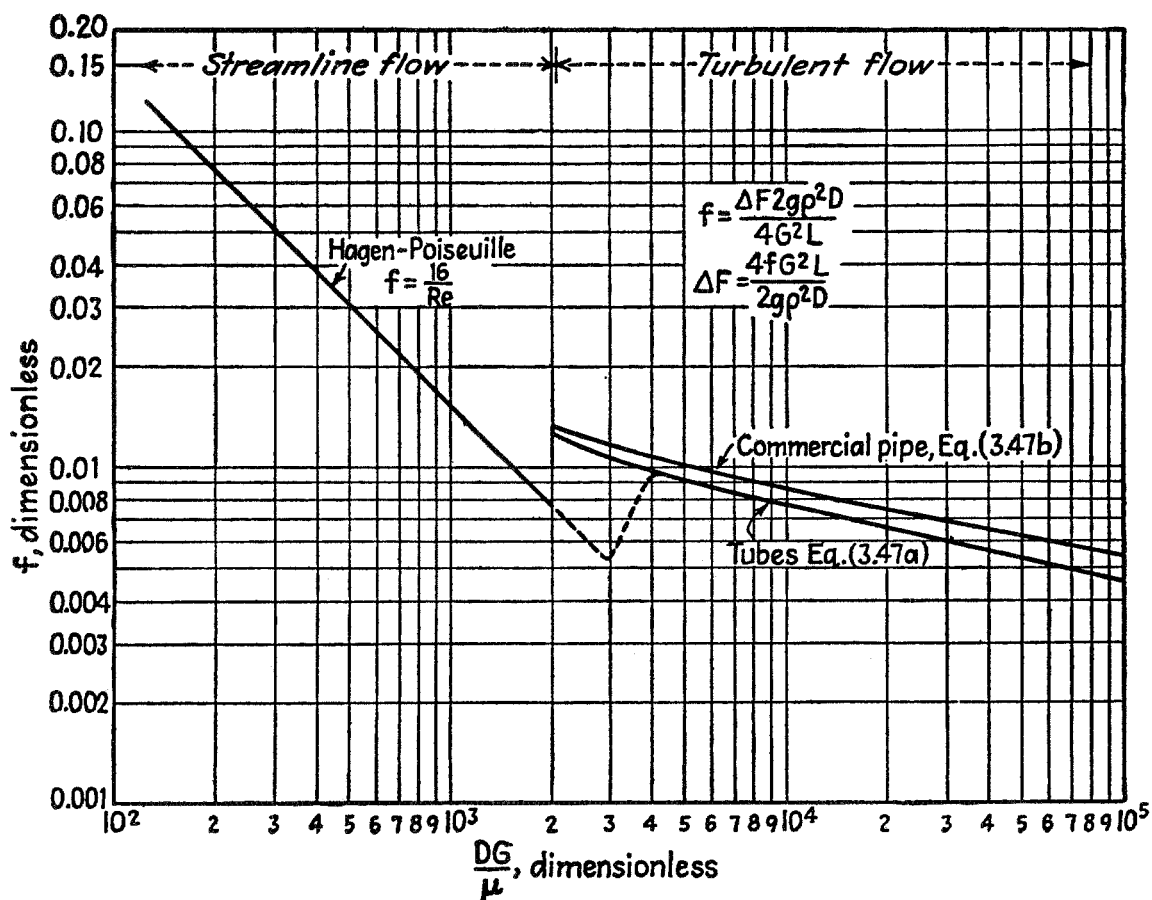


FIG. 3.11. Friction factors for flow in pipes and tubes.

McAdams, and Seltzer² within ± 10 per cent is

$$f = 0.0035 + \frac{0.264}{(DG/\mu)^{0.42}} \quad (3.47b)$$

It can be seen that, if the transition from streamline to turbulent flow is given by $Du\rho/\mu = 2300$, approximately, then the velocity at which a fluid in a pipe changes from streamline to turbulent flow is

$$u_{\text{crit}} = \frac{2300\mu}{D\rho}$$

For water flowing in a 1-in. IPS pipe at 100°F, the viscosity is 0.72 centi-

¹ Drew, T. B., E. C. Koo, and W. H. McAdams, *Trans. AIChE*, **28**, 56-72 (1932).

² Wilson, R. E., W. H. McAdams, and M. Seltzer, *Ind. Eng. Chem.*, **14**, 105-119 (1922).

poise or $0.72 \text{ gram-mass} \times 100 \text{ (cm)(sec)}$ or $0.72 \times 2.42 = 1.74 \text{ lb/(ft)(hr)}$, and the inside diameter of the pipe is 1.09 in., or $1.09/12 = 0.091 \text{ ft}$.

$$u_{\text{crit}} = \frac{2300 \times 1.74}{0.091 \times 62.3} = 707 \text{ ft/hr, or } 0.196 \text{ fps}$$

For air at 100°F the viscosity is $0.0185 \times 2.42 = 0.0447 \text{ lb/(ft)(hr)}$ and the density is approximately 0.075 lb/ft^3 . Then for the same pipe

$$u_{\text{crit}} = \frac{2300 \times 0.0447}{0.091 \times 0.075} = 15,100 \text{ ft/hr, or } 4.19 \text{ fps}$$

The Reynolds Analogy. Both heat transfer and fluid friction in turbulent flow have been treated empirically, whereas their streamline-

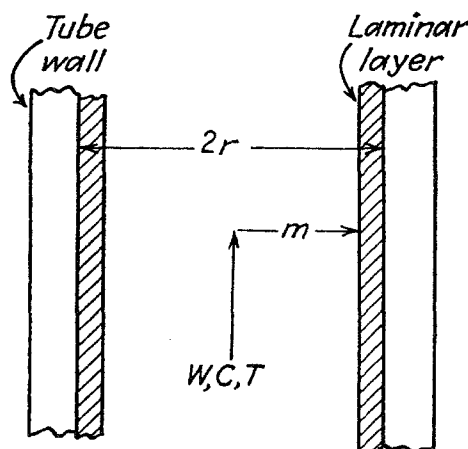


FIG. 3.12. Momentum transfer between a fluid and a boundary layer.

flow equivalents may be studied theoretically with reasonable accuracy. Turbulent flow is of greater importance to industry, yet this empiricism exists because of the lack of simple mathematics by which derivations might be obtained. As early as 1874, however, Osborne Reynolds¹ pointed out that there was probably a relationship between the transfer of heat and the fluid friction between a hot fluid and a surface.

There are several advantages which might accrue from an equation relating heat transfer and fluid friction. For the most part fluid-friction experiments are simpler to perform than are heat-transfer experiments, and the understanding of both fields could be increased by experimentation in either. The fundamentals of the mechanism of both might also be better understood if they were directly related. The analogy between the two is made possible by the fact that the transfer of heat and the transfer of fluid momentum can be related. A simplified proof follows.

Referring to Fig. 3.12 a fluid in amount $W \text{ lb/hr}$ and specific heat C flows through a tube of radius r . In the section of the pipe between L and

¹ Reynolds, *op. cit.*, pp. 81-85.

$L + dL$ the temperature of the fluid is T , and the temperature at the inside surface of the pipe wall is t_p . Of the total fluid flow along the axis of the pipe, assume that m lb/(hr)(ft²) impinges on the pipe wall where its velocity falls to zero and its temperature approaches the temperature of the pipe wall, t_p . Each particle of fluid which thus contacts the wall gives up its axial momentum and upon returning to the main body of the fluid has its axial momentum renewed at the expense of the energy of the main stream. The constant loss and renewal of momentum are the cause of the pressure drop. The traction or drag on unit area of the pipe wall is obtained by equating the drag on unit length of pipe wall to the product of the pressure gradient and the cross-sectional area of the pipe, which reduces to

$$\tau = \frac{f}{2} \rho u^2 \quad (3.48)$$

where τ is the drag. Since the drag is assumed to be equal to the loss of momentum of the fluid,

$$mu = \tau = \frac{f}{2} \rho u^2 \quad (3.49)$$

The rate of heat transfer between the fluid and the wall is given by

$$mC2\pi r dL(T - t_p) = WC dT \quad (3.50)$$

or from Eqs. (3.49) and (3.50)

$$m = \frac{WC dT}{2\pi r dL(T - t_p)} = \frac{\tau}{u} \quad (3.51)$$

In simple terms the last two members of Eq. (3.51) state

Heat actually given up to pipe wall

Total heat available to be given up

$$\begin{aligned} &= \frac{\text{momentum lost by skin friction to pipe wall}}{\text{total momentum available}} \\ &= \text{a constant} \end{aligned}$$

Rewriting Eq. (3.50) to include the heat-transfer coefficient,

$$WC dT = h_i 2\pi r dL(T - t_p) \quad (3.52)$$

$$h_i = \frac{WC dT}{2\pi r dL(T - t_p)} = \frac{\tau C}{u} = \frac{f}{2} \rho u C = \frac{fCG}{2} \quad (3.53)$$

or in dimensionless form,

$$\frac{h_i}{CG} = \frac{f}{2}$$

It is interesting to note in Eq. (3.53) that an equation has been obtained for the heat-transfer coefficient which involves the friction factor and

which can be determined from an experiment in which no heat was transferred. Like most derivations which require a number of assumptions, the use of Eq. (3.53) applies only to a small range of fluids, particularly permanent gases.

It was Reynolds as quoted by Stanton¹ who predicted that the coefficient of heat transfer obtained from Eq. (3.53) should be affected by the ratio of the thermal conductivity and viscosity of a specific fluid. While the presence of the thermal conductivity suggests the influence of conduction, it was observed experimentally that the entire cross section of a fluid in turbulent flow is not turbulent. Instead it has been found that a laminar layer exists near the pipe wall through which conduction must occur. Prandtl² and Taylor³ independently included this layer. If it is assumed that the layer has a thickness b and that the temperature of its inner circumference is t' , the heat flow per square foot of layer is given by

$$Q = \frac{k(t' - t_p)}{b} \quad (3.54)$$

where k is the mean conductivity for the layer. Assume that the transfer of heat and momentum is carried through this layer by molecular motion without disturbing its laminar flow. The inner surface of the layer moves with a velocity u' in streamline flow, and writing $(u - u')$ for u in Eq. (3.53),

$$Q = h_i(T - t') = \frac{\tau C(T - t')}{(u - u')} \quad (3.55)$$

From the definition of viscosity given by Eq. (3.4),

$$\tau = \mu \frac{u'}{b} \quad (3.56)$$

where μ is the viscosity of the fluid in the layer. From Eq. (3.54),

$$t' - t_p = \frac{Qb}{k} = \frac{\mu u' Q}{\tau k} \quad (3.57)$$

And from Eq. (3.55),

$$T - t' = \frac{Q(u - u')}{\tau C} \quad (3.58)$$

Combining Eqs. (3.57) and (3.58),

$$T - t_p = Q \left(\frac{\mu u'}{\tau k} + \frac{u}{\tau C} - \frac{u'}{\tau C} \right) \quad (3.59)$$

¹ Stanton, T. E., *Phil. Trans. Roy. Soc. (London)*, A 190, 67-88 (1897).

² Prandtl, L., *Physik. Z.*, 29, 487-489 (1928).

³ Taylor, G. I., *Brit. Adv. Comm. Aero., Rept. and Memo* 272 (1917).

The corrected value of h_i becomes

$$h_i = \frac{Q}{(T - t_p)} = \frac{\tau}{\frac{\mu u'}{k} + \frac{u}{C} - \frac{u'}{C}} \quad (3.60)$$

Substituting r' for the ratio u'/u and eliminating τ by means of Eq. (3.48),

$$h_i = \frac{f CG}{2 [1 - r' + r'(C\mu/k)]} \quad (3.61)$$

or in dimensionless form using c for C and h for h_i as usually given in the literature,

$$\frac{h}{cG} = \frac{f}{2} \frac{1}{1 - r' + r'(c\mu/k)} \quad (3.62)$$

Equation (3.61) is the Prandtl modification of the Reynolds analogy, which is sometimes called the *Prandtl analogy*. The dimensionless Prandtl group $c\mu/k$ has appeared earlier in this chapter, and when it is numerically equal to 1.0, Eq. (3.61) reduces to Eq. (3.53). This is approximately the case for permanent gases. While Eq. (3.61) is a notable extension of the Reynolds analogy, it too has definite limitations. Modern theory now presumes that the distribution of velocities no longer ends abruptly at the laminar layer but that there is instead a buffer layer within the laminar layer in which the transition occurs. Other extensions of the analogies also appear in the literature.

PROBLEMS

3.1. The heat-transfer coefficient h from a hot horizontal pipe to a gas by free convection has been found to be influenced by the specific heat c , thermal conductivity k , density ρ , viscosity μ , thermal coefficient of expansion β of the gas, pipe diameter D , gravitational constant g , and the temperature difference Δt between the surface of the pipe and the main body of the gas. Establish the form of a dimensionless equation for the heat-transfer coefficient.

3.2. The heat-transfer coefficient for the condensation of a vapor on a horizontal pipe has been found to be influenced by the diameter of the pipe D , acceleration of gravity g , the temperature difference Δt between the saturated vapor and the pipe wall, thermal conductivity k , latent heat of vaporization λ , viscosity μ , and the density ρ of the vapor. Establish a dimensionless expression for the heat-transfer coefficient.

3.3. The rate of cooling of a hot solid in still air has been found to be influenced by the specific heat c , thermal conductivity k , density ρ and viscosity μ of the gas, the length of the solid l , and the temperature difference Δt between the surface of the solid and the bulk temperature of the gas. Establish a dimensionless equation for the rate of cooling h .

3.4. When a fluid flows around a sphere, the force exerted by the fluid has been

found to be a function of the viscosity μ , density ρ and velocity u of the gas, and the diameter D of the sphere. Establish an expression for the pressure drop of the fluid as a function of the Reynolds number of the gas.

3.5. The heating of gas oil and straw oil in a pipe has been found for a $\frac{1}{2}$ -in. IPS pipe to follow Eq. (3.42).

$$\frac{h_i D}{k} = 0.0115 \left(\frac{DG}{\mu} \right)^{0.90} \left(\frac{c\mu}{k} \right)^{1/3}$$

It is desired to circulate 5600 lb/hr of the 36.8°API gas oil through a 1-in. IPS pipe while its temperature is raised from 110 to 130°F. In the absence of any additional heat-transfer data, calculate the heat-transfer coefficient in the 1-in. IPS pipe. How does this compare with the value of h_i when the same quantity of gas oil flows in a $\frac{1}{2}$ -in. IPS pipe as calculated by the same equation? The data will be found in the illustration in the chapter.

3.6. Using Eq. (3.42), it is desired to circulate 4000 lb/hr of amyl acetate through a $\frac{3}{4}$ -in. IPS pipe while its temperature is raised from 130 to 150°F. (a) From data available in the Appendix on the physical properties of amyl acetate calculate the heat-transfer coefficient. It may be necessary to extrapolate some of the data. (b) Do the same for 6000 lb/hr of ethylene glycol in the same pipe when heated from 170 to 200°F.

If only one point is given for a property, such as the thermal conductivity, and it is lower than the average temperature, its use will actually introduce a slight factor of safety.

3.7. On a $\frac{1}{2}$ -in. IPS pipe 10.125 ft long Morris and Whitman reported the following for heating water in a pipe while steam is circulated on the outside. Note that G' , the reported mass velocity, is not in consistent units.

G' , lb/(sec)(ft ²)	t_1 , °F	t_2 , °F	t_w , °F
58.6	91.6	181.5	198.4
60.5	92.7	180.3	198.0
84.3	102.2	175.3	196.5
115	103.1	171.3	194.5
118	103.4	168.2	194.9
145	105.0	165.9	194.0
168	107.2	163.3	192.4
171	106.7	164.6	191.1
200	108.5	160.1	190.0
214	106.3	158.9	188.3
216	110.1	160.2	190.2
247	107.6	158.2	186.3

Viscosities and conductivities can be found in the Appendix. The specific heat and gravity should be taken as 1.0. Establish an equation of the form of Eq. (3.26) using all the data. (*Hint.* To save time in the selection of the exponent of the Prandtl number, take three random points such as the first, last, and an intermediate one and solve algebraically.)

3.8. On cooling a 35.8°API gas oil with water Morris and Whitman reported the following for 10.125 ft of $\frac{1}{2}$ -in. IPS pipe:

G' , lb/(sec)(ft ²)	t_1 , °F	t_2 , °F	t_w , °F
82.6	150.3	125.5	66.0
115	138.7	118.7	68.0
164	130.5	113.9	70.0
234	141.2	124.2	79.2
253	210.0	179.8	115.4
316	197.6	173.3	122.9
334	132.3	119.4	82.1
335	191.1	168.4	116.3
413	194.4	173.0	121.6
492	132.5	122.1	89.5
562	200.2	182.4	139.4
587	188.6	171.9	127.7
672	190.0	175.1	140.7
682	191.8	176.4	139.5
739	132.6	124.6	97.2

The viscosity of the oil is 2.75 centipoises at 100°F and 1.05 centipoises at 200°F. To obtain intermediate values of the viscosity plot temperature vs. viscosity on logarithmic paper. Establish an equation of the form of Eq. (3.26) using all the data. (The hint given in Prob. 3.7 is applicable.)

3.9. On the cooling of the 29.4°API straw oil with water Morris and Whitman reported the following for 10.125 ft of ½-in. IPS pipe:

G' , lb/(sec)(ft ²)	t_1 , °F	t_2 , °F	t_w , °F
141	362.3	296.3	139.8
143	477.6	385.7	194.8
165	317.8	270.0	116.9
172	163.7	148.2	60.2
252	140.5	130.1	62.9
292	142.3	131.5	66.9
394	133.0	124.6	69.6
437	244.9	218.7	141.4
474	130.6	123.1	71.7
474	376.1	330.6	232.9
485	181.1	166.5	102.3
505	134.9	126.8	75.4
556	171.9	159.5	99.8
572	137.9	129.6	80.6
618	310.8	281.6	217.0
633	244.1	222.8	160.4
679	162.4	152.0	99.3
744	196.6	182.7	129.1
761	155.4	146.0	97.6

(a) Establish an equation of the form of Eq. (3.26) for all the data. (b) Combine this data with that of Prob. 3.8 to obtain one correlation for both oils. (c) Calculate j_H

when the exponent of the Prandtl number is $\frac{1}{3}$, and plot together with the data for the illustration in the text on heating. What conclusion may be drawn?

3.10. Sieder and Tate¹ obtained data on the cooling of a 21°API oil flowing on the inside of copper tube having an inside diameter of 0.62 in. and 5.1 ft long:

w , lb/hr	t_1 , °F	t_2 , °F	t_p
1306	136.85	135.15	73.0
1330	138.0	136.2	74.0
1820	160.45	158.5	76.5
1388	160.25	157.9	75.5
231	157.75	149.5	77.0
239	157.5	148.45	78.0
457	212.8	203.2	89.0
916	205.5	200.4	86.0
905	205.0	200.0	85.5
1348	206.35	202.9	87.5
1360	207.6	204.0	87.5
1850	206.9	203.7	88.5
1860	207.0	204.0	90.0
229	141.6	134.65	82.5
885	140.35	138.05	77.0
1820	147.5	146.0	79.8
473	79.6	84.75	118.5
469	80.2	86.5	136.0
460	80.0	82.0	137.0

The temperature t_p is for the inside surface of the tube. Inasmuch as these data fall below a Reynolds number of 2100, obtain an equation of the form of Eq. (3.32). The viscosity of the oil is 24.0 centipoises at 200°F and 250 centipoises at 100°F. Intermediate viscosities can be obtained by drawing a straight line on logarithmic paper as shown in Fig. 3.8. (The hint of Problem 3.7 is applicable.)

NOMENCLATURE FOR CHAPTER 3

A	Heat-transfer surface, ft ²
a	Fluid flow area, ft ²
b	Thickness of laminar layer, ft
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of pipe or tube, ft
F	Fundamental dimension of force, force-lb
ΔF	Pressure drop, ft of liquid
f	Friction factor in the Fanning equation, dimensionless
f'	Friction factor, dimensionless
G	Mass velocity, lb/(hr)(ft ²)

¹ *Ind. Eng. Chem.*, **28**, 1429-1435 (1936).

g	Acceleration of gravity, ft/hr ²
H	Heat unit, Btu
h	Heat-transfer coefficient in general, Btu/(hr)(ft ²)(°F)
h_i	Heat-transfer coefficient based on the inside pipe surface, Btu/(hr)(ft ²)(°F)
h_o	Heat-transfer coefficient based on the outside pipe surface, Btu/(hr)(ft ²)(°F)
j_H	Factor for heat transfer, dimensionless
K_H	Conversion factor between kinetic energy and heat, (mass-lb)(ft)/Btu
K_M	Conversion factor between force and mass, mass-lb/force-lb
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Length, ft
M	Fundamental dimension of mass, mass-lb
m	Mass velocity perpendicular to inside pipe surface, lb/(hr)(ft ²)
ΔP	Pressure drop, lb/ft ²
Q	Heat flow, Btu/hr
R_i	Thermal resistance on inside of pipe, (hr)(ft ²)(°F)/Btu
R_o	Thermal resistance on outside of pipe, (hr)(ft ²)(°F)/Btu
r	Radius, ft
r'	Ratio of u'/u , dimensionless
T	Temperature of hot fluid, °F
T_i	Temperature of hot fluid inside a pipe, °F
t'	Temperature of inner surface of laminar layer, °F
t_o	Temperature of cold fluid outside a pipe, °F
t_p	Temperature at inside surface of a pipe, °F
t_w	Temperature at outside surface of a pipe, °F
t_1, t_2	Inlet and outlet cold fluid temperatures, °F
Δt	Temperature difference for heat transfer, °F
Δt_i	Temperature difference between inside pipe fluid and inside pipe wall, °F
Δt_o	Temperature difference between outside pipe fluid and outside pipe wall, °F
u	Velocity in general, ft/hr
u'	Velocity of laminar layer, ft/hr
v	Specific volume, ft ³ /lb
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
x, y	Coordinates, ft (y is also used to indicate an ordinate)
Y	Distance, ft
α	Any of several proportionality constants, dimensionless
θ	Time, hr
μ	Viscosity, lb/(ft)(hr)
π	Dimensionless group
ρ	Density, lb/ft ³
τ	Shear stress, lb/ft ²
ϕ	Function

Superscripts p, q Constants**Subscripts (except as noted above)**

i Inside a pipe or tube
 o Outside a pipe or tube

CHAPTER 4

RADIATION

Introduction. All too often radiation is regarded as a phenomenon incidental only to hot, luminous bodies. In this chapter it will be seen that this is not the case and that radiation, as a third means of transferring heat, differs greatly from conduction and convection. In heat conduction through solids the mechanism consists of an energy transfer through a body whose molecules, except for vibrations, remain continuously in fixed positions. In convection the heat is first absorbed from a source by particles of fluid immediately adjacent to it and then transferred to the interior of the fluid by mixture with it. Both mechanisms require the presence of a medium to convey the heat from a source to a receiver. Radiant-heat transfer does not require an intervening medium, and heat can be transmitted by radiation across an absolute vacuum.

Wavelength and Frequency. It is convenient to mention the characteristics of radiant energy in transit before discussing the origins of radiant energy. Radiant energy is of the same nature as ordinary visible light. It is considered, in accordance with Maxwell's electromagnetic theory, to consist of an oscillating electric field accompanied by a magnetic field oscillating in phase with it. College physics texts usually treat the theory in detail.

The variation with time of the intensity of the electric field passing a given point can be represented by a sine wave having finite length from crest to crest and which is λ , the *wavelength*. The number of waves passing a given point in unit time is the *frequency* of the radiation, and the product of the frequency and wavelength is the velocity of the wave. For travel in a vacuum the velocity of propagation of radiation is very nearly 186,000 miles/sec. For travel through a medium the velocity is somewhat less, although the deviation is generally neglected.

The wavelength of radiation may be specified in any units of length, but the micron, 1×10^{-4} cm, is common. All the known waves included in the electromagnetic theory lie between the short wavelengths of cosmic rays, less than 1×10^{-6} micron, and the long wavelength of radio above 1×10^7 microns. Of these, only waves in the region between the near and far infrared with wavelengths of $\frac{3}{4}$ to 400 microns are of importance to radiant-heat transfer as found in ordinary industrial equipment.

The Origins of Radiant Energy. Radiant energy is believed to originate within the molecules of the radiating body, the atoms of such molecules vibrating in a simple harmonic motion as linear oscillators. The emission of radiant energy is believed to represent a decrease in the amplitudes of the vibrations within the molecules, while an absorption of energy represents an increase. In its essence the quantum theory postulates that for every frequency of radiation there is a small minimum pulsation of energy which may be emitted. This is the *quantum*, and a smaller quantity cannot be emitted although many such quanta may be emitted. The total radiation of energy of a given frequency emitted by a body is an integral number of quanta at that frequency. For different frequencies, the number of quanta and thus the total energy may be different. Planck showed that the energy associated with a quantum is proportional to the frequency of the vibration or, if the velocity of all radiation is considered constant, inversely proportional to the wavelength. Thus radiant energy of a given frequency may be pictured as consisting of successive pulses of radiant energy, each pulse of which has the value of the quantum for a given frequency.

The picture of the atom proposed by Bohr is helpful to a clearer understanding of one possible origin of radiant energy. Electrons are presumed to travel about the nucleus of an atom in elliptical orbits at varying distances from the nucleus. The outermost orbital electrons possess definite energies comprising their kinetic and potential energies, by virtue of their rotation about the nucleus. The potential energy is the energy required to remove an electron from its orbit to an infinite distance from the nucleus. A given electron in an orbit at a given distance from the nucleus will have a certain energy. Should a disturbance occur such as the collision of the atom with another atom or electron, the given electron may be displaced from its orbit and may (1) return to its original orbit, (2) pass to another orbit whose electrons possess a different energy, or (3) entirely leave the system influenced by the nucleus. If the transition is from an orbit of high energy to one of lower energy, the readjustment is affected by the radiation of the excess energy.

Another origin of radiant energy may be attributed to the changes in the energies of atoms and molecules themselves without reference to their individual electrons. If two or more nuclei of the molecule are vibrating with respect to each other, a change in the amplitude or amplitudes of vibration will cause a change in energy content. A decrease in amplitude is the result of an emission of radiant energy, while an increase is the result of the absorption of radiant energy. The energy of a molecule may be changed by an alteration of its kinetic energy of translation or rotation, and this will likewise result in the emission of radiant energy. A decrease

in velocity corresponds to the emission of radiant energy, while an increase corresponds to the absorption of radiant energy.

Since temperature is a measure of the average kinetic energy of molecules, the higher the temperature the higher the average kinetic energy both of translation and of vibration. It can therefore be expected that the higher the temperature the greater the quantity of radiant energy emitted from a substance. Since molecular movement ceases completely only at the absolute zero of temperature, it may be concluded that all substances will emit or absorb radiant energy provided the temperature of the substances is above absolute zero.

For radiant energy to be emitted from the interior of a solid it must penetrate the surface of the solid without being dissipated by producing other energy changes within its molecules. There is little probability that radiant energy generated in the interior of a solid will reach its surface without encountering other molecules, and therefore all radiant energy emitted from the surfaces of solid bodies is generated by energy-level changes in molecules near and on their surfaces. The quantity of radiant energy emitted by a solid body is consequently a function of the surface of the body, and conversely, radiation incident on solid bodies is absorbed at the surface. The probability that internally generated radiant energy will reach the surface is far greater for hot radiating gases than for solids, and the radiant energy emitted by a gas is a function of the gas volume rather than the surface of the gas shape. In liquids the situation is intermediate between gases and solids, and radiation may originate somewhat below the surface, depending on the nature of the liquid.

The Distribution of Radiant Energy. A body at a given temperature will emit radiation of a whole range of wavelengths and not a single wavelength. This is attributed to the existence of an infinite variety of linear oscillators. The energy emitted at each wavelength can be determined through the use of a dispersing prism and thermopiles. Such measurements on a given body will produce curves as shown in Fig. 4.1 for each given temperature. The curves are plots of the intensities of the radiant energy I_λ Btu/(hr)(ft²)(micron) against the wavelengths in microns λ as determined at numerous wavelengths and connecting points. For any given temperature each curve possesses a wavelength at which the amount of spectral energy given off is a maximum. For the same body at a lower temperature the maximum intensity of radiation is obviously less, but it is also significant that the wavelength at which the maximum occurs is longer. Since the curve for a single temperature depicts the amount of energy emitted for a single wavelength, the area under the curve must equal the sum of all the energy radiated by the body at all its wavelengths. The maximum intensity falls between $\frac{3}{4}$ and 400 microns, indicating that

red heat is a far better source of energy than white heat. Were it not for this fact, the near-white incandescent lamp would require more energy for illumination and give off uncomfortable quantities of heat.

When dealing with the properties of radiation, it is necessary to differentiate between two kinds of properties: monochromatic and total. A monochromatic property, such as the maximum values of I_λ in Fig. 4.1, refers to a single wavelength. A total property indicates that it is the

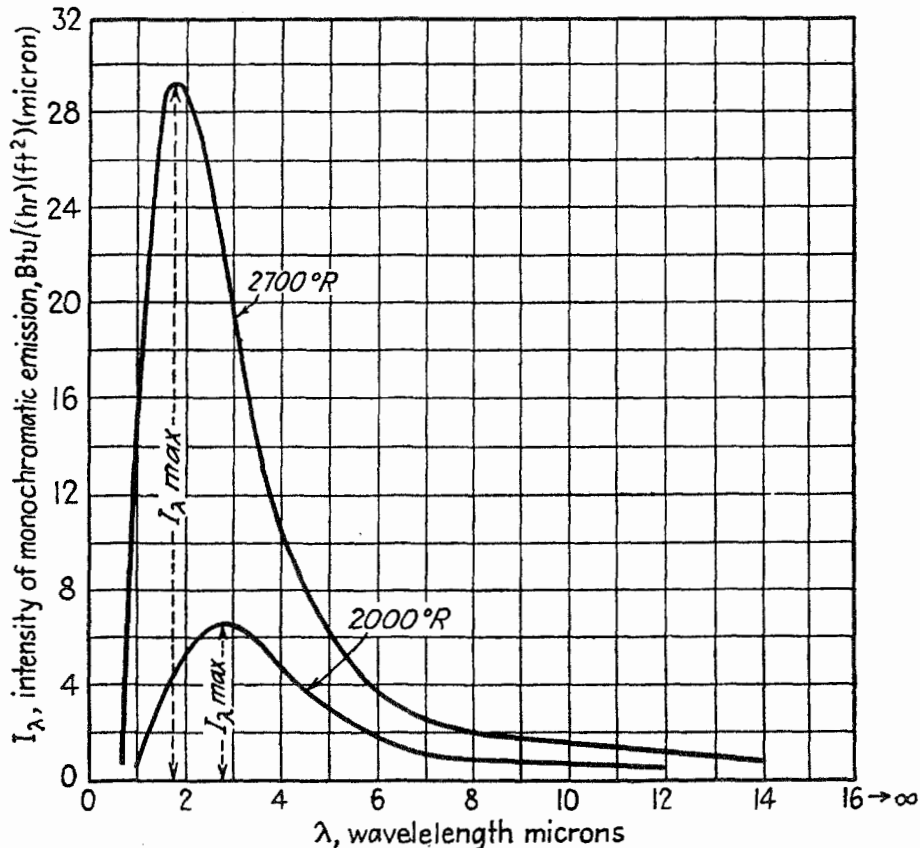


FIG. 4.1. Intensity of monochromatic radiation for a hot body at different temperatures.

algebraic sum of the monochromatic values of the property. Monochromatic radiation literally means "one color" or one wavelength, but experimentally it actually refers to a group or band of wavelengths, since wavelengths cannot be resolved individually. Monochromatic values are not important to the direct solution of engineering problems but are necessary for the derivation of basic radiation relationships.

The Emissive Power. The total quantity of radiant energy of all wavelengths emitted by a body per unit area and time is the *total emissive power* E , $\text{Btu}/(\text{hr})(\text{ft}^2)$. If the intensity of the radiant energy at any wavelength in Fig. 4.1 is I_λ $\text{Btu}/(\text{hr})(\text{ft}^2)(\text{micron})$, the total emissive power is the area under the curve and may be computed by

$$E = \int_0^\infty I_\lambda d\lambda \quad (4.1)$$

A relationship between I_λ and λ was the subject of many investigations both experimental and mathematical during the nineteenth century. Planck was the first to recognize the quantum nature of radiant energy and developed an equation which fits the spectral energy curve of Fig. 4.1 at any temperature. It is given by

$$I_\lambda = \frac{C_1 \lambda^{-5}}{e^{C_2/\lambda T} - 1} \quad (4.2)$$

where I_λ = monochromatic intensity of emission, Btu/(hr)(ft²)
(micron)

λ = wavelength, microns

C_1 and C_2 = constants with values 1.16×10^8 and 25,740

T = temperature of the body, °R

Wien postulated another law known as Wien's displacement law, which states that the product of the wavelength of the maximum value of the monochromatic intensity of emission and the absolute temperature is a constant, or

$$\lambda T = 2884 \text{ micron } ^\circ\text{R} \quad (4.3)$$

Equation (4.3) may be derived from Eq. (4.2) as follows,

$$dI_\lambda = d\left(\frac{C_1 \lambda^{-5}}{e^{C_2/\lambda T} - 1}\right) = 0 \quad (4.4)$$

$$(-5C_1 \lambda^{-6})(e^{C_2/\lambda T} - 1) + C_1 \lambda^{-5}(e^{C_2/\lambda T}) \frac{C_2}{\lambda^2 T} = 0$$

$$\left(-5 + \frac{C_2}{\lambda T}\right) e^{C_2/\lambda T} + 5 = 0 \quad (4.5)$$

By trial and error, the first term equals -5 when $\lambda T = 2884$.

Spectral measurements of the radiation received on the earth's surface from the sun and allowing for absorption by the atmosphere indicate that the peak I_λ lies approximately at 0.25 micron, which is well in the ultraviolet. This accounts for the high ultraviolet content of the sun's rays and the predominance of blue in the visible portion of the spectrum. The location of the peak allows an estimation of the sun's temperature from Eq. (4.3) at 11,000°F.

The Incidence of Radiant Energy: The Black Body. The preceding discussion has dealt with the generation of radiant energy. What happens when radiant energy falls upon a body? In the simple case of light it may be all or partially absorbed or reflected. If the receiving medium is transparent to the radiation, it will transmit some of the energy through itself. The same effects are applicable to radiant energy, and an energy balance about a receiver on which the total incident energy is unity is

given by

$$a + r + \tau = 1 \quad (4.6)$$

where the *absorptivity* a is the fraction absorbed, the *reflectivity* r the fraction reflected, and the *transmissivity* τ the fraction transmitted. The majority of engineering materials are opaque substances having zero transmissivities, but there are none which completely absorb or reflect all the incident energy. The substances having nearly complete or unit absorptivities are lampblack, platinum black, and bismuth black, absorbing 0.98 to 0.99 of all incident radiation.

If an ordinary body emits radiation to another body, it will have some of the emitted energy returned to itself by reflection. When Planck developed Eq. (4.2), he assumed that none of the emitted energy was returned, this was equivalent to an assumption that bodies having zero transmissivity also had zero reflectivity. This is the concept of the perfect *black body* and for which $a = 1.0$.

Relationship between Emissivity and Absorptivity: Kirchhoff's Law. Consider a body of given size and shape placed within a hollow sphere of constant temperature, and assume that the air has been evacuated. After thermal equilibrium has been reached, the temperature of the body and that of the enclosure will be the same, inferring that the body is absorbing and radiating heat at identical rates. Let the intensity of radiation falling on the body be I Btu/(hr)(ft²), the fraction absorbed a_1 , and the total emissive power E_1 Btu/(hr)(ft²). Then the energy emitted by the body of total surface A_1 is equal to that received, or

$$E_1 A_1 = I a_1 A_1 \quad (4.7)$$

$$E_1 = I a_1 \quad (4.8)$$

If the body is replaced by another of identical shape and equilibrium is again attained,

$$E_2 = I a_2 \quad (4.9)$$

If a third body, a black body, is introduced, then

$$E_b = I a_b \quad (4.10)$$

But by definition the absorptivity of a black body is 1.0.

$$\frac{E_1}{a_1} = \frac{E_2}{a_2} = E_b \quad (4.11)$$

or at thermal equilibrium the ratio of the total emissive power to the absorptivity for all bodies is the same. This is known as Kirchhoff's law. Since the maximum absorptivity of the black body is taken as 1.0 from Eq. (4.6), its reflectivity must be zero. Absolute values of the total

emissive power are not obtainable, but

$$E_1 = a_1 E_b \quad (4.12)$$

$$E_2 = a_2 E_b \quad (4.13)$$

$$\frac{E_1}{E_b} = a_1 = \epsilon_1 \quad (4.14)$$

$$\frac{E_2}{E_b} = a_2 = \epsilon_2 \quad (4.15)$$

The use of the ratio of the actual emissive power to the black-body emissive power under identical conditions is the *emissivity* ϵ . Since it is the reference, the emissivity of the black body is unity. The emissivities of common materials cover a large range and are tabulated in Table 4.1. Emissivities are influenced by the finish or polish of the surface and increase with its temperature. Highly polished and white surfaces generally have lower values than rough or black surfaces. From Eq. (4.12) it can be seen that any body having a high emissivity as a radiator will have a high absorptivity when acting as a receiver. The usual statement is as follows: Good radiators make good absorbers.

Experimental Determination of the Emissivity. The experimental determination of the emissivities of materials is particularly difficult at high temperatures. The problems of maintaining a system free of conduction, convection, and a radiation-absorbing medium require careful analysis. A method is given here which is satisfactory for the measurement of emissivities in the range of room temperatures and might be applicable to the calculation of such problems as the loss of heat from a pipe to air by radiation alone. A hollow, opaque, internally blackened cylinder is maintained in a constant-temperature bath as shown in Fig. 4.2. A total radiation receiver is mounted by a bracket to the wall of the cylinder. The radiation receiver consists of a copper cylinder a , which is blackened on the inside and highly polished on the outside. Two extremely thin, blackened, and highly conducting copper discs b and b' are mounted in the receiver for the purpose of absorbing radiation. By mounting the discs at equal distances from the top and bottom of the small cylinder, the angles α_1 and α_2 are equal and the discs have equal areas for receiving radiation. The lower disc receives radiation from the blackened constant-temperature walls of the vessel. The upper disc receives radiation from a plate of specimen material c which is electrically maintained at a fixed temperature. The two discs are wired together by a sensitive thermocouple so that they oppose each other, and only net differences in the quantity of radiation are measured by the galvanometer. By wiring them to oppose each other, any effects within the receiver itself are also canceled. If the galvanometer deflection for the specimen non-

black body is measured and then c is replaced by a perfect black body, the ratio of the two galvanometer deflections is the emissivity of the specimen. Data obtained in this manner are the *normal total emissivity* as given in Table 4.1. They may also be used in the solution of problems having hemispherical radiation except in the presence of highly polished surfaces.

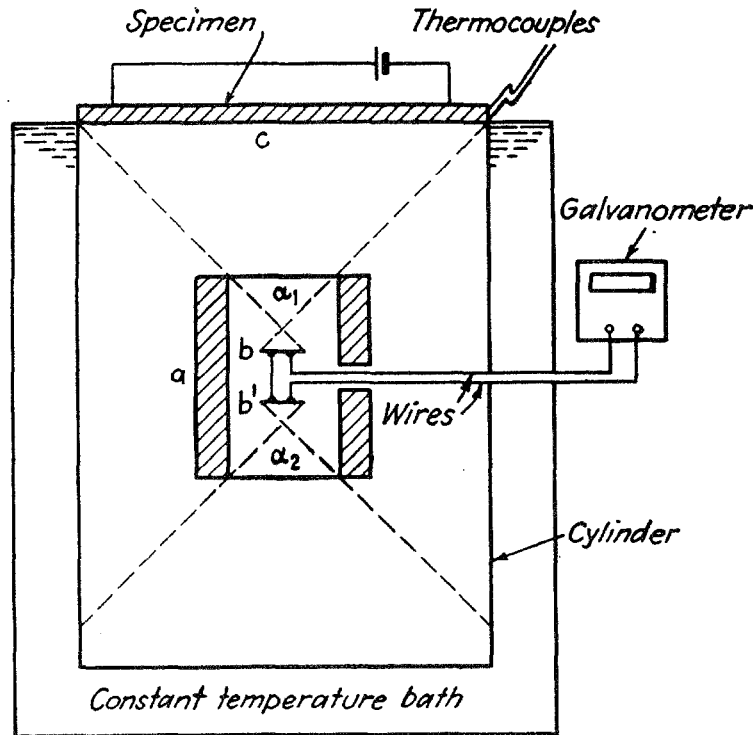


FIG. 4.2. Apparatus for measuring emissivities.

Influence of Temperature on the Emissive Power: Stefan-Boltzmann Law. If a perfect black body radiates energy, the total radiation may be determined from Planck's law. Starting with the monochromatic black-body equation

$$I_{\lambda} = \frac{C_1 \lambda^{-5}}{e^{C_2/\lambda T} - 1}$$

it may be applied to sum up all the energy by integration of the area under a curve in Fig. 4.1 or at a fixed temperature

$$E = \int_0^{\infty} \frac{C_1 \lambda^{-5}}{e^{C_2/\lambda T} - 1} d\lambda \quad (4.16)$$

Let $x = C_2/\lambda T$, $\lambda = C_2/Tx$, $d\lambda = (-C_2/Tx^2) dx$, from which

$$E_b = -\frac{C_1 T^4}{C_2^4} \int_0^{\infty} x^3 (e^x - 1)^{-1} dx \quad (4.17)$$

TABLE 4.1. THE NORMAL TOTAL EMISSIVITY OF VARIOUS SURFACES (HOTTEL)
A. Metals and Their Oxides

Surface	$t, ^\circ\text{F}^*$	Emissivity*
Aluminum		
Highly polished plate, 98.3 % pure.....	440-1070	0.039-0.057
Polished plate.....	73	0.040
Rough plate.....	78	0.055
Oxidized at 1110°F.....	390-1110	0.11-0.19
Al-surfaced roofing.....	100	0.216
Calorized surfaces, heated at 1110°F		
Copper.....	390-1110	0.18-0.19
Steel.....	390-1110	0.52-0.57
Brass		
Highly polished.....		
73.2 % Cu, 26.7 % Zn.....	476-674	0.028-0.031
62.4 % Cu, 36.8 % Zn, 0.4 % Pb, 0.3 % Al.....	494-710	0.033-0.037
82.9 % Cu, 17.0 % Zn.....	530	0.030
Hard rolled, polished, but direction of polishing		
visible.....	70	0.038
but somewhat attacked....	73	0.043
but traces of stearin from		
polish left on.....	75	0.053
Polished.....	100-600	0.096-0.096
Rolled plate, natural surface.....	72	0.06
Rubbed with coarse emery.....	72	0.20
Dull plate.....	120-660	0.22
Oxidized by heating at 1110°F.....	390-1110	0.61-0.59
Chromium (see Nickel alloys for Ni-Cr steels).....	100-1000	0.08-0.26
Copper		
Carefully polished electrolytic copper.....	176	0.018
Commercial emiered, polished, but pits remaining	66	0.030
Commercial, scraped shiny but not mirrorlike.....	72	0.072
Polished.....	242	0.023
Plate, heated long time, covered with thick oxide		
layer.....	77	0.78
Plate heated at 1110°F.....	390-1110	0.57-0.57
Cuprous oxide.....	1470-2010	0.66-0.54
Molten copper.....	1970-2330	0.16-0.13
Gold		
Pure, highly polished.....	440-1160	0.018-0.035
Iron and steel		
Metallic surfaces (or very thin oxide layer)		
Electrolytic iron, highly polished.....	350-440	0.052-0.064
Polished iron.....	800-1880	0.144-0.377
Iron freshly emiered.....	68	0.242
Cast iron, polished.....	392	0.21
Wrought iron, highly polished.....	100-480	0.28
Cast iron, newly turned.....	72	0.435
Polished steel casting.....	1420-1900	0.52-0.56
Ground sheet steel.....	1720-2010	0.55-0.61
Smooth sheet iron.....	1650-1900	0.55-0.60
Cast iron, turned on lathe.....	1620-1810	0.60-0.70

TABLE 4.1. THE NORMAL TOTAL EMISSIVITY OF VARIOUS SURFACES (HOTTEL).—
(Continued)

Surface	$t, ^\circ\text{F}^*$	Emissivity*
Iron and steel—(Continued)		
Oxidized surfaces		
Iron plate, pickled, then rusted red.....	68	0.612
Completely rusted.....	67	0.685
Rolled sheet steel.....	70	0.657
Oxidized iron.....	212	0.736
Cast iron, oxidized at 1100°F.....	390-1110	0.64-0.78
Steel, oxidized at 1100°F.....	390-1110	0.79-0.79
Smooth oxidized electrolytic iron.....	260-980	0.78-0.82
Iron oxide.....	930-2190	0.85-0.89
Rough ingot iron.....	1700-2040	0.87-0.95
Sheet steel, strong, rough oxide layer.....	75	0.80
Dense, shiny oxide layer.....	75	0.82
Cast plate, smooth.....	73	0.80
Rough.....	73	0.82
Cast iron, rough, strongly oxidized.....	100-480	0.95
Wrought iron, dull oxidized.....	70-680	0.94
Steel plate, rough.....	100-700	0.94-0.97
High-temperature alloy steels (see Nickel alloys)		
Molten metal		
Cast iron.....	2370-2550	0.29-0.29
Mild steel.....	2910-3270	0.28-0.28
Lead		
Pure (99.96 %), unoxidized.....	260-440	0.057-0.075
Gray oxidized.....	75	0.281
Oxidized at 390°F.....	390	0.63
Mercury.....	32-212	0.09-0.12
Molybdenum filament.....	1340-4700	0.096-0.292
Monel metal, oxidized at 1110°F.....	390-1110	0.41-0.46
Nickel		
Electroplated on polished iron, then polished.....	74	0.045
Technically pure (98.9 % Ni, + Mn), polished.....	440-710	0.07-0.087
Electroplated on pickled iron, not polished.....	68	0.11
Wire.....	368-1844	0.096-0.186
Plate, oxidized by heating at 1110°F.....	390-1110	0.37-0.48
Nickel oxide.....	1200-2290	0.59-0.86
Nickel alloys		
Chromnickel.....	125-1894	0.64-0.76
Nickelin (18-32 Ni; 55-68 Cu; 20 Zn), gray oxidized	70	0.262
KA-2S alloy steel (8 % Ni; 18 % Cr), light silvery,		
rough, brown, after heating.....	420-914	0.44-0.36
after 42 hr. heating at 980°F.....	420-980	0.62-0.73
NCT-3 alloy (20 Ni; 25 Cr). Brown, splotted,		
oxidized from service.....	420-980	0.90-0.97
NCT-6 alloy (60 Ni; 12 Cr). Smooth, black,		
firm adhesive oxide coat from service.....	520-1045	0.89-0.82
Platinum		
Pure, polished plate.....	440-1160	0.054-0.104
Strip.....	1700-2960	0.12-0.17
Filament.....	80-2240	0.036-0.192
Wire.....	440-2510	0.073-0.182

TABLE 4.1. THE NORMAL TOTAL EMISSIVITY OF VARIOUS SURFACES (HOTTEL).—
(Continued)

Surface	$t, ^\circ\text{F}^*$	Emissivity*
Silver		
Polished, pure.....	440–1160	0.0198–0.0324
Polished.....	100–700	0.0221–0.0312
Steel (see Iron)		
Tantalum filament.....	2420–5430	0.194–0.31
Tin, bright tinned iron sheet.....	76	0.043 and 0.064
Tungsten		
Filament, aged.....	80–6000	0.032–0.35
Filament.....	6000	0.39
Zinc		
Commercial 99.1 % pure, polished.....	440–620	0.045–0.053
Oxidized by heating at 750°F.....	750	0.11
Galvanized sheet iron, fairly bright.....	82	0.228
Galvanized sheet iron, gray oxidized.....	75	0.276
B. Refractories, Building Materials, Paints, and Miscellaneous		
Asbestos		
Board.....	74	0.96
Paper.....	100–700	0.93–0.945
Brick		
Red, rough, but no gross irregularities.....	70	0.93
Silica, unglazed, rough.....	1832	0.80
Silica, glazed, rough.....	2012	0.85
Grog brick, glazed.....	2012	0.75
(See Refractory materials)		
Carbon		
T-carbon (Gebruder Siemens) 0.9 % ash.....	260–1160	0.81–0.79
This started with emissivity at 260°F of 0.72, but on heating changed to values given.....		
Carbon filament.....	1900–2560	0.526
Candle soot.....	206–520	0.952
Lampblack-water-glass coating.....	209–362	0.959–0.947
Same.....	260–440	0.957–0.952
Thin layer on iron plate.....	69	0.927
Thick coat.....	68	0.967
Lampblack, 0.003 in. or thicker.....	100–700	0.945
Enamel, white fused, on iron.....	66	0.897
Glass, smooth.....	72	0.937
Gypsum, 0.02 in. thick on smooth or blackened plate.....	70	0.903
Marble, light gray, polished.....	72	0.931
Oak, planed.....	70	0.895
Oil layers on polished nickel (lub. oil).....	68	
Polished surface, alone.....		0.045
+0.001 in. oil.....		0.27
+0.002 in. oil.....		0.46
+0.005 in. oil.....		0.72
∞ thick oil layer.....		0.82

TABLE 4.1. THE NORMAL TOTAL EMISSIVITY OF VARIOUS SURFACES (HOTTEL).—
(Continued)

Surface	t , °F*	Emissivity*
Oil layers on aluminum foil (linseed oil).....		
Al foil.....	212	0.087†
+1 coat oil.....	212	0.561
+2 coats oil.....	212	0.574
Paints, lacquers, varnishes		
Snow-white enamel varnish on rough iron plate...	73	0.906
Black shiny lacquer, sprayed on iron.....	76	0.875
Black shiny shellac on tinned iron sheet.....	70	0.821
Black matte shellac.....	170-295	0.91
Black lacquer.....	100-200	0.80-0.95
Flat black lacquer.....	100-200	0.96-0.98
White lacquer.....	100-200	0.80-0.95
Oil paints, 16 different, all colors.....	212	0.92-0.96
Aluminum paints and lacquers		
10 % Al, 22 % lacquer body, on rough or smooth surface.....	212	0.52
26 % Al, 27 % lacquer body, on rough or smooth surface.....	212	0.3
Other Al paints, varying age and Al content.....	212	0.27-0.67
Al lacquer, varnish binder, on rough plate.....	70	0.39
Al paint, after heating to 620°F.....	300-600	0.35
Paper, thin		
Pasted on tinned iron plate.....	66	0.924
Rough iron plate.....	66	0.929
Black lacquered plate.....	66	0.944
Plaster, rough lime.....	50-190	0.91
Porcelain, glazed.....	72	0.924
Quartz, rough, fused.....	70	0.932
Refractory materials, 40 different.....	1110-1830	
Poor radiators.....		$\left[\begin{array}{l} 0.65 \\ 0.70 \end{array} \right\} - 0.75$
Good radiators.....		$\left[\begin{array}{l} 0.80 \\ 0.85 \end{array} \right\} - \left\{ \begin{array}{l} 0.85 \\ 0.90 \end{array} \right.$
Roofing paper.....	69	0.91
Rubber		
Hard, glossy plate.....	74	0.945
Soft, gray, rough (reclaimed).....	76	0.859
Serpentine, polished.....	74	0.900
Water.....	32-212	0.95-0.963

NOTE. The results of many investigators have been omitted because of obvious defects in experimental method.

* When two temperatures and two emissivities are given, they correspond, first to first and second to second, and linear interpolation is permissible.

† Although this value is probably high, it is given for comparison with the data, by the same investigator, to show the effect of oil layers (see Aluminum, part A of this table).

Expanding the term in parentheses,

$$E_b = -\frac{C_1 T^4}{C_2^4} \int_0^\infty x^3 (e^{-x} + e^{-2x} + e^{-3x} + e^{-4x} + \dots) dx \quad (4.18)$$

Integrating each term and summing only the first four as significant,

$$E_b = \frac{C_1 T^4}{C_2^4} \times 6.44 \quad (4.19)$$

Evaluating constants,

$$E_b = 0.173 \times 10^{-8} T^4 \quad (4.20)$$

Equation (4.20) being the area under a curve in Fig. 4.1 from $\lambda = 0$ to $\lambda = \infty$ states that the total radiation from a perfect black body is proportional to the fourth power of the absolute temperature of the body. This is known as the Stefan-Boltzmann law. The constant 0.173×10^{-8} Btu/(hr)(ft²)(°R⁴) is known as the Stefan-Boltzmann constant, usually designated by σ . The equation was also deduced by Boltzmann from the second law of thermodynamics. Equation (4.20) serves as the principal relationship for the calculation of radiation phenomena and is to radiation what $Q = hA \Delta t$ is to convection. However, Eq. (4.20) was derived for a perfect black body. From Eq. (4.14) if a body is nonblack, the emissivity is the ratio E/E_b and E can be written $E = E_b \epsilon$. Equation (4.20) becomes

$$E = \epsilon \sigma T^4 \quad (4.21)$$

and

$$\frac{Q}{A} = \epsilon \sigma T^4 \quad (4.22)$$

Exchange of Energy between Two Large Parallel Planes. Quantitative considerations so far have dealt with the energy change when radiation occurs only from a single body, and it has been assumed that energy once radiated is no longer returned to its source. This is true only if one black body radiates to another black body with no medium between them or no absorption occurs by the medium between them. Of the gases, chlorine, hydrogen, oxygen, and nitrogen are classified as nonabsorbing. Carbon monoxide, carbon dioxide, and organic gases and vapors are absorbing to greater or lesser extents. From earlier discussions it should be conceded that radiation from a small plate proceeds outwardly in a hemisphere with the plate as center and that the radiation incident upon unit area of a body at great distance is very small. In radiation it is necessary to qualify the condition under which all the radiation from the source is completely received by the receiver. It will occur if two plates

or radiant planes are infinitely large, so that the amount of radiation which fans out from the edges of the source and over the edges of the receiver is insignificant. If both of the plates or planes are black bodies, the energy from the first is $E_{b1} = \sigma T_1^4$ and from the second $E_{b2} = \sigma T_2^4$. By the definition of a black body, all the energy it receives is absorbed and the net exchange per square foot between two planes maintained at constant temperatures is

$$\frac{Q}{A} = E_{b1} - E_{b2} = \sigma(T_1^4 - T_2^4) \quad (4.23)$$

$$= 0.173 \left[\left(\frac{T_1}{100} \right)^4 - \left(\frac{T_2}{100} \right)^4 \right] \quad (4.24)$$

Example 4.1. Radiation between Two Large Planes. Two very large walls are at the constant temperature of 800 and 1000°F. Assuming they are black bodies, how much heat must be removed from the colder wall to maintain a constant temperature?

Solution:

$$T_1 = 1000 + 460 = 1460^\circ\text{R} \quad T_2 = 800 + 460 = 1260^\circ\text{R}$$

$$\frac{Q}{A} = 0.173 [(14.6)^4 - (12.6)^4] = 3,500 \text{ Btu}/(\text{hr})(\text{ft}^2)$$

Exchange of Energy between Two Parallel Planes of Different Emissivity. The preceding discussion applied to black bodies. If the two planes are not black bodies and have different emissivities, the net exchange of energy will be different. Some of the energy emitted from the first plane will be absorbed, and the remainder radiated back to the source. For two walls of infinite size the radiation of each wall can be traced. Thus if energy is emitted by the first wall per square foot of an amount E_1 and emissivity ϵ_1 , the second wall will absorb $E_1\epsilon_2$ and reflect $1 - \epsilon_2$ of it. The first wall will then again radiate but in the amount $E_1(1 - \epsilon_2)(1 - \epsilon_1)$. The changes at the two planes are

Hot plane

Radiated: E_1

Returned: $E_1(1 - \epsilon_2)$

Radiated: $E_1(1 - \epsilon_2)(1 - \epsilon_1)$

Returned: $E_1(1 - \epsilon_2)(1 - \epsilon_1)(1 - \epsilon_2)$

Cold plane

Radiated: E_2

Returned: $E_2(1 - \epsilon_1)$

Radiated: $E_2(1 - \epsilon_1)(1 - \epsilon_2)$

Returned: $E_2(1 - \epsilon_1)(1 - \epsilon_2)(1 - \epsilon_1)$

and

$$\begin{aligned} \frac{Q}{A} = & [E_1 - E_1(1 - \epsilon_2) - E_1(1 - \epsilon_1)(1 - \epsilon_2)^2 \\ & - E_1(1 - \epsilon_2)(1 - \epsilon_1)^2(1 - \epsilon_2)^2 + \dots] - [\epsilon_1 E_2 \\ & - \epsilon_1 E_2(1 - \epsilon_1)(1 - \epsilon_2) - \epsilon_1 E_2(1 - \epsilon_1)^2(1 - \epsilon_2)^2 + \dots] \quad (4.25) \end{aligned}$$

E_1 is given by $\epsilon_1 \sigma T_1^4$, E_2 by $\epsilon_2 \sigma T_2^4$, and Eq. (4.25) is a series whose solution is

$$\frac{Q}{A} = \frac{\sigma}{(1/\epsilon_1) + (1/\epsilon_2) - 1} (T_1^4 - T_2^4) \quad (4.26)$$

Example 4.2. Radiation between Planes with Different Emissivities. If the two walls in Example 4.1 have emissivities of 0.6 and 0.8, respectively, what is the net exchange?

Solution:

$$\frac{Q}{A} = \frac{0.173}{(1/0.6) + (1/0.8) - 1} [(14.6)^2 - (12.6)^2] = 1825 \text{ Btu/(hr)(ft}^2\text{)}$$

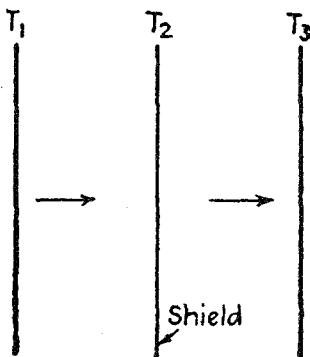


FIG. 4.3. Radiation with a shield.

For perfect black bodies the value was 3500 Btu/(hr)(ft²).

Radiation Intercepted by a Shield. Suppose two infinite and parallel planes are separated by a third plane which is opaque to direct radiation between the two and which is extremely thin (or has infinite thermal conductivity) as shown in Fig. 4.3. The net exchange between the two initial planes is given by Eq. (4.26).

$$Q = \frac{\sigma A}{(1/\epsilon_1) + (1/\epsilon_3) - 1} (T_1^4 - T_3^4) \quad (4.27)$$

If $\epsilon_1 = \epsilon_3$ but $\epsilon_1 \neq \epsilon_2$, the net exchange from 1 to 3 is given by

$$Q_1 = \frac{\sigma A}{(1/\epsilon_1) + (1/\epsilon_2) - 1} (T_1^4 - T_2^4) = \frac{\sigma A}{(1/\epsilon_2) + (1/\epsilon_3) - 1} (T_2^4 - T_3^4) \quad (4.28)$$

from which

$$T_2^4 = \frac{1}{2}(T_1^4 + T_3^4) \quad (4.29)$$

Then

$$Q_1 = \frac{\sigma A}{(1/\epsilon_1) + (1/\epsilon_2) - 1} \frac{1}{2} (T_1^4 - T_3^4) \quad (4.30)$$

When $\epsilon_1 = \epsilon_2$, $Q_1 = \frac{1}{2}Q$, and for the simple case where n shields are employed, each having the same emissivities as the initial planes,

$$Q_n = \frac{1}{n+1} Q$$

where Q is the exchange if the initial planes were not separated.

Spheres or Cylinders with Spherical or Cylindrical Enclosures. The radiation between a sphere and an enclosing sphere of radii r_1 and r_2 may be treated in the same manner as Eq. (4.26). The radiation emitted initially by the inner sphere is $E_1 A_1$, all of which falls on A_2 . Of this total, however, $(1 - \epsilon_2)E_1 A_1$ is reflected of which $\left(\frac{r_1}{r_2}\right)^2 (1 - \epsilon_2)E_1 A_1$ falls on A_1 and $\left[1 - \left(\frac{r_1}{r_2}\right)^2\right] (1 - \epsilon_2)E_1 A_1$ falls on A_2 . If this analysis is continued as before, the energy exchange will again be represented by a geometrical series and the net exchange between the inner and outer sphere is given by

$$Q = \frac{\sigma A_1}{\frac{1}{\epsilon_1} + \left(\frac{r_1}{r_2}\right)^2 \left(\frac{1}{\epsilon_2} - 1\right)} (T_1^4 - T_2^4) = \frac{\sigma A_1}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)} (T_1^4 - T_2^4) \quad (4.31)$$

The same relation will be seen to hold for infinitely long concentric cylinders except that A_1/A_2 is r_1/r_2 instead of r_1^2/r_2^2 .

Radiation of Energy to a Completely Absorbing Receiver. In Example 2.5, calculations were made on the loss of heat from a pipe to air. When a heat source is small by comparison with the enclosure it is customary to make the simplifying assumption that none of the heat radiated from the source is reflected to it. In such cases, Eq. (4.26) reduces to

$$\frac{Q}{A_1} = \epsilon_1 \sigma (T_1^4 - T_2^4) \quad (4.32)$$

Sometimes it is convenient to represent the net effect of the radiation in the same form employed in convection; namely,

$$Q = h_r A_1 (T_1 - T_2) \quad (4.33)$$

where h_r is a fictitious film coefficient representing the rate at which the radiation passes from the surface of the radiator. The values of Q in Eqs. (4.32) and (4.33) are identical, but the value in Eq. (4.32) is related to the mechanism by which the heat was transferred. Eq. (4.33) is a statement of the heat balance as applied before in the Fourier equation to conduction and convection. Fishenden and Saunders¹ have treated a number of interesting aspects of the subject.

¹ Fishenden, M., and O. A. Saunders, "The Calculation of Heat Transmission," His Majesty's Stationery Office, London, 1932.

Example 4.3. Calculation of Radiation from a Pipe. In Example 2.5 the outside temperature of a lagged pipe carrying steam at 300°F was 125°F and the surrounding atmosphere was at 70°F. The heat loss by free convection and radiation was 103.2 Btu/(hr)(lin ft), and the combined coefficient of heat transfer was 2.10 Btu/(hr)(ft²)(°F). How much of the heat loss was due to radiation, and what was the equivalent coefficient of heat transfer for the radiation alone?

$$\text{Area/lin ft} = \pi \times \frac{3.375}{12} \times 1 = 0.88 \text{ ft}^2$$

From Table 4.1B, the emissivity is approximately 0.90.

$$\begin{aligned} T_1 &= 125 + 460 = 585^\circ\text{R} & T_2 &= 70 + 460 = 530^\circ\text{R} \\ q &= 0.90 \times 0.88 \times 0.173[(585/100)^4 - (530/100)^4] = 52.5 \text{ Btu/(hr)(lin ft)} \\ h_r &= \frac{Q}{A(T_1 - T_2)} = \frac{52.5}{0.88(125 - 70)} = 1.08 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \end{aligned}$$

Exchange of Energy between Any Source and Any Receiver. The three preceding illustrations have been extremely limited. The study of two planes was directed only to sources and receivers which were infinitely large so that every point on one plane could be connected with every point on the second and no radiation from the one "leaked" past the other and out of the system. A slightly more complex arrangement can be achieved between two concentric spheres or two concentric cylinders. In either of these all the radiation from the source may be seen to fall on the receiver. But this is very rarely the case in practical engineering problems, particularly in the design of furnaces. The receiving surface, such as the banks of tubes, is cylindrical and may partially obscure some of the surface from "seeing" the source.

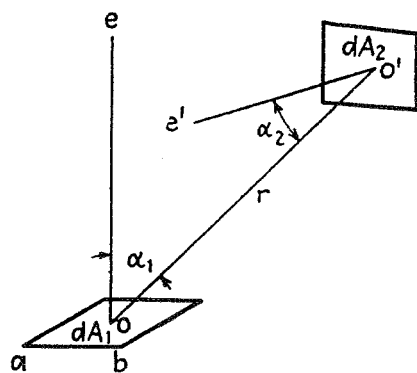


FIG. 4.4. Radiation between two plates.

In a system composed of walls and pipes running in different arrangements the geometry by which the radiation falls on the surfaces and the manner in which these surfaces reflect their energy are difficult to evaluate. The simplest elements are treated here, but many practical applications require the empirical methods of Chap. 19.

Consider the arrangement of two radiating plates at temperatures T_1 and T_2 as shown in Fig. 4.4. The two plates are not face-to-face and therefore have only an askance view of each other. The lower plate, represented isometrically in the horizontal plane, radiates in all directions upward and outward. Some of the radiation from the surface of the hot plate, dA_1 falls upon the second plate but not perpendicular to it. The second plate dA_2 will

reflect some of the incident energy, but only a part of it will return to the first plate. What is the net exchange of energy between the two?

The lines eo and $e'o'$ are perpendiculars to the two plates, respectively, on their mutually exposed surfaces. The length of the shortest line joining the two plates at their centers is r . When viewed from o' , a foreshortened picture of dA_1 is obtained. Instead of isometrically, the plate dA_1 may be viewed from its end as in Fig. 4.5, where dA_2 is assumed perpendicular to the plane of the paper for simplicity, the line ab represents a side of dA_1 , and the line $a'b$ represents the width of ab in the view obtained from o' . Since eo and oo' are mutually perpendicular to its

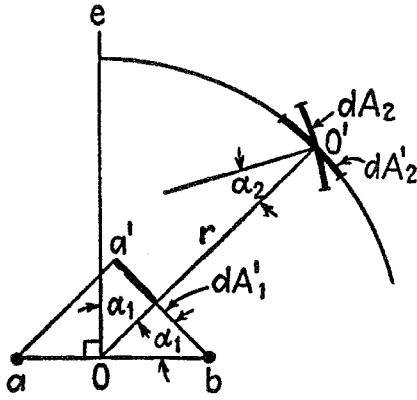


FIG. 4.5. Section view between two plates.

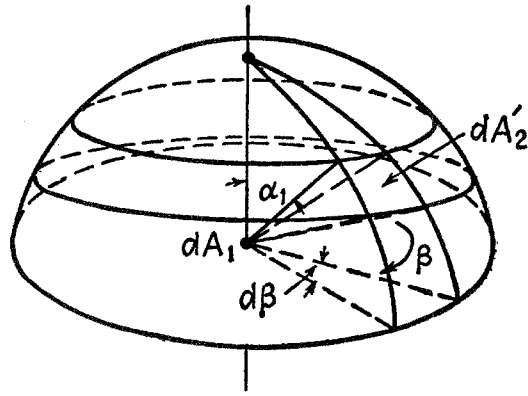


FIG. 4.6. Solid angle.

sides, angle aba' must equal α_1 and the side $a'b$ corresponding to dA'_1 is $dA'_1 = dA_1 \cos \alpha_1$. For the second plate, $dA'_2 = dA_2 \cos \alpha_2$. The center of surface dA'_2 may be considered to lie in the hemisphere receiving radiation from dA_1 , and the quantity falling on a surface in the hemisphere consequently diminishes with the square of the radius. If both are black bodies, the radiation from plate 1 to plate 2 is proportional to the normal surfaces exposed to each other and inversely to the square of the distance between them.

$$dQ_{1-2} = \frac{I_1}{r^2} dA'_1 dA'_2 \quad (4.34)$$

where I_1 is the proportionality constant dimensionally equal to the intensity of radiation. Substituting the original surfaces,

$$dQ_{1-2} = \frac{I_1}{r^2} \cos \alpha_1 \cos \alpha_2 dA_1 dA_2 \quad (4.35)$$

An important relationship exists between the intensity I_1 and the emissive power E . In Fig. 4.6 let $d\omega_1$ be the solid angle which is by definition the intercepted area on a sphere divided by r^2 . dA_1 is a small plate in

the center of the isometric plane of the base. Then

$$\begin{aligned} d\omega_1 &= \frac{dA'_2}{r^2} = \frac{r \sin \alpha \, d\beta \, r \, d\alpha}{r^2} \\ &= \sin \alpha \, d\alpha \, d\beta \end{aligned} \quad (4.36)$$

From Eqs. (4.35), (4.35a), and (4.36),

$$\begin{aligned} \frac{dQ}{dA} &= E_b = I_1 \int_0^{\pi/2} \sin \alpha \cos \alpha \, d\alpha \int_0^{2\pi} d\beta \\ &= I_1 \pi \end{aligned} \quad (4.37)$$

$$I_1 = \frac{E_b}{\pi} = \frac{\sigma T^4}{\pi} \quad (4.38)$$

Substituting Eq. (4.38) in Eq. (4.35) the net exchange between T_1 and T_2 is

$$dQ = \frac{\cos \alpha_1 \cos \alpha_2 \, dA_1 \, dA_2}{\pi r^2} \sigma (T_1^4 - T_2^4) \quad (4.39)$$

If $\cos \alpha_1 \cos \alpha_2 \, dA_2 / \pi r^2$ is written F_A , F_A is known as the configuration or geometric factor. For some systems it is very difficult to derive, but for

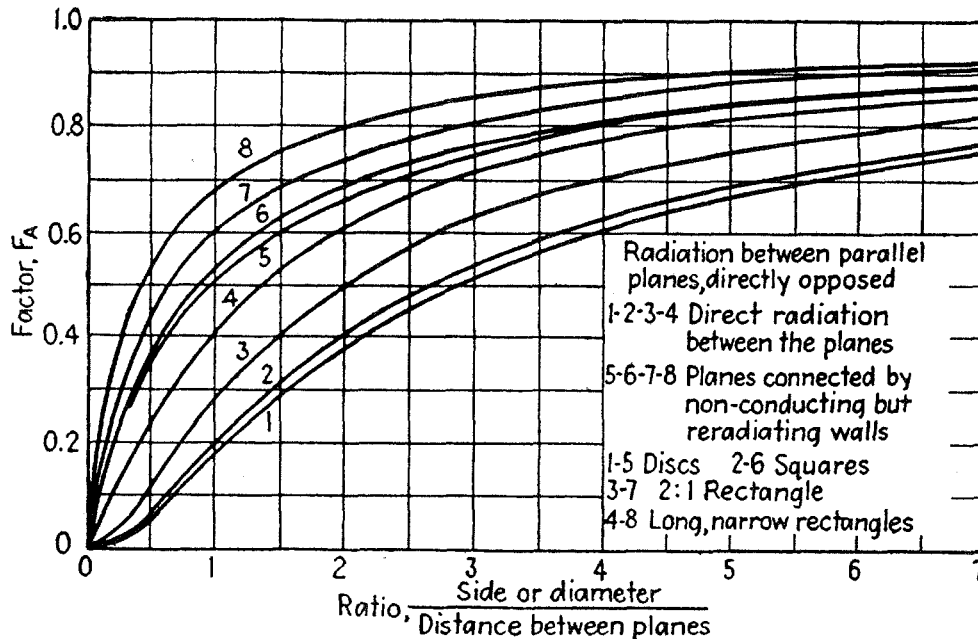


FIG. 4.7. Radiation between parallel planes. (Hottel.)

several basic arrangements it is fairly simple. Hottel¹ has integrated a number of cases, the commonest of which are plotted in Figs. 4.7, 4.8 and 4.9. Equation (4.39) can thus be written in integrated form

$$Q = F_A A_1 \sigma (T_1^4 - T_2^4) \quad (4.40)$$

¹ Hottel, H. C., *Mech. Eng.*, **52**, 699 (1930).

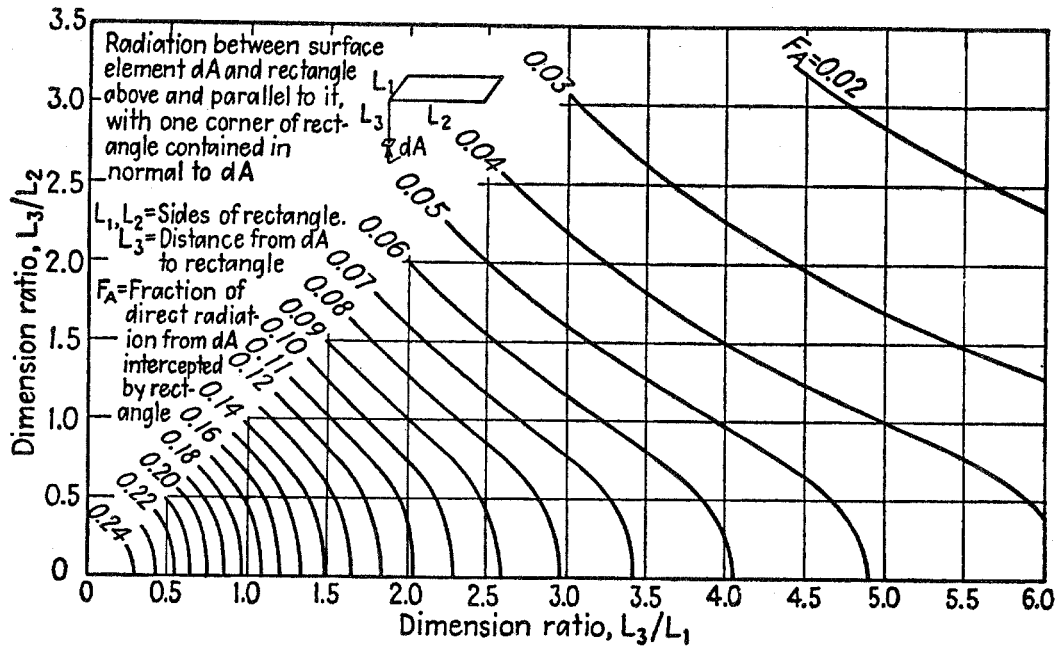


FIG. 4.8. Radiation between an element and a parallel plane. (Hottel.)

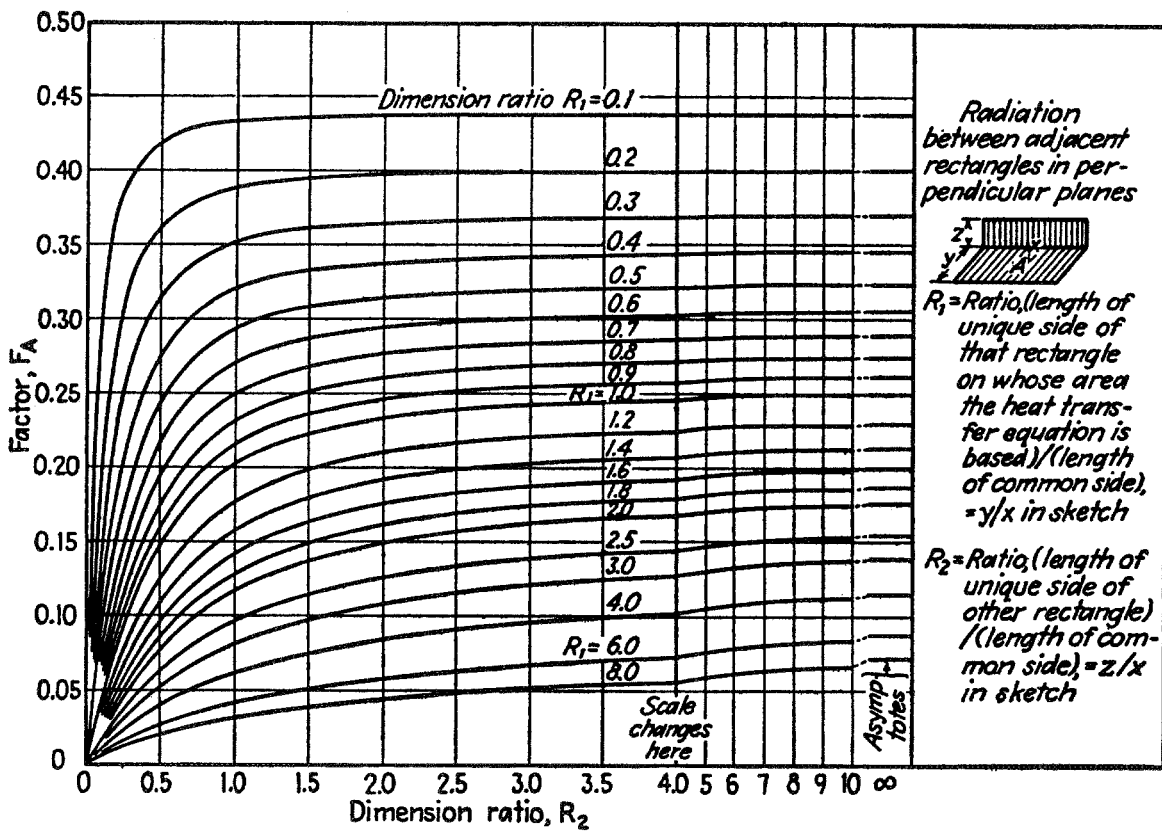


FIG. 4.9. Radiation between perpendicular planes. (Hottel.)

If the two surfaces are gray and therefore nonblack, from Eq. (4.26)

$$Q = \frac{F_A A_1 \sigma}{(1/\epsilon_1) + (1/\epsilon_2) - 1} (T_1^4 - T_2^4) \quad (4.41)$$

Writing F_e for the emissivity correction, Eq. (4.41) becomes

$$Q = F_A F_e A_1 \sigma (T_1^4 - T_2^4) \quad (4.42)$$

The summary in Table 4.2 gives the values of F_A and F_e for a number of the common cases derived here and elsewhere.

TABLE 4.2. VALUES OF F_A AND F_e

	F_A	F_e
(a) Surface A_1 small compared with the totally enclosing surface A_2	1	ϵ_1
(b) Surfaces A_1 and A_2 of parallel discs squares, 2:1 rectangles, long rectangles	Fig. 4.7	$\epsilon_1 \epsilon_2$
(c) Surface dA_1 and parallel rectangular surface A_2 with one corner of rectangle above dA_1	Fig. 4.8	$\epsilon_1 \epsilon_2$
(d) Surfaces A_1 or A_2 of perpendicular rectangles having a common side	Fig. 4.9	$\epsilon_1 \epsilon_2$
(e) Surfaces A_1 and A_2 of infinite parallel planes or surface A_1 of a completely enclosed body is small compared with A_2	1	$\frac{1}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2}\right) - 1}$
(f) Concentric spheres or infinite concentric cylinders with surfaces A_1 and A_2	1	$\frac{1}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)}$

Example 4.4. Radiation from a Pipe to a Duct. Calculate the radiation from a 2-in. IPS steel pipe carrying steam at 300°F and passing through the center of a 1-by 1-ft galvanized sheet-iron duct at 75°F and whose outside is insulated.

Solution. From Table 11 in the Appendix $A = 0.622$ ft² of external surface per linear foot of pipe. The emissivity of oxidized steel from Table 4.1 is $\epsilon_1 = 0.79$.

The surface of the duct is $A_2 = 4$ (1 × 1) = 4.0 ft²/lin ft.

The surface of the pipe is not negligible by comparison with that of the duct, and (f) of Table 4.2 applies most nearly.

$$\epsilon_2 = 0.276 \text{ (oxidized zinc in Table 4.1)}$$

$$F_A = 1$$

(Table 4.2)

$$F_e = \frac{1}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)} = \frac{1}{\frac{1}{0.79} + \frac{0.622}{4.0} \left(\frac{1}{0.276} - 1\right)} = 0.60 \quad (4.41)$$

$$\begin{aligned} Q &= F_A F_e A \sigma (T_1^4 - T_2^4) \\ &= 1 \times 0.60 \times 0.622 \times 0.173 \times 10^{-8} (760^4 - 535^4) \\ &= 164 \text{ Btu/(hr)(lin ft)} \end{aligned} \quad (4.42)$$

PROBLEMS

4.1. A 2-in. IPS steel pipe carries steam at 325°F through a room at 70°F. What decrease in radiation occurs if the bare pipe is coated with 26 per cent aluminum paint?

4.2. One wall of a corridor 8 by 28 by 4 ft wide between a drying chamber and an outside wall of the building is at 200°F, and the 8- by 28-ft wall of the building will be at 40°F during normal winter weather. What will be the heat passing across the corridor in winter if the drying chamber is surfaced with unglazed silica brick and the building wall is plastered?

4.3. A chamber for heat-curing steel sheets lacquered black on both sides operates by passing the sheets vertically between two steel plates 6 ft apart. One of the plates is at 600°F, and the other, exposed to the atmosphere, is at 80°F. What heat is transferred between the walls, and what is the temperature of the lacquered sheet when equilibrium has been reached?

4.4. A 3-in. IPS lagged pipe carries steam at 400°F through a room at 70°F. The lagging consists of a $\frac{1}{2}$ -in. thick layer of asbestos. The use of an overcoat of 26 per cent aluminum paint is to be investigated. What percentage savings in heat loss can be effected?

4.5. A molten organic compound is carried in the smaller of two concentric steel pipes 2- and 3-in. IPS. The annulus can be flooded with steam to prevent solidification, or the liquid can be heated somewhat and circulated without steam so that the annulus acts as an insulator. If the molten fluid is at 400°F and the outer pipe is at room temperature (80°F), what heat loss occurs from the molten material in 40 lin ft of pipe?

4.6. Calculate the radiant-heat loss from a furnace through a 2-in.-diameter peep door when the inside temperature is 1750°F and the outside temperature is 70°F. Consider the emission due to a black body.

4.7. A bare concrete pump house 10 by 20 by 10 ft high is to be heated by pipes laid in the concrete floor. Hot water is to be used as the heating medium to maintain a floor surface temperature of 78°F. The walls and ceiling are of such thickness as to maintain temperatures of 62°F on all inside surfaces in the winter. (a) What is the rate of radiation between the floor and the ceiling if the walls are considered nonconducting and reradiating? The pumps cover a negligible area of the floor. (b) What additional heat will be required if the floor area is doubled by enlarging the room to 20 by 20 ft?

4.8. A bath of molten zinc is located in the corner of the floor of a zinc dipping room 20 by 20 by 10 ft high. Zinc melts at 787°F, and the ceiling could be maintained at 90°F in the summer by conduction through the roof. (a) What heat will be radiated from a 1-ft² bath? (b) If the bath is moved to the center of the room, what heat will be radiated?

NOMENCLATURE FOR CHAPTER 4

A	Heat-transfer of emitting or absorbing surface, ft ²
A'	Effective surface, ft ²
a	Absorptivity, dimensionless
C_1, C_2	Constants in Planck's law
E	Emissive power, Btu/(hr)(ft ²)
F_A	Geometric factor, dimensionless
F_ϵ	Emissivity factor, dimensionless
h_r	Heat-transfer coefficient equivalent to radiation, Btu/(hr)(ft ²)(°F)

I	Intensity of radiation, Btu/(hr)(ft ²)
I_{λ}	Monochromatic intensity of emission, Btu/(hr)(ft ²)(micron)
n	Number of radiation shields
Q	Heat flow or net heat exchange, Btu/(hr)
r	Reflectivity, dimensionless; radius, ft
T	Temperature, °R
α, β, ω	Angle, deg
ϵ	Emissivity, dimensionless
λ	Wavelength, microns
σ	Stefan-Boltzmann constant, 0.173×10^{-8} Btu/(hr)(ft ²)(°R ⁴)
τ	Transmissivity, dimensionless

Subscripts

b	Black body
1	Source
2	Receiver

CHAPTER 5

TEMPERATURE

The Temperature Difference. A temperature difference is the driving force by which heat is transferred from a source to a receiver. Its influence upon a heat-transfer system which includes both a source and a receiver is the immediate subject for study.

In the experimental data of Chap. 3 the temperature of the inside pipe wall t_p was calculated from the reported value of the outside pipe-wall temperature t_w . The logarithmic mean of the differences $t_p - t_1$ and $t_p - t_2$ was used to calculate Δt_i . The reported temperature of the pipe was the average of a number of thermocouples which were not actually constant over the length of the pipe. It is not ordinarily possible in industrial equipment to measure the average pipe-wall temperature. Only the inlet and outlet temperatures of the hot and cold fluids are known or can be measured, and these are referred to as the *process temperatures*.¹

Plots of temperature vs. pipe length, t vs. L , for a system of two concentric pipes in which the annulus fluid is cooled sensibly and the pipe fluid heated sensibly are shown in Figs. 5.1 and 5.2. When the two fluids travel in opposite directions along a pipe as in Fig. 5.1, they are in *counterflow*. Figure 5.1 can be compared with Fig. 3.6 to which it is similar except that one is a plot of t vs. L and the other is a plot of t vs. Q , the heat transferred. When the fluids travel in the same direction as in Fig. 5.2, they are in *parallel* flow. The temperature of the inner pipe fluid in either case varies according to one curve as it proceeds along the length of the pipe, and the temperature of the annulus fluid varies according to

¹ In the remainder of this book the subscript 1 always denotes the inlet and subscript 2 the outlet. The cold terminal difference Δt_1 or Δt_c is given by $T_2 - t_1$, and the hot terminal difference Δt_2 or Δt_h by $T_1 - t_2$. There are two additional terms which are often used in industry. These are the *range* and the *approach*. By "range" is meant the actual temperature rise or fall, which for the hot fluid is $T_1 - T_2$ and for the cold fluid $t_2 - t_1$. The "approach" has two different meanings depending upon whether counterflow equipment such as concentric pipes or other types of equipment are intended. For counterflow the approach is the number of degrees between the hot-fluid inlet and cold-fluid outlet, $T_1 - t_2$, or hot-fluid outlet and cold fluid inlet, $T_2 - t_1$, whichever is smaller. Thus a close approach means that one terminal difference will be very small, a significant factor in heat transfer. The definition of approach for other types of equipment will be discussed in Chap. 7.

another. The temperature difference at any length from the origin where $L = 0$ is the vertical distance between the two curves.

Overall Coefficient of Heat Transfer. The concentric pipes in Figs. 5.1 and 5.2 bring together two streams each having a particular film coefficient and whose temperatures vary from inlet to outlet. For convenience, the method of calculating the temperature difference between the two should employ only the process temperatures, since these alone are generally known. To establish the temperature difference in this manner between some general temperature T of a hot fluid and some general temperature t of a cold fluid, it is necessary to account also for all the resistances between the two temperatures. In the case of two con-

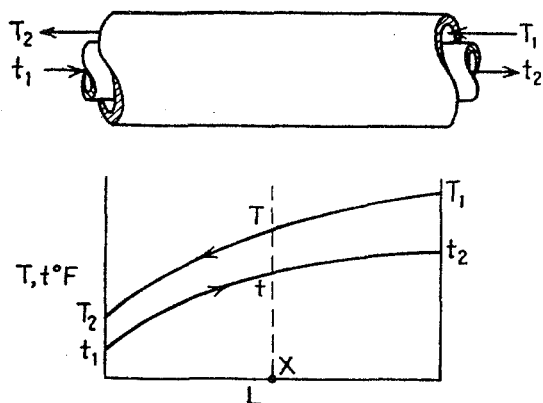


FIG. 5.1. Counterflow.

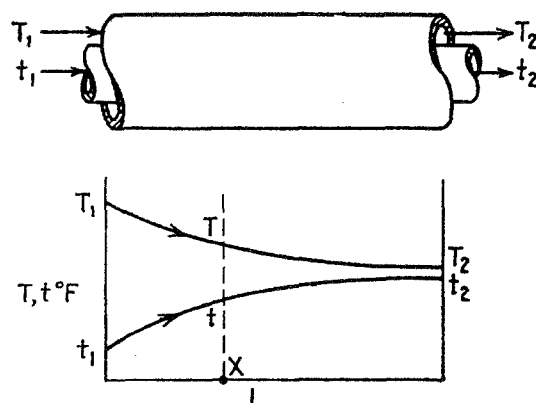


FIG. 5.2. Parallel flow.

centric pipes, the inner pipe being very thin, the resistances encountered are the pipe fluid-film resistance, the pipe-wall resistance, L_m/k_m , and the annulus fluid-film resistance. Since Q is equal to $\Delta t/\Sigma R$ as before,

$$\Sigma R = \frac{1}{h_i} + \frac{L_m}{k_m} + \frac{1}{h_o} \quad (5.1)$$

where ΣR is the overall resistance. It is customary to substitute $1/U$ for ΣR where U is called the *overall coefficient of heat transfer*. Inasmuch as a real pipe has different areas per linear foot on its inside and outside surfaces, h_i and h_o must be referred to the same heat-flow area or they will not coincide per unit length. If the outside area A of the inner pipe is used, then h_i must be multiplied by A_i/A to give the value that h_i would have if originally calculated on the basis of the larger area A instead of A_i . For a pipe with a thick wall Eq. (5.1) becomes

$$\frac{1}{U} = \frac{1}{h_i(A_i/A)} + \frac{2.3D_o}{2k_m} \log \frac{D_o}{D_i} + \frac{1}{h_o} = \frac{1}{h_i(D_i/D_o)} + \frac{2.3D_o}{2k_m} \log \frac{D_o}{D_i} + \frac{1}{h_o} \quad (5.2)$$

The integrated steady-state modification of Fourier's general equation may then be written

$$Q = UA \Delta t \quad (5.3)$$

where Δt is the temperature difference between the two streams for the entire surface A . Using the simplification that the thin metal pipe-wall resistance is negligible, Eq. (5.2) becomes

$$\frac{1}{U} = \frac{1}{h_i(A_i/A)} + \frac{1}{h_o} \quad (5.4)$$

Hereafter Eq. (5.3) will be referred to simply as the *Fourier equation*. Just as h_i was obtained from $h_i = Q/A_i \Delta t_i$ in Eq. (3.2) using thermocouples, so U can be obtained from $U = Q/A \Delta t$ using process temperatures alone. In experiments involving sensible heat transfer between two fluids, Eqs. (5.2) and (5.4) can be used to obtain either individual film coefficient from the overall coefficient U only if some supplementary means are available for computing the other film coefficient. Fortunately, the condensation of steam can provide a negligible resistance, so that h_i or h_o can usually be determined individually with suitable accuracy from an experiment using either of the fluids and steam.

Equation (5.3) is of particular value in design when the individual film coefficients can both be calculated through the use of equations of the types obtained by dimensional analysis such as Eq. (3.26) or (3.32), and U can be solved for accordingly. Then Eq. (5.3) is used in computing the total area or length of path required when Q is given and Δt is calculated from the process temperatures. When the process temperatures of the two respective streams are given, the total heat transfer Q Btu/hr is also given, being computed from $Q = wc(t_2 - t_1) = WC(T_1 - T_2)$.

The Controlling Film Coefficient. When the resistance of the pipe metal is small by comparison with the sum of the resistances of both film coefficients, and it usually is, it may be neglected. If one film coefficient is small and the other very large, the small coefficient provides the major resistance and the overall coefficient of heat transfer for the apparatus is very nearly the reciprocal of the major resistance. Suppose $h_i(A_i/A) = 10$ and $h_o = 1000$ Btu/(hr)(ft²)(°F) $R_i = 1/10 = 0.1$, $R_o = 1/1000 = 0.001$, and $\Sigma R = 0.101$. A variation of 50 per cent in R_o does not materially influence Q , since a value of $h_o = 500$ will change ΣR only from 0.101 to 0.102. When a significant difference exists, the smaller coefficient is the *controlling film coefficient*.

Logarithmic Mean Temperature Difference: Counterflow. Generally both fluids undergo temperature variations which are not straight lines when temperatures are plotted against length as in Figs. 5.1 and 5.2. At every point $T - t$ between the two streams differs, yet this should lead to the same result as Eq. (3.36), in which the logarithmic mean temperature difference was obtained from a study of $T - t$ vs. Q . However, there is an advantage to a derivation based on $T - t$ vs. L , since it permits

identification of the temperature difference anywhere along the pipe length. Later, when more complex flow patterns are encountered, this information will be essential. Although two fluids may transfer heat in a concentric pipe apparatus in either counterflow or parallel flow, *the relative direction* of the two fluids influences the value of the temperature difference. This point cannot be overemphasized: Any flow pattern formed by two fluids must be identified with its unique temperature difference. For the derivation of the temperature difference between the two fluids of Fig. 5.1 in counterflow, the following assumptions must be made:

1. The overall coefficient of heat transfer U is constant over the entire length of path.
2. The pounds per hour of fluid flow is constant, obeying the steady-state requirement.
3. The specific heat is constant over the entire length of path.
4. There are no partial phase changes in the system, *i.e.*, vaporization or condensation. The derivation is applicable for sensible-heat changes and when vaporization or condensation is isothermal over the whole length of path.
5. Heat losses are negligible.

Applying the differential form of the steady-state equation,

$$dQ = U(T - t)a'' dL \quad (5.5)$$

where a'' is the square feet of surface per foot of pipe length or

$$a'' dL = dA.$$

From a differential heat balance

$$dQ = WC dT = wc dt \quad (5.6)$$

where Q is the limit as dQ varies from 0 to Q . At any point in the pipe from left to right the heat gained by the cold fluid is equal to that given up by the hot fluid. Taking a balance from $L = 0$ to $L = X$

$$WC(T - T_2) = wc(t - t_1) \quad (5.7)$$

from which

$$T = T_2 + \frac{wc}{WC} (t - t_1) \quad (5.8)$$

From Eqs. (5.5) and (5.6) substituting for T ,

$$dQ = wc dt = U \left[T_2 + \frac{wc}{WC} (t - t_1) - t \right] a'' dL$$

t and L are the only variables. Collecting terms of t and L ,

$$\int \frac{Ua'' dL}{wc} = \int \frac{dt}{T_2 - \frac{wc}{WC} t_1 + \left(\frac{wc}{WC} - 1\right) t} \quad (5.9)$$

This right-hand term is of the form

$$\int \frac{dt}{a_1 + b_1 t} = \frac{1}{b_1} \log (a_1 + b_1 t)$$

Integrating dL between 0 and L and dt between t_1 and t_2 ,

$$\frac{UA}{wc} = \frac{1}{\left(\frac{wc}{WC} - 1\right)} \ln \frac{T_2 - \frac{wc}{WC} t_1 + \left(\frac{wc}{WC} - 1\right) t_2}{T_2 - \frac{wc}{WC} t_1 + \left(\frac{wc}{WC} - 1\right) t_1} \quad (5.10)$$

To simplify this expression substitute for T_2 in the numerator the expression from Eq. (5.7), expand the denominator, and cancel terms

$$\frac{UA}{wc} = \frac{1}{(wc/WC) - 1} \ln \frac{T_1 - t_2}{T_2 - t_1} \quad (5.11)$$

Substitute for wc/WC the expression from Eq. (5.7)

$$\begin{aligned} \frac{UA}{wc} &= \frac{1}{(T_1 - T_2)/(t_2 - t_1) - 1} \ln \frac{T_1 - t_2}{T_2 - t_1} \\ &= \frac{t_2 - t_1}{(T_1 - t_2) - (T_2 - t_1)} \ln \frac{T_1 - t_2}{T_2 - t_1} \end{aligned} \quad (5.12)$$

Since $wc(t_2 - t_1) = Q$ and substituting Δt_2 and Δt_1 for the hot and cold terminal temperature differences $T_1 - t_2$ and $T_2 - t_1$,

$$Q = UA \left(\frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1} \right) \quad (5.13)$$

If the difference between the two terminals $\Delta t_2 - \Delta t_1$ is written so as to be positive then the ratio of the two terminals taken in the same order is numerically greater than unity and any confusion due to negative signs is eliminated. The expression in parentheses in Eq. (5.13) again is the logarithmic mean or *log mean temperature difference* and is abbreviated LMTD. Equation (5.13) for counterflow may be written

$$Q = UA \Delta t = UA \times \text{LMTD} \quad (5.13a)$$

and

$$\Delta t = \text{LMTD} = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln (T_1 - t_2)/(T_2 - t_1)} = \frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1} \quad (5.14)$$

Parallel Flow. Referring to Fig. 5.2 for the case where both fluids flow in the same direction, the basic equations are essentially the same. For the steady state,

$$dQ = U(T - t)a'' dL$$

but

$$dQ = WC dT = -wc dt$$

since t declines in the direction of increasing values of T . Taking the heat balance between X and the left end,

$$W C (T - T_2) = w c (t_2 - t)$$

Again considering the hot terminal difference $\Delta t_2 = T_1 - t_1$ as the *greater temperature difference* in parallel flow and $\Delta t_1 = T_2 - t_2$ the *lesser temperature difference*, the result is

$$Q = UA \frac{(T_1 - t_1) - (T_2 - t_2)}{\ln (T_1 - t_1)/(T_2 - t_2)} = UA \frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1} \quad (5.15)$$

Relation between Parallel Flow and Counterflow. It may appear from the final form of the derivations for the two flow arrangements that there is little to choose between the two. The examples which follow demonstrate that except where one fluid is isothermal (such as condensing steam) there is a distinct thermal *disadvantage* to the use of parallel flow.

Example 5.1. Calculation of the LMTD. A hot fluid enters a concentric-pipe apparatus at a temperature of 300°F and is to be cooled to 200°F by a cold fluid entering at 100°F and heated to 150°F. Shall they be directed in parallel flow or counterflow?

Solution. It is convenient to write the temperatures in the form employed here and to realize that the log mean is always somewhat less than the arithmetic mean $(\Delta t_2 + \Delta t_1)/2$.

(a) *Counterflow:*

$$\begin{array}{rcl} \text{Hot fluid} & \text{Cold fluid} & \\ (T_1) 300 & - & 150 (t_2) = 150 (\Delta t_2) \\ (T_2) 200 & - & 100 (t_1) = 100 (\Delta t_1) \\ & & \hline & & 50 (\Delta t_2 - \Delta t_1) \end{array}$$

$$\text{LMTD} = \frac{\Delta t_2 - \Delta t_1}{2.3 \log \Delta t_2 / \Delta t_1} = \frac{50}{2.3 \log 150/100} = 123.5^\circ\text{F}$$

(b) *Parallel flow:*

$$\begin{array}{rcl} \text{Hot fluid} & \text{Cold fluid} & \\ (T_1) 300 & - & 100 (t_1) = 200 (\Delta t_2) \\ (T_2) 200 & - & 150 (t_2) = 50 (\Delta t_1) \\ & & \hline & & 150 (\Delta t_2 - \Delta t_1) \end{array}$$

$$\text{LMTD} = \frac{150}{2.3 \log 200/50} = 108^\circ\text{F}$$

The LMTD for the same process temperatures in parallel flow is lower than for counterflow.

Example 5.2. Calculation of the LMTD with Equal Outlet Temperatures. A hot fluid enters a concentric-pipe apparatus at 300°F and is to be cooled to 200°F by a cold fluid entering at 150°F and heated to 200°F.

(a) *Counterflow*:

$$\begin{array}{rclcl}
 \text{Hot fluid} & \text{Cold fluid} & & & \\
 300 & - & 200 & = & 100 \ (\Delta t_2) \\
 200 & - & 150 & = & 50 \ (\Delta t_1) \\
 & & & & \hline
 & & & & 50 \ (\Delta t_2 - \Delta t_1) \\
 \text{LMTD} & = & \frac{50}{2.3 \log \frac{100}{50}} & = & 72
 \end{array}$$

(b) *Parallel flow*:

$$\begin{array}{rclcl}
 \text{Hot fluid} & \text{Cold fluid} & & & \\
 300 & - & 150 & = & 150 \ (\Delta t_2) \\
 200 & - & 200 & = & 0 \ (\Delta t_1) \\
 & & & & \hline
 & & & & 150 \ (\Delta t_2 - \Delta t_1) \\
 \text{LMTD} & = & \frac{150}{2.3 \log \frac{150}{0}} & = & 0
 \end{array}$$

In parallel flow the lowest temperature theoretically attainable by the hot fluid is that of the outlet temperature of the cold fluid, t_2 . If this temperature were attained, the LMTD would be zero. In the Fourier equation $Q = UA \Delta t$, since Q and U are finite, the heat-transfer surface A would have to be infinite. The last is obviously not feasible.

The inability of the hot fluid in parallel flow to fall below the outlet temperature of the cold fluid has a marked effect upon the ability of parallel-flow apparatus to *recover* heat. Suppose it is desired to recover as much heat as possible from the hot fluid in Example 5.1 by using the same quantities of hot and cold fluid as before but by assuming that more heat-transfer surface is available. In a counterflow apparatus it is possible to have the hot-fluid outlet T_2 fall to within perhaps 5 or 10° of the cold-fluid inlet t_1 , say 110°F. In parallel-flow apparatus the heat transfer would be restricted by the cold-fluid outlet temperature rather than the cold-fluid inlet and the difference would be the loss in recoverable heat. Parallel flow is used for cold viscous fluids, however, since the arrangement may enable a higher value of U to be obtained.

Consider next the case where the hot terminal difference (Δt_2) in the preceding examples is not the greater temperature difference.

Example 5.3. Calculation of the LMTD When $\Delta t_c > \Delta t_h$. While a hot fluid is cooled from 300 to 200°F in counterflow, a cold fluid is heated from 100 to 275°F.

Counterflow:

$$\begin{array}{rclcl}
 \text{Hot fluid} & \text{Cold fluid} & & & \\
 300 & - & 275 & = & 25 \ (\Delta t_h) \\
 200 & - & 100 & = & 100 \ (\Delta t_c) \\
 & & & & \hline
 & & & & 75 \ (\Delta t_c - \Delta t_h) \\
 \text{LMTD} & = & \frac{75}{2.3 \log \frac{100}{25}} & = & 54.3^\circ\text{F}
 \end{array}$$

Lastly, when one of the fluids proceeds through the apparatus isothermally (condensing steam), parallel flow and counterflow yield identical temperature differences.

Example 5.4. Calculation of the LMTD with One Isothermal Fluid. A cold fluid is heated from 100 to 275°F by steam at 300°F.

(a) *Counterflow*:

<i>Hot fluid</i>		<i>Cold fluid</i>	
300	—	275	= 25
300	—	100	= 200

(b) *Parallel flow*:

<i>Hot fluid</i>		<i>Cold fluid</i>	
300	—	100	= 200
300	—	275	= 25

These are identical.

Hereafter, unless specifically qualified, all temperature arrangements will be assumed in counterflow. Many industrial types of equipment are actually a compromise between parallel flow and counterflow and receive additional study in later chapters.

Heat Recovery in Counterflow. Very often a counterflow apparatus is available which has a given length L and therefore a fixed surface A . Two process streams are available with inlet temperatures T_1, t_1 and flow rates and specific heats W, C and w, c . What outlet temperatures will be attained in the apparatus?

This problem requires an estimate of U which can be checked by the methods of succeeding chapters for different types of counterflow heat-transfer equipment. Rewriting Eq. (5.12),

$$wc(t_2 - t_1) = UA \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln (T_1 - t_2)/(T_2 - t_1)}$$

Rearranging

$$\ln \frac{T_1 - t_2}{T_2 - t_1} = \frac{UA}{wc} \left(\frac{T_1 - T_2}{t_2 - t_1} - 1 \right) \quad (5.16)$$

Since $WC(T_1 - T_2) = wc(t_2 - t_1)$, $wc/WC = (T_1 - T_2)/(t_2 - t_1)$. This means that the ratio of the temperature ranges can be established without recourse to actual working temperatures. Calling this unique ratio R without a subscript

$$R = \frac{wc}{WC} = \frac{T_1 - T_2}{t_2 - t_1}$$

Substituting in Eq. (5.16) and removing logarithms,

$$\frac{T_1 - t_2}{T_2 - t_1} = e^{(UA/wc)(R-1)} \quad (5.17)$$

To obtain an expression for T_2 alone.

$$t_2 = t_1 + \frac{T_1 - T_2}{R}$$

Substituting in Eq. (5.17) and solving,

$$T_2 = \frac{(1 - R)T_1 + [1 - e^{(UA/wc)(R-1)}]Rt_1}{1 - Re^{(UA/wc)(R-1)}} \quad (5.18)$$

For parallel flow it becomes

$$T_2 = \frac{[R + e^{(UA/wc)(R+1)}]T_1 + [e^{(UA/wc)(R+1)} - 1]Rt_1}{(R + 1)e^{(UA/wc)(R+1)}} \quad (5.19)$$

t_2 may be obtained from T_2 by applying the heat balance

$$WC(T_1 - T_2) = wc(t_2 - t_1)$$

The Caloric or Average Fluid Temperature. Of the four assumptions used in the derivation of Eq. (5.14) for the LMTD, the one which is subject to the largest deviation is that of a constant overall heat-transfer coefficient U . In the calculations of Chap. 3 the film coefficient was computed for the properties of the fluid at the arithmetic mean temperature between inlet and outlet, although the correctness of this calculation was not verified. In fluid-fluid heat exchange the hot fluid possesses a viscosity on entering which becomes greater as the fluid cools. The cold counterflow fluid enters with a viscosity which decreases as it is heated. There is a hot terminal $T_1 - t_2$ and a cold terminal $T_2 - t_1$, and the values of h_o and $h_i(A_i/A)$ vary over the length of the pipe to produce a larger U at the hot terminal than at the cold terminal. As a simple example, take the case of an individual transfer coefficient at the inlet and outlet as obtained from the data of Morris and Whitman through use of Eq. (3.42).

Example 5.5. Calculation of h_1 and h_2 . Calculation of point B-6:

$$t_1 = 99.1^\circ \quad t_2 = 129.2^\circ\text{F}$$

Inlet at 99.1°F :

$$c = 0.478 \text{ Btu/lb} \quad \left(\frac{DG}{\mu}\right)^{0.9} = \left(\frac{0.622 \times 854,000}{12 \times 2.95 \times 2.42}\right)^{0.9} = 2570$$

$$\mu = 2.95 \text{ cp}$$

$$k = 0.078 \text{ Btu/(hr)(ft)(}^\circ\text{F/ft)} \quad \left(\frac{c\mu}{k}\right)^{1/4} = \left(\frac{0.478 \times 2.95 \times 2.42}{0.078}\right)^{1/4} = 3.52$$

$$h_i = 0.078 \times \frac{12}{0.622} \times 0.0115 \times 2570 \times 3.52 = 156 \text{ at inlet}$$

Outlet at 129.2°F:

$$c = 0.495 \quad \left(\frac{DG}{\mu}\right)^{0.9} = \left(\frac{0.622 \times 854,000}{12 \times 2.20 \times 2.42}\right)^{0.9} = 3390$$

$$\mu = 2.20$$

$$k = 0.078 \quad \left(\frac{c\mu}{k}\right)^{1/4} = \left(\frac{0.495 \times 2.20 \times 2.42}{0.078}\right)^{1/4} = 3.23$$

$$h_2 = 0.078 \times \frac{12}{0.622} \times 0.0115 \times 3390 \times 3.23 = 190 \text{ at outlet}$$

At the arithmetic mean (114.3°F) $h_a = 174.5$, which is only 3.6 per cent in error of the experimental value of 181 but the variations against h_i at the arithmetic mean are $\left(\frac{156 - 174.5}{174.5}\right) 100 = -10.6$ per cent and $\left(\frac{190 - 174.5}{174.5}\right) 100 = +8.9$ per cent.

From the above it is seen that under actual conditions the variation of U may be even greater than that of h_i alone, since the outside film coefficient h_o will vary at the same time and in the same direction as h_i . The variation of U can be taken into account by numerical integration of dQ , the heat transferred over incremental lengths of the pipe $a''dL = dA$, and using the average values of U from point to point in the differential equation $dQ = U_{av} dA \Delta t$. The summation from point to point then gives $Q = UA \Delta t$ very closely. This is a time-consuming method, and the increase in the accuracy of the result does not warrant the effort. Colburn¹ has undertaken the solution of problems with varying values of U by assuming the variation of U to be linear with temperature and by deriving an expression for the true temperature difference accordingly. The ratio of the LMTD for constant U and the true temperature difference for varying U is then used as the basis for establishing a single overall coefficient which is the *true mean* rather than the arithmetic mean.

Assume:

1. The variation of U is given by the expression $U = a'(1 + b't)$
2. Constant weight flow
3. Constant specific heat
4. No partial phase changes

Over the whole transfer path

$$Q = WC(T_1 - T_2) = wc(t_2 - t_1)$$

Since $R = wc/WC = (T_1 - T_2)/(t_2 - t_1)$ or generalized as in Fig. 5.1,

$$R = \frac{(T - T_2)}{(t - t_1)}$$

The heat balance for the differential area dA is given by

$$dQ = U(T - t) dA = wc dt$$

¹ Colburn, A. P., *Ind. Eng. Chem.*, **35**, 873-877 (1933).

where U is the average value for the increment or

$$\frac{dt}{U(T-t)} = \frac{dA}{wc}$$

Since $U = a'(1 + b't)$, substitute for U .

$$\frac{dt}{a'(1 + b't)(T-t)} = \frac{dA}{wc}$$

From the heat balance obtain the expression for T in terms of t and separate into parts.

$$\frac{1}{a'(R-1-b'T_2+b'Rt_1)} \int_{t_1}^{t_2} \left[\frac{(R-1) dt}{T_2 - Rt_1 + (R-1)t} - \frac{b' dt}{1 + b't} \right] = \int \frac{dA}{wc} \quad (5.20)$$

Integrating,

$$\frac{1}{a'(R-1-b'T_2+b'Rt_1)} \left[\ln \frac{T_2 - Rt_1 + (R-1)t_2}{T_2 - Rt_1 + (R-1)t_1} - \ln \frac{1 + b't_2}{1 + b't_1} \right] = \frac{A}{wc} \quad (5.21)$$

Using the subscript 1 to indicate the cold terminal and 2 the hot terminal as heretofore

$$U_1 = a'(1 + b't_1) \quad U_2 = a'(1 + b't_2)$$

As before,

$$\Delta t_1 = T_2 - t_1 \quad \Delta t_2 = T_1 - t_2$$

and factoring Eq. (5.21)

$$\frac{t_2 - t_1}{U_1 \Delta t_2 - U_2 \Delta t_1} \ln \frac{U_1 \Delta t_2}{U_2 \Delta t_1} = \frac{A}{wc} \quad (5.22)$$

Combining with $Q = wc(t_2 - t_1)$,

$$\frac{Q}{A} = \frac{U_1 \Delta t_2 - U_2 \Delta t_1}{\ln U_1 \Delta t_2 / U_2 \Delta t_1} \quad (5.23)$$

Equation (5.23) is a modification of Eq. (5.13) which accounts for the variation of U by replacing it with U_1 and U_2 , where $A = 0$ and $A = A$, respectively. This is still unsatisfactory, however, since it requires twice calculating both individual film coefficients to obtain U_1 and U_2 . Colburn chose to obtain a *single* overall coefficient, U_x , at which all the surface can be regarded to be transferring heat at the LMTD. U_x is then defined by

$$\frac{Q}{A} = \frac{U_1 \Delta t_2 - U_2 \Delta t_1}{\ln U_1 \Delta t_2 / U_2 \Delta t_1} = U_x \left(\frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1} \right) \quad (5.24)$$

Substituting $U_x = a'(1 + b't_c)$,

$$U_x = a'(1 + b't_c) = \frac{\frac{a'(1 + b't_1) \Delta t_2 - a'(1 + b't_2) \Delta t_1}{\ln [(1 + b't_1) \Delta t_2] / [(1 + b't_2) \Delta t_1]}}{\frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1}} \quad (5.25)$$

U_x will now be identified by finding t_c , the *temperature* of the properties at which h_i and h_o are computed and at which such a value of U_x exists. Let F_c be a fraction. By multiplying the temperature rise of the *controlling* (film) *stream* by F_c and adding the resulting fractional rise to the lower terminal temperature of the stream, a temperature is obtained at which to evaluate heat transfer properties and calculate h_i , h_o , and U_x .

$$F_c = \frac{t_c - t_1}{t_2 - t_1} \quad (5.26)$$

t_c is the *caloric temperature* of the cold stream. By definition, let

$$K_c = \frac{t_2 - t_1}{1/b' + t_1} = \frac{U_2 - U_1}{U_1} \quad r = \frac{\Delta t_1}{\Delta t_2} = \frac{\Delta t_c}{\Delta t_h}$$

and substituting the equivalents in Eq. (5.25),

$$\frac{1 + b't_2}{1 + b't_1} = K_c + 1 \quad \frac{1 + b't_c}{1 + b't_1} = K_c F_c + 1$$

from which

$$F_c = \frac{(1/K_c) + [r/(r - 1)]}{1 + \frac{\ln (K_c + 1)}{\ln r}} - \frac{1}{K_c} \quad (5.27)$$

Equation (5.27) has been plotted in Fig. 17 in the Appendix with

$$K_c = \frac{U_2 - U_1}{U_1} = \frac{U_h - U_c}{U_c}$$

as the parameter, where c and h refer to the cold and hot terminals, respectively. The caloric fraction F_c can be obtained from Fig. 17 by computing K_c from U_h and U_c and $\Delta t_c/\Delta t_h$ for the process conditions. The caloric temperature of the hot fluid T_c is

$$T_c = T_2 + F_c(T_1 - T_2) \quad (5.28)$$

and for the cold fluid

$$t_c = t_1 + F_c(t_2 - t_1) \quad (5.29)$$

Colburn has correlated in the insert of Fig. 17 the values of K_c where the *controlling* film is that of a petroleum cut. A correlation of this type can

be made in any industry which deals with a particular group of fluids by obtaining a' and b' from properties and eliminates the calculation of U_h and U_c . If an apparatus transfers heat between two petroleum cuts, the cut giving the largest value of K_c is controlling and can be used directly to establish F_c for both streams from the figure. Thus, whenever there is a sizeable difference between U_h and U_c the LMTD is not the true temperature difference for counterflow. The LMTD may be retained, however, if a suitable value of U is employed to compensate for its use in Eq. (5.13).

Example 5.6. Calculation of the Caloric Temperature. A 20°API crude oil is cooled from 300 to 200°F by heating cold 60°API gasoline from 80 to 120°F in a counterflow apparatus. At what fluid temperatures should U be evaluated?

Solution:

Shell 20°API crude: 60°API gasoline		Tubes		
242.5	Caloric temp.	97		
250	Mean	100		
300	Higher temp.	120	180	Δt_2
200	Lower temp.	80	120	Δt_1
100	Diff	40		

Crude, $T_1 - T_2 = 300 - 200 = 100^\circ\text{F}$, $K_c = 0.68$ from Fig. 17 insert

Gasoline, $t_2 - t_1 = 120 - 80 = 40^\circ\text{F}$, $K_c \leq 0.10$

The larger value of K_c corresponds to the controlling heat-transfer coefficient which is assumed to establish the variation of U with temperature. Then

$$\frac{\Delta t_c}{\Delta t_h} = \frac{200 - 80}{300 - 120} = 0.667$$

$$F_c = 0.425 \text{ from Fig. 17}$$

Caloric temperature of crude, $T_c = 200 + 0.425(300 - 200) = 242.5^\circ\text{F}$.

Caloric temperature of gasoline, $t_c = 80 + 0.425(120 - 80) = 97.0^\circ\text{F}$.

It should be noted that there can be but one caloric mean and that the factor F_c applies to both streams but is determined by the controlling stream.

The Pipe-wall Temperature. The temperature of the pipe wall can be computed from the caloric temperatures when both h_i and h_o are known. Referring to Fig. 5.3 it is customary to neglect the temperature difference across the pipe metal $t_w - t_p$ and to consider the entire pipe to be at the temperature of the outside surface of the wall t_w . If the outside caloric temperature is T_c and the inside caloric temperature t_c and $1/R_{io} = h_{io} = h_i(A_i/A) = h_i \times (\text{ID}/\text{OD})$, where the subscript *io* refers

to the value of the coefficient inside the pipe referred to the outside surface of the pipe.

$$Q = \frac{\Delta t}{\Sigma R} = \frac{T_c - t_c}{R_o + R_{io}} = \frac{t_w - t_c}{R_{io}} \quad (5.30)$$

Replacing the resistances in the last two terms by film coefficients,

$$\frac{T_c - t_c}{1/h_o + 1/h_{io}} = \frac{t_w - t_c}{1/h_{io}}$$

Solving for t_w

$$t_w = t_c + \frac{h_o}{h_{io} + h_o} (T_c - t_c) \quad (5.31)$$

and

$$t_w = T_c - \frac{h_{io}}{h_{io} + h_o} (T_c - t_c) \quad (5.32)$$

When the hot fluid is inside the pipe these become

$$t_w = t_c + \frac{h_{io}}{h_{io} + h_o} (T_c - t_c) \quad (5.31a)$$

and

$$t_w = T_c - \frac{h_o}{h_{io} + h_o} (T_c - t_c) \quad (5.32a)$$

Isothermal Representation of Heating and Cooling. In streamline flow when a fluid flows isothermally, the velocity distribution is assumed to be parabolic. When a given quantity of liquid is heated as it travels along a pipe, the viscosity near the pipe wall is lower than that of the bulk of the fluid. The fluid near the wall travels at a faster velocity than it would in isothermal flow and modifies the parabolic velocity

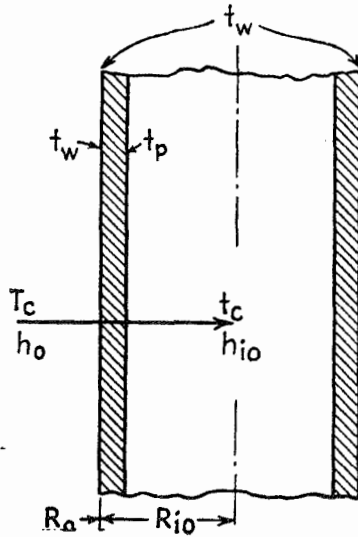


FIG. 5.3. Pipe-wall temperature.

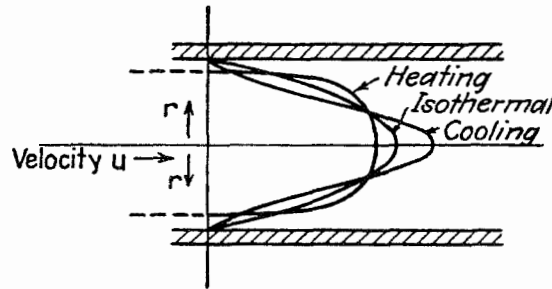


FIG. 5.4. Heating, cooling, and isothermal streamline flow.

distribution as indicated by the heating curve in Fig. 5.4. If the liquid is cooled, the reverse occurs: The fluid near the wall flows at a lower velocity than in isothermal flow, producing the velocity distribution indicated for cooling. For the liquid to flow more rapidly at the wall during heating some of the liquid near the center axis of the pipe must

flow outward toward the wall to maintain the increased velocity. This is a radial velocity component which actually modifies the nature of the streamline flow. If data for heating an oil in a given temperature range are plotted as in Fig. 3.10 together with data for cooling the oil in the same temperature range, two families of points are obtained. The data on heating give higher heat-transfer coefficients than those on cooling. Colburn¹ undertook to convert both heating and cooling data to a single isothermal line. He was able to employ a basic equation of the form of Eq. (3.32) by multiplying the right-hand term by the dimensionless ratio $(\mu/\mu_f)^{r'}$ where μ is the viscosity at the caloric temperature and μ_f is the viscosity at an arbitrary film temperature defined for streamline flow by

$$t_f = t_{av} + \frac{1}{4}(t_w - t_{av}) \quad (5.33)$$

and for turbulent flow by

$$t_f = t_{av} + \frac{1}{2}(t_w - t_{av}) \quad (5.34)$$

Sieder and Tate² undertook the correlation of a large quantity of data in tubes, rather than pipes, and obtained a dimensionless factor $(\mu/\mu_w)^{r''}$ where μ_w is the viscosity at the tube-wall temperature t_w . Using the Sieder and Tate correction, Eq. (3.32) for streamline flow becomes

$$\frac{h_i D}{k} = \alpha \left[\left(\frac{DG}{\mu} \right) \left(\frac{c\mu}{k} \right) \left(\frac{D}{L} \right) \right]^{d+1} \left(\frac{\mu}{\mu_w} \right)^{r''} \quad (5.35)$$

Equation (3.26) for turbulent flow becomes

$$\frac{h_i D}{k} = \alpha \left(\frac{DG}{\mu} \right)^p \left(\frac{c\mu}{k} \right)^q \left(\frac{\mu}{\mu_w} \right)^{r'''} \quad (5.36)$$

By incorporating the correlation factor for heating and cooling in this manner a single curve is obtained for both heating and cooling, since the value of μ/μ_w is greater than 1.0 for liquid heating and lower than 1.0 for liquid cooling. Inasmuch as the viscosities of gases increase rather than decrease with higher temperature, the deviations from the isothermal velocity distribution are the reverse of liquids.

PROBLEMS

5.1. For a concentric-pipe heat-transfer apparatus having a 1-in. IPS inner pipe the film coefficient h_i has been computed to be 10.0 Btu/(hr)(ft²)(°F). By suitable calculation three different fluids, when circulated through the annulus, will have film coefficients of 10.0, 20.0, and 200, respectively. Neglecting the resistance of the pipe, how does the value of the annulus coefficient affect the value of the overall coefficient?

5.2. (a) For values of $h_i = 100$ and $h_o = 500$, what error results in the calculated value of U for a concentric-pipe heat-transfer apparatus having a 2-in. IPS inner pipe when the metal resistance is obtained from $R_m = L_m/k_m A$ instead of $(2.3/2\pi k_m)$

¹ Colburn, A. P., *Trans. AIChE*, **29**, 174-210 (1933).

² Sieder, E. N., and G. E. Tate, *Ind. Eng. Chem.*, **28**, 1429-1436 (1936).

$\log D_2/D_1$ and when the pipe resistance is omitted entirely. (b) What are the errors when the coefficients are $h_i = 10$ and $h_o = 50$?

5.3. Calculate the LMTD for counterflow in the following cases in which the hot fluid is cooled from 200 to 100°F and the cold fluid, heated through an equal range in each case, is (a) 90 to 140°F, (b) 80 to 130°F, and (c) 60 to 110°F. Observe the nature of the deviation of the LMTD from the arithmetic means of the two terminal differences in each case.

5.4. A hot fluid is cooled from 245 to 225°F in each case. Compare the advantage of counterflow over parallel flow in the size of the LMTD when a cold fluid is to be heated from (a) 135 to 220°F, (b) 125 to 210°F, and (c) 50 to 135°F.

5.5. 10,000 lb/hr of cold benzene is heated under pressure from 100°F by cooling 9000 lb/hr of nitrobenzene at a temperature of 220°F. Heat transfer will occur in a concentric-pipe apparatus having a 1¼-in. IPS inner pipe 240 ft long. Tests on similar equipment transferring heat between the same liquids indicates that a value of $U = 120$ based on the outside surface of the inner pipe may be expected. (a) What outlet temperatures may be expected in counterflow? (b) What outlet temperatures may be expected in parallel flow? (c) If part of the concentric pipe is removed leaving only 160 lin ft, what outlet temperatures may be expected in counterflow?

5.6. Benzene is to be heated in a concentric-pipe apparatus having a 1¼-in. IPS inner pipe from 100 to 140°F by 8000 lb/hr of nitrobenzene having an initial temperature of 180°F. A value of $U = 100$ may be expected based on the outside surface of the pipe. How much cold benzene can be heated in 160 lin ft of concentric pipe (a) in counterflow, (b) in parallel flow? (*Hint.* Trial-and-error solution.)

5.7. Aniline is to be cooled from 200 to 150°F in a concentric-pipe apparatus having 70 ft² of external pipe surface by 8600 lb/hr of toluene entering at 100°F. A value of $U = 75$ may be anticipated. How much hot aniline can be cooled in counterflow?

5.8. In a counterflow concentric-pipe apparatus a liquid is cooled from 250 to 200°F by heating another from 100 to 225°F. The value of U_1 , at the cold terminal, is calculated to be 50.0 from the properties at the cold terminal, and U_2 at the hot terminal is calculated to be 60.0. At what fluid temperatures should U be computed to express the overall heat transfer for the entire apparatus?

5.9. In a counterflow concentric-pipe apparatus a liquid is cooled from 250 to 150°F by heating another from 125 to 150°F. The value of U_1 at the cold terminal is 52 and at the hot terminal U_2 is 58. At what liquid temperatures should U for the overall transfer be computed?

5.10. The calculation of the caloric temperatures can be accomplished directly by evaluating a' and b' in $U = a'(1 + b't)$ for a given temperature range. If the hot liquid in Prob. 5.8 always provides the controlling film coefficient, what are the numerical values of the constants a' and b' ?

5.11. A 40°API kerosene is cooled from 400 to 200°F by heating 34°API crude oil from 100 to 200°F. Between what caloric temperatures is the heat transferred, and how do these deviate from the mean?

5.12. A 35° API distillate used as a heating oil is cooled from 400 to 300°F by fresh 35°API distillate heated from 200 to 300°F. Between what caloric temperatures is the heat transferred, and how do these deviate from the mean?

NOMENCLATURE FOR CHAPTER 5

A	Heat-transfer surface or outside surface of pipes, ft²
a''	External pipe surface per foot of length, ft
a', b'	Constants in the equation $U = a'(1 + b't)$

a_1, b_1	Constants
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of pipe, ft
F_c	Caloric fraction, dimensionless
G	Mass velocity, lb/(hr)(ft ²)
h	Heat-transfer coefficient, Btu/(hr)(ft ²)(°F)
h_i, h_o	Inside and outside film coefficients, Btu/(hr)(ft ²)(°F)
h_{io}	$h_i A_i / A$, inside film coefficient referred to outside surface, Btu/(hr)(ft ²)(°F)
K_c	Caloric factor, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Length, ft
LMTD	Log mean temperature difference, °F
Q	Heat flow, Btu/hr
R	Ratio of $w_c/WC = (T_1 - T_2)/(t_2 - t_1)$, dimensionless
ΣR	Overall resistance to heat flow, (hr)(ft ²)(°F)/Btu
r	Ratio of cold to hot terminal differences, dimensionless
T	Temperature of the hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
T_1, T_2	Hot-fluid inlet and outlet temperatures, respectively, °F
t	Temperature of the cold fluid, °F
t_c	Caloric temperature of cold fluid, °F
t_f	Film temperature in Eqs. (5.33) and (5.34), °F
t_p	Inside pipe-wall temperature, °F
t_w	Outside pipe-wall temperature, °F
t_1, t_2	Cold-fluid inlet and outlet temperatures, respectively, °F
Δt	Temperature difference at a point or mean over an area, °F
$\Delta t_c, \Delta t_h$	Temperature difference at the cold and hot terminals, respectively, °F
Δt_i	Logarithmic mean of $t_p - t_1$ and $t_p - t_2$, °F
U	Overall coefficient of heat transfer in general, Btu/(hr)(ft ²)(°F)
U_c, U_h	Overall coefficient of heat transfer at cold and hot terminals, Btu/(hr)(ft ²)(°F)
U_1, U_2	Value of U at t_c , Btu/(hr)(ft ²)(°F)
U_x	Value of U at t_c , Btu/(hr)(ft ²)(°F)
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
α	Proportionality constant, dimensionless
μ	Viscosity at mean or caloric temperature, lb/(ft)(hr)
μ_f, μ_w	Viscosity at the film and pipe-wall temperatures, respectively, lb/(ft)(hr)

Superscripts d, p, q, r', r'', r''' Constants**Subscripts (except as noted above)**

i Inside a pipe or tube
 o Outside a pipe or tube
 io Value based on inside of a pipe or tube referred to the outside of the tube.

CHAPTER 6

COUNTERFLOW: DOUBLE PIPE EXCHANGERS

Definitions. Heat-transfer equipment is defined by the function it fulfills in a process. *Exchangers* recover heat between two process streams. Steam and cooling water are utilities and are not considered in the same sense as recoverable process streams. *Heaters* are used primarily to heat process fluids, and steam is usually employed for this purpose, although in oil refineries hot recirculated oil serves the same purpose. *Coolers* are employed to cool process fluids, water being the main cooling medium. *Condensers* are coolers whose primary purpose is the removal of latent instead of sensible heat. The purpose of *reboilers* is to supply the heat requirements of a distillation process as latent heat. *Evaporators* are employed for the concentration of a solution by the evaporation of water. If any other fluid is vaporized besides water, the unit is a *vaporizer*.

Double Pipe Exchangers. For the derivations in Chap. 5, a concentric-pipe heat-transfer apparatus was employed. The industrial counter-

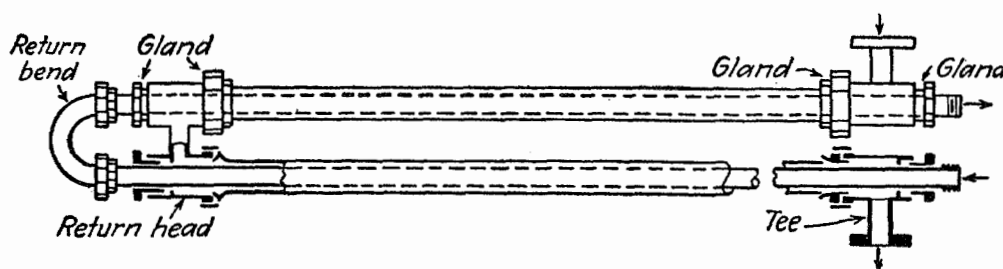


FIG. 6.1. Double pipe exchanger.

part of this apparatus is the double pipe exchanger shown in Fig. 6.1. The principal parts are two sets of concentric pipes, two connecting Tees, and a return head and a return bend. The inner pipe is supported within the outer pipe by packing glands, and the fluid enters the inner pipe through a threaded connection located outside the exchanger section proper. The Tees have nozzles or screwed connections attached to them to permit the entry and exit of the annulus fluid which crosses from one leg to the other through the return head. The two lengths of inner pipe are connected by a return bend which is usually exposed and does not provide effective heat-transfer surface. When arranged in two legs as in Fig. 6.1, the unit is a *hairpin*.

The double pipe exchanger is extremely useful because it can be assembled in any pipe-fitting shop from standard parts and provides inexpensive heat-transfer surface. The standard sizes of Tees and return heads are given in Table 6.1.

TABLE 6.1. DOUBLE PIPE EXCHANGER FITTINGS

Outer pipe, IPS	Inner pipe, IPS
2	1¼
2½	1¼
3	2
4	3

Double pipe exchangers are usually assembled in 12-, 15-, or 20-ft effective lengths, the effective length being the distance in *each* leg over which heat transfer occurs and excludes inner pipe protruding beyond the exchanger section. When hairpins are employed in excess of 20 ft in length corresponding to 40 effective linear feet or more of double pipe, the inner pipe tends to sag and touch the outer pipe, thereby causing a poor flow distribution in the annulus. The principal disadvantage to the use of double pipe exchangers lies in the small amount of heat-transfer surface contained in a single hairpin. When used with distillation equipment on an industrial process a very large number are required. These require considerable space, and each double pipe exchanger introduces no fewer than 14 points at which leakage might occur. The time and expense required for dismantling and periodically cleaning are prohibitive compared with other types of equipment. However, the double pipe exchanger is of greatest use where the total required heat-transfer surface is small, 100 to 200 ft² or less.

Film Coefficients for Fluids in Pipes and Tubes. Equation (3.42) was obtained for heating several oils in a pipe based on the data of Morris and Whitman. Sieder and Tate¹ made a later correlation of both heating and cooling a number of fluids, principally petroleum fractions, in horizontal and vertical tubes and arrived at an equation for streamline flow where $DG/\mu < 2100$ in the form of Eq. (5.35).

$$\frac{h_i D}{k} = 1.86 \left[\left(\frac{DG}{\mu} \right) \left(\frac{c\mu}{k} \right) \left(\frac{D}{L} \right) \right]^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14} = 1.86 \left(\frac{4}{\pi} \frac{wc}{kL} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (6.1)$$

where L is the total length of the heat-transfer path before mixing occurs. Equation (6.1) gave maximum mean deviations of approximately ± 12 per cent from $Re = 100$ to $Re = 2100$ except for water. Beyond the transition range, the data may be extended to turbulent flow in the form of Eq. (5.36).

$$\frac{h_i D}{k} = 0.027 \left(\frac{DG}{\mu} \right)^{0.8} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (6.2)$$

¹ Sieder, E. N., and G. E. Tate, *Ind. Eng. Chem.*, **28**, 1429-1436 (1936).

Equation (6.2) gave maximum mean deviations of approximately +15 and -10 per cent for the Reynolds numbers above 10,000. While Eqs. (6.1) and (6.2) were obtained for tubes, they will also be used indiscriminately for pipes. Pipes are rougher than tubes and produce more turbulence for equal Reynolds numbers. Coefficients calculated from tube-data correlations are actually lower and safer than corresponding calculations based on pipe data and there are no pipe correlations in the literature so extensive as tube correlations. Equations (6.1) and (6.2) are applicable for organic liquids, aqueous solutions, and gases. They are not conservative for water, and additional data for water will be given later. In

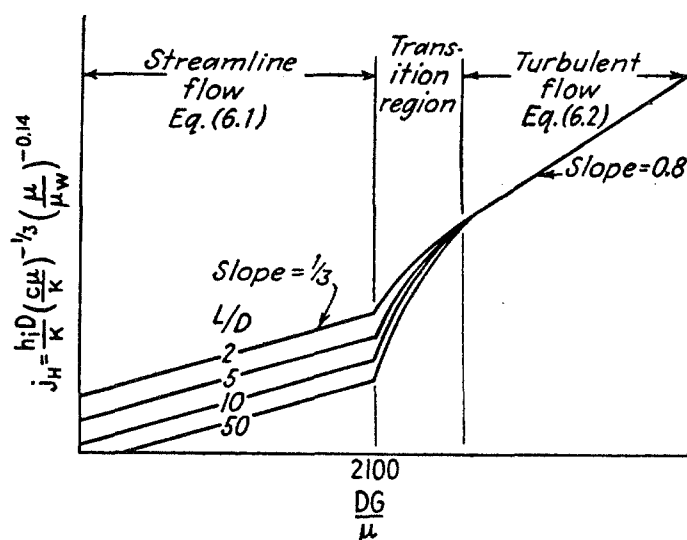


FIG. 6.2. Tube-side heat-transfer curve.

order to permit the graphical representation of both equations on a single pair of coordinates, refer to Fig. 6.2. Using the ordinate

$$j_H = \left(\frac{h_i D}{k} \right) \left(\frac{c \mu}{k} \right)^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14}$$

and the abscissa (DG/μ) only Eq. (6.2) can be shown. By using D/L or L/D as a parameter, Eq. (6.1) can also be included. The transition region joins the two. Working plots of Eqs. (6.1) and (6.2) are given in Fig. 24 of the Appendix together with a line of slope 0.14 to facilitate the solution of the ratio $\phi = (\mu/\mu_w)^{0.14}$.

Fluids Flowing in Annuli: The Equivalent Diameter. When a fluid flows in a conduit having other than a circular cross section, such as an annulus, it is convenient to express heat-transfer coefficients and friction factors by the same types of equations and curves used for pipes and tubes. To permit this type of representation for annulus heat transfer it has been found advantageous to employ an *equivalent diameter* D_e . The equivalent diameter is four times the hydraulic radius, and the

hydraulic radius is, in turn, the radius of a pipe equivalent to the annulus cross section. The hydraulic radius is obtained as the ratio of the flow area to the wetted perimeter. For a fluid flowing in an annulus as shown in Fig. 6.3 the flow area is evidently $(\pi/4)(D_2^2 - D_1^2)$ but the wetted perimeters for heat transfer and pressure drops are different. For heat transfer the wetted perimeter is the outer circumference of the inner pipe with diameter D_1 , and for heat transfer in annuli

$$D_e = 4r_h = \frac{4 \times \text{flow area}}{\text{wetted perimeter}} = \frac{4\pi(D_2^2 - D_1^2)}{4\pi D_1} = \frac{D_2^2 - D_1^2}{D_1} \quad (6.3)$$

In pressure-drop calculations the friction not only results from the resistance of the outer pipe but is also affected by the outer surface of the inner pipe. The total wetted perimeter is $\pi(D_2 + D_1)$, and for the pressure drop in annuli

$$D'_e = \frac{4 \times \text{flow area}}{\text{frictional wetted perimeter}} = \frac{4\pi(D_2^2 - D_1^2)}{4\pi(D_2 + D_1)} = D_2 - D_1 \quad (6.4)$$

This leads to the anomalous result that the Reynolds numbers for the same flow conditions, w , G , and μ , are different for heat transfer and pressure drop since D_e might be above 2100 while D'_e is below 2100. Actually both Reynolds numbers should be considered only approximations, since the sharp distinction between streamline and turbulent flow at the Reynolds number of 2100 is not completely valid in annuli.

Film Coefficients for Fluids in Annuli. When the equivalent diameter from Eq. (6.3) is substituted for D , Fig. 24 in the Appendix may be considered a plot of $D_e G / \mu$ vs. $(h_o D_e / k)(c \mu / k)^{-1/2} (\mu / \mu_w)^{-0.14}$. h_o is the outside or annulus coefficient and is obtained in the same manner as h_i by multiplication of the ordinate. Even though D differs from D_e , h_o is effective at the outside diameter of the inner pipe. In double pipe exchangers it is customary to use the outside surface of the inner pipe as the reference surface in $Q = UA \Delta t$, and since h_i has been determined for A_i and not A , it must be corrected. h_i is based on the area corresponding to the inside diameter where the surface per foot of length is $\pi \times \text{ID}$. On the outside of the pipe the surface per foot of length is $\pi \times \text{OD}$; and again letting h_{io} be the value of h_i referred to the outside diameter,

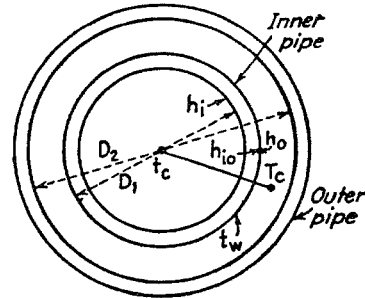


FIG. 6.3. Annulus diameters and location of coefficients.

$$h_{io} = h_i \frac{A_i}{A} = h_i \frac{\text{ID}}{\text{OD}} \quad (6.5)$$

Fouling Factors. The overall coefficient of heat transfer *required* to fulfill the process conditions may be determined from the Fourier equation when the surface A is known and Q and Δt are calculated from the process conditions. Then $U = Q/A \Delta t$. If the surface is not known U can be obtained independently of the Fourier equation from the two film coefficients. Neglecting the pipe-wall resistance,

$$\frac{1}{U} = R_{io} + R_o = \frac{1}{h_{io}} + \frac{1}{h_o} \quad (6.6)$$

or

$$U = \frac{h_{io}h_o}{h_{io} + h_o} \quad (6.7)$$

The locations of the coefficients and temperatures are shown in Fig. 6.3. When U has been obtained from values of h_{io} and h_o and Q and Δt are calculated from the process conditions, the surface A required for the process can be computed. The calculation of A is known as *design*.

When heat-transfer apparatus has been in service from some time, however, dirt and scale deposit on the inside and outside of the pipe, adding two more resistances than were included in the calculation of U by Eq.

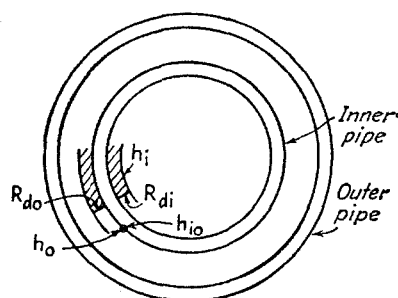


FIG. 6.4. Location of fouling factors and heat-transfer coefficients.

(6.6). The additional resistances reduce the original value of U , and the required amount of heat is no longer transferred by the original surface A ; T_2 rises above and t_2 falls below the desired outlet temperatures, although h_i and h_o remain substantially constant. To overcome this eventuality, it is customary in designing equipment to anticipate the deposition of dirt and scale by introducing a resistance R_d called the *dirt, scale, or fouling factor*, or resistance. Let R_{di} be the dirt factor for the

inner pipe fluid at its inside diameter and R_{do} the dirt factor for the annulus fluid at the outside diameter of the inner pipe. These may be considered very thin for dirt but may be appreciably thick for scale, which has a higher thermal conductivity than dirt. The resistances are shown in Fig. 6.4. The value of U obtained in Eq. (6.7) only from $1/h_{io}$ and $1/h_o$ may be considered the *clean overall coefficient* designated by U_c to show that dirt has not been taken into account. The coefficient which includes the dirt resistance is called the *design or dirty overall coefficient* U_D . The value of A corresponding to U_D rather than U_c provides the basis on which equipment is ultimately built. The relationship between the two overall coefficients U_c and U_D is

$$\frac{1}{U_D} = \frac{1}{U_c} + R_{di} + R_{do} \quad (6.8)$$

or setting

$$R_{di}^* + R_{do} = R_d \quad (6.9)$$

$$\frac{1}{U_D} = \frac{1}{U_C} + R_d \quad (6.10)$$

Thus suppose that for a double pipe exchanger, h_{io} and h_o have been computed to be 300 and 100 respectively, then

$$\frac{1}{U_C} = \frac{1}{h_{io}} + \frac{1}{h_o} = 0.0033 + 0.01 = 0.0133,$$

or $U_C = 1/0.0133 = 75.0$ Btu/(hr)(ft²)(°F). From experience, let us say, it has been found that a thermal dirt resistance of $R_{di} = 0.001$ (hr)(ft²)(°F)/Btu will deposit annually inside the pipe and $R_{do} = 0.0015$ will deposit on the outside of the pipe. For what overall coefficient should the surface be calculated so that the apparatus need be cleaned only once a year? Then $R_d = R_{di} + R_{do} = 0.0025$, and

$$\frac{1}{U_D} = \frac{1}{U_C} + R_d = \frac{1}{75.0} + 0.0025 = 0.0158 \text{ (hr)(ft}^2\text{)(°F)/Btu}$$

or

$$U_D = \frac{1}{0.0158} = 63.3 \text{ Btu/(hr)(ft}^2\text{)(°F)}$$

The Fourier equation for surface on which dirt will be deposited becomes

$$Q = U_D A \Delta t \quad (6.11)$$

If it is desired to obtain A , then h_{io} and h_o must first be calculated from equations such as Eq. (6.1) and (6.2) which are independent of the extent of the surface but dependent upon its form, such as the diameter and fluid flow area. With these, U_C is obtained from Eq. (6.6) and U_D is obtained from U_C using Eq. (6.10). Sometimes, however, it is desirable to study the rates at which dirt accumulates on a known surface A . U_C will remain constant if the scale or dirt deposit does not alter the mass velocity by constricting the fluid flow area. U_D and Δt will obviously change as the dirt accumulates because the temperatures of the fluids will vary from the time the surface is freshly placed in service until it becomes fouled. If Δt is calculated from *observed* temperatures instead of process temperatures then Eq. (6.11) may be used to determine R_d

* Actually R_{di} should be referred to the outside diameter as $R_{dio} = R_i(A/A_i)$. When a cylinder is very thin compared with its diameter, as a layer of dirt, its resistance is nearly the same as that through a flat wall. As shown by Eqs. (2.34) and (2.35), failure to correct to the outside will introduce a negligible error, usually well below 1 per cent. For thick scale, however, the error may be appreciable.

for a given fouling period. Then from Eq. (6.10)

$$R_d = \frac{1}{U_D} - \frac{1}{U_C} \quad (6.12)$$

which can also be written

$$R_d = \frac{U_C - U_D}{U_C U_D} \quad (6.13)$$

When R_d (deposited) $> R_d$ (allowed), as after a period of service, the apparatus no longer delivers a quantity of heat equal to the process requirements and must be cleaned.

Numerical values of the dirt or fouling factors for a variety of process services are provided in Appendix Table 12. The tabulated fouling factors are intended to protect the exchanger from delivering less than the required process heat load for a period of about a year to a year and a half. Actually the purpose of the tabulated fouling factors should be considered from another point of view. In designing a process plant containing many heat exchangers but without alternate or spare pieces of heat-transfer equipment, the process must be discontinued and the equipment cleaned as soon as the first exchanger becomes fouled. It is impractical to shut down every time one exchanger or another is fouled, and by using the tabulated fouling factors, it can be arranged so that all the exchangers in the process become dirty at the same time regardless of service. At that time all can be dismantled and cleaned during a single shutdown. The tabulated values may differ from those encountered by experience in particular services. If too frequent cleaning is necessary, a greater value of R_d should be kept in mind for future design.

It is to be expected that heat-transfer equipment will transfer more heat than the process requirements when newly placed in service and that it will deteriorate through operation, as a result of dirt, until it just fulfills the process requirements. The calculation of the temperatures delivered initially by a clean exchanger whose surface has been designed for U_D but which is operating without dirt and which is consequently oversurfaced is not difficult. Referring to Eqs. (5.18) and (5.19) use U_C for U and the actual surface of the exchanger A (which is based on U_D). This calculation is also useful in checking whether or not a clean exchanger will be able to deliver the process heat requirements when it becomes dirty.

Pressure Drop in Pipes and Pipe Annuli. The *pressure-drop* allowance in an exchanger is the static fluid pressure which may be expended to drive the fluid through the exchanger. The pump selected for the circulation of a process fluid is one which develops sufficient head at the desired capacity to overcome the frictional losses caused by connecting

pipings, fittings, control regulators, and the pressure drop in the exchanger itself. To this head must be added the static pressure at the end of the line such as the elevation or pressure of the final receiving vessel. Once a definite pressure drop allowance has been designated for an exchanger as a part of a pumping circuit, it should always be utilized as completely as possible in the exchanger, since it will otherwise be blown off or expanded through a pressure reducer. Since in Eq. (3.44)

$$\Delta F \propto G^2 \quad \left(\text{nearly, since } f \text{ varies somewhat with } \frac{DG}{\mu} \right)$$

and in Eq. (6.2) for turbulent flow

$$h_i \propto G^{0.8} \quad (\text{nearly})$$

the best use of available pressure is to increase the mass velocity which also increases h_i and lessens the size and cost of the apparatus. It is customary to allow a pressure drop of 5 to 10 psi for an exchanger or battery of exchangers fulfilling a single process service except where the flow is by gravity. For each pumped stream 10 psi is fairly standard. For gravity flow the allowable pressure drop is determined by the elevation of the storage vessel above the final outlet z in feet of fluid. The feet of fluid may be converted to pounds per square inch by multiplying z by $\rho/144$.

The pressure drop in pipes can be computed from the Fanning equation [Eq. (3.44)], using an appropriate value of f from Eq. (3.46) or Eq. (3.47b), depending upon the type of flow. For the pressure drop in fluids flowing in annuli, replace D in the Reynolds number by D'_e to obtain f . The Fanning equation may then be modified to give

$$\Delta F = \frac{4fG^2L}{2g\rho^2D'_e} \quad (6.14)$$

Where several double pipe exchangers are connected in series, annulus to annulus and pipe to pipe as in Fig. 6.5, the length in Eq. (3.44) or (6.14) is the total for the entire path.

The pressure drop computed by Eq. (3.44) or (6.14) does not include the pressure drop encountered when the fluid enters or leaves exchangers. For the inner pipes of double pipe exchangers connected in series, the entrance loss is usually negligible, but for annuli it may be significant. The allowance of a pressure drop of one velocity head, $V^2/2g'$, per hairpin will ordinarily suffice. Suppose water flows in an annulus with a mass velocity of 720,000 lb/(hr)(ft²). Since $\rho = 62.5$ lb/ft³ (approximately),

$$V = \frac{G}{3600\rho} = \frac{720,000}{3600 \times 62.5} = 3.2 \text{ fps.}$$

The pressure drop per hairpin will be $3.2^2/(2 \times 32.2) = 0.159$ ft of water or 0.07 psi. Unless the velocity is well above 3 fps, the entrance and exit losses may be neglected. Values of $V^2/2g'$ are plotted directly against the mass velocity for a fluid with a specific gravity of 1.0 in Fig. 27 in the Appendix.

The Calculation of a Double Pipe Exchanger. All the equations developed previously will be combined to outline the solution of a double pipe exchanger. The calculation consists simply of computing h_o and h_{io} to obtain U_c . Allowing a reasonable fouling resistance, a value of U_D is calculated from which the surface can be found with the use of the Fourier equation $Q = U_D A \Delta t$.

Usually the first problem is to determine which fluid should be placed in the annulus and which in the inner pipe. This is expedited by establishing the relative sizes of the flow areas for both streams. For equal allowable pressure drops on both the hot and cold streams, the decision rests in the arrangement producing the most nearly equal mass velocities and pressure drops. For the standard arrangements of double pipes the flow areas are given in Table 6.2.

TABLE 6.2. FLOW AREAS AND EQUIVALENT DIAMETERS IN DOUBLE PIPE EXCHANGERS

Exchanger, IPS	Flow area, in. ²		Annulus, in.	
	Annulus	Pipe	d_e	d'_e
2 $\times$ 1 $\frac{1}{4}$	1.19	1.50	0.915	0.40
2 $\frac{1}{2}$ $\times$ 1 $\frac{1}{4}$	2.63	1.50	2.02	0.81
3 $\times$ 2	2.93	3.35	1.57	0.69
4 $\times$ 3	3.14	7.38	1.14	0.53

In the outline below, hot- and cold-fluid temperatures are represented by upper and lower case letters, respectively. All fluid properties are indicated by lower case letters to eliminate the requirement for new nomenclature.

Process conditions required:

Hot fluid: T_1, T_2, W, c, s or $\rho, \mu, k, \Delta P, R_{do}$ or R_{di}

Cold fluid: t_1, t_2, w, c, s or $\rho, \mu, k, \Delta P, R_{di}$ or R_{do}

The diameter of the pipes must be given or assumed.

A convenient order of calculation follows:

- (1) From T_1 , T_2 , t_1 , t_2 check the heat balance, Q , using c at T_{mean} and t_{mean} .

$$Q = WC(T_1 - T_2) = wc(t_2 - t_1)$$

Radiation losses from the exchanger are usually insignificant compared with the heat load transferred in the exchanger.

- (2) LMTD, assuming counterflow. (5.14)
- (3) T_c and t_c : If the liquid is neither a petroleum fraction nor a hydrocarbon the caloric temperatures cannot be determined through the use of Fig. 17 and Eqs. (5.28) and (5.29). Instead, the calculation of U_c must be performed for the hot and cold terminals giving U_h and U_c from which one may obtain K_c . F_c is then gotten from Fig. 17 or Eq. (5.27). If neither of the liquids is very viscous at the cold terminal, say not more than 1.0 centipoise, if the temperature ranges do not exceed 50 to 100°F, and if the temperature difference is less than 50°F, the arithmetic means of T_1 and T_2 and t_1 and t_2 may be used in place of T_c and t_c for evaluating the physical properties. For nonviscous fluids $\phi = (\mu/\mu_w)^{0.14}$ may be taken as 1.0 as assumed below.

Inner pipe:

- (4) Flow area, $a_p = \pi D^2/4$, ft².
- (5) Mass velocity, $G_p = w/a_p$, lb/(hr)(ft²).
- (6) Obtain μ at T_c or t_c depending upon which flows through the inner pipe. μ , lb/(ft)(hr) = centipoise $\times 2.42$.
From D ft, G_p lb/(hr)(ft²), μ lb/(ft)(hr) obtain the Reynolds number, $Re_p = DG_p/\mu$.
- (7) From Fig. 24 in which $j_H = (h_i D/k)(c\mu/k)^{-1/3}(\mu/\mu_w)^{-0.14}$ vs. DG_p/μ obtain j_H .
- (8) From c Btu/(lb)(°F), μ lb/(ft)(hr), k Btu/(hr)(ft²)(°F/ft), all obtained at T_c or t_c compute $(c\mu/k)^{1/3}$.
- (9) To obtain h_i multiply j_H by $(k/D)(c\mu/k)^{1/3}$ ($\phi = 1.0$) or

$$\frac{h_i D}{k} \left(\frac{c\mu}{k} \right)^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/3} \times 1.0 = h_i \text{ Btu/(hr)(ft}^2\text{)(°F)} \quad (6.15a)$$

- (10) Convert h_i to h_{io} ; $h_{io} = h_i(A_i/A) = h_i \times \text{ID/OD}$. (6.5)

Annulus:

- (4') Flow area, $a_a = \pi(D_2^2 - D_1^2)/4$, ft²

$$\text{Equivalent diameter } D_e = \frac{4 \times \text{flow area}}{\text{wetted perimeter}} = \frac{D_2^2 - D_1^2}{D_1} \quad \text{ft} \quad (6.3)$$

- (5') Mass velocity, $G_a = w/a_a$, lb/(hr)(ft²)
 (6') Obtain μ at T_c or t_c , lb/(ft)(hr) = centipoise $\times 2.42$. From D_e ft, G_a lb/(hr)(ft²), μ lb/(ft)(hr) obtain the Reynolds number,

$$Re_a = \frac{D_e G_a}{\mu}$$

- (7') From Fig. 24 in which $j_H = (h_o D_e / k)(c\mu/k)^{-1/3}(\mu/\mu_w)^{-0.14}$ vs. $D_e G_a / \mu$ obtain j_H .
 (8') From c , μ , and k , all obtained at T_c or t_c compute $(c\mu/k)^{1/3}$.
 (9') To obtain h_o multiply j_H by $(k/D_e)(c\mu/k)^{1/3}$ ($\phi = 1.0$) or

$$\frac{h_o D_e}{k} \left(\frac{c\mu}{k} \right)^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \times 1.0 = h_o \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.15b)$$

Overall coefficients:

- (11) Compute $U_c = h_{io} h_o / (h_{io} + h_o)$, Btu/(hr)(ft²)(°F). (6.7)
 (12) Compute U_D from $1/U_D = 1/U_c + R_a$. (6.10)
 (13) Compute A from $Q = U_D A \Delta t$ which may be translated into length. If the length should not correspond to an integral number of hairpins, a change in the dirt factor will result. The recalculated dirt factor should equal or exceed the required dirt factor by using the next larger integral number of hairpins.

Calculation of ΔP . This requires a knowledge of the total length of path satisfying the heat-transfer requirements.

Inner pipe:

- (1) For Re_p in (6) above obtain f from Eq. (3.46) or (3.47b).
 (2) $\Delta F_p = 4fG^2L/2g\rho^2D$, ft. (3.45)
 $\Delta F_p \rho / 144 = \Delta P_p$, psi.

Annulus:

- (1') Obtain $D'_e = \frac{4\pi(D_2^2 - D_1^2)}{4\pi(D_2 + D_1)} = (D_2 - D_1)$. (6.4)

Compute the frictional Reynolds number, $Re'_a = D'_e G_a / \mu$. For Re'_a obtain f from Eq. (3.46) or (3.47b).

- (2') $\Delta F_a = 4fG^2L/2g\rho^2D'_e$, ft. (6.14)
 (3') Entrance and exit losses, one velocity head per hairpin:

$$\Delta F_i = \frac{V^2}{2g'} \text{ ft/hairpin}$$

$$(\Delta F_a + \Delta F_i) \rho / 144 = \Delta P_a, \text{ psi.}$$

There is an advantage if both fluids are computed side by side, and the use of the outline in this manner will be demonstrated in Example 6.1.

Example 6.1. Double Pipe Benzene-Toluene Exchanger. It is desired to heat 9820 lb/hr of cold benzene from 80 to 120°F using hot toluene which is cooled from 160 to 100°F. The specific gravities at 68°F are 0.88 and 0.87, respectively. The other fluid properties will be found in the Appendix. A fouling factor of 0.001 should be provided for each stream, and the allowable pressure drop on each stream is 10.0 psi.

A number of 20-ft hairpins of 2- by 1¼-in. IPS pipe are available. How many hairpins are required?

Solution:

(1) Heat balance:

$$\text{Benzene, } t_{av} = \frac{80 + 120}{2} = 100^\circ\text{F} \quad c = 0.425 \text{ Btu/(lb)}(^\circ\text{F}) \quad (\text{Fig. 2})$$

$$Q = 9820 \times 0.425(120 - 80) = 167,000 \text{ Btu/hr}$$

$$\text{Toluene, } T_{av} = \frac{160 + 100}{2} = 130^\circ\text{F} \quad c = 0.44 \text{ Btu/(lb)}(^\circ\text{F}) \quad (\text{Fig. 2})$$

$$W = \frac{167,000}{0.44(160 - 100)} = 6330 \text{ lb/hr}$$

(2) LMTD, (see the method of Chap. 3):

Hot fluid		Cold fluid	Diff.	
160	Higher temp	120	40	Δt_2
100	Lower temp	80	20	Δt_1
			20	$\Delta t_2 - \Delta t_1$

$$\text{LMTD} = \frac{\Delta t_2 - \Delta t_1}{2.3 \log \Delta t_2 / \Delta t_1} = \frac{20}{2.3 \log 4/2} = 28.8^\circ\text{F} \quad (5.14)$$

(3) Caloric temperatures: A check of both streams will show that neither is viscous at the cold terminal (the viscosities less than 1 centipoise) and the temperature ranges and temperature difference are moderate. The coefficients may accordingly be evaluated from properties at the arithmetic mean, and the value of $(\mu/\mu_w)^{0.14}$ may be assumed equal to 1.0.

$$T_{av} = \frac{1}{2}(160 + 100) = 130^\circ\text{F} \quad t_{av} = \frac{1}{2}(120 + 80) = 100^\circ\text{F}$$

Proceed now to the inner pipe. A check of Table 6.2 indicates that the flow area of the inner pipe is greater than that of the annulus. Place the larger stream, benzene in the inner pipe.

Hot fluid: annulus, toluene

(4') Flow area,
 $D_2 = 2.067/12 = 0.1725 \text{ ft}$
 $D_1 = 1.66/12 = 0.138 \text{ ft}$
 $a_a = \pi(D_2^2 - D_1^2)/4$
 $= \pi(0.1725^2 - 0.138^2)/4 = 0.00826 \text{ ft}^2$
 Equiv diam, $D_e = (D_2^2 - D_1^2)/D_1 \text{ ft}$
 [Eq (6.3)]
 $D_e = (0.1725^2 - 0.138^2)/0.138$
 $= 0.0762 \text{ ft}$

Cold fluid: inner pipe, benzene

(4) $D = 1.38/12 = 0.115 \text{ ft}$
 Flow area, $a_p = \pi D^2/4$
 $= \pi \times 0.115^2/4 = 0.0104 \text{ ft}^2$

Hot fluid: annulus, toluene

$$(5') \text{ Mass vel, } G_a = W/a_a = 6330/0.00826 = 767,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6') \text{ At } 130^\circ\text{F, } \mu = 0.41 \text{ cp [Fig. 14]} \\ = 0.41 \times 2.42 = 0.99 \text{ lb/(ft)(hr)}$$

$$\text{Reynolds no., } Re_a = \frac{D_e G_a}{\mu} \\ = 0.0762 \times 767,000/0.99 = 59,000$$

$$(7') j_H = 167 \quad [\text{Fig. 24}]$$

$$(8') \text{ At } 130^\circ\text{F, } c = 0.44 \text{ Btu/(lb)(}^\circ\text{F)} \quad [\text{Fig. 2}]$$

$$k = 0.085 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Table 4}]$$

$$\left(\frac{c\mu}{k}\right)^{1/4} = \left(\frac{0.44 \times 0.99}{0.085}\right)^{1/4} = 1.725$$

$$(9') h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/4} \left(\frac{\mu}{\mu_w}\right)^{0.14} \quad [\text{Eq. (6.15b)}] \\ = 167 \times \frac{0.085}{0.0762} \times 1.725 \times 1.0 \\ = 323 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Cold fluid: inner pipe, benzene

$$(5) \text{ Mass vel, } G_p = w/a_p = 9820/0.0104 = 943,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6) \text{ At } 100^\circ\text{F, } \mu = 0.50 \text{ cp [Fig. 14]} \\ = 0.50 \times 2.42 = 1.21 \text{ lb/(ft)(hr)}$$

$$\text{Reynolds no., } Re_p = \frac{D G_p}{\mu} \\ = 0.115 \times 943,000/1.21 = 89,500$$

$$(7) j_H = 236 \quad [\text{Fig. 24}]$$

$$(8) \text{ At } 100^\circ\text{F, } c = 0.425 \text{ Btu/(lb)(}^\circ\text{F)} \quad [\text{Fig. 2}]$$

$$k = 0.091 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Table 4}]$$

$$\left(\frac{c\mu}{k}\right)^{1/4} = \left(\frac{0.425 \times 1.21}{0.091}\right)^{1/4} = 1.78$$

$$(9) h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/4} \left(\frac{\mu}{\mu_w}\right)^{0.14} \quad [\text{Eq. (6.15a)}] \\ = 236 \times \frac{0.091}{0.115} \times 1.78 \times 1.0 \\ = 333 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(10) Correct h_i to the surface at the OD

$$h_{io} = h_i \times \frac{ID}{OD} \quad [\text{Eq. (6.5)}] \\ = 333 \times \frac{1.38}{1.66} = 276$$

Now proceed to the annulus.

(11) Clean overall coefficient, U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{276 \times 323}{276 + 323} = 149 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.7)$$

(12) Design overall coefficient, U_D :

$$\frac{1}{U_D} = \frac{1}{U_C} + R_d \quad (6.10)$$

$$R_d = 0.002 \text{ (required by problem)}$$

$$\frac{1}{U_D} = \frac{1}{149} + 0.002$$

$$U_D = 115 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Summary

323	h outside	276
U_C	149	
U_D	115	

(13) Required surface:

$$Q = U_D A \Delta t \quad A = \frac{Q}{U_D \Delta t}$$

$$\text{Surface} = \frac{167,000}{115 \times 28.8} = 50.5 \text{ ft}^2$$

From Table 11 for 1½-in. IPS standard pipe there are 0.435 ft² of external surface per foot length.

$$\text{Required length} = \frac{50.5}{0.435} = 116 \text{ lin ft}$$

This may be fulfilled by connecting three 20-ft hairpins in series.

(14) The surface supplied will actually be $120 \times 0.435 = 52.2 \text{ ft}^2$. The dirt factor will accordingly be greater than required. The actual design coefficient is

$$U_D = \frac{167,000}{52.2 \times 28.8} = 111 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

$$R_d = \frac{U_c - U_D}{U_c U_D} = \frac{149 - 111}{149 \times 111} = 0.0023 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Pressure Drop

(1') D'_e for pressure drop differs from D_e for heat transfer.

$$D'_e = (D_2 - D_1) \quad [\text{Eq. (6.4)}]$$

$$= (0.1725 - 0.138) = 0.0345 \text{ ft}$$

$$Re'_a = \frac{D'_e G_a}{\mu}$$

$$= 0.0345 \times 767,000 / 0.99 = 26,800$$

$$f = 0.0035 + \frac{0.264}{26,800^{0.42}} = 0.0071 \quad [\text{Eq. (3.47b)}]$$

$$s = 0.87, \rho = 62.5 \times 0.87 = 54.3 \quad [\text{Table 6}]$$

$$(2') \Delta F_a = \frac{4fG_a^2 L}{2g\rho^2 D'_e}$$

$$= \frac{4 \times 0.0071 \times 767,000^2 \times 120}{2 \times 4.18 \times 10^8 \times 54.3^2 \times 0.0345}$$

$$= 23.5 \text{ ft}$$

$$(3') V = \frac{G}{3600\rho} = \frac{767,000}{3600 \times 54.3} = 3.92 \text{ fps}$$

$$F_l = 3 \left(\frac{V^2}{2g'} \right) = 3 \times \frac{3.92^2}{2 \times 32.2} = 0.7 \text{ ft}$$

$$\Delta P_a = \frac{(23.5 + 0.7)54.3}{144} = 9.2 \text{ psi}$$

Allowable $\Delta P_a = 10.0 \text{ psi}$

(1) For $Re_p = 89,500$ in (6) above

$$f = 0.0035 + \frac{0.264}{(DG/\mu)^{0.42}} \quad [\text{Eq. (3.47b)}]$$

$$= 0.0035 + \frac{0.264}{89,500^{0.42}} = 0.0057$$

$$s = 0.88, \rho = 62.5 \times 0.88 = 55.0 \quad [\text{Table 6}]$$

$$(2) \Delta F_p = \frac{4fG_p^2 L}{2g\rho^2 D}$$

$$= \frac{4 \times 0.0057 \times 943,000^2 \times 120}{2 \times 4.18 \times 10^8 \times 55.0^2 \times 0.115}$$

$$= 8.3 \text{ ft}$$

$$\Delta P_p = \frac{8.3 \times 55.0}{144} = 3.2 \text{ psi}$$

Allowable $\Delta P_p = 10.0 \text{ psi}$

A check of U_h and U_c gives 161 and 138, respectively, and $K_c = 0.17$. From Fig. 17 for $\Delta t_c/\Delta t_h = 2/40 = 0.5$, $F_c = 0.43$, whereas in the solution above the arithmetic mean temperatures were used. The arithmetic mean assumes $F_c = 0.50$. However, since the ranges are small for both fluids, the error is too small to be significant. If the ranges of the fluids or their viscosities were large, the error might be considerable for $F_c = 0.43$.

Double Pipe Exchangers in Series-parallel Arrangements. Referring to Example 6.1, it is seen that a calculated pressure drop of 9.2 psi is

obtained against an allowable pressure drop of 10.0 psi. Suppose, however, that the calculated pressure drop were 15 or 20 psi and exceeded the available head. How then might the heat load be transferred with the available pressure head? One possibility is the use of a by-pass so that only three-quarters or two-thirds of the fluid flows through the exchanger and the remainder through the by-pass. This does not provide an ideal solution, since the reduced flow causes several unfavorable changes in the design. (1) The reduced flow through the exchanger reduces the mass velocity G_a and the film coefficient h_o . Since both of the coefficients are nearly alike, 323 vs. 276, any sizable reduction in G_a alone decreases U_c by nearly $G_a^{0.8}$. (2) If less liquid circulates through the annulus, it has to be cooled over a longer range than from 160 to 100°F so that, upon

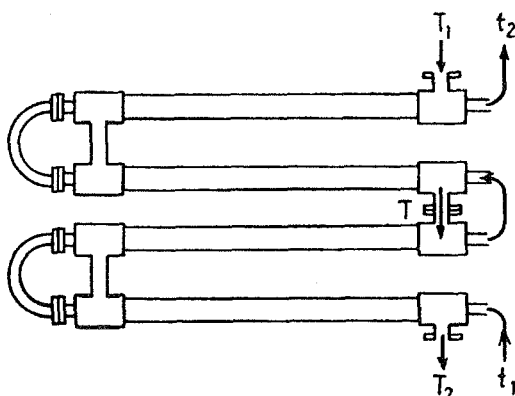


FIG. 6.5. Double pipe exchangers in series.

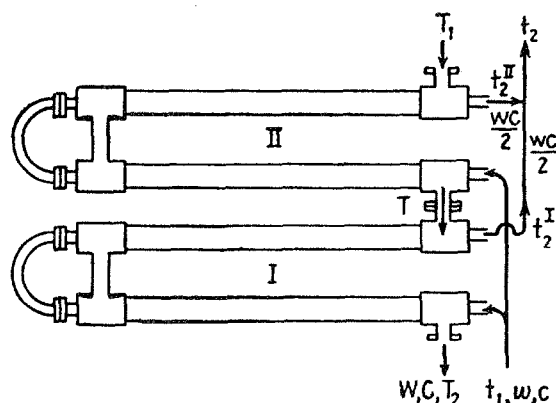


FIG. 6.6. Series—parallel arrangement.

mixing with the by-pass fluid, the process outlet temperature of 100°F results. As an example, the portion circulating through the annulus might have to be cooled over the range from 160 to 85°F depending upon the percentage by-passed. The outlet temperature of 85°F is closer to the inner pipe inlet of 80°F than originally, and the new cold-terminal difference Δt_1 of only 5°F greatly decreases the LMTD. The two effects, decreased U_c and LMTD, increase the required number of hairpins greatly even though the heat load is constant. Reversing the location of the streams by placing the benzene in the annulus does not provide a solution in this case, since the benzene stream is larger than the toluene stream. The possibility of reversing the location of the streams should always be examined first whenever the allowable pressure drop cannot be met.

A solution is still possible, however, even when all the above have failed. When two double pipe exchangers are connected in series, the arrangement is shown in Fig. 6.5. Suppose that the stream which is too large to be accommodated in several exchangers in series is divided in half and each half traverses but one exchanger through the inner pipes in

Fig. 6.6. Dividing a stream in half while keeping the flow area constant produces about *one-eighth* of the series pressure drop, since G and L will be half and the product of G^2L in Eq. (6.14) will be one-eighth. While the film coefficient will also be reduced, the unfavorable temperature difference of by-passing can be circumvented. Where there is a substantial unbalance between the weight flow of the two streams because one operates over a long range and the other over a very short range, the large stream may be divided in three, four, or more parallel streams. In larger services each parallel stream may also flow through several exchangers in series in each parallel bank. The term "parallel streams" should not be confused with "parallel flow." The former refers to the division of the flow of one fluid, while the latter refers to the direction of flow between two fluids.

The True Temperature Difference for Series-parallel Arrangements. The LMTD calculated from T_1 , T_2 , t_1 , and t_2 for the series arrangement will not be the same for a series-parallel arrangement. Half of the pipe fluid enters the upper exchanger II in Fig. 6.6 where the annulus fluid is hot, and half enters the lower exchanger I in which the annulus fluid has already been partially cooled. While exchangers in series do not transfer equal quantities of heat, the series-parallel relationship is even more adverse, the lower exchanger accounting for relatively less of the total heat transfer. If the true temperature difference is called Δt , it will not be identical with the LMTD for the process conditions although both of the exchangers operate in counterflow.

Consider the two exchangers in Fig. 6.6 designated by I and II. The intermediate temperature is T , and the outlets of the parallel streams are designated by t_2^I and t_2^I . Their mixed temperature is t_2 .

For exchanger I, containing half the surface,

$$Q_I = WC(T - T_2) = \frac{UA}{2} \times \text{LMTD}_I \quad (6.16)$$

and

$$\text{LMTD}_I = \frac{(T - t_2^I) - (T_2 - t_1)}{\ln (T - t_2^I)/(T_2 - t_1)} \quad (6.17)$$

Substituting in Eq. (6.16),

$$\frac{UA}{2WC} = \frac{(T - T_2)}{(T - t_2^I) - (T_2 - t_1)} \ln \frac{T - t_2^I}{T_2 - t_1}$$

Rearranging,

$$\begin{aligned} \frac{UA}{2WC} &= \frac{(T - T_2)}{(T - T_2) - (t_2^I - t_1)} \ln \frac{(T - t_2^I)}{(T_2 - t_1)} \\ &= \frac{1}{1 - (t_2^I - t_1)/(T - T_2)} \ln \frac{T - t_2^I}{T_2 - t_1} \end{aligned} \quad (6.18)$$

Let

$$R^I = \frac{(T - T_2)}{(t_2^I - t_1)} = \frac{wc}{2WC}$$

$$\frac{UA}{2WC} = \frac{R^I}{R^I - 1} \ln \frac{T - t_2^I}{T_2 - t_1} \quad (6.19)$$

Similarly for exchanger II

$$Q_{II} = WC(T_1 - T) = \frac{UA}{2} \times \text{LMTD}_{II} \quad (6.20)$$

$$\text{LMTD}_{II} = \frac{(T_1 - t_2^{II}) - (T - t_1)}{\ln (T_1 - t_2^{II}) / (T - t_1)} \quad (6.21)$$

Let

$$R^{II} = \frac{T_1 - T}{t_2^{II} - t_1} = \frac{wc}{2WC}$$

$$\frac{UA}{2WC} = \frac{R^{II}}{R^{II} - 1} \ln \frac{T_1 - t_2^{II}}{T - t_1} \quad (6.22)$$

Since c and C were assumed constant,

$$R^I = R^{II} = R' = \frac{wc}{2WC} \quad (6.23)$$

Let

$$S^I = \frac{t_2^I - t_1}{T - t_1} \quad M^I = \frac{T - T_2}{T - t_1}$$

$$M^I = R'S^I$$

Similarly let

$$S^{II} = \frac{t_2^{II} - t_1}{T_1 - t_1} \quad M^{II} = \frac{T_1 - T}{T_1 - t_1}$$

R' and S are ratios which recur frequently in obtaining the true temperature difference Δt from the LMTD. S is the ratio of the cold fluid range to the maximum temperature span, the latter being the difference between both inlet temperatures, T_1 and t_1 .

But

$$M^{II} = R'S^{II}$$

$$1 - S^I = \frac{T - t_2^I}{T - t_1} = \frac{T - t_1}{T - t_1} - \frac{t_2^I - t_1}{T - t_1}$$

$$\frac{T - t_2^I}{T_2 - t_1} = \frac{1 - S^I}{1 - R'S^I}$$

and from Eq. (6.19)

$$\frac{UA}{2WC} = \frac{R'}{R' - 1} \ln \frac{1 - S^I}{1 - R'S^I} \quad (6.24)$$

and from Eq. (6.22)

$$\frac{UA}{2WC} = \frac{R'}{R' - 1} \ln \frac{1 - S^{II}}{1 - R'S^{II}} \quad (6.25)$$

and equating Eqs. (6.24) and (6.25),

$$\frac{1 - S^I}{1 - R'S^I} = \frac{1 - S^{II}}{1 - R'S^{II}}$$

Therefore

$$\begin{aligned} S^I &= S^{II} \\ M^I &= M^{II} \end{aligned}$$

Adding Eqs. (6.24) and (6.25),

$$\frac{UA}{WC} = \frac{2R'}{R' - 1} \ln \frac{1 - S^I}{1 - R'S^I} = \frac{2R'}{R' - 1} \ln \frac{T - t_2^I}{T_2 - t_1} \quad (6.26)$$

in which T is the only unknown, and since $M^I = M^{II}$,

$$\begin{aligned} \frac{T_1 - T}{T_1 - t_1} &= \frac{T - T_2}{T - t_1} \\ T^2 - 2t_1T + t_1(T_1 + T_2) - T_1T_2 &= 0 \end{aligned} \quad (6.27)$$

Equation (6.27) is a quadratic whose solution is

$$\begin{aligned} T &= \frac{2t_1 \pm \sqrt{4t_1^2 - 4t_1[(T_1 + T_2) - 4T_1T_2]}}{2} \\ &= t_1 \pm \sqrt{(T_1 - t_1)(T_2 - t_1)} \end{aligned} \quad (6.28)$$

The minus sign applies when the heating medium is in the pipes. The plus sign applies when the cooling medium is in the pipes.

Substituting for T in Eq. (6.26),

$$\begin{aligned} \frac{UA}{WC} &= \frac{2R'}{R' - 1} \ln \left[\frac{(R' - 1)(T_1 - t_1) + \sqrt{(T_1 - t_1)(T_2 - t_1)}}{R' \sqrt{(T_1 - t_1)(T_2 - t_1)}} \right] \\ &= \frac{2R'}{R' - 1} \ln \left[\left(\frac{R' - 1}{R'} \right) \left(\frac{T_1 - t_1}{T_2 - t_1} \right)^{1/2} + \frac{1}{R'} \right] \end{aligned} \quad (6.29)$$

Δt is the single value for the entire series-parallel arrangement; thus

$$Q = UA \Delta t = WC(T_1 - T_2) \quad (6.30)$$

$$\Delta t = \frac{Q}{UA} = \frac{WC}{UA} (T_1 - T_2) \quad (6.31)$$

It is convenient in this derivation to employ a definition for the true temperature difference in terms of the maximum temperature span $T_1 - t_1$:

$$\Delta t = \gamma(T_1 - t_1) \quad (6.32)$$

Equating (6.31) and (6.32),

$$\begin{aligned} \frac{WC}{UA} (T_1 - T_2) &= \gamma(T_1 - t_1) \\ \gamma &= \frac{WC(T_1 - T_2)}{UA(T_1 - t_1)} \end{aligned}$$

Since $M = (T_1 - T_2)/(T_1 - t_1)$, define $P' = (T_2 - t_1)/(T_1 - t_1)$ and $UA/WC = M/\gamma$; then

$$P' + M = 1 \quad \text{or} \quad M = 1 - P'$$

Substituting in Eq. (6.29),

$$\frac{UA}{WC} = \frac{2R'}{R' - 1} \ln \left[\left(\frac{R' - 1}{R'} \right) \left(\frac{1}{P'} \right)^{1/2} + \frac{1}{R'} \right] \quad (6.33)$$

or

$$\frac{1 - P'}{\gamma} = 2 \left(\frac{R'}{R' - 1} \right) \ln \left[\left(\frac{R' - 1}{R'} \right) \left(\frac{1}{P'} \right)^{1/2} + \frac{1}{R'} \right] \quad (6.34)$$

If developed in a generalized manner it can be shown that, for *one series hot stream* and *n parallel cold streams*, Eq. (6.34) becomes

$$\frac{1 - P'}{\gamma} = 2.3 \frac{nR'}{R' - 1} \log \left[\left(\frac{R' - 1}{R'} \right) \left(\frac{1}{P'} \right)^{1/n} + \frac{1}{R'} \right] \quad (6.35a)$$

where

$$R' = \frac{T_1 - T_2}{n(t_2 - t_1)}$$

For *one series cold stream* and *n parallel hot streams*,

$$\frac{1 - P''}{\gamma} = 2.3 \frac{n}{1 - R''} \log \left[(1 - R'') \left(\frac{1}{P''} \right)^{1/n} + R'' \right] \quad (6.35b)$$

where

$$P'' = \frac{T_1 - t_2}{T_1 - t_1} \quad \text{and} \quad R'' = \frac{n(T_1 - T_2)}{t_2 - t_1}$$

Example 6.2. Calculation of the True Temperature Difference. A bank of double pipe exchangers operates with the hot fluid in series from 300 to 200°F and the cold fluid in six parallel streams from 190 to 220°F. What is the true temperature difference Δt ?

$$P' = \frac{T_2 - t_1}{T_1 - t_1} = \frac{200 - 190}{300 - 190} = 0.091 \quad R' = \frac{T_1 - T_2}{n(t_2 - t_1)} = \frac{300 - 200}{6(220 - 190)} = 0.558$$

Substituting in Eq. (6.35a) and solving, $\gamma = 0.242$.

$$\Delta t = 0.242(300 - 190) = 26.6^\circ\text{F} \quad (6.32)$$

The LMTD would be 33.7°F, and an error of 27 per cent would be introduced by its use.

Exchangers with a Viscosity Correction, ϕ . For heating or cooling fluids, the use of Fig. 24 with an assumed value of $(\mu/\mu_w)^{0.14} = 1.0$ also assumes a negligible deviation of fluid properties from isothermal flow. For nonviscous fluids the deviation from isothermal flow during heating

or cooling does not introduce an appreciable error in the calculation of the heat-transfer coefficient. When the pipe-wall temperature differs appreciably from the caloric temperature of the controlling fluid and the controlling fluid is viscous, the actual value of $\phi = (\mu/\mu_w)^{0.14}$ must be taken into account. To include the correction, t_w may be determined by Eq. (5.31) or by (5.32) from uncorrected values of h_o/ϕ_a and h_{io}/ϕ_p , which are then corrected accordingly by multiplication by ϕ_a and ϕ_p respectively. The corrected coefficients where $\phi \neq 1.0$ are

$$h_o = \left(\frac{h_o}{\phi_a} \right) \phi_a \quad (6.36)$$

$$h_{io} = \left(\frac{h_{io}}{\phi_p} \right) \phi_p \quad (6.37)$$

Similarly for two resistances in series employing the viscosity corrections for deviation from the isothermal the clean overall coefficient is again

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} \quad (6.38)$$

Example 6.3. Double Pipe Lube Oil-Crude Oil Exchanger. 6,900 lb/hr of a 26°API lube oil must be cooled from 450 to 350°F by 72,500 lb/hr of 34°API mid-continent crude oil. The crude oil will be heated from 300 to 310°F.

A fouling factor of 0.003 should be provided for each stream, and the allowable pressure drop on each stream will be 10 psi.

A number of 20-ft hairpins of 3- by 2-in. IPS pipe are available. How many must be used, and how shall they be arranged? The viscosity of the crude oil may be obtained from Fig. 14. For the lube oil, viscosities are 1.4 centipoises at 500°F, 3.0 at 400°F, and 7.7 at 300°F. These viscosities are great enough to introduce an error if $(\mu/\mu_w)^{0.14} = 1$ is assumed.

Solution:

(1) Heat Balance:

Lube oil, $Q = 6900 \times 0.62(450 - 350) = 427,000$ Btu/hr

Crude oil, $Q = 72,500 \times 0.585(310 - 300) = 427,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.	
450	Higher temp	310	140	Δt_2
350	Lower temp	300	50	Δt_1
			90	$\Delta t_2 - \Delta t_1$

It will be impossible to put the 72,500 lb/hr of crude into a single pipe or annulus, since the flow area of each is too small. Assume, as a trial, that it will be employed in two parallel streams.

$$\Delta t = 87.5^\circ\text{F} \quad (6.35a)$$

(3) Caloric temperatures:

$$\frac{\Delta t_c}{\Delta t_h} = \frac{50}{140} = 0.357 \quad K_c = 0.43 \quad F_c = 0.395 \quad (\text{Fig. 17})$$

$$T_c = 350 \times 0.395(450 - 350) = 389.5^\circ\text{F} \quad (5.28)$$

$$t_c = 300 \times 0.395(310 - 300) = 304^\circ\text{F} \quad (5.29)$$

Proceed now to the inner pipe.

Hot fluid: annulus, lube oil

$$(4') \text{ Flow area, } D_2 = 3.068/12 = 0.256 \text{ ft}$$

$$D_1 = 2.38/12 = 0.199 \text{ ft}$$

$$a_a = \pi(D_2^2 - D_1^2)/4$$

$$= \pi(0.256^2 - 0.199^2)/4 = 0.0206 \text{ ft}^2$$

$$\text{Equiv diam, } D_e = (D_2^2 - D_1^2)/D_1$$

[Eq. (6.3)]

$$= (0.256^2 - 0.199^2)/0.199 = 0.13 \text{ ft}$$

$$(5') \text{ Mass vel, } G_a = W/a_a$$

$$= 6900/0.0206 = 335,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6') \text{ At } 389.5^\circ\text{F, } \mu = 3.0 \text{ cp}$$

$$= 3.0 \times 2.42 = 7.25 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$Re_a = D_e G_a / \mu$$

[Eq. (3.6)]

$$= 0.13 \times 335,000 / 7.25 = 6,000$$

If only 2 hairpins in series are required,
 L/D will be $2 \times 40/0.13 = 614$. Use
 $L/D = 600$.

$$(7') j_H = 20.5$$

[Fig. 24]

$$(8') \text{ At } T_c = 389.5^\circ\text{F, } c = 0.615 \text{ Btu}/(\text{lb})(^\circ\text{F})$$

[Fig. 4]

$$k = 0.067 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

[Fig. 1]

$$\left(\frac{c\mu}{k}\right)^{1/3} = \left(\frac{0.615 \times 7.25}{0.067}\right)^{1/3} = 4.05$$

$$(9') h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/3} \phi_a \quad [\text{Eq. (6.15)}]$$

$$\frac{h_o}{\phi_a} = \frac{20.5 \times 0.067 \times 4.05}{0.13}$$

$$= 42.7 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$t_w = t_c + \frac{h_o/\phi_a}{h_{io}/\phi_p + h_o/\phi_a} (T_c - t_c)$$

[Eq. (5.31)]

$$= 304 + \frac{42.7}{297 + 42.7} (389.5 - 304)$$

$$= 314^\circ\text{F}$$

$$\mu_w = 6.6 \times 2.42 = 16.0 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$\phi_a = (\mu/\mu_w)^{0.14}$$

$$= (7.25/16.0)^{0.14} = 0.90$$

[Fig. 24]

$$h_o = \frac{h_o}{\phi_a} \phi_a$$

[Eq. (6.36)]

$$= 42.7 \times 0.90 = 38.4$$

Cold fluid: inner pipe, crude oil

$$(4) \text{ Flow area, } D = 2.067/12 = 0.172 \text{ ft}$$

$$a_p = \pi D^2/4$$

$$= \pi \times 0.172^2/4 = 0.0233 \text{ ft}^2$$

Since two parallel streams have been assumed,
 $w/2$ lb/hr will flow in each pipe.

$$(5) \text{ Mass vel, } G_p = w/a_p$$

$$= \frac{72,500}{2 \times 0.0233}$$

$$= 1,560,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6) \text{ At } 304^\circ\text{F, } \mu = 0.83 \text{ cp}$$

$$= 0.83 \times 2.42 = 2.01 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$Re_p = D G_p / \mu$$

$$= 0.172 \times 1,560,000 / 2.01 = 133,500$$

$$(7) j_H = 320$$

[Fig. 24]

$$(8) \text{ At } t_c = 304^\circ\text{F, } c = 0.585 \text{ Btu}/(\text{lb})(^\circ\text{F})$$

[Fig. 4]

$$k = 0.073 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

[Fig. 1]

$$\left(\frac{c\mu}{k}\right)^{1/3} = \left(\frac{0.585 \times 2.01}{0.073}\right)^{1/3} = 2.52$$

$$(9) h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/3} \phi_p \quad [\text{Eq. (6.15a)}]$$

$$\frac{h_i}{\phi_p} = \frac{320 \times 0.073 \times 2.52}{0.172}$$

$$= 342 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$(10) \frac{h_{io}}{\phi_p} = \frac{h_i}{\phi_p} \times \frac{ID}{OD}$$

$$= 342 \times 2.067/2.38 = 297$$

Now proceed from (4') to (9') to obtain t_w .

$$\mu_w = 0.77 \times 2.42 = 1.86$$

[Fig. 14]

$$\phi_p = (\mu/\mu_w)^{0.14}$$

$$= (2.01/1.86)^{0.14} = 1.0 \text{ nearly}$$

[Fig. 24]

$$h_{io} = \frac{h_{io}}{\phi_p} \phi_p$$

[Eq. (6.37)]

$$= 297 \times 1.0 = 297$$

(11) Clean overall coefficient, U_c :

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{297 \times 38.4}{297 + 38.4} = 34.0 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(12) Design overall coefficient, U_D :

$$\frac{1}{U_D} = \frac{1}{U_c} + R_d \quad (6.10)$$

$$R_d = 0.003 + 0.003 = 0.006 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu}$$

$$U_D = 28.2$$

Summary

38.4	h outside	297
U_c	34.0	
U_D	28.2	

(13) Surface:

$$A = \frac{Q}{U_D \Delta t} = \frac{427,000}{28.2 \times 87.5} = 173 \text{ ft}^2$$

$$\text{External surface/lin ft, } a'' = 0.622 \text{ ft} \quad (\text{Table 11})$$

$$\text{Required length} = \frac{173}{0.622} = 278 \text{ lin ft}$$

This is equivalent to more than six 20-ft hairpins or 240 lin feet. Since two parallel streams are employed, use eight hairpins or 320 lin. feet. The hairpins should have the annuli connected in series and the tubes in two parallel banks of four exchangers.

The corrected U_D will be $U_D = Q/A \Delta t = 427,000/320 \times 0.622 \times 87.5 = 24.5$.

The corrected dirt factor will be $R_d = 1/U_D - 1/U_c = 1/24.5 - 1/34.0 = 0.0114$.

Pressure Drop

$$(1') D'_c = (D_2 - D_1) \quad [\text{Eq. (6.4)}]$$

$$= (0.256 - 0.198) = 0.058 \text{ ft}$$

$$Re'_a = D'_c G_a / \mu$$

$$= 0.058 \times 335,000 / 7.25 = 2680$$

$$f = 0.0035 + \frac{0.264}{2680^{0.42}} = 0.0132 \quad [\text{Eq. (3.47b)}]$$

$$s = 0.775, \rho = 62.5 \times 0.775 = 48.4 \quad [\text{Fig. 6}]$$

$$(2') \Delta F_a = \frac{4fG_a^2 L_a}{2g\rho^2 D'_c}$$

$$= \frac{4 \times 0.0132 \times 335,000^2 \times 320}{2 \times 4.18 \times 10^8 \times 48.4^2 \times 0.058}$$

$$= 16.7 \text{ ft}$$

$$(3') V = \frac{G_a}{3600\rho} = \frac{335,000}{3600 \times 48.4} = 1.9 \text{ fps}$$

$$\Delta F_t = 8 \left(\frac{V^2}{2g} \right) = 8 \left(\frac{1.9^2}{2 \times 32.2} \right) = 0.45 \text{ ft}$$

$$\Delta P_a = \frac{(16.7 + 0.45) \times 48.4}{144} = 5.8 \text{ psi}$$

$$\text{Allowable } \Delta P_a = 10.0 \text{ psi}$$

(1) For $Re_p = 133,500$ in (6) above

$$f = 0.0035 + \frac{0.264}{133,500^{0.42}} = 0.005375 \quad [\text{Eq. (3.47b)}]$$

$$s = 0.76, \rho = 62.5 \times 0.76 = 47.5 \quad [\text{Fig. 6}]$$

Halves of the tube fluid will flow through only four exchangers.

$$(2) \Delta F_p = \frac{4fG_p^2 L_p}{2g\rho^2 D}$$

$$= \frac{4 \times 0.005375 \times 1,560,000^2 \times 160}{2 \times 4.18 \times 10^8 \times 47.5^2 \times 0.172}$$

$$= 25.7 \text{ ft}$$

$$\Delta P_p = \frac{25.7 \times 47.5}{144} = 8.5 \text{ psi}$$

Allowable $\Delta P_p = 10.0 \text{ psi}$

If the flow had not been divided, the pressure drop would be nearly eight times as great, or about 60 psi.

PROBLEMS

6.1. What is the fouling factor when (a) $U_c = 30$ and $U_D = 20$, (b) $U_c = 60$ and $U_D = 50$, and (c) $U_c = 110$ and $U_D = 100$? Which do you consider reasonable to specify between two moderately clean streams?

6.2. A double pipe exchanger was oversized because no data were available on the rate at which dirt accumulated. The exchanger was originally designed to cool 13,000 lb/hr of 100 per cent acetic acid from 250 to 150°F by heating 19,000 lb/hr of butyl alcohol from 100 to 157°F. A design coefficient $U_D = 85$ was employed, but during initial operation a hot-liquid outlet temperature of 117°F was obtained. It rose during operation at the average rate of 3°F per month. What dirt factor should have been specified for a 6-month cleaning cycle?

6.3. O-xylene coming from storage at 100°F is to be heated to 150°F by cooling 18,000 lb/hr of butyl alcohol from 170 to 140°F. Available for the purpose are five 20-ft hairpin double pipe exchangers with annuli and pipes each connected in series. The exchangers are 3- by 2-in. IPS. What is (a) the dirt factor, (b) the pressure drops? (c) If the hot and cold streams in (a) are reversed with respect to the annulus and inner pipe, how does this justify or refute your initial decision where to place the hot stream?

6.4. 10,000 lb/hr of 57°API gasoline is cooled from 150 to 130°F by heating 42°API kerosene from 70 to 100°F. Pressure drops of 10 psi are allowable with a minimum dirt factor of 0.004. (a) How many 2½- by 1¼-in. IPS hairpins 20 ft long are required? (b) How shall they be arranged? (c) What is the final fouling factor?

6.5. 12,000 lb/hr 26°API lube oil (see Example 6.3 in text for viscosities) is to be cooled from 450 to 350°F by heating 42°API kerosene from 325 to 375°F. A pressure drop of 10 psi is permissible on both streams, and a minimum dirt factor of 0.004 should be provided. (a) How many 20-ft hairpins of 2½- by 1¼-in. IPS double pipe are required? (b) How shall they be arranged, and (c) what is the final dirt factor?

6.6. 7,000 lb/hr of aniline is to be heated from 100 to 150°F by cooling 10,000 lb/hr of toluene with an initial temperature of 185°F in 2- by 1-in. IPS double pipe hairpin exchangers 15 ft long. Pressure drops of 10 psi are allowable, and a dirt factor of 0.005 is required. (a) How many hairpin sections are required? (b) How shall they be arranged? (c) What is the final dirt factor?

6.7. 24,000 lb/hr of 35°API distillate is cooled from 400 to 300°F by 50,000 lb/hr of 34°API crude oil heated from an inlet temperature of 250°F. Pressure drops of 10 psi are allowable, and a dirt factor of 0.006 is required. Using 20-ft hairpins of 4- by 3-in. IPS (a) how many are required, (b) how shall they be arranged, and (c) what is the final fouling factor?

6.8. A liquid is cooled from 350 to 300°F by another which is heated from 290 to 315°F. How does the true temperature difference deviate from the LMTD if (a) the hot fluid is in series and the cold fluid flows in two parallel counterflow paths, (b) the hot fluid is in series and the cold fluid flows in three parallel-flow-counterflow paths, (c) The cold-fluid range in (a) and (b) is changed to 275 to 300°F.

6.9. A fluid is cooled from 300 to 275°F by heating a cold fluid from 100 to 290°F. If the hot fluid is in series, how is the true temperature difference affected by dividing the hot stream into (a) two parallel streams and (b) into three parallel streams?

6.10. 6330 lb/hr of toluene is cooled from 160 to 100°F by heating amyl acetate from 90 to 100°F using 15-ft hairpins. The exchangers are 2- by 1¼-in. IPS. Allowing 10 psi pressure drops and providing a minimum dirt factor of 0.004 (a) how many hairpins are required, (b) how shall they be arranged, and (c) what is the final dirt factor?

6.11. 13,000 lb/hr of 26°API gas oil (see Example 6.3 in text for viscosities) is cooled from 450 to 350°F by heating 57°API gasoline under pressure from 220 to 230°F in as many 3- by 2-in. IPS double pipe 20-ft hairpins as are required. Pressure drops of 10 psi are permitted along with a minimum dirt factor of 0.004. (a) How many hairpins are required? (b) How shall they be arranged? (c) What is the final dirt factor?

6.12. 100,000 lb/hr of nitrobenzene is to be cooled from 325 to 275°F by benzene heated from 100 to 300°F. Twenty-foot hairpins of 4- by 3-in. IPS double pipe will be employed, and pressure drops of 10 psi are permissible. A minimum dirt factor of 0.004 is required. (a) How many hairpins are required? (b) How shall they be arranged? (c) What is the final dirt factor?

NOMENCLATURE FOR CHAPTER 6

A	Heat-transfer surface, ft^2
a	Flow area, ft^2
a''	External surface per linear foot of pipe, ft
C	Specific heat of hot fluid in derivations, $\text{Btu}/(\text{lb})(^\circ\text{F})$
C_1	A constant
c	Specific heat of cold fluid in derivations or either fluid in calculations $\text{Btu}/(\text{lb})(^\circ\text{F})$
D	Inside diameter, ft
D_1, D_2	For annuli D_1 is the outside diameter of inner pipe, D_2 is the inside diameter of the outer pipe, ft
D_e, D'_e	Equivalent diameter for heat-transfer and pressure drop, ft
d_e, d'_e	Equivalent diameter for heat-transfer and pressure drop, in.
D_o	Outside diameter, ft
F_c	Caloric fraction, dimensionless
ΔF	Pressure drop, ft
f	Friction factor, dimensionless
G	Mass velocity, $\text{lb}/(\text{hr})(\text{ft}^2)$
g	Acceleration of gravity $4.18 \times 10^8 \text{ ft/hr}^2$
g'	Acceleration of gravity 32.2 ft/sec^2
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
h_{io}	Value of h_i when referred to the pipe outside diameter, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
ID	Inside diameter, ft or in.
j_H	Heat-transfer factor, dimensionless
K_c	Caloric factor, dimensionless
k	Thermal conductivity, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$
L	Pipe length or length of path, ft
M	Temperature group $(T_1 - T_2)/(T_1 - t_1)$, dimensionless
n	Number of parallel streams
OD	Outside diameter, ft or in.
P'	Temperature group $(T_2 - t_1)/(T_1 - t_1)$, dimensionless
P''	Temperature group $(T_1 - t_2)/(T_1 - t_1)$, dimensionless
ΔP	Pressure drop, psi
Q	Heat flow, Btu/hr
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R'	Temperature group $(T_1 - T_2)/n(t_2 - t_1)$, dimensionless
R''	Temperature group $n(T_1 - T_2)/(t_2 - t_1)$, dimensionless

R_d, R_i, R_o	Combined dirt factor, inside dirt factor, outside dirt factor, (hr)(ft ²)(°F)/Btu
Re, Re'	Reynolds number for heat transfer and pressure drop, dimensionless
r_h	Hydraulic radius, ft
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
s	Specific gravity, dimensionless
T, T_1, T_2	Hot-fluid temperature in general, inlet and outlet of hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
t, t_1, t_2	Cold-fluid temperature in general, inlet and outlet of cold fluid, °F
t_c	Caloric temperature of cold fluid, °F
Δt	True or effective temperature difference in $Q = U_D A \Delta t$
$\Delta t_c, \Delta t_h$	Cold- and hot-terminal temperature differences, °F
U, U_C, U_D	Overall coefficient of heat transfer, clean coefficient, design coefficient, Btu/(hr)(ft ²)(°F)
V	Velocity, fps
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
z	Height, ft
γ	A constant, dimensionless
μ	Viscosity at the caloric temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at the pipe-wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³
ϕ	$(\mu/\mu_w)^{0.14}$

Subscripts and Superscripts

a	Annulus
l	Loss
p	Pipe
I	First of two exchangers
II	Second of two exchangers

CHAPTER 7

1-2 PARALLEL-COUNTERFLOW: SHELL-AND-TUBE EXCHANGERS

INTRODUCTION

The Tubular Element. The fulfillment of many industrial services requires the use of a large number of double pipe hairpins. These consume considerable ground area and also entail a large number of points at which leakage may occur. Where large heat-transfer surfaces are required, they can best be obtained by means of shell-and-tube equipment.

Shell-and-tube equipment involves expanding a tube into a tube sheet and forming a seal which does not leak under reasonable operating con-

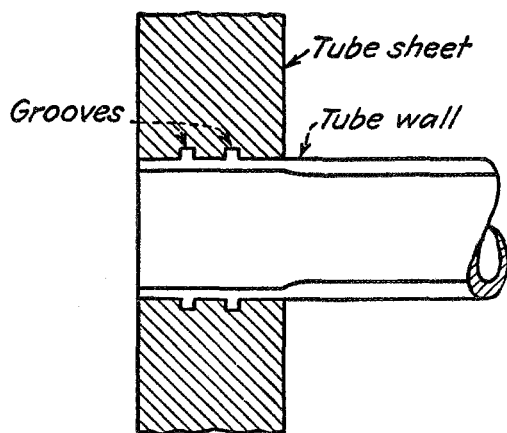


FIG. 7.1. Tube roll.

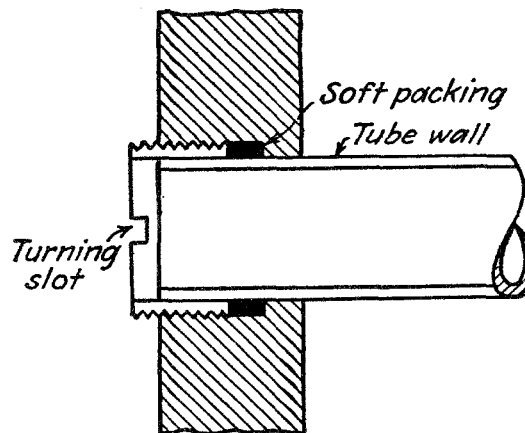


FIG. 7.2. Ferrule.

ditions. A simple and common example of an expanded tube is shown in Fig. 7.1. A tube hole is drilled in a tube sheet with a slightly greater diameter than the outside diameter of the tube, and two or more grooves are cut in the wall of the hole. The tube is placed inside the tube hole, and a tube roller is inserted into the end of the tube. The roller is a rotating mandril having a slight taper. It is capable of exceeding the elastic limit of the tube metal and transforming it into a semiplastic condition so that it flows into the grooves and forms an extremely tight seal. Tube rolling is a skill, since a tube may be damaged by rolling to paper thinness and leaving a seal with little structural strength.

In some industrial uses it is desirable to install tubes in a tube sheet so that they can be removed easily as shown in Fig. 7.2. The tubes are

actually packed in the tube sheet by means of ferrules using a soft metal packing ring.

Heat-exchanger Tubes. Heat-exchanger tubes are also referred to as condenser tubes and should not be confused with steel pipes or other types of pipes which are extruded to iron pipe sizes. The outside diameter of heat exchanger or condenser tubes is the actual outside diameter in inches within a very strict tolerance. Heat-exchanger tubes are available in a variety of metals which include steel, copper, admiralty, Muntz metal, brass, 70-30 copper-nickel, aluminum bronze, aluminum, and the stainless steels. They are obtainable in a number of different wall thicknesses defined by the Birmingham wire gage, which is usually referred to as the BWG or *gage* of the tube. The sizes of tubes which are generally available are listed in Table 10 of the Appendix of which the $\frac{3}{4}$ outside diam-

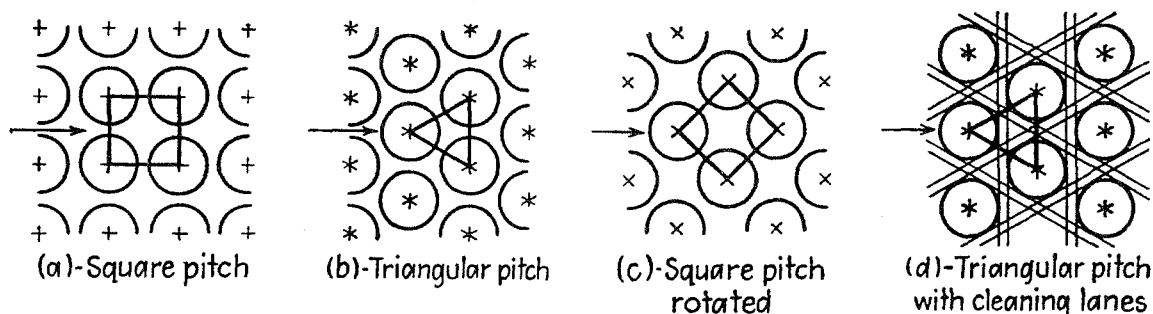


FIG. 7.3. Common tube layouts for exchangers.

eter and 1 in. OD are most common in heat-exchanger design. The data of Table 10 have been arranged in a manner which will be most useful in heat-transfer calculations.

Tube Pitch. Tube holes cannot be drilled very close together, since too small a width of metal between adjacent tubes structurally weakens the tube sheet. The shortest distance between two adjacent tube holes is the *clearance* or *ligament*, and these are now fairly standard. Tubes are laid out on either square or triangular patterns as shown in Fig. 7.3a and b. The advantage of square pitch is that the tubes are accessible for external cleaning and cause a lower pressure drop when fluid flows in the direction indicated in Fig. 7.3a. The *tube pitch* P_T is the shortest center-to-center distance between adjacent tubes. The common pitches for square layouts are $\frac{3}{4}$ in. OD on 1-in. square pitch and 1 in. OD on $1\frac{1}{4}$ -in. square pitch. For triangular layouts these are $\frac{3}{4}$ in. OD on $1\frac{5}{16}$ -in. triangular pitch, $\frac{3}{4}$ in. OD on 1-in. triangular pitch, and 1-in. OD on $1\frac{1}{4}$ -in. triangular pitch. In Fig. 7.3c the square-pitch layout has been rotated 45° , yet it is essentially the same as in Fig. 7.3a. In Fig. 7.3d a mechanically cleanable modification of triangular pitch is

shown. If the tubes are spread wide enough, it is possible to allow the cleaning lanes indicated.

Shells. Shells are fabricated from steel pipe with nominal IPS diameters up to 12 in. as given in Table 11. Above 12 and including 24 in. the actual outside diameter and the nominal pipe diameter are the same. The standard wall thickness for shells with inside diameters from 12 to 24 in. inclusive is $\frac{3}{8}$ in., which is satisfactory for shell-side operating pressures up to 300 psi. Greater wall thicknesses may be obtained for greater pressures. Shells above 24 in. in diameter are fabricated by rolling steel plate.

Stationary Tube-sheet Exchangers. The simplest type of exchanger is the *fixed* or *stationary tube-sheet* exchanger of which the one shown in Fig. 7.4 is an example. The essential parts are a shell (1), equipped with two nozzles and having tube sheets (2) at both ends, which also serve as flanges for the attachment of the two channels (3) and their respective

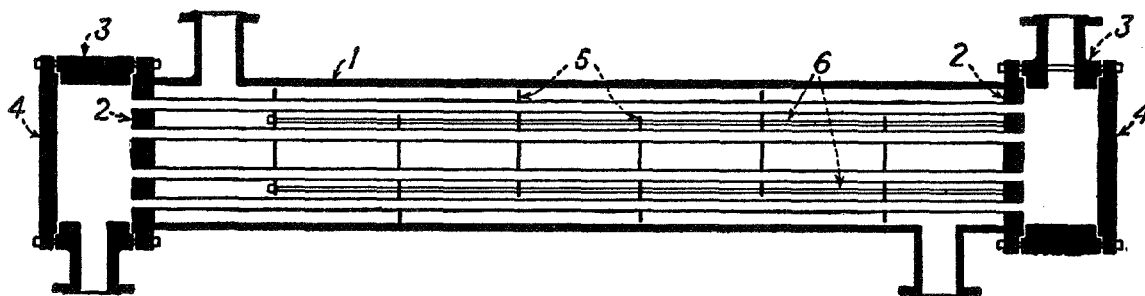


FIG. 7.4. Fixed-head tubular exchanger.

channel covers (4). The tubes are expanded into both tube sheets and are equipped with transverse baffles (5) on the shell side. The calculation of the effective heat-transfer surface is frequently based on the distance between the inside faces of the tube sheets instead of the overall tube length.

Baffles. It is apparent that higher heat-transfer coefficients result when a liquid is maintained in a state of turbulence. To induce turbulence outside the tubes it is customary to employ baffles which cause the liquid to flow through the shell at right angles to the axes of the tubes. This causes considerable turbulence even when a small quantity of liquid flows through the shell. The center-to-center distance between baffles is called the *baffle pitch* or *baffle spacing*. Since the baffles may be spaced close together or far apart, the mass velocity is not entirely dependent upon the diameter of the shell. The baffle spacing is usually not greater than a distance equal to the inside diameter of the shell or closer than a distance equal to one-fifth the inside diameter of the shell. The baffles are held securely by means of baffle spacers (6) as shown in Fig. 7.4, which consist of through-bolts screwed into the tube sheet and a number of

smaller lengths of pipe which form shoulders between adjacent baffles. An enlarged detail is shown in Fig. 7.5.

There are several types of baffles which are employed in heat exchangers, but by far the most common are segmental baffles as shown in Fig. 7.6. Segmental baffles are drilled plates with heights which are generally 75 per cent of the inside diameter of the shell. These are known

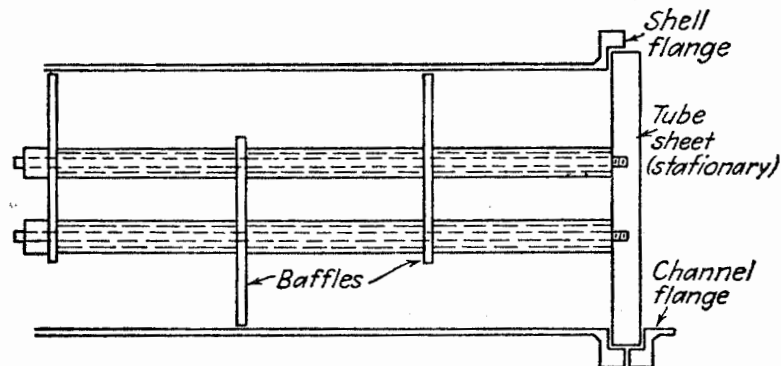


FIG. 7.5. Baffle spacer detail (enlarged).

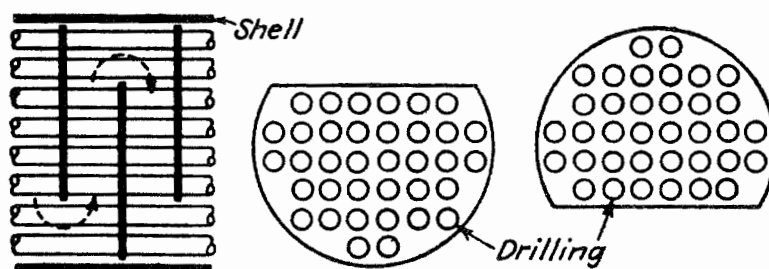


FIG. 7.6. Segmental baffle detail.

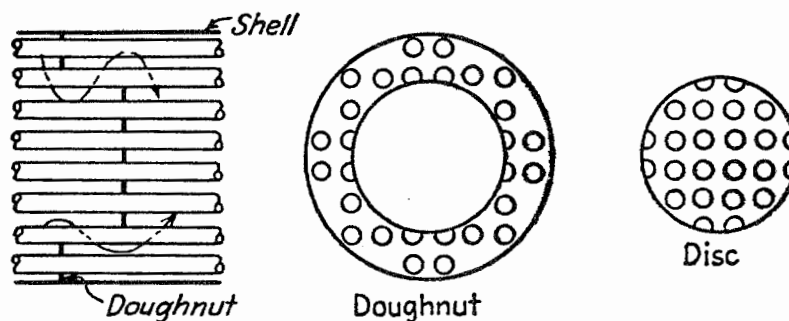


FIG. 7.7. Disc and doughnut baffle.

as 25 per cent cut baffles and will be used throughout this book although other fractional baffle cuts are also employed in industry. An excellent review of the influence of the baffle cut on the heat-transfer coefficient has been presented by Donohue.¹ They may be arranged, as shown, for "up-and-down" flow or may be rotated 90° to provide "side-to-side" flow, the latter being desirable when a mixture of liquid and gas flows

¹ Donohue, D. A., *Ind. Eng. Chem.*, **41**, 2499-2510 (1949).

through the shell. The baffle pitch and not the 25 per cent cut of the baffles, as shown later, determines the effective velocity of the shell fluid.

Other types of baffles are the *disc* and *doughnut* of Fig. 7.7 and the *orifice* baffle in Fig. 7.8. Although additional types are sometimes employed, they are not of general importance.

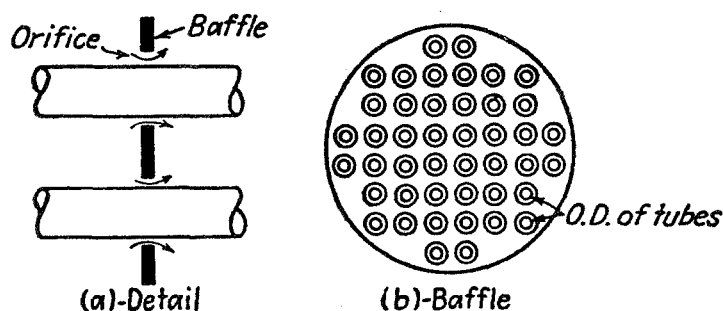


FIG. 7.8. Orifice baffle.

Fixed-tube-sheet Exchanger with Integral Channels. Another of several variations of the fixed-tube-sheet exchanger is shown in Fig. 7.9, in which the tube sheets are inserted into the shell, forming channels which are integral parts of the shell. In using stationary tube-sheet exchangers it is often necessary to provide for differential thermal expansion between the tubes and the shell during operation, or thermal stresses will develop across the tube sheet. This can be accomplished by the use of an *expansion joint* on the shell, of which a number of types of flexible joints are available.

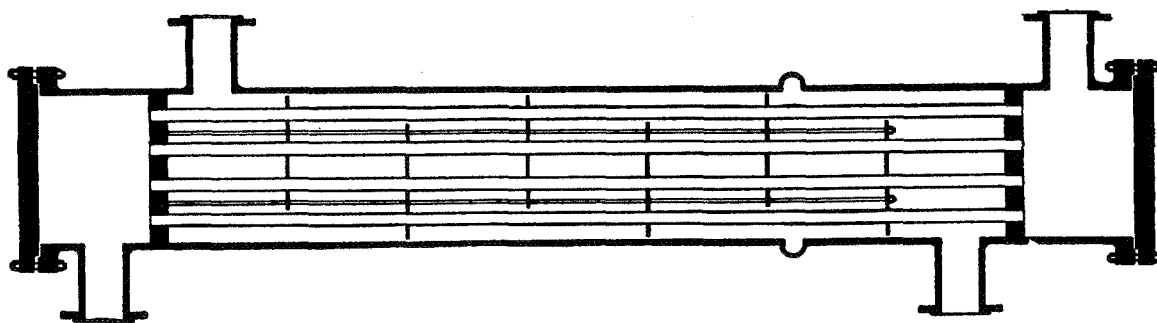


FIG. 7.9. Fixed-tube-sheet exchanger with integral channels. (Patterson Foundry & Machine Co.)

Fixed-tube-sheet 1-2 Exchanger. Exchangers of the type shown in Figs. 7.4 and 7.9 may be considered to operate in counterflow, notwithstanding the fact that the shell fluid flows across the outside of the tubes. From a practical standpoint it is very difficult to obtain a high velocity when one of the fluid flows through all the tubes in a single pass. This can be circumvented, however, by modifying the design so that the tube fluid is carried through fractions of the tubes consecutively. An example of a *two-pass* fixed-tube-sheet exchanger is shown in Fig. 7.10, in which all the tube fluid flows through the two halves of the tubes successively.

The exchanger in which the shell-side fluid flows in one shell pass and the tube fluid in two or more passes is the 1-2 *exchanger*. A single channel is employed with a *partition* to permit the entry and exit of the tube fluid from the same channel. At the opposite end of the exchanger a bonnet is provided to permit the tube fluid to cross from the first to the second pass. As with all fixed-tube-sheet exchangers, the outsides of the tubes

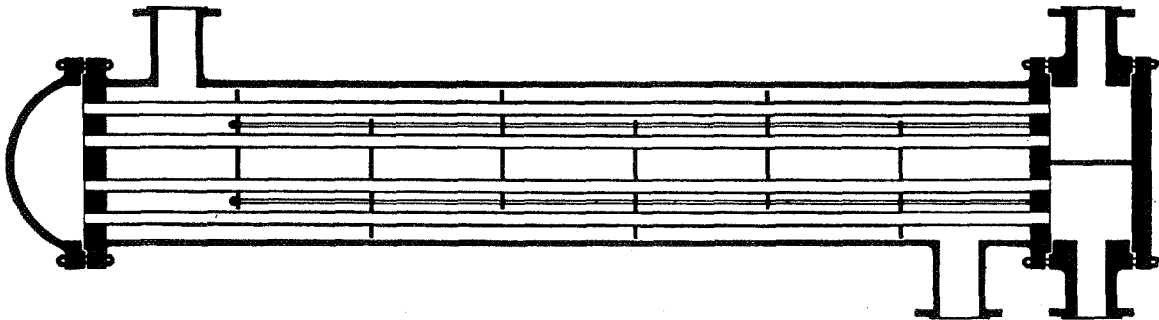


FIG. 7.10. Fixed-head 1-2 exchanger. (Patterson Foundry & Machine Co.)

are inaccessible for inspection or mechanical cleaning. The insides of the tubes can be cleaned in place by removing only the channel cover and using a rotary cleaner or a wire brush. Expansion problems are extremely critical in 1-2 fixed-tube-sheet exchangers, since both passes, as well as the shell itself, tend to expand differently and cause stress on the stationary tube sheets.

Removable-bundle Exchangers. In Fig. 7.11 is shown a counterpart of the 1-2 exchanger having a tube bundle which is removable from the

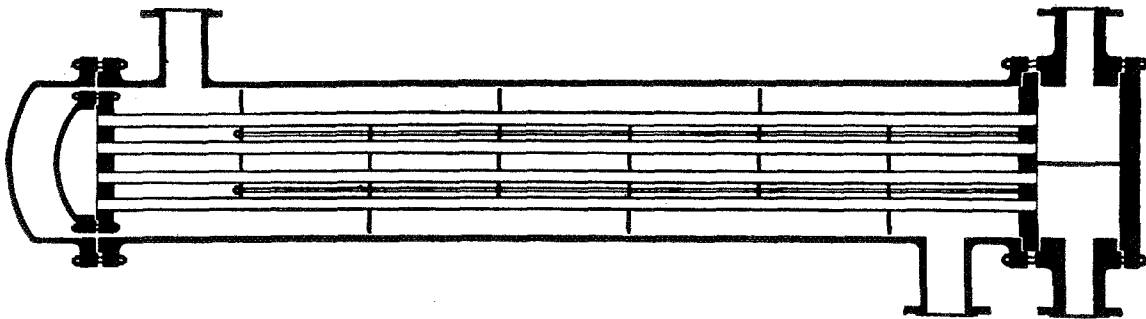


FIG. 7.11. Pull-through floating-head 1-2 exchanger. (Patterson Foundry & Machine Co.)

shell. It consists of a stationary tube sheet, which is clamped between the single channel flange and a shell flange. At the opposite end of the bundle the tubes are expanded into a freely riding *floating tube sheet* or *floating head*. A *floating-head cover* is bolted to the tube sheet, and the entire bundle can be withdrawn from the channel end. The shell is closed by a shell bonnet. The floating head illustrated eliminates the differential expansion problem in most cases and is called a *pull-through floating head*.

The disadvantage to the use of a pull-through floating head is one of simple geometry. To secure the floating-head cover it is necessary to bolt it to the tube sheet, and the bolt circle requires the use of space where it would be possible to insert a great number of tubes. The bolting not only reduces the number of tubes which might be placed in the tube bundle but also provides an undesirable flow channel between the bundle and the shell. These objections are overcome in the more conventional

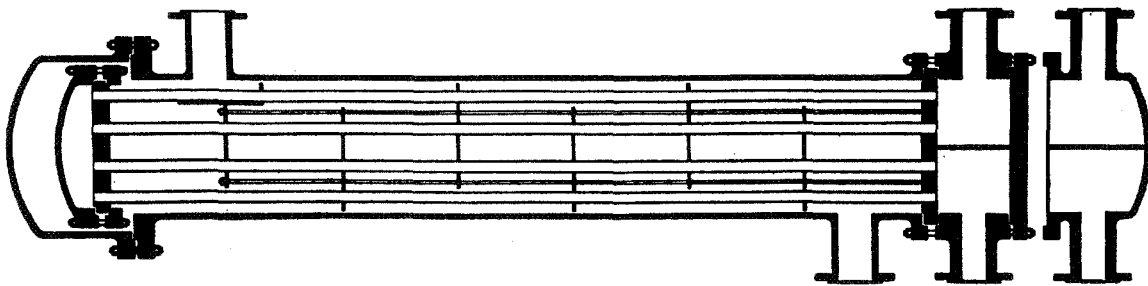


FIG. 7.12. Floating-head 1-2 exchanger. (Patterson Foundry & Machine Co.)

split-ring floating-head 1-2 exchanger shown in Fig. 7.12. Although it is relatively expensive to manufacture, it does have a great number of mechanical advantages. It differs from the pull-through type by the use of a split-ring assembly at the floating tube sheet and an oversized shell cover which accommodates it. The detail of a split ring is shown in Fig. 7.13. The floating tube sheet is clamped between the floating-head cover and a clamp ring placed in back of the tube sheet which is split in half to permit dismantling. Different manufacturers have different

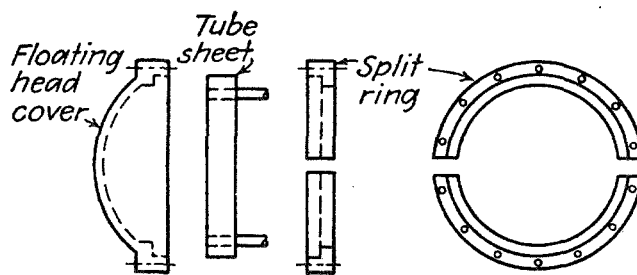


FIG. 7.13. Split-ring assembly.

modifications of the design shown here, but they all accomplish the purpose of providing increased surface over the pull-through floating head in the same size shell. Cast channels with nonremovable channel covers are also employed as shown in Fig. 7.12.

Tube-sheet Layouts and Tube Counts. A typical example of the layout of tubes for an exchanger with a split-ring floating head is shown in Fig. 7.14. The actual layout is for a $13\frac{1}{4}$ in. ID shell with 1 in. OD tubes on $1\frac{1}{4}$ -in. triangular pitch arranged for six tube passes. The partition

arrangement is also shown for the channel and floating-head cover along with the orientation of the passes. Tubes are not usually laid out symmetrically in the tube sheet. Extra entry space is usually allowed in the shell by omitting tubes directly under the inlet nozzle so as to minimize the contraction effect of the fluid entering the shell. When tubes are

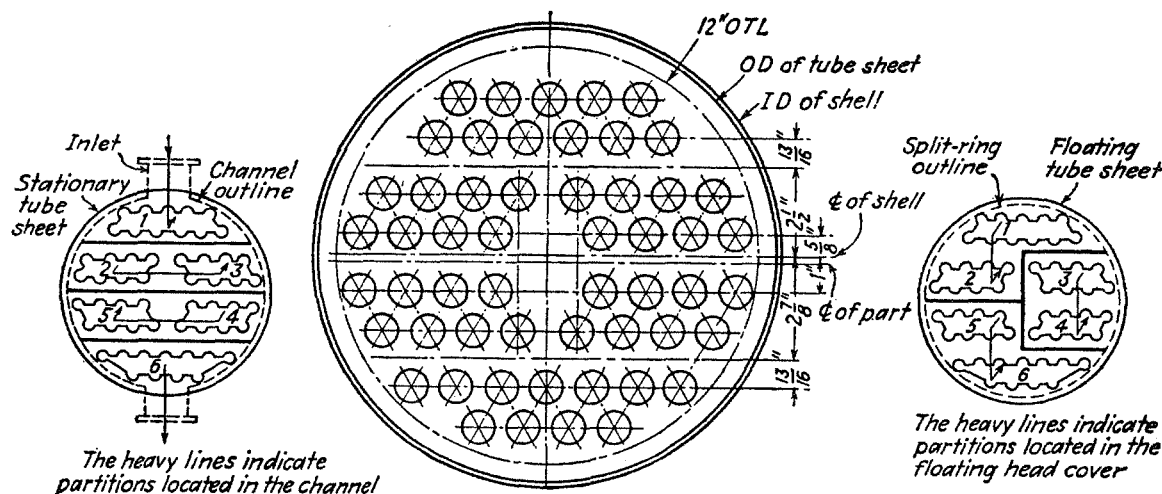


FIG. 7.14. Tube-sheet layout for a $13\frac{1}{4}$ in. ID shell employing 1 in. OD tubes on $1\frac{1}{4}$ -in. triangular pitch with six tube passes.

laid out with minimum space allowances between partitions and adjoining tubes and within a diameter free of obstruction called the *outer tube limit*, the number of tubes in the layout is the *tube count*. It is not always possible to have an equal number of tubes in each pass, although in large exchangers the unbalance should not be more than about 5 per cent. In Appendix Table 9 the tube counts for $\frac{3}{4}$ and 1 in. OD tubes are given for one pass shells and one, two, four, six, and eight tube pass arrangements.

TABLE 7.1. TUBE COUNT ENTRY ALLOWANCES

Shell ID, in.	Nozzle, in.
Less than 12.....	2
12-17 $\frac{1}{4}$	3
19 $\frac{1}{4}$ -21 $\frac{1}{4}$	4
23 $\frac{1}{4}$ -29.....	6
31 -37.....	8
Over 39.....	10

These tube counts include a free entrance path below the inlet nozzle equal to the cross-sectional area of the nozzles shown in Table 7.1. When a larger inlet nozzle is used, extra entry space can be obtained by flaring the inlet nozzle at its base or removing the tubes which ordinarily lie close to the inlet nozzle.

Packed Floating Head. Another modification of the floating-head 1-2 exchanger is the packed floating-head exchanger shown in Fig. 7.15.

This exchanger has an extension on the floating tube sheet which is confined by means of a packing gland. Although entirely satisfactory for shells up to 36 in. ID, the larger packing glands are not recommended for higher pressures or services causing vibration.

U-bend Exchangers. The 1-2 exchanger shown in Fig. 7.16 consists of tubes which are bent in the form of a U and rolled into the tube sheet.

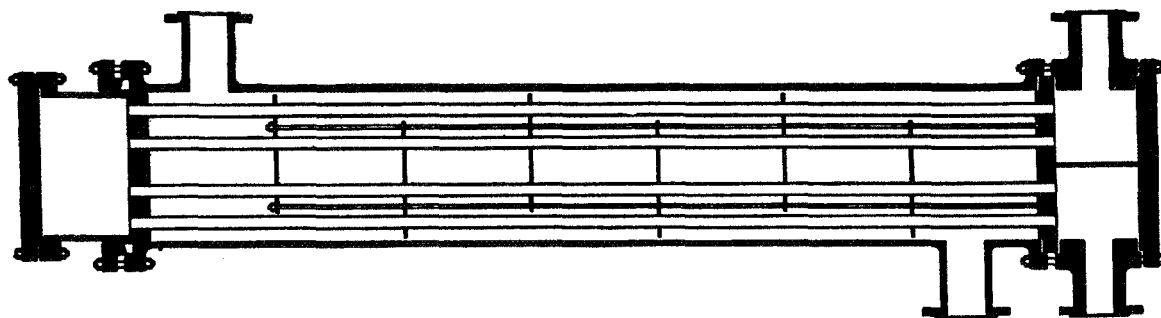


FIG. 7.15. Packed floating head 1-2 exchanger. (*Patterson Foundry & Machine Co.*)

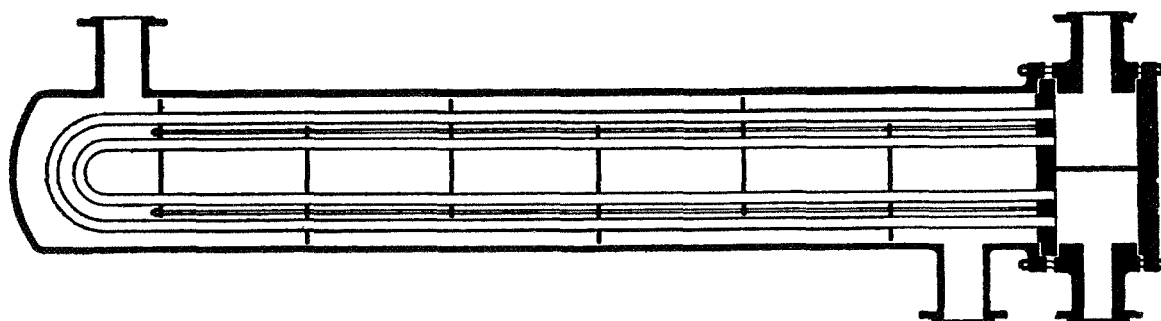


FIG. 7.16. U-bend 1-2 exchanger. (*Patterson Foundry & Machine Co.*)

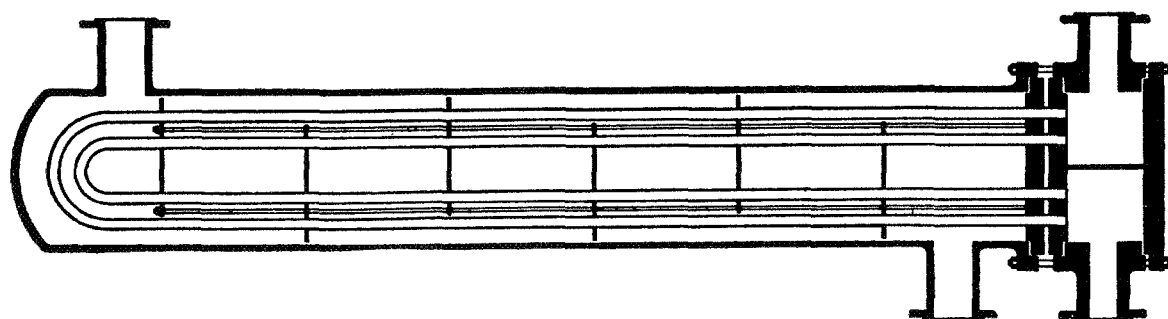


FIG. 7.17. U-bend double-tube-sheet exchanger. (*Patterson Foundry & Machine Co.*)

The tubes can expand freely, eliminating the need for a floating tube sheet, floating-head cover, shell flange, and removable shell cover. Baffles may be installed in the conventional manner on square or triangular pitch. The smallest diameter U-bend which can be turned without deforming the outside diameter of the tube at the bend has a diameter of three to four times the outside diameter of the tubing. This means that it will usually be necessary to omit some tubes at the center of the bundle, depending upon the layout.

An interesting modification of the U-bend exchanger is shown in Fig. 7.17. It employs a double stationary tube sheet and is used when the leakage of one fluid stream into the other at the tube roll can cause serious corrosion damage. By using two tube sheets with an air gap between them, either fluid leaking through its adjoining tube sheet will escape to the atmosphere. In this way neither of the streams can contaminate the other as a result of leakage except when a tube itself corrodes. Even tube failure can be prevented by applying a pressure shock test to the tubes periodically.

THE CALCULATION OF SHELL-AND-TUBE EXCHANGERS

Shell-side Film Coefficients. The heat-transfer coefficients outside tube bundles are referred to as shell-side coefficients. When the tube bundle employs baffles directing the shell-side fluid across the tubes from top to bottom or side to side, the heat-transfer coefficient is higher than for undisturbed flow along the axes of the tubes. The higher transfer

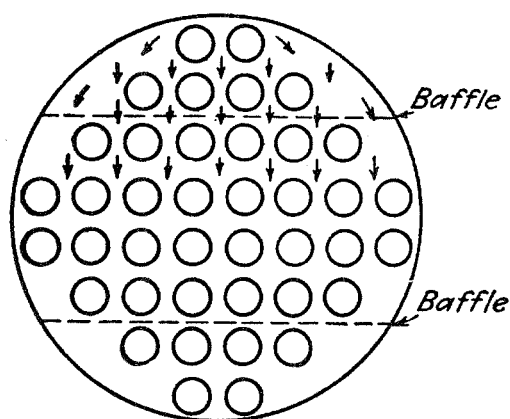


FIG. 7.18. Flow across a bundle.

coefficients result from the increased turbulence. In square pitch, as seen in Fig. 7.18, the velocity of the fluid undergoes continuous fluctuation because of the constricted area between adjacent tubes compared with the flow area between successive rows. In triangular pitch even greater turbulence is encountered because the fluid flowing between adjacent tubes at high velocity impinges directly on the succeeding row. This would indicate that, when the pressure drop and cleanability are of little consequence, triangular pitch is superior for the attainment of high shell-side film coefficients. This is actually the case, and under comparable conditions of flow and tube size the coefficients for triangular pitch are roughly 25 per cent greater than for square pitch.

Several factors not treated in preceding chapters influence the rate of heat transfer on the shell side. Suppose the length of a bundle is divided by six baffles. All the fluid travels across the bundle seven times. If ten baffles are installed in the same length of bundle, it would require that the bundle be crossed a total of eleven times, the closer spacing causing the greater turbulence. In addition to the effects of the baffle spacing the shell-side coefficient is also affected by the type of pitch, tube size, clearance, and fluid-flow characteristics. Furthermore, there is no true flow area by which the shell-side mass velocity can be computed,

since the flow area varies *across* the diameter of the bundle with the different number of tube clearances in each longitudinal row of tubes. *The correlation obtained for fluids flowing in tubes is obviously not applicable to fluids flowing over tube bundles with segmental baffles*, and this is indeed borne out by experiment. However, in establishing a method of correlation the form of the heat-transfer factor $j_H = (hD/k)(c\mu/k)^{-1/3}(\mu/\mu_w)^{-0.14}$ vs. DG/μ has been retained, in agreement with the suggestion of McAdams,¹ but using fictitious values for the equivalent diameter D_e and the mass velocity G_s as discussed below.

Figure 28 in the Appendix is a correlation of industrial data which gives satisfactory results for the hydrocarbons, organic compounds, water, aqueous solutions, and gases when the bundle employs baffles with acceptable clearances between baffles and tubes and between baffles and shells.² It is not the mean curve through the data but a safe curve such that the deviation of the test points from the curve ranges from 0 to approximately 20 per cent high. Inasmuch as the line expressing the equation possesses a curvature, it cannot be evaluated in the simple form of Eq. (3.42), since the proportionality constant and the exponent of the Reynolds number actually vary. For values of Re from 2000 to 1,000,000, however, the data are closely represented by the equation

$$\frac{h_o D_e}{k} = 0.36 \left(\frac{D_e G_s}{\mu} \right)^{0.55} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

where h_o , D_e and G_s are as defined below. Calculations using Fig. 28 agree very well with the methods of Colburn³ and Short⁴ and the test data of Breidenbach⁵ and O'Connell on a number of commercial heat exchangers. It will be observed in Fig. 28 that there is no discontinuity at a Reynolds number of 2100 such as occurs for fluids in tubes. The different equivalent diameters used in the correlation of shell and tube data precludes comparison between fluids flowing in tubes and across tubes on the basis of the Reynolds number alone. All the data in Fig. 28 refer to turbulent flow.

Shell-side Mass Velocity. The linear and mass velocities of the fluid change continuously across the bundle, since the width of the shell and the number of tubes vary from zero at the top and bottom to maxima at the center of the shell. The width of the flow area in the correlation

¹ McAdams, W. H., "Heat Transmission," 2d ed., p. 217, McGraw-Hill Book Company, Inc., New York, 1942.

² For mechanical details and standards see *Standards of the Tubular Exchanger Manufacturers' Association*, New York (1949).

³ Colburn, A. P., *Trans. AIChE*, **29**, 174-210 (1933).

⁴ Short, B. E., *Univ. Texas Pub.* 3819 (1938).

⁵ Breidenbach, E. P., and H. E. O'Connell, *Trans. AIChE*, **42**, 761-776 (1946).

represented by Fig. 28 was taken at the hypothetical tube row possessing the maximum flow area and corresponding to the center of the shell. The length of the flow area was taken equal to the baffle spacing B . The tube pitch is the sum of the tube diameter and the clearance C' . If the inside diameter of the shell is divided by the tube pitch, it gives a fictitious, but not necessarily integral, number of tubes which may be assumed to exist at the center of the shell. Actually in most layouts there is no row of tubes through the center but instead two equal maximum rows on either side of it having fewer tubes than computed for the center. These deviations are neglected. For each tube or fraction there are considered to be $C' \times 1 \text{ in.}^2$ of crossflow area per inch of baffle space. The shell-side or bundle crossflow area a_s is then given by

$$a_s = \frac{\text{ID} \times C' B}{P_T \times 144} \quad \text{ft}^2 \quad (7.1)$$

and as before, the mass velocity is

$$G_s = \frac{W}{a_s} \quad \text{lb}/(\text{hr})(\text{ft}^2) \quad (7.2)$$

Shell-side Equivalent Diameter. By definition, the hydraulic radius corresponds to the area of a circle equivalent to the area of a noncircular flow channel and consequently in a plane at right angles to the direction of flow. The hydraulic radius employed for correlating shell-side coefficients for bundles having baffles is not the true hydraulic radius. The direction of flow in the shell is partly along and partly at right angles to the long axes of the tubes of the bundle. The flow area at right angles to the long axes is variable from tube row to tube row. A hydraulic radius based upon the flow area across any one row could not distinguish between square and triangular pitch. In order to obtain a simple correlation combining both the size and closeness of the tubes and their type of pitch, excellent agreement is obtained if the hydraulic radius is calculated *along* instead of across the long axes of the tubes. The equivalent diameter for the shell is then taken as four times the hydraulic radius obtained for the pattern as layed out on the tube sheet. Referring to Fig. 7.19, where the crosshatch covers the free area¹ for *square pitch*

$$D_e = \frac{4 \times \text{free area}}{\text{wetted perimeter}} \quad \text{ft} \quad (7.3)$$

or

$$d_e = \frac{4 \times (P_T^2 - \pi d_0^2/4)}{\pi d_0} \quad \text{in.} \quad (7.4)$$

¹ The expression free area is used to avoid confusion with the free-flow area, an actual entity in the hydraulic radius.

where P_T is the tube pitch in inches and d_o the tube outside diameter in inches. For triangular pitch as shown in Fig. 7.19 the wetted perimeter of the element corresponds to half a tube.

$$d_e = \frac{4 \times (\frac{1}{2}P_T \times 0.86P_T - \frac{1}{2}\pi d_o^2/4)}{\frac{1}{2}\pi d_o} \quad \text{in.} \quad (7.5)$$

The equivalent diameters for the common arrangements are included in Fig. 28.

It would appear that this method of evaluating the hydraulic radius and equivalent diameter does not distinguish between the relative percentage of right-angle flow to axial flow, and this is correct. It is possible, using the same shell, to have equal mass velocities, equivalent diameters, and

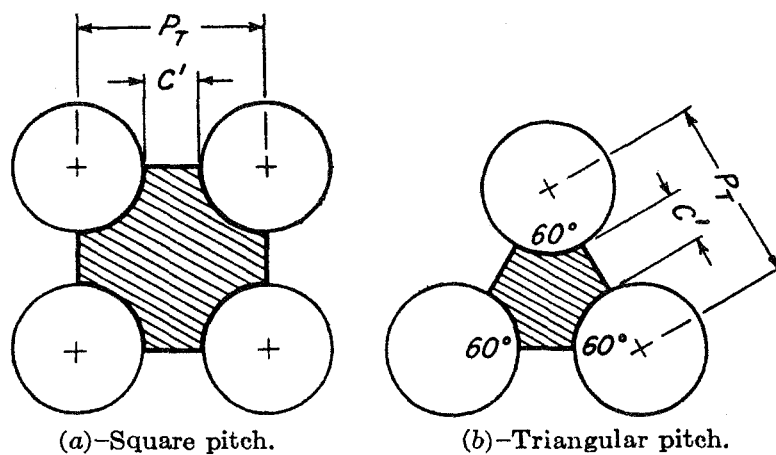


FIG. 7.19. Equivalent diameter.

Reynolds numbers using a large quantity of fluid and a large baffle pitch or a small quantity of fluid and a small baffle pitch, although the proportions of right-angle flow to axial flow differ. Apparently, where the range of baffle pitch is restricted between the inside diameter and one-fifth the inside diameter of the shell, the significance of the error is not too great to permit correlation.

Example 7.1. Compute the shell-side equivalent diameter for a $\frac{3}{4}$ in. OD tube on 1-in. square pitch. From Eq. (7.4)

$$d_e = \frac{4(1^2 - 3.14 \times 0.75^2/4)}{3.14 \times 0.75} = 0.95 \text{ in.}$$

$$D_e = \frac{0.95}{12} = 0.079 \text{ ft}$$

The True Temperature Difference Δt in a 1-2 Exchanger. A typical plot of temperature vs. length for an exchanger having one shell pass and two tube passes is shown in Fig. 7.20 for the nozzle arrangement indicated. Relative to the shell fluid, one tube pass is in counterflow and the other in parallel flow. Greater temperature differences have

been found, in Chap. 5, to result when the process streams are in counterflow and lesser differences for parallel flow. The 1-2 exchanger is a combination of both, and the LMTD for counterflow or parallel flow alone cannot be the true temperature difference for a parallel flow-counterflow arrangement. Instead it is necessary to develop a new equation for calculation of the effective or true temperature difference Δt to replace the counterflow LMTD. The method employed here is a modification of the derivation of Underwood¹ and is presented in the final

form proposed by Nagle² and Bowman, Mueller, and Nagle.³

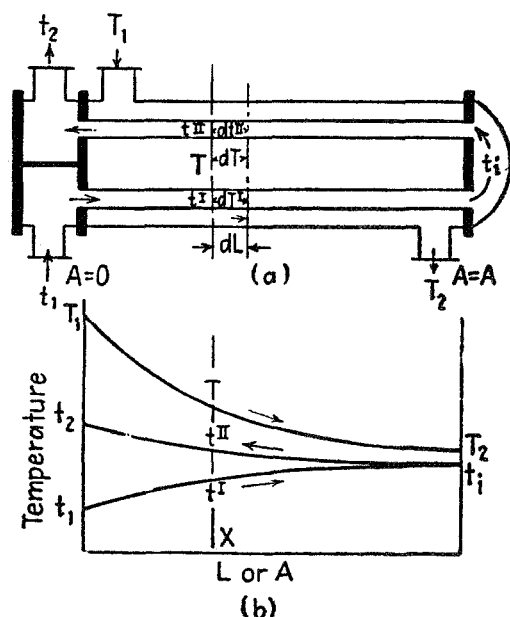


FIG. 7.20. Temperature relations in a 1-2 exchanger.

The temperature of the shell fluid may undergo either of two variations as it proceeds from the inlet to outlet, crossing the tube bundle several times in its progress: (1) So much turbulence is induced that the shell fluid is completely mixed at any length X from the inlet nozzle, or (2) so little turbulence is induced that there is a selective temperature atmosphere about the tubes of each tube pass individually. The baffles and turbulent nature of the flow components across the bundle appear to eliminate (2) so that (1) is taken as the first of the assumptions for the derivation of the true temperature

difference in a 1-2 exchanger. The assumptions are

1. The shell fluid temperature is an average isothermal temperature at any cross section.
2. There is an equal amount of heating surface in each pass.
3. The overall coefficient of heat transfer is constant.
4. The rate of flow of each fluid is constant.
5. The specific heat of each fluid is constant.
6. There are no phase changes of evaporation or condensation in a part of the exchanger.
7. Heat losses are negligible.

The overall heat balance, where Δt is the true temperature difference, is

$$Q = UA \Delta t = WC(T_1 - T_2) = wc(t_2 - t_1) \quad (7.6)$$

¹ Underwood, A. J. V., *J. Inst. Petroleum Technol.*, **20**, 145-158 (1934).

² Nagle, W. M., *Ind. Eng. Chem.*, **25**, 604-608 (1933).

³ Bowman, R. A., A. C. Mueller, and W. M. Nagle. *Trans. ASME*, **62**, 283-294 (1940).

From which

$$\Delta t = \left(\frac{T_1 - T_2}{UA/WC} \right)_{\text{true}} = \left(\frac{t_2 - t_1}{UA/wc} \right)_{\text{true}} \quad (7.7)$$

In Fig. 7.20a let T be the temperature of the shell fluid at any cross section of the shell $L = X$ between $L = 0$ and $L = L$. Let t^I and t^{II} represent temperatures in the first and second tube passes, respectively, and at the same cross section as T . Let a'' be the external surface per foot of length. In the incremental surface $dA = a''dL$ the shell temperature changes by $-dT$. Over the area dA

$$-WC dT = U \frac{dA}{2} (T - t^I) + U \frac{dA}{2} (T - t^{II}) \quad (7.8)$$

$$-WC dT = U dA \left(T - \frac{t^I + t^{II}}{2} \right) \quad (7.9)$$

$$- \int \frac{U dA}{WC} = \int \frac{dT}{T - (t^I + t^{II})/2} \quad (7.10)$$

But in this equation T , t^I , and t^{II} are dependent variables. The heat balance from $L = X$ to the hot-fluid inlet is

$$WC(T - T_2) = wc(t^{II} - t^I) \quad (7.11)$$

and the heat balance per pass

$$wc dt^I = U \frac{dA}{2} (T - t^I) \quad (7.12)$$

$$wc dt^{II} = -U \frac{dA}{2} (T - t^{II}) \quad (7.13)$$

Dividing Eq. (7.13) by Eq. (7.12),

$$\frac{dt^{II}}{dt^I} = - \frac{T - t^{II}}{T - t^I} \quad (7.14)$$

To eliminate t^{II} and dt^{II} from Eq. (7.11) and (7.13)

$$t^{II} = \frac{WC}{wc} (T_2 - T) + t^I \quad (7.15)$$

Differentiating Eq. (7.15), with the hot fluid inlet T_1 constant,

$$dt^{II} = - \frac{WC}{wc} dT + dt^I \quad (7.16)$$

Substituting in Eq. (7.14) and rearranging,

$$\frac{WC}{wc} \frac{dT}{dt^I} = 1 + \frac{T - t^I - (WC/wc)(T_2 - T)}{T - t^I} \quad (7.17)$$

The number of variables in Eq. (7.15) has been reduced from three (T, t^I, t^{II}) to two (T and t^I). For a solution it is necessary to eliminate either T or t^I . Simplifying by the use of parameters as in the case of the double pipe exchanger let

$$R = \frac{T_1 - T_2}{t_2 - t_1} = \frac{wc}{WC} \quad \text{and} \quad S = \frac{t_2 - t_1}{T_1 - t_1}$$

Rearranging Eq. (7.8),

$$WC \frac{dT}{dA} + \frac{U}{2} (T - t^I) + \frac{U}{2} (T - t^{II}) = 0 \quad (7.18)$$

Simplifying and substituting $WC = wc/R$,

$$\frac{dT}{dA} + \frac{URT}{wc} - \frac{UR}{2wc} (t^I + t^{II}) = 0 \quad (7.19)$$

Differentiating with respect to A ,

$$\frac{d^2T}{dA^2} + \frac{UR}{wc} \frac{dT}{dA} - \frac{UR}{2wc} \left(\frac{dt^I}{dA} + \frac{dt^{II}}{dA} \right) = 0 \quad (7.20)$$

Substituting Eqs. (7.12) and (7.13),

$$\frac{d^2T}{dA^2} + \frac{UR}{wc} \frac{dT}{dA} - \frac{U^2R}{(2wc)^2} (t^{II} - t^I) = 0 \quad (7.21)$$

Since the heat change is sensible, a direct proportionality exists between the percentage of the temperature rise (or fall) and Q .

$$\frac{T - T_2}{T_1 - T_2} = \frac{t^I - t^{II}}{t_1 - t_2} \quad (7.22)$$

or

$$t^{II} - t^I = \frac{T - T_2}{R} \quad (7.23)$$

$$\frac{d^2T}{dA^2} + \frac{UR}{wc} \frac{dT}{dA} - \frac{U^2T}{(2wc)^2} = - \frac{U^2T_2}{(2wc)^2} \quad (7.24)$$

Differentiating again with respect to A ,

$$\frac{d^3T}{dA^3} + \frac{UR}{wc} \frac{d^2T}{dA^2} - \frac{U^2}{(2wc)^2} \frac{dT}{dA} = 0 \quad (7.25)$$

The solution of this equation will be found in any standard differential-equations text. The equation is

$$T = K_1 + K_2 e^{-(UA/2wc)(R+\sqrt{R^2+1})} + K_3 e^{-(UA/2wc)(R-\sqrt{R^2+1})} \quad (7.26)$$

When $T = T_2$, A will have increased from 0 to A , and from the solution

of Eq. (7.24) $K_1 = T_2$ so that Eq. (7.26) becomes

$$-K_2 e^{-(UA/2wc)(R+\sqrt{R^2+1})} = K_3 e^{-(UA/2wc)(R-\sqrt{R^2+1})} \quad (7.27)$$

Taking logarithms of both sides and simplifying,

$$\frac{UA}{wc} = \frac{1}{\sqrt{R^2+1}} \ln \left(-\frac{K_2}{K_3} \right) \quad (7.28)$$

Differentiate Eq. (7.26):

$$\begin{aligned} \frac{dT}{dA} = & -K_2 \frac{U}{2wc} (R + \sqrt{R^2+1}) e^{-(UA/2wc)(R+\sqrt{R^2+1})} \\ & - K_3 \frac{U}{2wc} (R - \sqrt{R^2+1}) e^{-(UA/2wc)(R-\sqrt{R^2+1})} \end{aligned} \quad (7.29)$$

Substituting the value of dT/dA from Eq. (7.19) and since at $A = 0$, $t^I = t_1$, $t^{II} = t_2$, and $T = T_1$, $t^I + t^{II} = t_1 + t_2$.

$$\begin{aligned} R(t_1 + t_2) - 2RT_1 = & -K_2(R + \sqrt{R^2+1}) \\ & - K_3(R - \sqrt{R^2+1}) \end{aligned} \quad (7.30)$$

From Eq. (7.26) at $A = 0$ and $T = T_1$ and $K_1 = T_2$

$$T_1 - T_2 = K_2 + K_3 \quad (7.31)$$

Multiplying both sides of Eq. (7.31) by $(R + \sqrt{R^2+1})$,

$$\begin{aligned} (R + \sqrt{R^2+1})(T_1 - T_2) = & K_2(R + \sqrt{R^2+1}) \\ & + K_3(R + \sqrt{R^2+1}) \end{aligned} \quad (7.32)$$

Adding Eqs. (7.31) and (7.32) and solving for K_3 ,

$$K_3 = \frac{R(t_1 + t_2) + (T_1 - T_2)(R + \sqrt{R^2+1}) - 2RT_1}{2\sqrt{R^2+1}} \quad (7.33)$$

Returning to Eq. (7.31),

$$\begin{aligned} -K_2 = K_3 - (T_1 - T_2) = \\ \frac{(R + \sqrt{R^2+1})(T_1 - T_2) - 2\sqrt{R^2+1}(T_1 - T_2) - 2RT_1 + R(t_1 + t_2)}{2\sqrt{R^2+1}} \end{aligned} \quad (7.34)$$

Since $R = (T_1 - T_2)/(t_2 - t_1)$,

$$-\frac{K_2}{K_3} = \frac{(R - \sqrt{R^2+1})(t_1 - t_2) - (T_1 - t_1) - (T_1 - t_2)}{(R + \sqrt{R^2+1})(t_1 - t_2) - (T_1 - t_1) - (T_1 - t_2)} \quad (7.35)$$

Dividing by $T_1 - t_1$ and substituting $S = (t_2 - t_1)/(T_1 - t_1)$ and $1 - S = (T_1 - t_2)/(T_1 - t_1)$,

$$-\frac{K_2}{K_3} = \frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} \quad (7.36)$$

Substituting in Eq. (7.28),

$$\left(\frac{UA}{wc}\right)_{\text{true}} = \frac{1}{\sqrt{R^2 + 1}} \ln \frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} \quad (7.37)$$

Equation (7.37) is the relationship for the true temperature difference for 1-2 parallel flow-counterflow. How does this compare with the LMTD for counterflow employing the same process temperatures? For counterflow

$$Q = wc(t_2 - t_1) = UA \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln (T_1 - t_2)/(T_2 - t_1)} \quad (7.38)$$

from which

$$\left(\frac{UA}{wc}\right)_{\text{counterflow}} = \frac{t_2 - t_1}{\frac{(T_1 - t_2) - (T_2 - t_1)}{\ln (T_1 - t_2)/(T_2 - t_1)}} = \frac{\ln (1 - S)/(1 - RS)}{R - 1} \quad (7.39)$$

The ratio of the true temperature difference to the LMTD is

$$\frac{t_2 - t_1}{(UA/wc)_{\text{true}}} \bigg/ \frac{t_2 - t_1}{(UA/wc)_{\text{counterflow}}} = \frac{(UA/wc)_{\text{counterflow}}}{(UA/wc)_{\text{true}}} \quad (7.40)$$

Calling the fractional ratio of the true temperature difference to the LMTD F_T ,

$$F_T = \frac{\sqrt{R^2 + 1} \ln (1 - S)/(1 - RS)}{(R - 1) \ln \frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})}} \quad (7.41)$$

The Fourier equation for a 1-2 exchanger can now be written:

$$Q = UA \Delta t = UAF_T(\text{LMTD}) \quad (7.42)$$

To reduce the necessity of solving Eq. (7.37) or (7.41), correction factors F_T for the LMTD have been plotted in Fig. 18 in the Appendix as a function of S with R as the parameter. When a value of S and R is close to the vertical portion of a curve, it is difficult to read the figure and F_T should be computed from Eq. (7.41) directly. When an exchanger has one shell pass and four, six, eight, or more even-numbered tube passes such as a 1-4, 1-6, or 1-8 exchanger, Eq. (7.10) becomes for a 1-4 exchanger

$$-\int \frac{U dA}{WC} = \int \frac{dT}{T - (t^I + t^{II} + t^{III} + t^{IV})/4}$$

for a 1-6 exchanger

$$-\int \frac{U dA}{WC} = \int \frac{dT}{T - (t^I + t^{II} + t^{III} + t^{IV} + t^V + t^{VI})/6} \dots$$

It can be shown that the values of F_T for a 1-2 and 1-8 exchanger are less than 2 per cent apart in the extreme case and generally considerably less. It is therefore customary to describe any exchanger having one shell pass and *two or more even-numbered* tube passes in parallel flow-counterflow as a 1-2 exchanger and to use the value of F_T obtained from Eq. (7.41). The reason F_T will be less than 1.0 is naturally due to the fact that the tube passes in parallel with the shell fluid do not contribute so effective a temperature difference as those in counterflow with it.

There is an important limitation to the use of Fig. 18. Although any exchanger having a value of F_T above zero will theoretically operate, it is not practically true. The failure to fulfill in practice all the assumptions employed in the derivation, assumptions 1, 3, and 7 in particular, may cause serious discrepancies in the calculation of Δt . As a result of the discrepancies if the actual value of t_i in Fig. 7.20a at the end of the parallel pass is required to approach T_2 more closely than the derived value of t_i , it may impose a violation of the rule of parallel flow; namely, the outlet of the one stream t_i may not attain the outlet of the other, T_2 , without infinite surface. Accordingly it is not advisable or practical to use a 1-2 exchanger whenever the correction factor F_T is computed to be less than 0.75. Instead some other arrangement is required which more closely resembles counterflow.

The temperature relationship for the case where the orientation of the shell nozzles has been reversed is shown in Fig. 7.21 for the same inlet and outlet temperatures plotted in Fig. 7.20. Underwood¹ has shown that the values of F_T for both are identical.² Since a 1-2 exchanger is a combination of counterflow and parallel-flow passes, it may be expected that the outlet of one process stream cannot approach the inlet of the

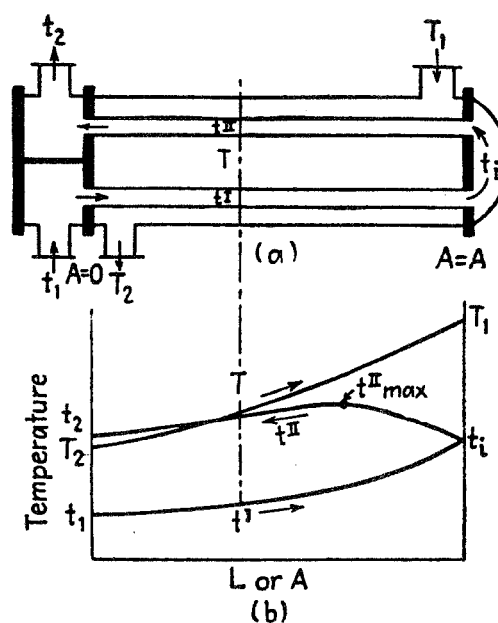


FIG. 7.21. Temperature relations in a 1-2 exchanger with conventional nozzle arrangement.

¹ Underwood, *op. cit.*

² The values of t_i , however, differ for both cases.

other very closely. In fact it is customary in parallel flow-counterflow equipment to call $T_2 - t_2$ the *approach*, and if $t_2 > T_2$, then $t_2 - T_2$ is called the *temperature cross*.

It is useful to investigate several typical process temperatures and to note the influence of different approaches and crosses upon the value of

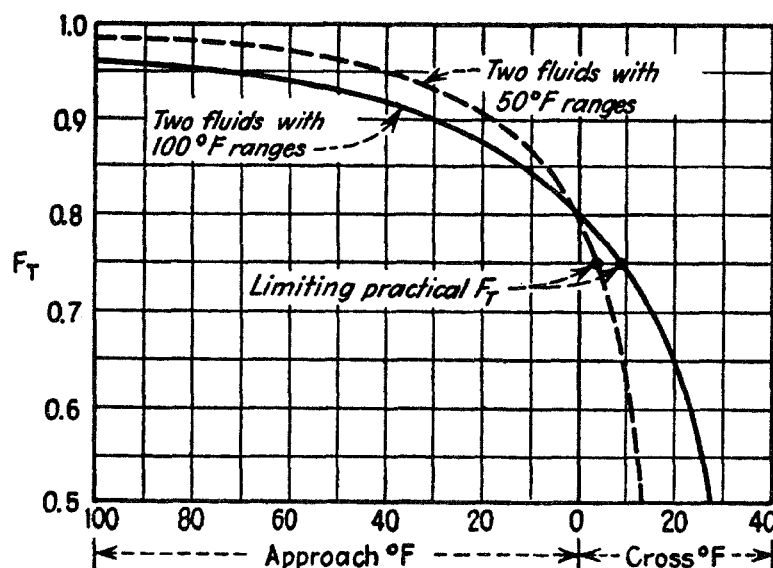


FIG. 7.22. Influence of approach temperature on F_T with fluids having equal ranges in a 1-2 exchanger.

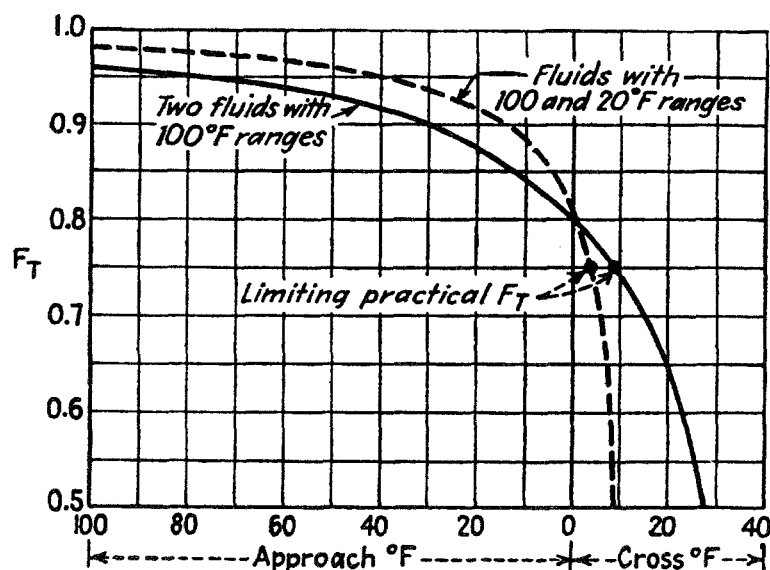


FIG. 7.23. Influence of approach temperature on F_T with fluids having unequal ranges in a 1-2 exchanger.

F_T . For a given service the reduction of F_T below unity in Eq. (7.42) is compensated for by increasing the surface. Thus if the process temperatures are fixed it may be inadvisable to employ a parallel flow-counterflow exchanger as against a counterflow exchanger, since it increases the cost of the equipment beyond the value of its mechanical advantages. In Fig. 7.22 two pairs of fluids each with equal ranges of 100 and 50°F

are studied. The operating temperatures of the cold fluid are fixed, while the hot-fluid temperatures are variable thereby changing the approach in each case. Note the conditions under which F_T shrinks rapidly, particularly the approach at the practical minimum $F_T = 0.75$ and the influence of the relationship between T_2 and t_2 . The calculation of several points is demonstrated.

Example 7.2. Calculation of F_T for Fluids with Equal Ranges

Point:	(a) 50° approach	(b) Zero approach	(c) 20° cross
	(T_1) 350 200 (t_2)	(T_1) 300 200 (t_2)	(T_1) 280 200 (t_2)
	(T_2) 250 100 (t_1)	(T_2) 200 100 (t_1)	(T_2) 180 100 (t_1)
	$\frac{100}{100}$ $\frac{100}{100}$	$\frac{100}{100}$ $\frac{100}{100}$	$\frac{100}{100}$ $\frac{100}{100}$
	$R = \frac{T_1 - T_2}{t_2 - t_1} = \frac{100}{100} = 1.0$	$R = 1.0$	$R = 1.0$
	$S = \frac{t_2 - t_1}{T_1 - t_1} = \frac{100}{350 - 100} = 0.40$	$S = 0.50$	$S = 0.555$
	$F_T = 0.925$ (Fig. 18)	$F_T = 0.80$	$F_T = 0.64$

In Fig. 7.23 are shown the results of the calculation when one fluid has a range five times as great as the other.

Shell-side Pressure Drop. The pressure drop through the shell of an exchanger is proportional to the number of times the fluid crosses the bundle between baffles. It is also proportional to the distance across the bundle each time it is crossed. Using a modification of Eq. (3.44) a correlation has been obtained using the product of the distance across the bundle, taken as the inside diameter of the shell in feet D_s , and the number of times the bundle is crossed $N + 1$, where N is the number of baffles. If L is the tube length in feet,

$$\begin{aligned} \text{Number of crosses, } N + 1 &= \text{tube length, in./baffle space, in.} \\ &= 12 \times L/B \quad (7.43) \end{aligned}$$

If the tube length is 16'0" and the baffles are spaces 18 in. apart, there will be 11 crosses or 10 baffles. There should always be an odd number of crosses if both shell nozzles are on opposite sides of the shell and an even number if both shell nozzles are on the same side of the shell. With close baffle spacings at convenient intervals such as 6 in. and under, one baffle may be omitted if the number of crosses is not an integer. The equivalent diameter used for calculating the pressure drop is the same as for heat transfer, the additional friction of the shell itself being neglected. The isothermal equation for the pressure drop of a fluid being heated or cooled and including entrance and exit losses is

$$\Delta P_s = \frac{fG_s^2 D_s (N + 1)}{2g\rho D_s \phi_s} = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad \text{psf} \quad (7.44)$$

where s is the specific gravity of the fluid. Equation (7.44) gives the pressure drop in pounds per square foot. The common engineering unit is pounds per square inch. To permit the direct solution for ΔP_t in psi *dimensional* shell-side friction factors, square foot per square inch, have been plotted in Fig. 29. To obtain the pressure drop in consistent units by Eq. (7.44) multiply f in Fig. 29 by 144.

Tube-side Pressure Drop. Equation (3.44) may be used to obtain the pressure drop in tubes, but it applies principally to an isothermal fluid. Sieder and Tate have correlated friction factors for fluids being heated or cooled in tubes. They are plotted in dimensional form in Fig. 26 and are used in the equation

$$\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_e s \phi_t} \quad \text{psf} \quad (7.45)$$

where n is the number of tube passes, L the tube length, and Ln is the total length of path in feet. The deviations are not given, but the curve has been accepted by the Tubular Exchanger Manufacturers Association. In flowing from one pass into the next at the channel and floating head the fluid changes direction abruptly by 180° , although the flow area provided in the channel and floating-head cover should not be less than the combined flow area of all the tubes in a single pass. The change of direction introduces an additional pressure drop ΔP_r , called the return loss and accounted for by allowing four velocity heads per pass. The velocity head $V^2/2g'$ has been plotted in Fig. 27 against the mass velocity for a fluid with a specific gravity of 1, and the return losses for any fluid will be

$$\Delta P_r = \frac{4n}{s} \frac{V^2}{2g'} \quad \text{psi} \quad (7.46)$$

where V = velocity, fps

s = specific gravity

g' = acceleration of gravity, ft/sec²

The total tube-side pressure drop ΔP_T will be

$$\Delta P_T = \Delta P_t + \Delta P_r \quad \text{psi} \quad (7.47)$$

The Analysis of Performance in an Existing 1-2 Exchanger. When all the pertinent equations are used to calculate the suitability of an existing exchanger for given process conditions, it is known as *rating* an exchanger. There are three significant points in determining the suitability of an existing exchanger for a new service.

1. What clean coefficient U_c can be "performed" by the two fluids as the result of their flow and individual film coefficients $h_{i,c}$ and $h_{o,c}$?

2. From the heat balance $Q = WC(T_1 - T_2) = wc(t_2 - t_1)$, known surface A , and the true temperature difference for the process temperatures a value of the design or dirty coefficient, U_D is obtained. U_c must exceed U_D sufficiently so that the dirt factor, which is a measure of the excess surface, will permit operation of the exchanger for a reasonable period of service.

3. The allowable pressure drops for the two streams may not be exceeded.

When these are fulfilled, an existing exchanger is suitable for the process conditions for which it has been rated. In starting a calculation the first point which arises is to determine whether the hot or cold fluid should be placed in the shell. There is no fast rule. One stream may be large and the other small, and the baffle spacing may be such that in one instance the shell-side flow area a_s will be larger. Fortunately any selection can be checked by switching the two streams and seeing which arrangement gives the larger value of U_c without exceeding the allowable pressure drop. Particularly in preparation for later methods there is some advantage, however, in starting calculations with the tube side, and it may be well to establish the habit. The detailed steps in the rating of an exchanger are outlined below. The subscripts s and t are used to distinguish between the shell and tubes, and for the outline the hot fluid has been assumed to be in the shell. By always placing the hot fluid on the left the usual method of computing the LMTD may be retained.

The Calculation of an Existing 1-2 Exchanger. Process conditions required

Hot fluid: $T_1, T_2, W, c, s, \mu, k, R_d, \Delta P$

Cold fluid: $t_1, t_2, w, c, s, \mu, k, R_d, \Delta P$

For the exchanger the following data must be known:

<i>Shell side</i>	<i>Tube side</i>
ID	Number and length
Baffle space	OD, BWG, and pitch
Passes	Passes

(1) Heat balance, $Q = WC(T_1 - T_2) = wc(t_2 - t_1)$

(2) True temperature difference Δt :

$$\text{LMTD, } R = \frac{T_1 - T_2}{t_2 - t_1}, \quad S = \frac{t_2 - t_1}{T_1 - t_1} \quad (5.14)$$

$$\Delta t = \text{LMTD} \times F_T \quad (F_T \text{ from Fig. 18}) \quad (7.42)$$

(3) Caloric temperature T_c and t_c :¹

(5.28), (5.29)

<i>Hot fluid: shell side</i>	<i>Cold fluid: tube side</i>
(4') Flow area, $a_s = ID \times C'B/144P_T$, ft ² [Eq. (7.1)]	(4) Flow area, a_t : Flow area per tube a'_t from Table 10, in. ² $a_t = \frac{\text{No. of tubes} \times \text{flow area/tube}}{\text{No. of passes}}$ $= N_t a'_t / 144n$, ft ² Eq. (7.48)
(5') Mass vel, $G_s = W/a_s$, lb/(hr)(ft ²) [Eq. (7.2)]	(5) Mass vel, $G_t = w/a_t$, lb/(hr)(ft ²)
(6') Obtain D_e from Fig. 28 or compute from Eq. (7.4). Obtain μ at T_c , lb/(ft)(hr) = cp $\times$ 2.42 $Re_s = D_e G_s / \mu$	(6) Obtain D from Table 10, ft. Obtain μ at t_c , lb/(ft)(hr) = cp $\times$ 2.42 $Re_t = D G_t / \mu$
(7') Obtain j_H from Fig. 28.	(7) Obtain j_H from Fig. 24.
(8') At T_c obtain c , Btu/(lb)(°F) and k , Btu/(hr)(ft ²)(°F/ft). Compute* $(c\mu/k)^{1/3}$.	(8) At t_c obtain c , Btu/(lb)(°F) and k , Btu/(hr)(ft ²)(°F/ft). Compute* $(c\mu/k)^{1/3}$.
(9') $h_o = j_H \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/3} \phi_s$ [Eq. (6.15b)]	(9) $h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/3} \phi_t$ [Eq. (6.15a)]
(10') Tube-wall temp, t_w $t_w = t_c + \frac{h_o/\phi_s}{h_{io}/\phi_t + h_o/\phi_s} (T_c - t_c)$ [Eq. (5.31)]	(10) $\frac{h_{io}}{\phi_t} = \frac{h_i}{\phi_t} \times \frac{ID}{OD}$ [Eq. (6.5)]
(11') Obtain μ_w and $\phi_s = (\mu/\mu_w)^{0.14}$. [Fig. 24]	(11) Obtain t_w from (10'). Obtain μ_w and $\phi_t = (\mu/\mu_w)^{0.14}$. [Fig. 24]
(12') Corrected coefficient, $h_o = \frac{h_o \phi_s}{\phi_s}$ [Eq. (6.36)]	(12) Corrected coefficient, $h_{io} = \frac{h_{io} \phi_t}{\phi_t}$ [Eq. (6.37)]
(13) Clean overall coefficient U_C : $U_C = \frac{h_{io} h_o}{h_{io} + h_o} \quad (6.38)$	
(14) Design overall coefficient U_D : Obtain external surface/lin ft a'' from Appendix Table 10. Heat-transfer surface, $A = a'' L N_t$, ft ² $U_D = \frac{Q}{A \Delta t} \quad \text{Btu/(hr)(ft}^2\text{)(°F)}$	
(15) Dirt factor R_d : $R_d = \frac{U_C - U_D}{U_C U_D} \quad (\text{hr})(\text{ft}^2)(\text{°F})/\text{Btu} \quad (6.13)$	

If R_d equals or exceeds the required dirt factor, proceed under the pressure drop.

¹ The use of caloric temperatures is in partial contradiction of the derivation for the 1-2 parallel flow-counterflow temperature difference in which U was assumed constant. The use of caloric temperatures presumes that a linear variation of U with t can be accounted for as the product of $U_{\text{caloric}} \Delta t$ where Δt is the true parallel flow-counterflow temperature difference when U is constant.

* A convenient graph of $k(c\mu/k)^{1/3}$ vs. μ for petroleum fractions is given in Fig. 16.

Pressure Drop

- | | |
|---|--|
| <p>(1') For Re_s in (6') obtain f, $\text{ft}^2/\text{in.}^2$
[Fig. 29]</p> <p>(2') No. of crosses, $N + 1 = 12L/B$
[Eq. (7.43)]</p> <p>(3') $\Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ psi
[Eq. (7.44)]</p> | <p>(1) For Re_t in (6) obtain f, $\text{ft}^2/\text{in.}^2$
[Fig. 26]</p> <p>(2) $\Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)]</p> <p>(3) $\Delta P_r = \frac{4n}{s} \frac{V^2}{2g'} \frac{62.5}{144}$ psi [Eq. (7.46)]</p> <p>$\Delta P_T = \Delta P_t + \Delta P_r$ psi [Eq. (7.47)]</p> |
|---|--|

Example 7.3. Calculation of a Kerosene-Crude Oil Exchanger. 43,800 lb/hr of a 42°API kerosene leaves the bottom of a distilling column at 390°F and will be cooled to 200°F by 149,000 lb/hr of 34°API Mid-continent crude coming from storage at 100°F and heated to 170°F. A 10 psi pressure drop is permissible on both streams, and in accordance with Table 12, a combined dirt factor of 0.003 should be provided.

Available for this service is a 21¼ in. ID exchanger having 158 1 in. OD, 13 BWG tubes 16'0" long and laid out on 1¼-in. square pitch. The bundle is arranged for four passes, and baffles are spaced 5 in. apart.

Will the exchanger be suitable; *i.e.*, what is the dirt factor?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 21¼ in.	Number and length = 158, 16'0"
Baffle space = 5 in.	OD, BWG, pitch = 1 in., 13 BWG, 1¼-in. square
Passes = 1	Passes = 4

(1) Heat balance:

Kerosene, $Q = 43,800 \times 0.605(390 - 200) = 5,100,000$ Btu/hr

Mid-continent crude, $Q = 149,000 \times 0.49(170 - 100) = 5,100,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
390	Higher Temp	170	220
200	Lower Temp	100	100
190	Differences	70	120

($\Delta t_2 - \Delta t_1$)

($T_1 - T_2$) ($t_2 - t_1$)

$$\text{LMTD} = 152.5^\circ\text{F} \quad (5.14)$$

$$R = \frac{190}{70} = 2.71$$

$$S = \frac{70}{390 - 100} = 0.241$$

$$F_T = 0.905 \quad (\text{Fig. 18})$$

$$\Delta t = 0.905 \times 152.5 = 138^\circ\text{F} \quad (7.42)$$

(3) T_c and t_c :

$$\frac{\Delta t_c}{\Delta t_h} = 0.455$$

(Fig. 17)

$$K_c = 0.20 \text{ (crude oil controlling)}$$

$$F_c = 0.42$$

$$T_c = 200 + 0.42 \times 190 = 280^\circ\text{F} \quad (5.28)$$

$$t_c = 100 + 0.42 \times 70 = 129^\circ\text{F} \quad (5.29)$$

Since the flow area of both the shell and tube sides will be nearly equal, assume the larger stream to flow in the tubes and start calculation on the tube side.

Hot fluid: shell side, kerosene

$$(4') \text{ Flow area, } a_s = \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= 21.25 \times 0.25 \times 5/144 \times 1.25 = 0.1475 \text{ ft}^2$$

$$(5') \text{ Mass vel, } G_s = W/a_s \quad [\text{Eq. (7.2)}]$$

$$= 43,800/0.1475 = 297,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6') Re_s = D_s G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$\text{At } T_c = 280^\circ\text{F, } \mu = 0.40 \times 2.42 = 0.97 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$D_s = 0.99/12 = 0.0825 \text{ ft} \quad [\text{Fig. 28}]$$

$$Re_s = 0.0825 \times 297,000/0.97 = 25,300$$

$$(7') j_H = 93 \quad [\text{Fig. 28}]$$

$$(8') \text{ At } T_c = 280^\circ\text{F,}$$

$$c = 0.59 \text{ Btu/(lb)(}^\circ\text{F)} \quad [\text{Fig. 4}]$$

$$k = 0.0765 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Fig. 1}]$$

$$(c\mu/k)^{1/3} = (0.59 \times 0.97/0.0765)^{1/3} = 1.95$$

$$(9') h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k} \right)^{1/3} \phi_s \quad [\text{Eq. (6.15b)}]$$

$$\frac{h_o}{\phi_s} = 93 \times \frac{0.0765}{0.0825} \times 1.95 = 169$$

(10') Tube-wall temperature:

$$t_w = t_c + \frac{h_o/\phi_s}{h_{io}/\phi_i + h_o/\phi_s} (T_c - t_c) \quad [\text{Eq. (5.31)}]$$

$$= 129 + \frac{169}{109 + 169} (280 - 129) = 221^\circ\text{F}$$

$$(11') \text{ At } t_w = 221^\circ\text{F, } \mu_w = 0.56 \times 2.42$$

$$= 1.36 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$\phi_s = (\mu/\mu_w)^{0.14} = (0.97/1.36)^{0.14}$$

$$= 0.96 \quad [\text{Fig. 24 insert}]$$

$$(12') \text{ Corrected coefficient, } h_o = \frac{h_o}{\phi_s} \phi_s \quad [\text{Eq. (6.36)}]$$

$$= 169 \times 0.96 = 162 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Cold fluid: tube side, crude oil

$$(4) \text{ Flow area, } a_t' = 0.515 \text{ in.}^2 \quad [\text{Table 10}]$$

$$a_t = N_t a_t' / 144n \quad [\text{Eq. (7.48)}]$$

$$= 158 \times 0.515/144 \times 4 = 0.141 \text{ ft}^2$$

$$(5) \text{ Mass vel, } G_t = w/a_t$$

$$= 149,000/0.141 = 1,060,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6) Re_t = D G_t / \mu$$

$$\text{At } t_c = 129^\circ\text{F, } \mu = 3.6 \times 2.42 = 8.7 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$D = 0.81/12 = 0.0675 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = 0.0675 \times 1,060,000/8.7 = 8,220$$

$$(7) L/D = 16/0.0675 = 237$$

$$j_H = 31 \quad [\text{Fig. 24}]$$

$$(8) \text{ At } t_c = 129^\circ\text{F,}$$

$$c = 0.49 \text{ Btu/(lb)(}^\circ\text{F)} \quad [\text{Fig. 4}]$$

$$k = 0.077 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Fig. 1}]$$

$$(c\mu/k)^{1/3} = (0.49 \times 8.7/0.077)^{1/3} = 3.81$$

$$(9) h_i = j_H \left(\frac{k}{D} \right) \left(\frac{c\mu}{k} \right)^{1/3} \phi_i \quad [\text{Eq. (6.15a)}]$$

$$\frac{h_i}{\phi_i} = 31 \times \frac{0.077}{0.0675} \times 3.81 = 135$$

$$(10) \frac{h_{io}}{\phi_i} = \frac{h_i}{\phi_i} \times \frac{\text{ID}}{\text{OD}} = 135 \times \frac{0.81}{1.0} = 109 \quad [\text{Eq. (6.5)}]$$

$$(11) \text{ At } t_w = 221^\circ\text{F, } \mu_w = 1.5 \times 2.42$$

$$= 3.63 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$\phi_i = (\mu/\mu_w)^{0.14} = (8.7/3.63)^{0.14}$$

$$= 1.11 \quad [\text{Fig. 24 insert}]$$

$$(12) \text{ Corrected coefficient, } h_{io} = \frac{h_{io}}{\phi_i} \phi_i \quad [\text{Eq. (6.37)}]$$

$$= 109 \times 1.11 = 121 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(13') Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{121 \times 162}{121 + 162} = 69.3 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Design overall coefficient U_D :

$$\begin{aligned} a'' &= 0.2618 \text{ ft}^2/\text{lin ft} && \text{(Table 10)} \\ \text{Total surface, } A &= 158 \times 16'0'' \times 0.2618 = 662 \text{ ft}^2 \\ U_D &= \frac{Q}{A\Delta t} = \frac{5,100,000}{662 \times 138} = 55.8 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \end{aligned}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{69.3 - 55.8}{69.3 \times 55.8} = 0.00348 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

162	h outside	121
U_C	69.3	
U_D	55.8	
R_d Calculated	0.00348	
R_d Required	0.00300	

Pressure Drop

(1') For $Re_s = 25,300$,

$$f = 0.00175 \text{ ft}^2/\text{in}^2 \quad [\text{Fig. 29}]$$

$$s = 0.73 \quad [\text{Fig. 6}]$$

$$D_s = 21.25/12 = 1.77 \text{ ft}$$

$$(2') \text{ No. of crosses, } N + 1 = 12L/B \quad [\text{Eq. (7.43)}]$$

$$= 12 \times 16/5 = 39$$

$$(3') \Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad [\text{Eq. (7.44)}]$$

$$= \frac{0.00175 \times 297,000^2 \times 1.77 \times 39}{5.22 \times 10^{10} \times 0.0825 \times 0.73 \times 0.96} = 3.5 \text{ psi}$$

$$\text{Allowable } \Delta P_s = 10.0 \text{ psi}$$

(1) For $Re_t = 8220$,

$$f = 0.000285 \text{ ft}^2/\text{in}^2 \quad [\text{Fig. 26}]$$

$$s = 0.83 \quad [\text{Fig. 6}]$$

$$(2) \Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.000285 \times 1,060,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0675 \times 0.83 \times 1.11} = 6.3 \text{ psi}$$

$$(3) G_t = 1,060,000, \frac{V^2}{2g'} = 0.15 \quad [\text{Fig. 27}]$$

$$\begin{aligned} \Delta P_r &= \frac{4n}{s} \frac{V^2}{2g'} && [\text{Eq. (7.46)}] \\ &= \frac{4 \times 4}{0.83} \times 0.15 = 2.9 \text{ psi} \end{aligned}$$

$$(4) \Delta P_T = \Delta P_t + \Delta P_r = 6.3 + 2.9 = 9.2 \text{ psi} \quad [\text{Eq. (7.47)}]$$

$$\text{Allowable } \Delta P_T = 10.0 \text{ psi}$$

It is seen that a dirt factor of 0.00348 will be obtained although only 0.003 will be required to provide reasonable maintenance period. The pressure drops have not been exceeded and the exchanger will be satisfactory for the service.

Exchangers Using Water. Cooling operations using water in tubular equipment are very common. Despite its abundance, the heat-transfer characteristics of water separate it from all other fluids. It is corrosive to steel, particularly when the tube-wall temperature is high and dissolved air is present, and many industrial plants use nonferrous tubes exclusively for heat-transfer services involving water. The commonest nonferrous tubes are admiralty, red brass, and copper, although in certain localities there is a preference for Muntz metal, aluminum bronze, and aluminum. Since shells are usually fabricated of steel, water is best handled in the tubes. When water flows in the tubes, there is no serious problem of corrosion of the channel or floating-head cover, since these parts are often made of cast iron or cast steel. Castings are relatively passive to water, and large corrosion allowances above structural requirements can be provided inexpensively by making the castings heavier. Tube sheets may be made of heavy steel plate with a corrosion allowance of about $\frac{1}{8}$ in. above the required structural thickness or fabricated of naval brass or aluminum without a corrosion allowance.

When water travels slowly through a tube, dirt and slime resulting from microorganic action adhere to the tubes which would be carried away if there were greater turbulence. As a standard practice, the use of cooling water at velocities less than 3 fps should be avoided, although in certain localities minimum velocities as high as 4 fps are required for continued operation. Still another factor of considerable importance is the deposition of mineral scale. When water of average mineral and air content is brought to a temperature in excess of 120°F, it is found that tube action becomes excessive, and for this reason an outlet water temperature above 120°F should be avoided.

Cooling water is rarely abundant or without cost. One of the serious problems facing the chemical and power industries today results from the gradual deficiency of surface and subsurface water in areas of industrial concentration. This can be partially overcome through the use of cooling towers (Chap. 17), which reuse the cooling water and reduce the requirement to only 2 per cent of the amount of water required in once-through use. River water may provide part of the solution to a deficiency of ground water, but it is costly and presupposes the proximity of a river. River water must usually be strained by moving screens and pumped considerable distances, and in some localities the water from rivers servicing congested industrial areas requires cooling in cooling towers before it can be used.

Many sizable municipalities have legislated against the use of public water supplies for large cooling purposes other than for make-up in cooling towers or spray-pond systems. Where available, municipal water

may average about 1 cent per 1000 gal., although it has the advantage of being generally available at from 30 to 60 psi pressure which is adequate for most process needs including the pressure drops in heat exchangers. Where a cooling tower is used, the cost of the water is determined by the cost of fresh water, pumping power, fan power, and write-off on the original investment.

The shell-side heat-transfer curve (Fig. 28) correlates very well for the flow of water across tube bundles. The high thermal conductivity of water results in relatively high film coefficients compared with organic fluids. The use of the tube-side curve (Fig. 24), however, gives coefficients which are generally high. In its place the data of Eagle and Ferguson¹ for water alone are given in Fig. 25 and are recommended whenever water flows in tubes. Since this graph deals only with water, it has been possible to plot film coefficients vs. velocity in feet per second with temperature as the parameter. The data have been plotted with the $\frac{3}{4}$ in., 16 BWG tube as the base, and the correction factor obtained from the insert in Fig. 25 should be applied when any other inside diameter is used.

In a water-to-water exchanger with individual film coefficients ranging from 500 to 1500 for both the shell and tube, the selection of the required dirt factor merits serious judgment. As an example, if film coefficients of 1000 are obtained on the shell and tube sides, the combined resistance is 0.002, or $U_c = 500$. If a fouling factor of 0.004 is required, the fouling factor becomes the controlling resistance. When the fouling factor is 0.004, U_D must be less than $1/0.004$ or 250. Whenever high coefficients exist on both sides of the exchanger, the use of an unnecessarily large fouling factor should be avoided.

The following heat-recovery problem occurs in powerhouses. Although it involves a moderate-size exchanger, the heat recovery is equivalent to nearly 1500 lb/hr of steam, which represents a sizable economy in the course of a year.

Example 7.4. Calculation of a Distilled-water-Raw-water Exchanger. 175,000 lb/hr of distilled water enters an exchanger at 93°F and leaves at 85°F. The heat will be transferred to 280,000 lb/hr of raw water coming from supply at 75°F and leaving the exchanger at 80°F. A 10 psi pressure drop may be expended on both streams while providing a fouling factor of 0.0005 for distilled water and 0.0015 for raw water when the tube velocity exceeds 6 fps.

Available for this service is a $15\frac{1}{4}$ in. ID exchanger having 160 $\frac{3}{4}$ in. OD, 18 BWG tubes 16'0" long and laid out on $1\frac{5}{16}$ -in. triangular pitch. The bundle is arranged for two passes, and baffles are spaced 12 in. apart.

Will the exchanger be suitable?

¹ Eagle, A., and R. M. Ferguson, *Pro. Roy. Soc.*, A127, 540-566 (1930).

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 15¼ in.	Number and length = 160, 16'0"
Baffle space = 12 in.	OD, BWG, pitch = ¾ in., 18 BWG, 1½ in. tri.
Passes = 1	Passes = 2

(1) Heat balance:

Distilled water, $Q = 175,000 \times 1(93 - 85) = 1,400,000$ Btu/hr

Raw water, $Q = 280,000 \times 1(80 - 75) = 1,400,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
93	Higher Temp	80	13
85	Lower Temp	75	10
8	Differences	5	3

$$\text{LMTD} = 11.4^\circ\text{F} \quad (5.14)$$

$$R = \frac{8}{5} = 1.6 \quad S = \frac{5}{93 - 75} = 0.278$$

$$F_T = 0.945$$

(Fig. 18)

$$\Delta t = 0.945 \times 11.4 = 10.75^\circ\text{F} \quad (7.42)$$

(3) T_a and t_a :

The average temperatures T_a and t_a of 89 and 77.5°F will be satisfactory for the short ranges and ϕ_s and ϕ_t taken as 1.0. Try hot fluid in shell as a trial, since it is the smaller of the two.

<i>Hot fluid: shell side, distilled water</i>	<i>Cold fluid: tube side, raw water</i>
(4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)]	(4) $a'_t = 0.334 \text{ in.}^2$ [Table 10]
$= 15.25 \times 0.1875 \times 12/$	$a_t = N a'_t / 144n$ [Eq. (7.48)]
$144 \times 0.9375 = 0.254 \text{ ft}^2$	$= 160 \times 0.334 / 144 \times 2 = 0.186 \text{ ft}^2$
(5') $G_s = W/a_s$ [Eq. (7.2)]	(5) $G_t = w/a_t$
$= 175,000/0.254$	$= 280,000/0.186$
$= 690,000 \text{ lb}/(\text{hr})(\text{ft}^2)$	$= 1,505,000 \text{ lb}/(\text{hr})(\text{ft}^2)$
	Vel, $V = G_t/3600\rho$
	$= 1,505,000/3600 \times 62.5 = 6.70 \text{ fps}$
(6') At $T_a = 89^\circ\text{F}$, $\mu = 0.81 \times 2.42$	(6) At $t_a = 77.5^\circ\text{F}$, $\mu = 0.92 \times 2.42$
$= 1.96 \text{ lb}/(\text{ft})(\text{hr})$	$= 2.23 \text{ lb}/(\text{ft})(\text{hr})$
[Fig. 14]	[Fig. 14]
$D_s = 0.55/12 = 0.0458 \text{ ft}$ [Fig. 28]	$D = 0.65/12 = 0.054 \text{ ft}$ (Re_t is for pres-
$Re_s = D_s G_s / \mu$ [Eq. (7.3)]	sure drop only) [Table 10]
$= 0.0458 \times 690,000 / 1.96 = 16,200$	$Re_t = D G_t / \mu$
(7') $j_H = 73$ [Fig. 28]	$= 0.054 \times 1,505,000 / 2.23 = 36,400$
(8') At $T_a = 89^\circ\text{F}$, $c = 1.0 \text{ Btu}/(\text{lb})(^\circ\text{F})$	
$k = 0.36 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ [Table 4]	
$(c\mu/k)^{1/4} = (1.0 \times 1.96/0.36)^{1/4} = 1.76$	

<i>Hot fluid: shell side, distilled water</i> (9') $h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \times 1$ [Eq. (6.15b)] $= 73 \times 0.36 \times 1.76/0.0458 = 1010$ (10') (11') (12') The small difference in the average temperatures eliminates the need for a tube-wall correction, and $\phi_s = 1$.	<i>Cold fluid: tube side, raw water</i> (9) $h_i = 1350 \times 0.99 = 1335$ [Fig. 25] $h_{io} = h_i \times ID/OD = 1335 \times 0.65/0.75 = 1155$ [Eq. (6.5)]
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(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1155 \times 1010}{1155 + 1010} = 537 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

When both film coefficients are high the thermal resistance of the tube metal is not necessarily insignificant as assumed in the derivation of Eq. (6.38). For a steel 18 BWG tube $R_m = 0.00017$ and for copper $R_m = 0.000017$.

(14) Design overall coefficient U_D :

$$\begin{aligned} \text{External surface/ft, } a'' &= 0.1963 \text{ ft}^2/\text{ft} \\ A &= 160 \times 16'0'' \times 0.1963 = 502 \text{ ft}^2 \\ U_D &= \frac{Q}{A\Delta t} = \frac{1,400,000}{502 \times 10.75} = 259 \end{aligned} \quad (5.3)$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C \times U_D} = \frac{537 - 259}{537 \times 259} = 0.0020 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

1010	h outside	1155
U_C	537	
U_D	259	
R_d Calculated 0.0020		
R_d Required 0.0020		

Pressure Drop

(1') For $Re_s = 16,200$, $f = 0.0019 \text{ ft}^2/\text{in.}^2$ [Fig. 29] (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 12 \times 16/12 = 16$ $D_s = 15.25/12 = 1.27 \text{ ft}$	(1) For $Re_t = 36,400$, $f = 0.00019 \text{ ft}^2/\text{in.}^2$ [Fig. 26] (2) $\Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)] $= \frac{0.00019 \times 1,505,000^2 \times 16 \times 2}{5.22 \times 10^{10} \times 0.054 \times 1.0 \times 1.0} = 4.9 \text{ psi}$
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Pressure Drop

$(3') \Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad [\text{Eq. (7.44)}]$ $= \frac{0.0019 \times 690,000^2 \times 1.27 \times 16}{5.22 \times 10^{10} \times 0.0458 \times 1.0 \times 1.0}$ $= 7.7 \text{ psi}$ Allowable $\Delta P_s = 10.0 \text{ psi}$	$(3) \Delta P_r: G_t = 1,505,000, V^2/2g' = 0.33 \quad [\text{Fig. 27}]$ $\Delta P_r = (4n/s)(V^2/2g') \quad [\text{Eq. (7.46)}]$ $= \frac{4 \times 2}{1} \times 0.33 = 2.3 \text{ psi}$ $(4) \Delta P_T = P_t + P_r \quad [\text{Eq. (7.47)}]$ $= 4.9 + 2.6 = 7.5 \text{ psi}$ Allowable $\Delta P_T = 10.0 \text{ psi}$
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It is seen that the overall coefficient for this problem is five times that of the oil-to-oil exchange of Example 7.3, the principal difference being due to the excellent thermal properties of water. The exchanger is satisfactory for the service.

Optimum Outlet-water Temperature. In using water as the cooling medium for a given duty it is possible to circulate a large quantity with a small temperature range or a small quantity with a large temperature range. The temperature range of the water naturally affects the LMTD. If a large quantity is used, t_2 will be farther from T_1 and less surface is required as a result of the larger LMTD. Although this will reduce the original investment and fixed charges, since depreciation and maintenance will also be smaller, the operating cost will be increased owing to the greater quantity of water. It is apparent that there must be an optimum between the two conditions: much water and small surface or little water and large surface.

In the following it is assumed that the line pressure on the water is sufficient to overcome the pressure drop in the exchanger and that the cost of the water is related only to the amount used. It is also assumed that the cooler operates in true counterflow so that $\Delta t = \text{LMTD}$. If the approach is small or there is a temperature cross, the derivation below requires an estimate of F_T by which the LMTD is multiplied.

The total annual cost of the exchanger to the plant will be the sum of the annual cost of water and the fixed charges, which include maintenance and depreciation.

If C_T is the total annual cost,

$$C_T = (\text{water cost/lb})(\text{lb/hr})(\text{annual hr}) + (\text{annual fixed charges/ft}^2)(\text{ft}^2)$$

$$Q = wc(t_2 - t_1) = UA(\text{LMTD}) \quad (7.49)$$

Substituting the heat-balance terms in Eq. (7.49), where $w = Q/[c(t_2 - t_1)]$ and the surface $A = Q/U(\text{LMTD})$

$$C_T = \frac{Q\theta C_w}{c(t_2 - t_1)} + \frac{C_F Q}{U(\text{LMTD})}$$

where θ = annual operating hours

C_w = water cost/lb

C_F = annual fixed charges/ft²

Assuming U is constant

$$\text{LMTD} = \frac{\Delta t_2 - \Delta t_1}{\ln \Delta t_2 / \Delta t_1}$$

Keeping all factors constant except the water-outlet temperature and consequently Δt_2 ,

$$C_T = \frac{Q\theta C_w}{c(t_2 - t_1)} + \frac{C_F Q}{U \left[\frac{T_1 - t_2 - \Delta t_1}{\ln (T_1 - t_2) / \Delta t_1} \right]} \quad (7.50)$$

The optimum condition will occur when the total annual cost is a minimum, thus when $dC_T/dt_2 = 0$.

Differentiating and equating the respective parts,

$$\frac{U\theta C_w}{C_F c} \left(\frac{T_1 - t_2 - \Delta t_1}{t_2 - t_1} \right)^2 = \ln \frac{T_1 - t_2}{\Delta t_1} - \left[1 - \frac{1}{(T_1 - t_2) / \Delta t_1} \right] \quad (7.51)$$

Equation (7.51) has been plotted by Colburn and is reproduced in Fig. 7.24.

Example 7.5. Calculation of the Optimum Outlet-water Temperature. A viscous fluid is cooled from 175 to 150°F with water available at 85°F. What is the optimum-outlet water temperature?

$$\begin{aligned} 175 - x &= \Delta t_2 \\ 150 - 85 &= \Delta t_1 = 65 \end{aligned}$$

It will first be necessary to assume a value of U . Since the material is viscous, assume $U = 15$. To evaluate the group $U\theta C_w/C_F c$:

$\theta = 8000$ operating hours annually

C_w computed at \$0.01/1000 gal. = 0.01/8300, dollars/lb

For annual charges assume 20 per cent repair and maintenance and 10 per cent depreciation. At a unit cost of \$4 per square foot the annual fixed charge is

$$\$4 \times 0.30 = \$1.20$$

The specific heat of water is taken as 1.0.

$$\begin{aligned} \frac{U\theta C_w}{C_F c} &= \frac{15 \times 8000}{1.20 \times 1.0} \left(\frac{0.01}{8300} \right) = 0.1205 \\ \frac{T_1 - T_2}{\Delta t_1} &= \frac{175 - 150}{150 - 85} = 0.39 \end{aligned}$$

From Figure 7.24,

$$\begin{aligned} \frac{\Delta t_2}{\Delta t_1} &= 0.96 \\ \Delta t_2 &= T_1 - t_2 = 0.96 \times 65 = 62.3^\circ\text{F} \\ t_2 &= 175 - 62.3 = 112.7^\circ\text{F} \end{aligned}$$

When the value of U is high or there is a large hot-fluid range, the optimum outlet-water temperature may be considerably above the upper limit of 120°F. This is not completely correct, since the maintenance cost will probably rise considerably above 20 per cent of the initial cost when the temperature rises above 120°F. Usually information is not available on the increase in maintenance cost with increased water-outlet

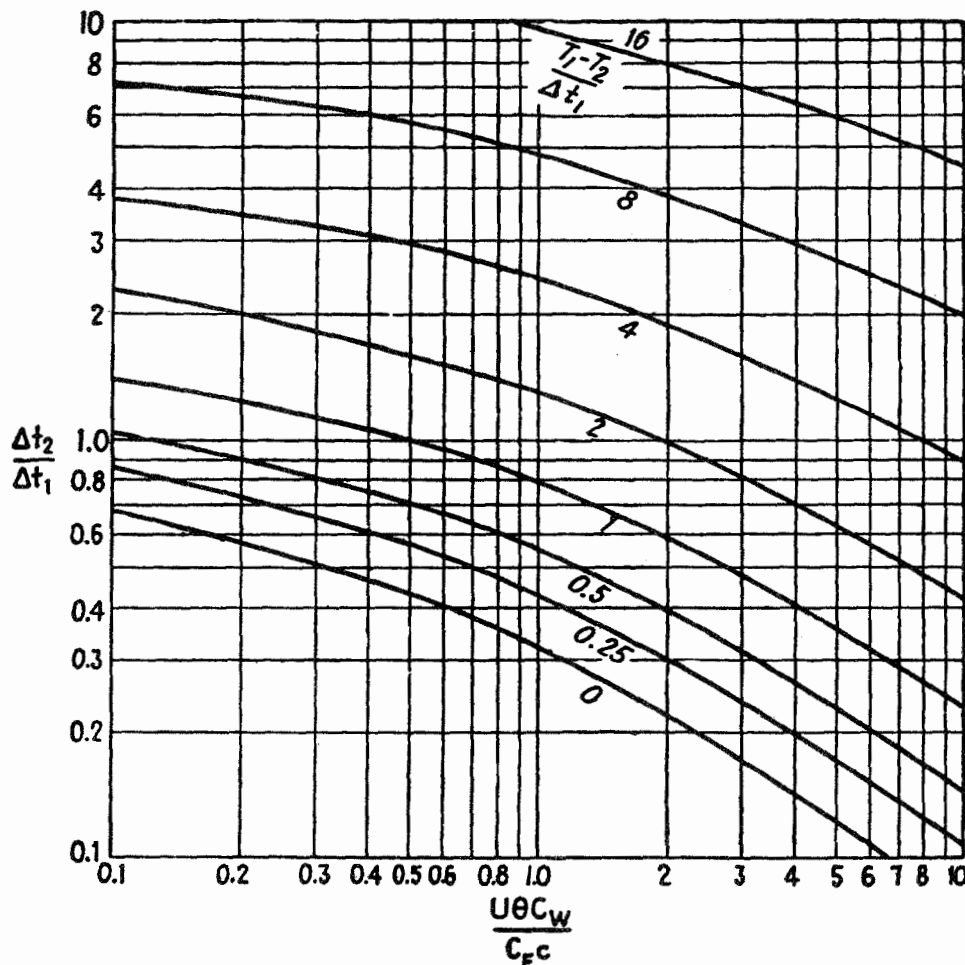


FIG. 7.24. Optimum outlet water temperature. (Perry, *Chemical Engineers' Handbook*, McGraw-Hill Book Company, Inc., New York, 1950.)

temperature, since such data entail not only destructive tests but records kept over a long period of time.

Solution Exchangers. One of the commonest classes of exchangers embraces the cooling or heating of solutions for which there is a paucity of physical data. This is understandable, since property vs. temperature plots are required not only for each combination of solute and solvent but for different concentrations as well. Some of the data available in the literature and other studies permit the formulation of rules for estimating the heat-transfer properties of solutions when the rules are used with considerable caution. They are given as follows:

Thermal conductivity:

Solutions of *organic liquids*: use the weighted conductivity.

Solutions of organic liquids and water: use 0.9 times the weighted conductivity.

Solutions of salts and water circulated through the shell: use 0.9 times the conductivity of water up to concentrations of 30 per cent.

Solutions of salts and water circulating through the tubes and not exceeding 30 per cent: use Fig. 24 with a conductivity of 0.8 that of water.

Colloidal dispersions: use 0.9 times the conductivity of the dispersion liquid.

Emulsions: use 0.9 times the conductivity of the liquid surrounding the droplets.

Specific heat:

Organic solutions: use the weighted specific heat.

Organic solutions in water: use the weighted specific heat.

Fusible salts in water: use the weighted specific heat where the specific heat of the salt is for the crystalline state.

Viscosity:

Organic liquids in organics: use the reciprocal of the sum of the terms, (weight fraction/viscosity) for each component.

Organic liquids in water: use the reciprocal of the sum of the terms, (weight fraction/viscosity) for each component.

Salts in water where the concentration does not exceed 30 per cent and where it is known that a sirup-type of solution does not result: use a viscosity twice that of water. A solution of sodium hydroxide in water under even very low concentrations should be considered sirupy and can not be estimated.

Wherever laboratory tests are available or data can be obtained, they will be preferable to any of the foregoing rules. The following demonstrates the solution of a problem involving an aqueous solution:

Example 7.6. Calculation of a Phosphate Solution Cooler. 20,160 lb/hr of a 30% K_3PO_4 solution, specific gravity at 120°F = 1.30, is to be cooled from 150 to 90°F using well water from 68 to 90°F. Pressure drops of 10 psi are allowable on both streams, and a total dirt factor of 0.002 is required.

Available for this service is a 10.02 in. ID 1-2 exchanger having 52 $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long laid out on 1-in. square pitch. The bundle is arranged for two passes, and the baffles are spaced 2 in. apart.

Will the exchanger be suitable?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 10.02 in.	Number and length = 52, 16'0"
Baffle space = 2 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1 in. square
Passes = 1	Passes = 2

(1) Heat balance:

Specific heat of phosphate solution = $0.3 \times 0.19 + 0.7 \times 1 = 0.757$ Btu/(lb)(°F)

30% K_2PO_4 solution, $Q = 20,160 \times 0.757(150 - 90) = 915,000$ Btu/hr

Water, $Q = 41,600 \times 1.0(90 - 68) = 915,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
150	Higher Temp	90	60
90	Lower Temp	68	22
60	Differences	22	38

LMTD = 37.9°F

(5.14)

$$R = \frac{60}{22} = 2.73 \quad S = \frac{22}{150 - 68} = 0.268$$

$F_T = 0.81$

(Fig. 18)

$\Delta t = 0.81 \times 37.9 = 30.7^\circ\text{F}$

(7.42)

(3) T_c and t_c : The average temperatures T_a and t_a of 120 and 79°F will be satisfactory.

Hot fluid: shell side, phosphate solution

$$(4') \quad a_s = ID \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= 10.02 \times 0.25 \times 2/144 \times 1$$

$$= 0.0347 \text{ ft}^2$$

$$(5') \quad G_s = w/a_s$$

$$= 20,160/0.0347$$

$$= 578,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6') \quad \text{At } T_a = 120^\circ\text{F},$$

$$\mu = 2\mu_{\text{water}} = 1.20 \times 2.42$$

$$= 2.90 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Fig. 14}]$$

$$D_s = 0.95/12 = 0.079 \text{ ft} \quad [\text{Fig. 28}]$$

$$Re_s = D_s G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.079 \times 578,000 / 2.90 = 15,750$$

$$(7') \quad j_H = 71 \quad [\text{Fig. 28}]$$

$$(8') \quad \text{At } T_a = 120^\circ\text{F}, k = 0.9k_{\text{water}}$$

$$= 0.9 \times 0.37$$

$$= 0.33 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

$$(c\mu/k)^{1/2} = (0.757 \times 2.90/0.33)^{1/2} = 1.88$$

$$(9') \quad h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k} \right)^{1/2} \times 1 \quad [\text{Eq. (6.15b)}]$$

$$= 71 \times 0.33 \times 1.88 / 0.079$$

$$= 558 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$(10') \quad (11') \quad (12') \quad \phi_s \text{ and } \phi_t = 1$$

Cold fluid: tube side, water

$$(4) \quad a_t' = 0.302 \text{ in.}^2 \quad [\text{Table 10}]$$

$$a_t = N_t a_t' / 144n$$

$$= 52 \times 0.302 / 144 \times 2 = 0.0545 \text{ ft}^2$$

$$(5) \quad G_t = w/a_t$$

$$= 41,600 / 0.0545$$

$$= 762,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$V = G_t / 3600\rho = 762,000 / 3600 \times 62.5$$

$$= 3.40 \text{ fps}$$

$$(6) \quad \text{At } t_a = 79^\circ\text{F}, \mu = 0.91 \times 2.42$$

$$= 2.20 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Table 14}]$$

$$D = 0.62/12 = 0.0517 \text{ ft} \quad [\text{Table 10}]$$

(Re_t is for pressure drop only)

$$Re_t = D G_t / \mu$$

$$= 0.0517 \times 762,000 / 2.20 = 17,900$$

$$(9) \quad h_i = 800 \times 1.0 = 800 \quad [\text{Fig. 25}]$$

$$h_{io} = h_i \times ID/OD = 800 \times 0.62/0.75$$

$$= 662 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad [\text{Eq. (6.5)}]$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_i h_o}{h_i + h_o} = \frac{662 \times 558}{662 + 558} = 303 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Design overall coefficient U_D :External surface/ft, $a'' = 0.1963 \text{ ft}$ (Table 10)

$$A = 52 \times 16'0'' \times 0.1963 = 163 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{915,000}{163 \times 30.7} = 183 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{303 - 183}{303 \times 183} = 0.00216 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

558	h outside	662
U_C	303	
U_D	183	
R_d Calculated	0.00216	
R_d Required	0.0020	

Pressure Drop

(1') For $Re_s = 15,750$, $f = 0.0019 \text{ ft}^2/\text{in.}^2$ [Fig. 29] (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 12 \times 16\frac{1}{2}$ $= 96$ $D_s = 10.02/12 = 0.833 \text{ ft}$ (3') $\Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ [Eq. (7.44)] $= \frac{0.0019 \times 578,000^2 \times 0.833 \times 96}{5.22 \times 10^{10} \times 0.079 \times 1.30 \times 1.0}$ $= 9.5 \text{ psi}$ Allowable $\Delta P_s = 10.0 \text{ psi}$	(1) For $Re_t = 17,900$, $f = 0.00023 \text{ ft}^2/\text{in.}^2$ [Fig. 26] (2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)] $= \frac{0.00023 \times 762,000^2 \times 16 \times 2}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0}$ $= 1.6 \text{ psi}$ (3) $G_t = 762,000$, $\frac{V^2}{2g'} = 0.08$ [Fig. 27] $\Delta P_r = \frac{4n}{s} \frac{V^2}{2g'}$ [Eq. (7.46)] $= \frac{4 \times 2}{1} \times 0.08 = 0.7 \text{ psi}$ $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)] $= 1.6 + 0.7 = 2.3 \text{ psi}$ Allowable $\Delta P_T = 10.0 \text{ psi}$
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The exchanger is satisfactory for the service.

Steam as a Heating Medium. Thus far none of the heat-transfer services studied has employed steam although it is by far the commonest

heating medium. Steam as a heating medium introduces several difficulties: (1) Hot steam condensate is fairly corrosive, and care must be exercised to prevent condensate from accumulating within an exchanger where continuous contact with metal will cause damage. (2) The condensate line must be connected with discretion. Suppose exhaust steam at 5 psig and 228°F is used to heat a cold fluid entering at a temperature of 100°F. The tube wall will be at a temperature between the two but nearer that of the steam, say 180°F, which corresponds to a saturation pressure of only 7.5 psia for the condensate at the tube wall. Although the steam entered at 5 psig, the pressure on the steam side may drop locally to a pressure below that of the atmosphere, so that the condensate will *not* run out of the heater. Instead it will remain and build up in the exchanger until it blocks off all the surface available for heat transfer. Without surface, the steam will not continue to condense and will retain its inlet pressure long enough to blow out some or all of the accumulated condensate so as to reexpose surface, depending upon the design. The heating operation will become cyclical and to overcome this difficulty and attain uniform flow, it may be necessary to employ a trap or suction for which piping arrangements will be discussed in Chap. 21.

The heat-transfer coefficients associated with the condensation of steam are very high compared with any which have been studied so far. It is customary to adopt a conventional and conservative value for the film coefficient, since it is never the controlling film, rather than obtain one by calculation. In this book in all heating services employing relatively air-free steam a value of 1500 Btu/(hr)(ft²)(°F) will be used for the condensation of steam without regard to its location. Thus $h_i = h_o = h_{io} = 1500$.

It is advantageous in heating to connect the steam to the tubes of the heater rather than the shell. In this way, since the condensate may be corrosive, the action can be confined to the tube side alone, whereas if the steam is introduced into the shell, both may be damaged. When steam flows through the tubes of a 1-2 exchanger, there is no need for more than two tube passes. Since steam is an isothermally condensing fluid, the true temperature difference Δt and the LMTD are identical.

When using superheated steam as a heating medium, except in desuperheaters, it is customary to disregard the temperature range of desuperheating and consider all the heat to be delivered at the saturation temperature corresponding to the operating pressure. A more intensive analysis of the condensation of steam will be undertaken in the chapters dealing with condensation.

Pressure Drop for Steam. When steam is employed in two passes on the tube side, the allowable pressure drop should be very small, less than 1.0 psi, particularly if there is a gravity return of condensate to the boiler. In a gravity-return system, condensate flows back to the boiler because of the difference in static head between a vertical column of steam and a vertical column of condensate. The pressure drop including entrance and end losses through an exchanger can be calculated by taking one-half of the pressure drop for steam as calculated in the usual manner by Eq. (7.45) for the inlet vapor conditions. The mass velocity is calculated from the inlet steam rate and the flow area of the first pass (which need not be equal to that of the second pass). The Reynolds number is based on the mass velocity and the viscosity of steam as found in Fig. 15. The specific gravity used with Eq. (7.45) is the density of the steam obtained from Table 7 for the inlet pressure divided by the density of water taken as 62.5 lb/ft³.

Quite apparently this calculation is an approximation. It is conservative inasmuch as the pressure drop per foot of length decreases successively with the square of the mass velocity while the approximation assumes a value more nearly the mean of the inlet and outlet.

The Optimum Use of Exhaust and Process Steam. Some plants obtain power from noncondensing turbines or engines. In such places there may be an abundance of exhaust steam at low pressures from 5 to 25 psig which is considered a by-product of the power cycles in the plant. While there are arbitrary aspects to the method of estimating the cost of exhaust steam, it will be anywhere from one-quarter to one-eighth of the cost of process or live steam. Although it possesses a high latent heat, exhaust steam is of limited process value, since the saturation temperature is usually about 215 to 230°F. If a liquid is to be heated to 250 or 275°F, it is necessary to use process steam at 100 to 200 psi developed at the powerhouse specially for process purposes.

When a fluid is to be heated to a temperature close to or above that of exhaust steam, all the heating can be done in a single shell using only process steam. As an alternative the heat load can be divided into two shells, one utilizing as much exhaust steam as possible and the other using as little process steam as possible. This leads to an optimum: If the outlet temperature of the cold fluid in the first exchanger is made to approach the exhaust steam temperature too closely, a small Δt and large first heater will result. On the other hand, if the approach is not close, the operating cost of the higher process steam requirement in the second heater increases so that the initial cost of two shells may not be justified.

In the following analysis it is assumed that the pressure drop, pumping

cost, and overall coefficient are identical in a single and double heater arrangement. It is assumed also that the fixed charges per square foot of surface is constant, although this too is not strictly true. The cost equation is taken as the sum of steam and fixed charges and because steam condenses isothermally, $\Delta t = \text{LMTD}$.

$$C_T = wc(t - t_1)\theta C_E + A_1 C_F + wc(t_2 - t)\theta C_P + A_2 C_F \quad (7.52)$$

where C_T = total annual cost, dollars

C_F = annual fixed charges, dollars/ft²

C_E = cost of exhaust steam, dollars/Btu

C_P = cost of process steam, dollars/Btu

T_E = temperature of exhaust steam, °F

T_P = temperature of process steam, °F

t = intermediate temperature between shells

θ = total annual operating hours

$$A_1 = \frac{Q_1}{U \Delta t_1} = \frac{wc}{U} \ln \frac{T_E - t_1}{T_E - t} \quad \text{and} \quad A_2 = \frac{Q_2}{U \Delta t_2} = \frac{wc}{U} \ln \frac{T_P - t}{T_P - t_2}$$

Substituting, differentiating Eq. (7.52) with respect to t , and setting equal to zero

$$(T_P - t)(T_E - t) = \frac{C_F(T_P - T_E)}{(C_P - C_E)U\theta} \quad (7.53)$$

Example 7.7. The Optimum Use of Exhaust and Process Steam. Exhaust steam at 5 psi ($\approx 228^\circ\text{F}$) and process steam at 85 psi ($\approx 328^\circ\text{F}$) are available to heat a liquid from 150 to 250°F . Exhaust steam is available at 5 cents per per 1000 lb, and process steam at 30 cents per 1000 lb. From experience an overall rate of 50 Btu/(hr)(ft²)(°F) may be expected. The assumption may be checked later. Use annual fixed charges of \$1.20 per square foot, 8000 annual hours, latent heats of 960.1 Btu/lb for exhaust, and 888.8 Btu/lb for process steam.

Solution:

$$(328 - t)(228 - t) = \frac{1.20(328 - 228)}{(0.30/1000 \times 888.8 - 0.05/1000 \times 960)50 \times 8000} \quad (7.53)$$

$$t = 218^\circ\text{F}$$

1-2 Exchangers without Baffles. Not all 1-2 exchangers have 25 per cent cut segmental baffles. When it is desired that a fluid pass through the shell with an extremely small pressure drop, it is possible to depart from the use of segmental baffles and use only support plates. These will usually be half-circle, 50 per cent cut plates which provide rigidity and prevent the tubes from sagging. Successive support plates overlap at the shell diameter so that the entire bundle can be supported by two

half circles which support one or two rows of tubes in common. These may be spaced farther apart than the outside diameter of the shell, but when they are employed, the shell fluid is considered to flow along the axis instead of across the tubes. When the shell fluid flows *along* the tubes or the baffles are cut more than 25 per cent, Fig. 28 no longer applies. The flow is then analogous to the annulus of a double pipe exchanger and can be treated in a similar manner, using an equivalent diameter based on the distribution of flow area and the wetted perimeter for the entire shell. The calculation of the shell-side pressure drop will also be similar to that for an annulus.

Example 7.8. Calculation of a Sugar-solution Heater without Baffles. 200,000 lb/hr of a 20 per cent sugar solution ($s = 1.08$) is to be heated from 100 to 122°F using steam at 5 psi pressure.

Available for this service is a 12 in. ID 1-2 exchanger without baffles having 76 $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long laid out on a 1-in. square pitch. The bundle is arranged for two passes.

Can the exchanger provide a 0.003 dirt factor without exceeding a 10.0 psi solution pressure drop?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 12 in.	Number and length = 76, 16'0"
Baffle space = half circles	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1 in. square
Passes = 1	Passes = 2

(1) Heat balance:

$$\text{Specific heat of 20 per cent sugar at } 111^{\circ}\text{F} = 0.2 \times 0.30 + 0.8 \times 1 = 0.86 \text{ Btu/(lb)}(^{\circ}\text{F})$$

$$\text{Sugar solution, } Q = 200,000 \times 0.86(122 - 100) = 3,790,000 \text{ Btu/hr}$$

$$\text{Steam } Q = 3950 \times 960.1 = 3,790,000 \text{ Btu/hr} \quad (\text{Table 7})$$

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
228	Higher Temp	122	106
228	Lower Temp	100	128
0	Differences	22	22

$$\text{When } R = 0, \Delta t = \text{LMTD} = 116.5^{\circ}\text{F}. \quad (5.14)$$

(3) T_c and t_c : The steam coefficient will be very great compared with that for the sugar solution, and the tube wall will be considerably nearer 228°F than the caloric temperature of the fluid. Obtain F_c from U_1 and U_2 . Failure to correct for wall effects, however, will keep the heater calculation on the safe side. Use 111°F as the average, t_a .

Hot fluid: tube side, steam

$$(4) a'_t = 0.302 \text{ in}^2 \quad [\text{Table 10}]$$

$$a_t = N_t a'_t / 144n \quad [\text{Eq. (7.48)}]$$

$$= 76 \times 0.302 / 144 \times 2 = 0.0797 \text{ ft}^2$$

$$(5) G_t \text{ (for pressure drop only)} = W/a_t$$

$$= 3950 / 0.0797 = 49,500 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6) \text{ At } T_a = 228^\circ\text{F,}$$

$$\mu_{\text{steam}} = 0.0128 \times 2.42$$

$$= 0.031 \text{ lb/(ft)(hr)} \quad [\text{Fig. 15}]$$

$$D = 0.62/12 = 0.0517 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = DG_t/\mu \quad [\text{Eq. (3.6)}]$$

$$= 0.0517 \times 49,500 / 0.031 = 82,500$$

Re_t is for pressure drop.

$$(9) \text{ Condensation of steam:}$$

$$h_{io} = 1500 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)}$$

$$(10) t_w^*:$$

$$t_w = t_c + \frac{h_{io}}{h_{io} + h_o} (T_a - t_a) \quad [\text{Eq. (5.31a)}]$$

$$= 111 + \frac{1500}{1500 + 278} (228 - 111)$$

$$= 210^\circ\text{F}$$

Cold fluid: shell side, sugar solution

$$(4') a_s = (\text{area of shell}) - (\text{area of tubes})$$

$$= \frac{1}{4} \pi 144 (\pi 12^2 / 4 - 76 \times \pi$$

$$\times 0.75^2 / 4)$$

$$= 0.55 \text{ ft}^2$$

$$(5') G_s = w/a_s \quad [\text{Eq. (7.2)}]$$

$$= 200,000 / 0.55$$

$$= 364,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$(6') \text{ At } t_a = 111^\circ\text{F, } \mu = 2\mu_{\text{water}}$$

$$= 1.30 \times 2.42 = 3.14 \text{ lb/(ft)(hr)}$$

[Fig. 14]

$$D_e = 4a_s/(\text{wetted perimeter}) \quad [\text{Eq. (6.3)}]$$

$$= 4 \times 0.55 / (76 \times \pi \times 0.75/12)$$

$$= 0.148 \text{ ft}$$

$$Re_s = D_e G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.148 \times 364,000 / 3.14 = 17,100$$

$$(7') \text{ From Fig. 24 (tube-side data)}$$

$$j_H = 61.5$$

$$(8') \text{ At } t_a = 111^\circ\text{F,}$$

$$k = 0.9 \times 0.37$$

$$= 0.333 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft)}$$

$$(c\mu/k)^{1/3} = (0.86 \times 3.14 / 0.333)^{1/3} = 2.0$$

$$(9') h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \phi_s \quad [\text{Eq. (6.15b)}]$$

$$\frac{h_o}{\phi_s} = 61.5 \times 0.333 \times 2.0 / 0.148 = 278$$

$$(11') \text{ At } t_w = 210^\circ\text{F, } \mu_w = 2\mu_{\text{water}}$$

$$= 0.51 \times 2.42 = 1.26 \text{ lb/(ft)(hr)}$$

[Fig. 14]

$$\phi_s = (\mu/\mu_w)^{0.14} = (3.14/1.26)^{0.14} = 1.12$$

$$(12') \text{ Corrected coefficient, } h_o = \frac{h_o \phi_s}{\phi_s}$$

[Eq. (6.36)]

$$= 278 \times 1.12 = 311 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)}$$

$$(13) \text{ Clean overall coefficient } U_C:$$

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{1500 \times 311}{1500 + 311} = 257 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)} \quad (6.38)$$

$$(14) \text{ Design overall coefficient } U_D:$$

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft}$$

$$A = 76 \times 16'0'' \times 0.1963 = 238 \text{ ft}^2 \quad (\text{Table 10})$$

$$U_D = \frac{Q}{A \Delta t} = \frac{3,790,000}{238 \times 116.5} = 137 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)}$$

$$(15) \text{ Dirt factor } R_d:$$

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{257 - 137}{257 \times 137} = 0.0034 \text{ (hr)(ft}^2\text{)}(^{\circ}\text{F)/Btu} \quad (6.13)$$

* Note h_{io} in the numerator.

Summary

1500	<i>h</i> outside	311
U_c	257	
U_d	137	
R_d Calculated 0.0034		
R_d Required 0.003		

Pressure Drop

(1) Specific vol of steam from Table 7:

$$v = 20.0 \text{ ft}^3/\text{lb}$$

$$s = \frac{1/20.0}{62.5} = 0.00080$$

$$Re_i = 82,500, f = 0.000155 \text{ ft}^2/\text{in.}^2$$

[Fig. 26]

$$\Delta P_i = \frac{1}{2} \times \frac{f G_i^2 L n}{5.22 \times 10^{10} D_s \phi_i} \quad [\text{Eq. (7.45)}]$$

$$= \frac{1}{2} \times \frac{0.000155 \times 49,500^2 \times 16 \times 2}{5.22 \times 10^{10} \times 0.0517}$$

$$\times 0.0008 \times 1.0$$

$$= 2.8 \text{ psi}$$

This is a relatively high pressure drop for steam with a gravity condensate return.

The exchanger is satisfactory.

(1') $D_e' = 4 \times \text{flow area}/\text{frictional}$

wetted perimeter [Eq. (6.4)]

$$= 4 \times 0.55 / (76 \times 3.14 \times 0.75 / 12 + 3.14 \times 1 \frac{1}{2} / 12) = 0.122 \text{ ft}$$

$$Re_s' = D_e' G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.122 \times 364,000 / 3.14 = 14,100$$

 f (from Fig. 26 for tube side)

$$= 0.00025 \text{ ft}^2/\text{in.}^2$$

$$(2') \Delta P_s = \frac{f G_s^2 L n}{5.22 \times 10^{10} D_e' s \phi_s} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00025 \times 364,000^2 \times 16 \times 1}{5.22 \times 10^{10} \times 0.122 \times 1.08}$$

$$\times 1.12$$

$$= 0.07 \text{ psi}$$

Heat Recovery in a 1-2 Exchanger. When an exchanger is clean, the hot-fluid outlet temperature is lower than the process outlet temperature and the cold-fluid outlet temperature is higher than the process outlet temperature. For counterflow it was possible to obtain the value of T_2 and t_2 for a clean exchanger from Eq. (5.18), starting with

$$wc(t_2 - t_1) = UA \times \text{LMTD}$$

For a 1-2 exchanger the outlet temperatures can be obtained starting with the expression $wc(t_2 - t_1) = UAF_T \times \text{LMTD}$, where the LMTD is defined in terms of parameters R and S by Eq. (7.39) and F_T is defined by Eq. (7.41).

Recognizing that F_T can be eliminated when UA/wc in Eq. (7.37) is plotted against S , Ten Broeck¹ developed the graph shown in Fig. 7.25. In an existing 1-2 exchanger A and wc are known. U can be computed

¹ Ten Broeck, H., *Ind. Eng. Chem.*, **30**, 1041-1042 (1938).

from the flow quantities and temperatures, and R can be evaluated from wC/WC . This permits S to be read directly from the graph. Since $S = (t_2 - t_1)/(T_1 - t_1)$ and T_1 and t_1 are known, it is then possible to obtain t_2 and from the heat balance $wC(t_2 - t_1) = WC(T_1 - T_2)$. The line designated *threshold* represents the initial points at which a temperature cross occurs. Values on this line correspond to $T_2 = t_2$.

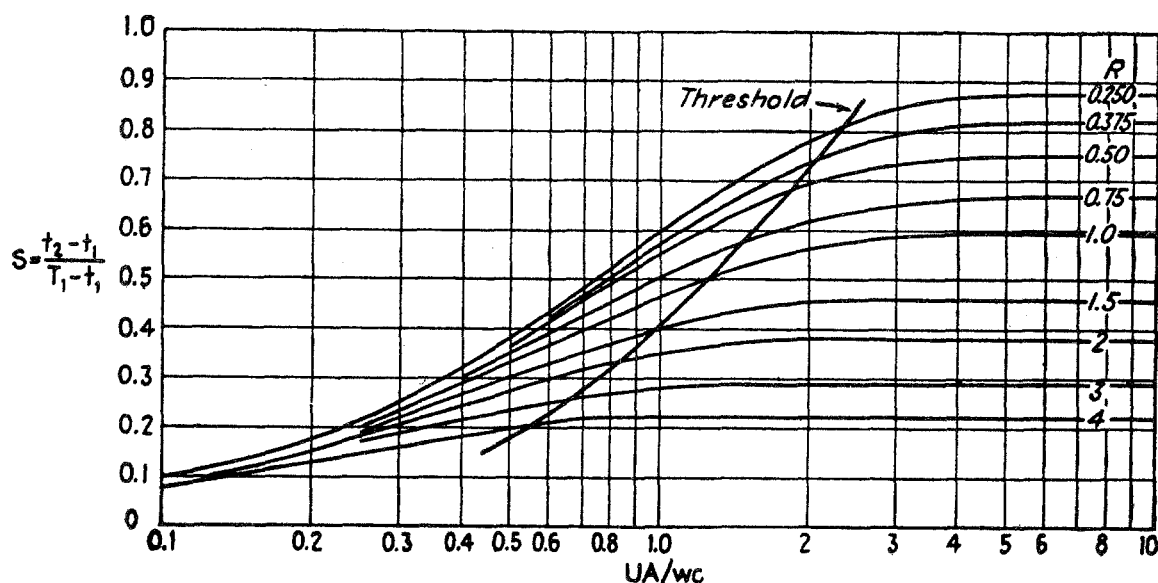


FIG. 7.25. Ten Broeck chart for determining t_2 when T_1 and t_1 are known in a 1-2 exchanger. (*Industrial & Engineering Chemistry*.)

Example 7.9. Outlet Temperatures for a Clean 1-2 Exchanger. In Example 7.3, kerosene-crude oil exchanger, what will the outlet temperatures be when the exchanger is freshly placed in service?

Solution:

$$U_C = 69.3 \quad A = 662 \quad w = 149,000 \quad c = 0.49$$

$$W = 43,800 \quad C = 0.60$$

$$\frac{UA}{wc} = \frac{69.3 \times 662}{149,000 \times 0.49} = 0.63$$

$$R = \frac{wc}{WC} = \frac{149,000 \times 0.49}{43,800 \times 0.60} = 2.78$$

From Figure 7.25,

$$S = \frac{t_2 - t_1}{T_1 - t_1} = 0.265$$

$$t_2 = t_1 + 0.265(T_1 - t_1) = 100 + 0.265(390 - 100) = 177^\circ\text{F}$$

$$T_2 = T_1 - R(t_2 - t_1) = 390 - 2.78(177 - 100) = 176^\circ\text{F}$$

The Efficiency of an Exchanger. In the design of many types of apparatus it is frequently desirable to establish a standard of maximum performance. The efficiency is then defined as the fractional performance of an apparatus delivering less than the standard. Dodge¹ gives the definition of the efficiency of an exchanger as the ratio of the quantity

¹Dodge, B. F., "Chemical Engineering Thermodynamics," McGraw-Hill Book Company, Inc., New York, 1944.

of heat removed from a fluid to the maximum which might have been removed. Using the usual nomenclature,

$$e = \frac{wc(t_2 - t_1)}{wc(T_1 - t_1)} = \frac{t_2 - t_1}{T_1 - t_1} \quad (7.54)$$

which is identical with the temperature group S and presumed that $t_2 = T_1$. Depending upon whether the hot or cold terminal approaches zero, the efficiency may also be expressed by

$$e = \frac{WC(T_1 - T_2)}{WC(T_1 - t_1)} \quad (7.55)$$

Although there is merit to this definition from the standpoint of thermodynamics there is a lack of realism in an efficiency definition which involves a terminal difference and a temperature difference of zero. It is the same as defining the efficiency as the ratio of the heat transferred by a real exchanger to an exchanger with infinite surface.

In process heat transfer there is another definition which is useful. The process temperatures are capable of providing a maximum temperature difference if arranged in counterflow. There appears to be some value in regarding the efficiency of an exchanger as the ratio of the temperature difference attained by any other exchanger to that for true counterflow. This is identical with F_T , which proportionately influences the surface requirements. It will be seen in the next and later chapters that other flow arrangements besides 1-2 parallel flow-counterflow can be attained in tubular equipment and by which the value of F_T may be increased for given process temperatures. These obviously entail flow patterns which approach true counterflow more closely than the 1-2 exchanger.

PROBLEMS

7.1. A 1-2 exchanger is to be used for heating 50,000 lb/hr of methyl ethyl ketone from 100 to 200°F using hot amyl alcohol available at 250°F. (a) What minimum quantity of amyl alcohol is required to deliver the desired heat load in a 1-2 exchanger? (b) If the amyl alcohol is available at 275°F, how does this affect the total required quantity?

7.2. A 1-2 exchanger has one shell and two tube passes. The passes do not have equal surfaces. Instead X per cent of the tubes are in the first pass and $(1 - X)$ per cent are in the second, but if the tube-side film coefficient is not controlling, the assumption of constant U is justifiable. (a) Develop an expression for the true temperature difference when X per cent of the tubes are in the colder of the two tube passes. (b) What is the true temperature difference when the hot fluid is cooled from 435 to 225°F by a noncontrolling cooling medium in the tubes which is heated from 100 to 150°F when 60 per cent of the tubes are in the colder tube pass and (c) when 40 per cent of the tubes are in the colder pass? How do these compare with the 1-2 true temperature difference with equal surfaces in each pass?

7.3. A double pipe exchanger has been designed for the nozzle arrangement shown in Fig. 7.26. If the hot stream is cooled from 275 to 205°F while the cold stream enters at 125°F and is heated to $t_2 = 190^\circ\text{F}$, what is the true temperature difference? (*Hint.* Establish an equation for the temperature difference with the nozzle arrangement shown and sufficient to allow a numerical trial-and-error solution.) How does this compare with the LMTD for counterflow?

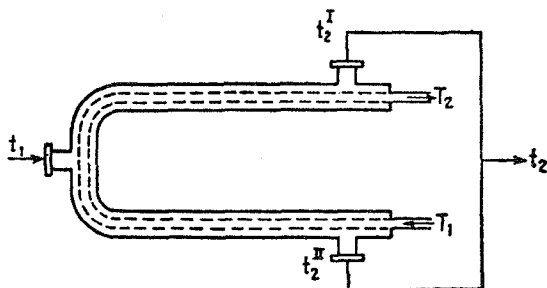


FIG. 7.26. Illustration for Prob. 7.3.

temperatures will be obtained? Calculate the outlet temperatures directly from F_T . How does the total heat load compare with that which could be delivered by a true counterflow exchanger assuming that the same U could be obtained?

7.5. It is necessary on a new installation to preheat 149,000 lb/hr of 34°API crude oil from 170°F to a temperature of 285°F, corresponding to that of the feed plate of a fractionating tower. There is a utility 33°API gas oil line running near the tower at 530°F of relatively unlimited quantity. Because the pumping cost for cold gas oil is prohibitive, the temperature of the gas oil from the heat exchanger, returning to the line, should not be less than 300°F.

Available on the site is a 25 in. ID 1-2 exchanger containing 252 tubes 1 in. OD, 13 BWG, 16'0" long arranged on a six-pass 1¼-in. triangular pitch layout. The shell baffles are spaced at 5-in. centers. A pumping head of 10 psi is allowable on the gas oil line and 15 psi on the feed line. Will the exchanger be acceptable if cleaned, and if so, what will the fouling factor be? For the gas oil the viscosities are 0.4 centipoise at 530°F and 0.7 centipoise at 300°F. For the crude oil the viscosities are 0.9 centipoise at 285°F and 2.1 centipoise at 170°F. (Interpolate by plotting °F vs. centipoise on logarithmic paper.)

7.6. 96,000 lb/hr of 35°API absorption oil in being cooled from 400 to 200°F is used to heat 35°API distillate from 100 to 200°F. Available for the service is a 29 in. ID 1-2 exchanger having 338 tubes 1 in. OD, 14 BWG, 16'0" long on 1¼-in. triangular pitch. Baffles are spaced 10 in. apart, and the bundle is arranged for four tube passes. What arrangement gives the more nearly balanced pressure drops, and what is the dirt factor? The viscosity of the absorption oil is 2.6 centipoise at 100°F and 1.15 centipoise at 210°F. (Plot on logarithmic paper °F vs. viscosity in centipoise, and extrapolate as a straight line.) The viscosity of the distillate is 3.1 centipoise at 100°F and 1.3 centipoise at 210°F.

7.7. 43,200 lb/hr of 35°API distillate is cooled from 250 to 120°F using cooling water from 85 to 120°F. Available for the service is a 19¼ in. ID 1-2 exchanger having 204 tubes ¾ in. OD, 16 BWG, 16'0" long on 1-in. square pitch. Baffles are spaced 5 in. apart, and the bundle is arranged for four passes. What arrangement gives the more nearly balanced pressure drops, and what is the dirt factor? What is the optimum outlet-water temperature? (Viscosities of the distillate are given in Prob. 7.6.)

7.8. 75,000 lb/hr of ethylene glycol is heated from 100 to 200°F using steam at 250°F. Available for the service is a 17¼ in. ID 1-2 exchanger having 224 tubes

How does this compare with the LMTD for counterflow?

7.4. 43,800 lb/hr of 42°API kerosene between 390 and 200°F is used to heat 149,000 lb/hr of 34°API Mid-continent crude from 100 to 170°F in a 662-ft² exchanger (Example 7.3). The clean coefficient is 69.3 Btu/(hr)(ft²)(°F). When the 1-2 exchanger is clean, what outlet

$\frac{3}{4}$ in. OD, 14 BWG, 16'0" long on $1\frac{5}{16}$ -in. triangular pitch. Baffles are spaced 7 in. apart, and there are two tube passes to accommodate the steam. What are the pressure drops, and what is the dirt factor?

7.9. 100,000 lb/hr of 20 per cent potassium iodide solution is to be heated from 80 to 200°F using steam at 15 psig. Available for the service is a 10 in. ID 1-2 exchanger without baffles having 50 tubes $\frac{3}{4}$ in. OD, 16 BWG, 16'0" long arranged for two passes on $1\frac{5}{16}$ -in. triangular pitch. What are the pressure drops and the dirt factor?

7.10. 78,359 lb/hr of isobutane (118°API) is cooled from 203 to 180°F by heating butane (111.5°API) from 154 to 177°F. Available for the service is a 17 $\frac{1}{4}$ in. ID 1-2 exchanger having 178 tubes $\frac{3}{4}$ in. OD, 14 BWG, 12'0" long on 1-in. triangular pitch. Baffles are spaced 6 in. apart, and the bundle is arranged for four passes. What are the pressure drops and the dirt factor?

7.11. A 1-2 exchanger recovers heat from 10,000 lb/hr of boiler blowdown at 135 psig by heating raw water from 70 to 96°F. Raw water flows in the tubes. Available for the service is a 10.02 in. ID 1-2 exchanger having 52 tubes $\frac{3}{4}$ in. OD, 16 BWG, 8'0" long. Baffles are spaced 2 in. apart, and the bundle is arranged for two tube passes. What are the pressure drops and fouling factors?

7.12. 60,000 lb/hr of a 25% NaCl solution is cooled from 150 to 100°F using water with an inlet temperature of 80°F. What outlet water temperature may be used? Available for the service is a 21 $\frac{1}{4}$ in. ID 1-2 exchanger having 302 tubes $\frac{3}{4}$ in. OD, 14 BWG, 16'0" long. Baffles are spaced 5 in. apart, and the bundle is arranged for two passes. What are the pressure drops and fouling factor?

NOMENCLATURE FOR CHAPTER 7

A	Heat-transfer surface, ft ²
a	Flow area, ft ²
a''	External surface per linear foot, ft
B	Baffle spacing, in.
C	Specific heat of hot fluid, in derivations, Btu/(lb)(°F)
C'	Clearance between tubes, in.
c	Specific heat of fluid, Btu/(lb)(°F)
C_E	Cost of exhaust steam, dollars/Btu
C_F	Annual fixed charges, dollars/ft ²
C_P	Cost of process steam, dollars/Btu
C_T	Total annual cost, dollars/year
C_W	Cost of water, dollars/lb
D	Inside diameter of tubes, ft
d_o	Outside diameter of tubes, in.
D_e, D'_e	Equivalent diameter for heat transfer and pressure drop, ft
d_e, d'_e	Equivalent diameter for heat transfer and pressure drop, in.
D_s	Inside diameter of shell, ft
e	Efficiency, dimensionless
F_c	Caloric fraction, dimensionless
F_T	Temperature difference factor, $\Delta t = F_T \times \text{LMTD}$, dimensionless
f	Friction factor, dimensionless; for ΔP in psi, ft ² /in ²
G	Mass velocity, lb/(hr)(ft ²)
g	Acceleration of gravity, ft/hr ²
g'	Acceleration of gravity, ft/sec ²
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, Btu/(hr)(ft ²)(°F)

h_{io}	Value of h_i when referred to the tube outside diameter, Btu/(hr)-(ft ²)(°F)
ID	Inside diameter, in.
j_H	Factor for heat transfer, dimensionless
K_c	Caloric constant, dimensionless
K_1, K_2	Numerical constants
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Tube length, ft
LMTD	Log mean temperature difference, °F
N	Number of shell-side baffles
N_t	Number of tubes
n	Number of tube passes
P_T	Tube pitch, in.
$\Delta P_T, \Delta P_t, \Delta P_r$	Total, tube side and return pressure drop, respectively, psi
Q	Heat flow, Btu/hr
R	Temperature group, $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_d, R_i, R_o	Combined, inside and outside dirt factors, respectively, (hr)(ft ²)(°F)/Btu
Re, Re'	Reynolds number for heat transfer and pressure drop, dimensionless
S	Temperature group, $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
s	Specific gravity
T, T_1, T_2	Temperature in general, inlet and outlet of hot fluid, °F
T_E, T_P	Saturation temperatures of exhaust and pressure steam, °F
T_a	Average temperature of hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
t, t_1, t_2	Temperature in general or outlet of the first of two heaters, inlet and outlet of cold fluid, °F
t^I, t^{II}	Temperatures in first and second passes, °F
t_a	Average temperature of cold fluid, °F
t_i	Temperature at end of first pass, °F
t_c	Caloric temperature of cold fluid, °F
t_w	Tube wall temperature, °F
Δt	True temperature difference in $Q = U_D A \Delta t$, °F
$\Delta t_c, \Delta t_h$	Temperature differences at the cold and hot terminals, °F
U, U_c, U_D	Overall coefficient of heat transfer, clean coefficient, design coefficient, Btu/(hr)(ft ²)(°F)
V	Velocity, fps
v	Specific volume, ft ³ /lb
W	Weight flow in general, weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
X	Length, ft
z	Height, ft
ϕ	The viscosity ratio $(\mu/\mu_w)^{0.14}$
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at tube-wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³

Subscripts (except as noted above)

s	Shell
t	Tubes

CHAPTER 8

FLOW ARRANGEMENTS FOR INCREASED HEAT RECOVERY

The Lack of Heat Recovery in Exchangers. The important limitation of the 1-2 exchangers treated in Chap. 7 lies in their inherent inability to provide effective heat recovery. The advantages of 1-2 exchangers have already been discussed. When a temperature cross occurs in a 1-2 exchanger, the value of F_T drops sharply, and the small extent to which the shell-outlet temperature can fall below the tube-outlet temperature eliminates them from consideration for high heat recoveries. Assume conditions in which the shell fluid is reduced from 200 to 140°F while the tube fluid rises from 80 to 160°F. All the heat in the hot fluid from 140 to 80°F is necessarily lost in a 1-2 exchanger because of the close approach required between the tube fluid at the end of the parallel pass and the shell fluid outlet T_2 , as in Figs. 7.20 and 7.21.

This chapter deals with shell-and-tube equipment and the methods by which the temperature cross of two fluid streams $t_2 - T_2$ can be increased with an accompanying increase in heat recovery. Consider an exchanger similar to a 1-2 exchanger except that it is equipped with a longitudinal baffle (heavy line) as shown in Fig. 8.1. In this exchanger the fluid enters the shell through one of the nozzles adjacent to the tube sheet and travels the length of the shell before reversing its direction about the longitudinal baffle and returning to an exit nozzle also adjacent to the tube sheet. Assume that the tube bundle contains four or more passes with equal surface in each pass. Such an exchanger is a 2-4 exchanger. A generalized sketch of temperature vs. length for a 2-4 exchanger is shown in Fig. 8.2a. In a 1-2 exchanger operating with the identical temperatures and shown in Fig. 8.2b, it is seen that a cross exists so that hot fluid leaving the shell at 140°F is forced to pass over tubes carrying heated cold fluid having a temperature of 160°F. Thus the shell fluid may be cooled at some point to a temperature lower than its outlet, and

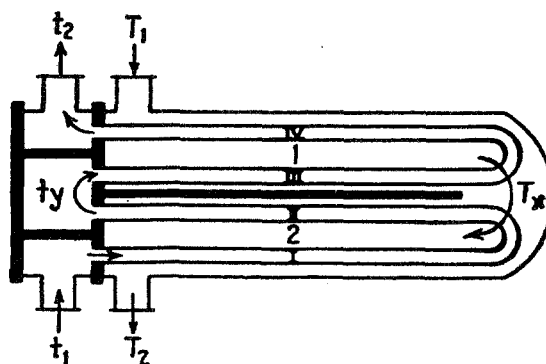


FIG. 8.1. Pass arrangements in a 2-4 exchanger.

the tube fluid may be heated to a temperature above its outlet. When the two fluids are near their outlets, the shell fluid, being cooled, is actually heated and the tube fluid is actually cooled. In exchangers this is called *reheating*.

The True Temperature Difference Δt in a 2-4 Exchanger. In the 2-4 exchanger the longitudinal baffle reduces reheating as shown in Fig. 8.2a so that the 140°F shell-

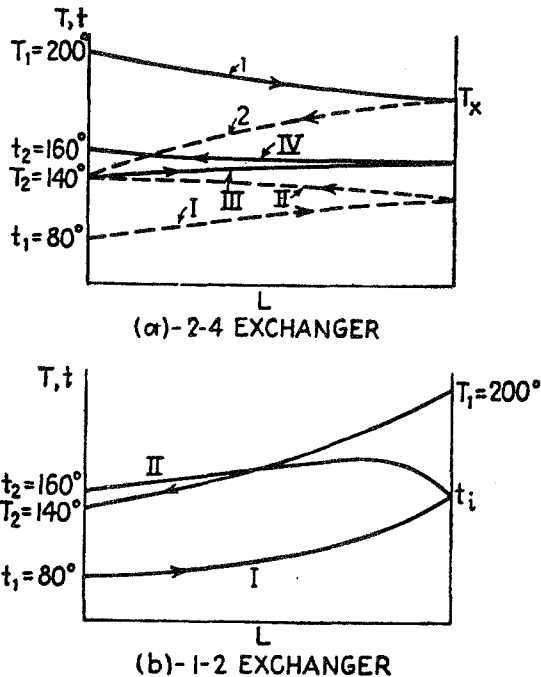


FIG. 8.2. Temperature relations in 1-2 and 2-4 exchangers.

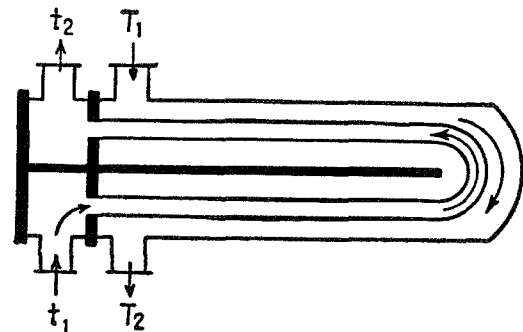


FIG. 8.3. The 2-2 true counterflow exchanger.

outlet fluid need never come in contact with the 160°F tube-outlet fluid. Passes I and II are in contact only with 2 and passes III and IV are in contact only with 1. If there are two shell passes and but two tube passes,

they can be arranged in true counterflow as shown in Fig. 8.3. However, where the shell contains two shell passes and the tube bundle contains four or more tube passes, the flow pattern differs from any encountered heretofore. The derivation of a factor F_T for the 2-4 exchanger can be readily established.

In the 2-4 exchanger it is assumed that there is no fluid leakage between the longitudinal shell baffle and the shell and that no heat is transferred across it, although this may lead to an error of 10 to 15 per

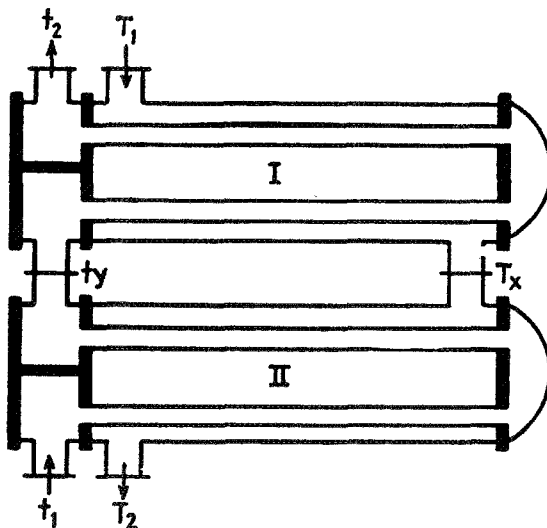


FIG. 8.4. Two 1-2 exchangers connected in series.

cent when a large temperature difference exists between the average temperatures of the shell fluid in the two shell passes. The assumptions for the 1-2 exchanger also apply. Referring to the 2-4 exchanger in Fig. 8.1,

the shell fluid temperature is T_x where it reverses direction after the first shell pass and the tube temperature is t_y where it reverses direction after the second tube pass. The entire 2-4 exchanger may then be considered the same as two 1-2 exchangers in series as shown in Fig. 8.4 with intermediate temperatures T_x and t_y . Calling these exchangers I and II in Fig. 8.4, the heat balances are, respectively,

$$\text{I.} \quad WC(T_1 - T_x) = wc(t_2 - t_y) \quad (8.1)$$

and

$$\text{II.} \quad WC(T_x - T_2) = wc(t_y - t_1) \quad (8.2)$$

The quantities of heat transferred in I and II are obviously not the same. Equation (7.37) may be written for each of the exchangers:

$$\text{I.} \quad \frac{UA}{wc} = \frac{1}{2\sqrt{R^2 + 1}} \ln \frac{T_1 + T_x - t_y - t_2 + (T_1 - T_x)\sqrt{R^2 + 1}}{T_1 + T_x - t_y - t_2 - (T_1 - T_x)\sqrt{R^2 + 1}} \quad (8.3)$$

$$\text{II.} \quad \frac{UA}{wc} = \frac{1}{2\sqrt{R^2 + 1}} \ln \frac{T_x + T_2 - t_1 - t_y + (T_x - T_2)\sqrt{R^2 + 1}}{T_x + T_2 - t_1 - t_y - (T_x - T_2)\sqrt{R^2 + 1}} \quad (8.4)$$

By algebraically eliminating T_x and t_y in Eqs. (8.3) and (8.4) through the use of S and the heat balances in Eqs. (8.1) and (8.2), F_T is given by

$$F_T = \frac{[\sqrt{R^2 + 1}/2(R - 1)] \ln (1 - S)/(1 - RS)}{\ln \frac{2/S - 1 - R + (2/S)\sqrt{(1 - S)(1 - RS)} + \sqrt{R^2 + 1}}{2/S - 1 - R + (2/S)\sqrt{(1 - S)(1 - RS)} - \sqrt{R^2 + 1}}} \quad (8.5)$$

Equation (8.5) has been plotted in Fig. 19 and will be used for 2-4 exchangers in the Fourier equation $Q = UA \Delta t = UAF_T \times \text{LMTD}$. A comparison has been made between the values of F_T in 1-2 and 2-4 exchangers as shown in Fig. 8.5, where both exchangers employ fluids operating with identical temperature ranges. The advantage of the 2-4 flow arrangement is apparent from the large temperature crosses permissible. In Fig. 8.6 the hot fluid at each point has a 100° range with varying approach and the cold fluid a 20° fixed range from 180 to 200°F . With a 5° cross the 1-2 exchanger has a value of $F_T = 0.70$ compared with $F_T = 0.945$ for a 2-4 exchanger. It is general that the greater the number of shell passes in an exchanger the greater the cross or the greater the heat recovery which may be obtained. Mechanically, however, it is impractical to design single items of heat-transfer equipment with removable bundles having more than two shell passes, although it is seen that the 2-4 exchanger is thermally identical with two 1-2 exchangers in series. Greater crosses than those possible in a 2-4 exchanger can be achieved by using three 1-2 exchangers in series (3-6 arrangement) or two 2-4 exchang-

ers in series (4-8 arrangement). Values of F_T for arrangements up to six shell passes and twelve tube passes are plotted in Figs. 18 through 23. A Ten Broeck chart for heat-recovery calculations in a 2-4 exchanger is given in Fig. 8.7 and is used in the same manner as Fig. 7.25.

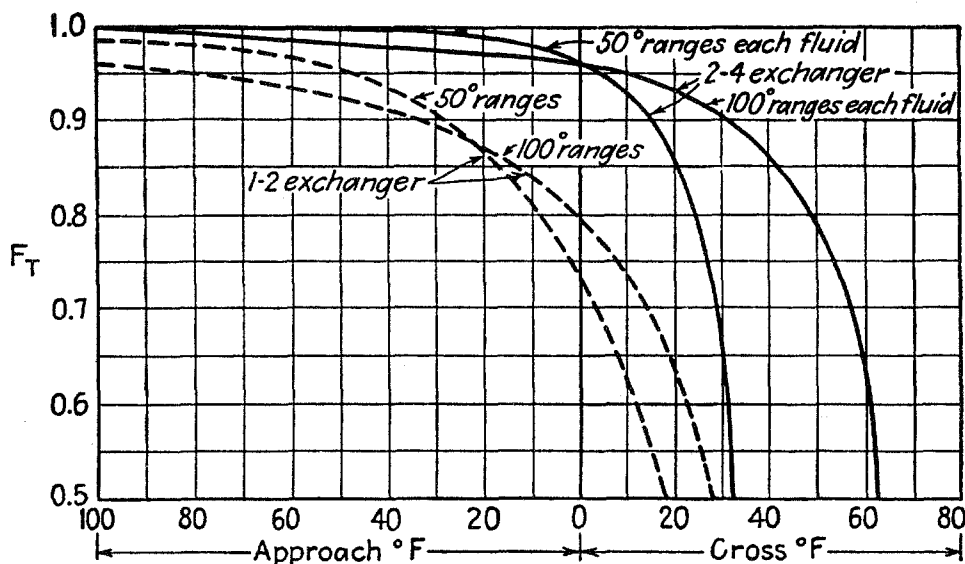


FIG. 8.5. Comparison of efficiency of 2-4 and 1-2 exchangers with equal fluid temperature ranges.

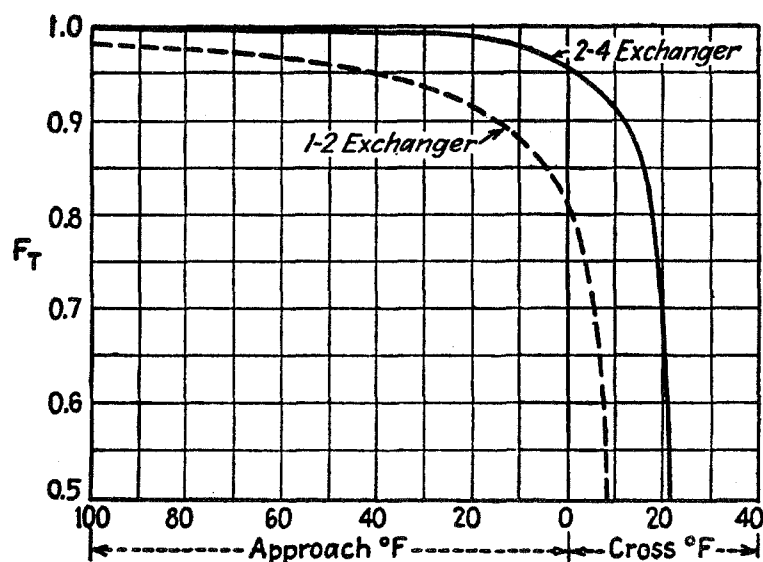


FIG. 8.6. Influence of approach temperature on F_T for unequal fluid temperature ranges.

Exchangers having odd numbers of tube passes have not been treated here because they may create mechanical problems in stationary tube-sheet exchangers and are not often employed. Fischer¹ has calculated and plotted the values of F_T for several odd-numbered tube arrangements. Naturally for a maximum value of F_T the odd tube-pass arrangements must be piped so that the majority of the tube passes are in counterflow with the shell fluid rather than in parallel flow.

¹ Fischer, F. K., *Ind. Eng. Chem.*, **30**, 377-383 (1938).

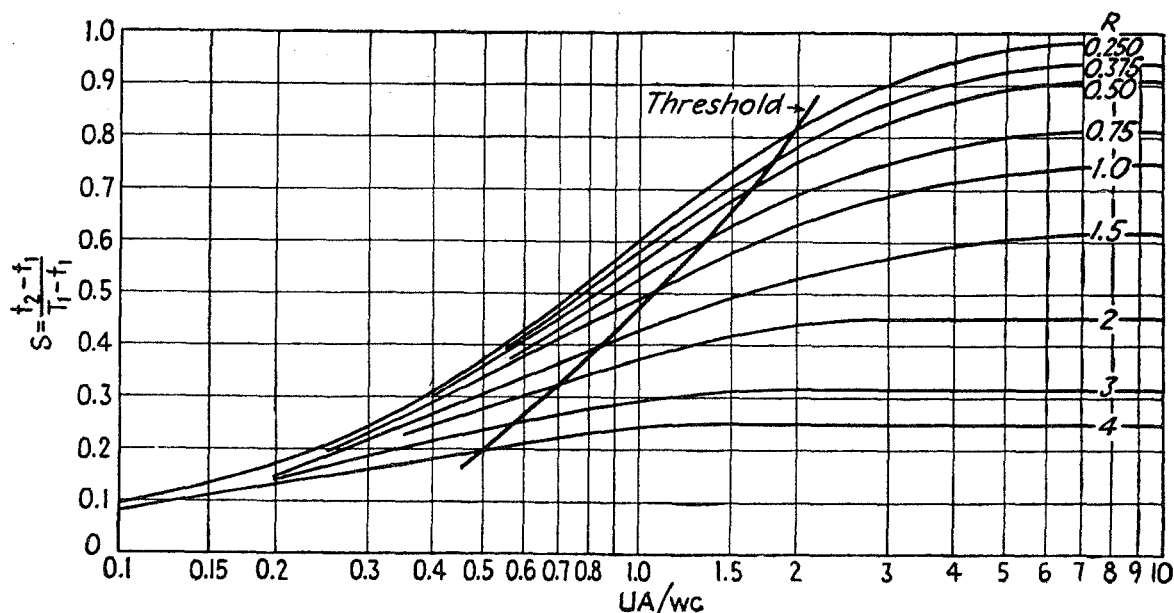


FIG. 8.7. Ten Broeck chart for determining t_2 in a 2-4 exchanger. (*Industrial and Engineering Chemistry*.)

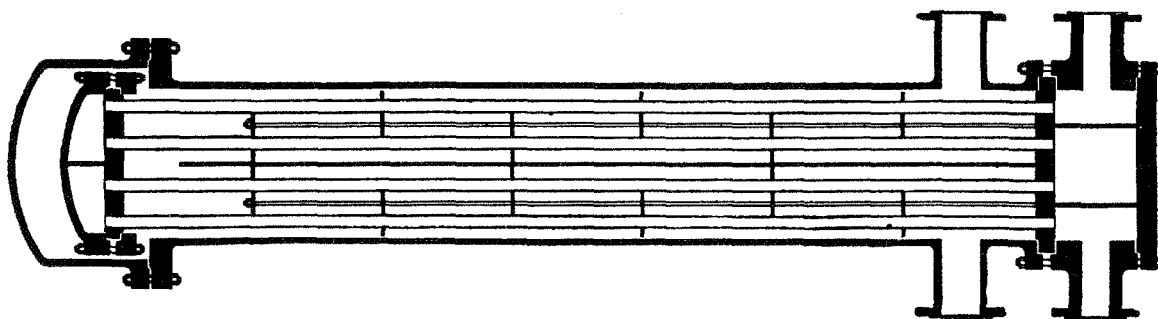


FIG. 8.8. Removable baffle 2-4 floating-head exchanger. (*Patterson Foundry & Machine Co.*)

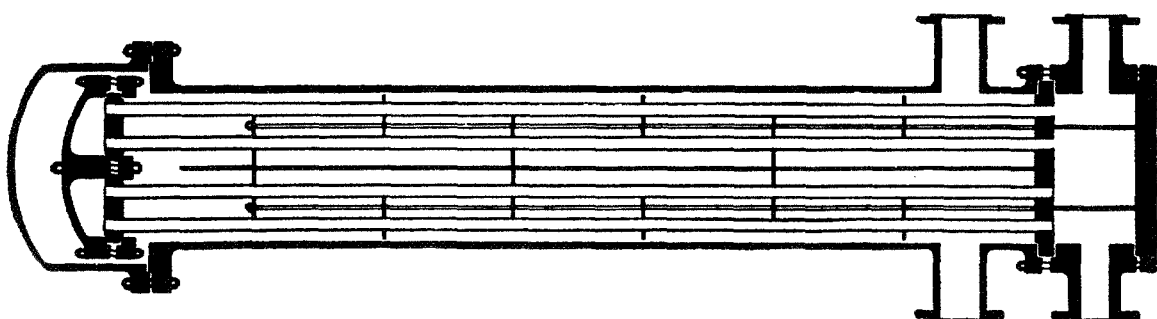


FIG. 8.9. Welded baffle 2-4 floating-head exchanger. (*Patterson Foundry & Machine Co.*)

2-4 Tubular Exchangers. Two methods by which the 2-4 temperature arrangement is achieved in tubular exchangers are shown in Figs. 8.8 and 8.9. The exchanger in Fig. 8.8 is similar to an ordinary 1-2 exchanger except that both of the shell nozzles are adjacent to the stationary tube sheet. Through the use of split segmental baffles a *removable* longitudinal baffle is inserted in the tube bundle. Usually some type of seal is provided between the longitudinal baffle and the shell, since any

appreciable leakage between the two shell passes invalidates the calculated value of F_T for the 2-4 exchanger.

A more expensive but more positive form of the 2-4 exchanger is shown in Fig. 8.9. In this exchanger the baffle is welded to the shell. This requires cutting smaller inside diameter shells in half and welding the baffle from the outside of the shell. In larger shells the baffle is installed internally. Furthermore, in order that the bundle be removable it is necessary to make the floating tube sheet in two halves which are joined by a single floating-head cover and a backing piece to prevent the two

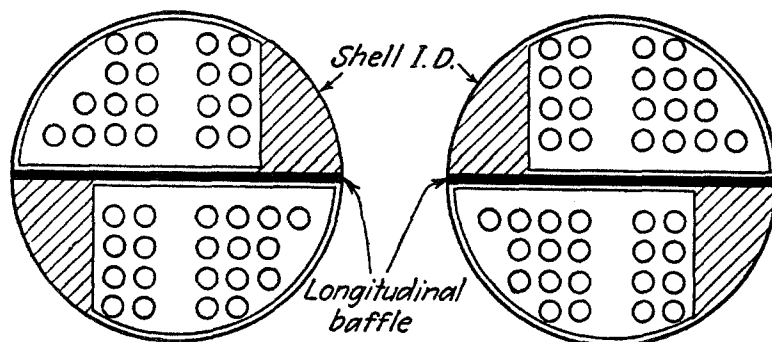


FIG. 8.10. Vertically cut segmental baffles.

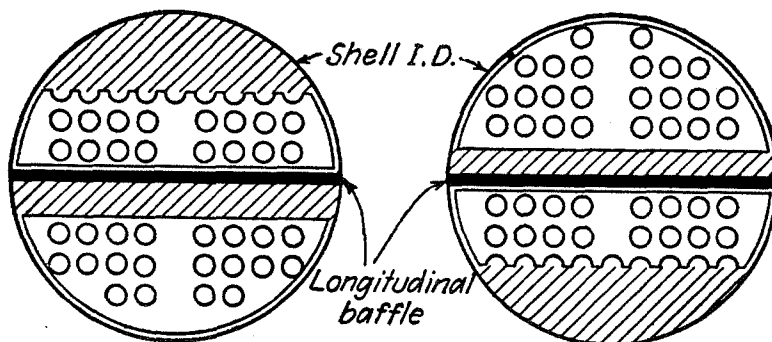


FIG. 8.11. Horizontally cut segmental baffles.

halves from leaking at the center line. The space allowance for the welded longitudinal baffle at the center line of the tube sheet also permits fewer tubes to be accommodated in the shell than for a 1-2 exchanger.

To permit the introduction of the longitudinal baffle, the segmental baffles may possess either of the two forms shown in Fig. 8.10 and 8.11. Those in Fig. 8.10 are *vertically cut* baffles and are similar to the segmental baffles found in 1-2 exchangers. The crosshatch section is the flow area. The flow distribution is nearly identical with that which exists in 1-2 exchangers with but one-half the flow area per inch of baffle spacing. Consequently the mass velocity for any given weight flow of fluid, W or w , will be twice that of a 1-2 exchanger of the same inside diameter and baffle spacing. The baffles shown in Fig. 8.11 are *horizontally cut* baffles in which the cut-out portions are equal to those of ordinary segmental

baffles. The mass velocities in this case are the same per inch of baffle spacing as for 1-2 exchangers for a given weight flow although the fluid traverses a depth but half the inside diameter of the shell. Horizontally cut baffles are seldom used, since most fluids involved in large temperature crosses necessitating 2-4 exchangers also have long temperature ranges and relatively small flow quantities. The reduced flow area and increased mass velocity and film coefficient afforded by vertically cut baffles are usually desirable.

The Calculation of a 2-4 Exchanger. A 2-4 exchanger can be used when the process temperatures give a correction factor F_T of less than 0.75 for a 1-2 exchanger. If the factor F_T obtained from Fig. 19 for a 2-4 exchanger exceeds 0.90 with a removable longitudinal baffle or 0.85 with a welded longitudinal baffle, a single 2-4 exchanger will be adequate. If the value of F_T is below these limits, it will then be necessary to use a larger number of shell passes until an arrangement is found in which F_T approximates these values.

The calculation of a 2-4 exchanger differs in only three minor respects from the calculation of a 1-2 exchanger as outlined in Chap. 7. (1) F_T will be read from Fig. 19, (2) the flow area for vertically cut baffles will be half the values computed from Eq. (7.44), and (3) the number of crosses used for computing the pressure drop will be double, since one set of baffles is above and one is below the longitudinal baffle.

Example 8.1. Calculation of a 2-4 Oil Cooler. A 33.5°API oil has a viscosity of 1.0 centipoise at 180°F and 2.0 centipoise at 100°F. 49,600 lb/hr of oil leaves a distilling column at 358°F and is to be used in an absorption process at 100°F. Cooling will be achieved by water from 90 to 120°F. Pressure-drop allowances of 10 psi may be used on both streams along with a combined dirt factor of 0.004.

Available for this service from a discontinued operation is a 35 in. ID 2-4 exchanger having 454 1 in. OD, 11 BWG tubes 12'0" long and laid out on 1¼-in. square pitch. The bundle is arranged for six tube passes, and vertically cut baffles are spaced 7 in. apart. The longitudinal baffle is welded to the shell.

Is it necessary to use a 2-4 exchanger? Will the available exchanger fulfill the requirements?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 35 in.	Number and length = 454, 12'0"
Baffle space = 7 in. (vert. cut)	OD, BWG, pitch = 1 in., 11 BWG, 1¼-in. square
Passes = 2	Passes = 6

(1) Heat balance:

Oil, $Q = 49,600 \times 0.545(358 - 100) = 6,980,000$ Btu

Water, $Q = 233,000 \times 1.0(120 - 90) = 6,980,000$ Btu

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
358	Higher Temp	120	238
100	Lower Temp	90	10
258	Differences	30	228

$$\text{LMTD} = 72.0^\circ\text{F} \quad (5.14)$$

$$R = \frac{258}{30} = 8.6 \quad S = \frac{30}{358 - 90} = 0.112$$

R and S do not intersect on Fig. 18, making a 2-4 exchanger imperative.

From Fig. 19 for a 2-4 exchanger, $F_T = 0.93$

$$\Delta t = F_T \times \text{LMTD} = 0.93 \times 72.0 = 66.9^\circ\text{F} \quad (7.42)$$

(3) T_c and t_c :

$$\frac{\Delta t_c}{\Delta t_h} = \frac{10}{238} = 0.042 \quad K_c = 0.47 \quad F_c = 0.25 \quad (\text{Fig. 17})$$

$$T_c = 100 + 0.25(258) = 165^\circ\text{F} \quad (5.28)$$

$$t_c = 90 + 0.25(30) = 98^\circ\text{F} \quad (5.29)$$

Water will flow in the tubes to prevent corrosion of the shell.

<i>Hot fluid: shell side, 33.5° API oil</i> (4') $a_s = \frac{1}{2}(\text{ID} \times C'B)/144P_T$ (modified for a 2-4 exchanger) [Eq. (7.1)] $= \frac{1}{2} \times 35 \times 0.25 \times 7/144 \times 1.25 = 0.17 \text{ ft}^2$ (5') $G_s = w/a_s$ [Eq. (7.2)] $= 49,600/0.17 = 292,000 \text{ lb}/(\text{hr})(\text{ft}^2)$ (6') At $T_c = 165^\circ\text{F}$, By obtaining X and Y in Fig. 14 from original data, $\mu = 1.12 \text{ cp}$ $= 1.12 \times 2.42 = 2.71 \text{ lb}/(\text{ft})(\text{hr})$ $D_e = 0.99/12 = 0.0825 \text{ ft}$ [Fig. 28] $Re_s = D_e G_s / \mu$ [Eq. (7.3)] $= 0.0825 \times 292,000 / 2.71 = 8900$ (7') $j_H = 52.5$ [Fig. 28] (8') At $T_c = 165^\circ\text{F}$, when $\mu = 1.12 \text{ cp}$ $k(c\mu/k)^{1/2} = 0.20 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$ [Fig. 16] (9') $h_o = j_H \frac{k}{D_e} (c\mu/k)^{1/2} \phi_s$ [Eq. (6.15b)] $\frac{h_o}{\phi_s} = 52.5 \times 0.20 / 0.0825 = 127$	<i>Cold fluid: tube side, water</i> (4) $a'_t = 0.455 \text{ in.}^2$ [Table 10] $a_t = N a'_t / 144n$ $= 454 \times 0.455 / 144 \times 6 = 0.239 \text{ ft}^2$ (5) $G_t = w/a_t$ $= 233,000 / 0.239 = 975,000 \text{ lb}/(\text{hr})(\text{ft}^2)$ $V = G_t / 3600\rho = 975,000 / 3600 \times 62.5 = 4.33 \text{ fps}$ (6) At $t_c = 98^\circ\text{F}$, $\mu = 0.73 \times 2.42 = 1.77 \text{ lb}/(\text{ft})(\text{hr})$ [Fig. 14] $D = 0.76/12 = 0.0633 \text{ ft}$ [Table 10] (Re_t is for pressure drop only) $Re_t = DG_t / \mu$ $= 0.0633 \times 975,000 / 1.77 = 34,900$ (9) $h_i = 1010 \times 0.96 = 970 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$ [Fig. 25]
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Hot fluid: shell side, 33.5° API oil
 (10') Tube-wall temp:

$$t_w = t_c + \frac{h_o}{h_{io} + h_o} (T_c - t_c) \quad [\text{Eq. (5.31)}]$$

$$= 98 + \frac{127}{737 + 127} (165 - 98) = 108^\circ\text{F}$$
 (11') At t_w ,
 $\mu_w = 1.95 \times 2.42 = 4.72 \text{ lb/(ft)(hr)}$ [Fig. 14]
 $\phi_s = (\mu/\mu_w)^{0.14} = (2.71/4.72)^{0.14}$
 $= 0.92$ [Fig. 24 insert]
 (12') Corrected coefficient h_o :
 $h_o = 127 \times 0.92 = 117 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$ [Eq. (6.36)]
 (13') Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{737 \times 117}{737 + 117} = 101 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14') Design overall coefficient U_D :

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft} \quad (\text{Table 10})$$

$$\text{Total surface, } A = 454 \times 12'0'' \times 0.2618 = 1425 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{6,980,000}{1425 \times 66.9} = 73.3 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{101 - 73.3}{101 \times 73.3} = 0.0038 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

117	h outside	737
U_C	101	
U_D	73.3	
R_d Calculated	0.0038	
R_d Required	0.0040	

Pressure Drop

(1') For $Re_s = 8900$, $f = 0.00215 \text{ ft}^2/\text{in.}^2$ [Fig. 29] | (1) For $Re_t = 34,900$,
 $f = 0.000195 \text{ ft}^2/\text{in.}^2$ [Fig. 26]
 (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] | (2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)]
 $= 12 \times 12/7 = 20.1$
 Say 21 per pass or 42 for bundle
 $D_s = 3\frac{5}{12} = 2.92 \text{ ft}$
 $s = 0.82$ [Fig. 6]

$$= \frac{0.000195 \times 975,000^2 \times 12 \times 6}{5.22 \times 10^{10} \times 0.0633 \times 1.0 \times 1.0}$$

$$= 4.0 \text{ psi}$$

Pressure Drop

$$\begin{aligned}
 (3') \Delta P_s &= \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad [\text{Eq. (7.44)}] \\
 &= \frac{0.00215 \times 292,000^2 \times 2.92 \times 42}{5.22 \times 10^{10} \times 0.0825 \times 0.82 \times 0.92} \\
 &= 7.0 \text{ psi}
 \end{aligned}$$

Allowable $\Delta P_s = 10.0$ psi

$$(3) G_t = 975,000, \frac{V^2}{2g'} = 0.13 \quad [\text{Fig. 27}]$$

$$\begin{aligned}
 \Delta P_r &= \frac{4n}{s} \frac{V^2}{2g'} \quad [\text{Eq. (7.46)}] \\
 &= \frac{4 \times 6}{1} \times 0.13 = 3.2 \text{ psi}
 \end{aligned}$$

$$\begin{aligned}
 (4) \Delta P_T &= \Delta P_t + \Delta P_r \quad [\text{Eq. (7.47)}] \\
 &= 4.0 + 3.2 = 7.2 \text{ psi}
 \end{aligned}$$

Allowable $\Delta P_T = 10.0$ psi

The exchanger will have a slightly low dirt factor but is otherwise satisfactory.

Exchangers in Series. In plants where large numbers of exchangers are used, certain size standards (total number of tubes, pass arrangements, baffle spacing) are established for 1-2 exchangers so that a majority of future services can be fulfilled by an arrangement of a number of standard exchangers in series or parallel. Although this may sometimes cause awkwardness because of the impossibility of utilizing the equipment most efficiently, it has the great advantage of reducing the type and number of replacement parts, tubes, and tools. In these plants when a process has become obsolete, it is customary to find a number of exchangers of identical size available for other uses. If the tube bundle is simply retubed, the exchanger will frequently be as serviceable as when new. When two exchangers are connected in series on *both* the shell and tube sides, they form a temperature arrangement which has been shown to be identical with the 2-4 exchanger. When a temperature cross involves a correction factor for an arrangement which approximates true counterflow more closely than that possible in a 1-2 exchanger, it can be met by a series arrangement of a number of 1-2 exchangers. The 2-4, 3-6, 4-8, etc., arrangements are all based upon shells and channels being connected in series. Any arrangement which is an even multiple of two shell passes such as 2-4, 4-8, etc., may be fulfilled by a number of 1-2 exchangers or half as many 2-4 exchangers.

The calculation for process conditions requiring more than one shell pass follows the method used for 1-2 exchangers except that the entire group of exchangers is treated as a unit.

Example 8.2. Calculation of an Acetone-Acetic Acid Exchanger. Acetone ($s = 0.79$) at 250°F is to be sent to storage at 100°F and at a rate of 60,000 lb/hr. The heat will be received by 185,000 lb/hr of 100 per cent acetic acid ($s = 1.07$) coming from storage at 90°F and heated to 150°F . Pressure drops of 10.0 psi are available for both fluids, and a combined dirt factor of 0.004 should be provided.

Available for the service are a large number of 1-2 exchangers having $21\frac{1}{4}$ in. ID shells with 270 tubes $\frac{3}{4}$ in. OD, 14 BWG, 16'0" long and laid out on 1-in. square

pitch. The bundles are arranged for two tube passes with segmental baffles spaced 5 in. apart.

How many of the 1-2 exchangers should be installed in series?

Solution:

Exchanger:

Shell side	Tube side
ID = 21¼ in.	Number and length = 270, 16'0"
Baffle space = 5 in.	OD, BWG, pitch = ¾ in., 14 BWG, 1 in. square
Passes = 1	Passes = 2

(1) Heat balance:

$$\text{Acetone, } Q = 60,000 \times 0.57(250 - 100) = 5,130,000 \text{ Btu/hr} \quad (\text{Fig. 2})$$

$$\text{Acetic acid, } Q = 168,000 \times 0.51(150 - 90) = 5,130,000 \text{ Btu/hr} \quad (\text{Fig. 2})$$

(2) Δt :

Hot Fluid		Cold Fluid		Diff.
250	Higher Temp	150		100
100	Lower Temp	90		10
150	Difference	60		90

$$\text{LMTD} = 39.1^\circ\text{F} \quad (5.14)$$

$$R = 15\%_0 = 2.5; \quad S = 60/(250 - 90) = 0.375 \quad (7.18)$$

$$F_T: \text{1-2 exchanger, } F_T = \text{not possible} \quad (\text{Fig. 18})$$

$$\text{2-4 exchanger, } F_T = 0.67 \text{ (too low)} \quad (\text{Fig. 19})$$

$$\text{3-6 exchanger, } F_T = 0.88 \quad (\text{Fig. 20})$$

$$\Delta t = 0.88 \times 39.1 = 34.4^\circ\text{F} \quad (7.42)$$

To permit the transfer of heat with the temperatures given by the process, a minimum of three shell passes is required. If the sum of the surfaces in three shells is insufficient, a greater number will be required.

(3) T_c and t_c : These liquids are not viscous, and the viscosity correction will be negligible, $\phi_s = \phi_t = 1$. Average temperatures may be used.

Hot fluid: shell side, acetone	Cold fluid: tube side, acetic acid
(4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)]	(4) $a'_t = 0.268 \text{ in.}^2$ [Table 10]
$= 21.25 \times 0.25 \times 5/144 \times 1.0$	$a_t = N_t a'_t/144n$ [Eq. (7.48)]
$= 0.185 \text{ ft}^2$	$= 270 \times 0.268/144 \times 2 = 0.251 \text{ ft}^2$
(5') $G_s = W/a_s$ [Eq. (7.2)]	(5) $G_t = w/a_t$
$= 60,000/0.185$	$= 168,000/0.251$
$= 324,000 \text{ lb}/(\text{hr})(\text{ft}^2)$	$= 670,000 \text{ lb}/(\text{hr})(\text{ft}^2)$
(6') At $T_a = 175^\circ\text{F}$,	(6) At $t_a = 120^\circ\text{F}$,
$\mu = 0.20 \times 2.42 = 0.49 \text{ lb}/(\text{ft})(\text{hr})$	$\mu = 0.85 \times 2.42 = 2.06 \text{ lb}/(\text{ft})(\text{hr})$
[Fig. 14]	[Fig. 14]
$D_e = 0.95/12 = 0.079 \text{ ft}$ [Fig. 28]	$D = 0.584/12 = 0.0487 \text{ ft}$
$Re_s = D_e G_s/\mu$ [Eq. (7.3)]	$Re_t = D G_t/\mu$
$= 0.079 \times 324,000/0.49 = 52,200$	$= 0.0487 \times 670,000/2.06 = 15,800$
(7') $j_H = 137$ [Fig. 28]	(7) $j_H = 55$ [Fig. 24]

Hot fluid: shell side, acetone	Cold fluid: tube side, acetic acid
(8') At $T_a = 175^\circ\text{F}$, $c = 0.63 \text{ Btu}/(\text{lb})(^\circ\text{F})$ [Fig. 2] $k = 0.095 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ [Table 4]	(8) At $t_a = 120^\circ\text{F}$, $c = 0.51 \text{ Btu}/(\text{lb})(^\circ\text{F})$ [Fig. 2] $k = 0.098 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ [Table 4]
$(c\mu/k)^{1/3} = (0.63 \times 0.49/0.095)^{1/3} = 1.47$	$(c\mu/k)^{1/3} = (0.51 \times 2.06/0.098)^{1/3} = 2.21$
(9') $h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/3} \times 1$ [Eq. (6.15b)] $= 137 \times 0.095 \times 1.47/0.079$ $= 242 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$	(9) $h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/3} \times 1$ [Eq. (6.15a)] $= 56 \times 0.098 \times 2.21/0.0487$ $= 249 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
(10') (11') (12') The viscosity correction will be very small.	(10) $h_{io} = h_i \times \text{ID}/\text{OD}$ [Eq. (6.5)] $= 249 \times \frac{0.584}{0.75} = 194$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{242 \times 194}{242 + 194} = 107.5 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Design overall coefficient U_D :

$$\begin{aligned} a'' &= 0.1963 \text{ ft}^2/\text{lin ft} && \text{(Table 10)} \\ \text{Total surface, } A &= 3(270 \times 16 \text{ 0''} \times 0.1963) = 2540 \text{ ft}^2 \\ U_D &= \frac{Q}{A \Delta t} = \frac{5,130,000}{2540 \times 34.4} = 58.8 \text{ (Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \end{aligned}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{107.5 - 58.8}{107.5 \times 58.8} = 0.0077 \text{ (hr)(ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Summary

242	h outside	194
U_C	107.5	
U_D	58.8	
R_d Calculated	0.0077	
R_d Required	0.0040	

Pressure Drop

(1') For $Re_s = 52,200$, $f = 0.00155 \text{ ft}^2/\text{in.}^2$ [Fig. 29]	(1) For $Re_t = 158,000$, $f = 0.00024 \text{ ft}^2/\text{in.}^2$ [Fig. 26] $s = 1.07$ [Table 6]
(2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 16 \times 12/5 = 39$ Total for 3 exchangers $= 39 \times 3 = 117$ $D_s = 21.25/12 = 1.78 \text{ ft}$ $s = 0.79$ [Table 6]	(2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)] $= \frac{0.00024 \times 670,000^2 \times 16 \times 2 \times 3}{5.22 \times 10^{10} \times 0.0487 \times 1.07 \times 1.0}$ $= 3.8 \text{ psi}$

Pressure Drop

$$\begin{aligned} (3') \Delta P_s &= \frac{fG_s^2 D_s (N+1)}{5.22 \times 10^{10} D_s s \phi_s} \\ &= \frac{0.00155 \times 324,000^2 \times 1.78 \times 117}{5.22 \times 10^{10} \times 0.079 \times 0.79 \times 1.0} \\ &= 10.4 \text{ psi} \end{aligned}$$

Allowable $\Delta P_s = 10.0$ psi

$$\begin{aligned} (3) G_t &= 670,000, \\ V^2/2g' &= 0.063 \quad [\text{Fig. 27}] \\ \Delta P_r &= 3(4n/s)(V^2/2g') \quad [\text{Eq. (7.46)}] \\ &= \frac{3 \times 4 \times 2}{1.07} \times 0.063 = 1.4 \text{ psi} \end{aligned}$$

$$\begin{aligned} (4) \Delta P_t &= \Delta P_i + \Delta P_r \quad [\text{Eq. (7.47)}] \\ &= 3.8 + 1.4 = 5.2 \text{ psi} \end{aligned}$$

Allowable $\Delta P_T = 10.0$ psi

The three exchangers are more than adequate for heat transfer, even though the pressure drop is insignificantly high. Fewer exchangers cannot be used.

The 1-1 True Counterflow Exchanger. There are instances in which the temperature cross is so great that the only solution lies in the use of

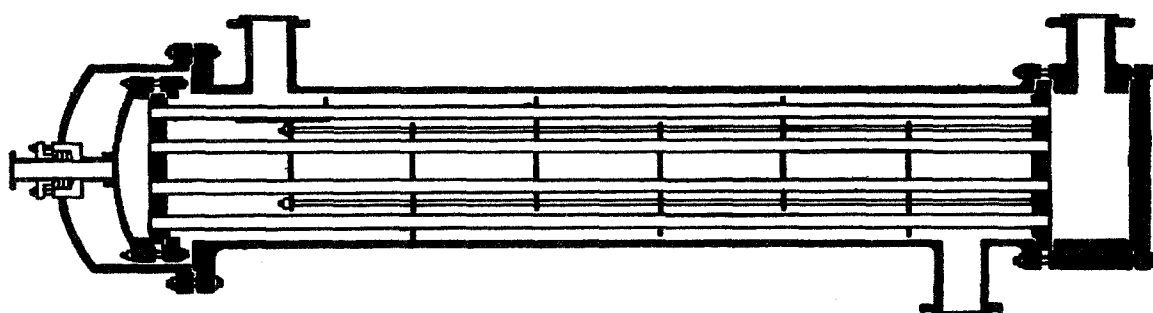


FIG. 8.12. The 1-1 floating-head exchanger. (Patterson Foundry & Machine Co.)

true counterflow. In fixed-tube-sheet equipment this is readily accomplished, but in floating-head equipment it is slightly more difficult. It can be achieved as shown in Fig. 8.12. An extension forming the tube fluid-outlet nozzle protrudes through the shell bonnet by means of a packing gland. This type of exchanger is also used when there is a great quantity of tube-side fluid.

PROBLEMS

8.1. 33,114 lb/hr of *n*-butyl alcohol at 210°F is to be cooled to 105°F using water from 95 to 115°F. Available for the purpose is a 19¼ in. ID, two-pass shell exchanger with 204 tubes ¾ in. OD, 16 BWG, 16'0" long on 1-in. square pitch arranged for four passes. Vertically cut baffles are spaced 5 in. apart. Pressure drops of 10 psi are allowable. What is the dirt factor?

8.2. 62,000 lb/hr of hot 26°API quench oil having viscosities equivalent to a 42°API kerosene (Fig. 14) is cooled from 425 to 304°F by heating 27,200 lb/hr of 35°API distillate (Fig. 14) with an inlet temperature of 100°F. Available for the service are 15¼ in. ID 1-2 exchangers with 108 tubes ¾ in. OD, 16 BWG, 16'0" long arranged for six passes. Baffles are spaced 6 in. apart. Total pressure drops of 10 psi are allowable. How many 1-2 exchangers are required, and what dirt factor do they yield?

8.3. 36,000 lb/hr of ethyl acetate is cooled from 190 to 100°F using water from 80 to 120°F. Available for the service is a 23¼ in. ID, two-pass shell exchanger with

292 $\frac{3}{4}$ in. OD, 14 BWG, 12'0" long tubes arranged for eight passes. Vertically cut baffles are spaced 8 in. apart. What pressure drops and dirt factor will be obtained?

8.4. 55,400 lb/hr of nitrobenzene coming from a reactor under pressure at 365°F is to be cooled to 150°F by preheating benzene under pressure from 100° to the highest outlet temperature attainable. Available for the service are two 25 in. ID, two-pass shell exchangers with 356 tubes $\frac{3}{4}$ in. OD, 13 BWG, 16'0" long arranged for six tube passes. Vertically cut baffles are spaced 8 in. apart. What are the pressure drops and the dirt factor?

NOMENCLATURE FOR CHAPTER 8

A	Heat-transfer surface, ft ²
a''	External surface per lin ft, ft
a	Flow area, ft ²
B	Baffle spacing, in.
C'	Clearance between tubes, in.
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
c	Specific heat of fluid, Btu/(lb)(°F)
D	Inside diameter of tubes, ft
D_e	Equivalent diameter for heat transfer and pressure drop, ft
D_s	Inside diameter of shell, ft
d	Inside diameter of tubes, in.
d_e	Equivalent diameter for heat transfer and pressure drop, in.
F_c	Caloric fraction, dimensionless
F_T	Temperature difference factor, $\Delta t = F_T \times \text{LMTD}$, dimensionless
f	Friction factor, ft ² /in. ²
G	Mass velocity, lb/(hr)(ft ²)
g'	Acceleration of gravity, ft/sec ²
h_i, h_o	Heat-transfer coefficient for inside and outside fluids, respectively, Btu/(hr)(ft)(°F)
h_{io}	Value of h_i when referred to the tube outside diameter, Btu/(hr)(ft ²)(°F)
ID	Inside diameter, in.
j_H	Factor for heat transfer, dimensionless
K_c	Caloric constant, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Tube length, ft
LMTD	Log mean temperature difference, °F
N	Number of shell-side baffles
N_t	Number of tubes
n	Number of tube passes
OD	Outside diameter of tube, in.
ΔP	Pressure drop, psi
$\Delta P_T, \Delta P_t, \Delta P_r$	Total, tube and return pressure drop, psi
P_T	Tube pitch, in.
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_d	Combined dirt factor, (hr)(ft ²)(°F)/Btu
Re	Reynolds number, dimensionless
S	Temperature group, $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
s	Specific gravity, dimensionless
T, T_1, T_2	Temperature in general, inlet and outlet of hot fluid, °F

T_a	Average temperature of hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
T_s	Temperature of shell fluid between first and second passes, °F
t_1, t_2	Inlet and outlet temperatures of cold fluid, °F
t_a	Average temperature of hot fluid, °F
t_c	Caloric temperature of cold fluid, °F
t_i	Temperature at end of first pass, °F
t_w	Tube-wall temperature, °F
t_y	Temperature of tube fluid between second and third passes, °F
Δt	True temperature difference in $Q = U_D A \Delta t$, °F
$\Delta t_c, \Delta t_h$	Temperature differences at the cold and hot terminals, °F
U_C, U_D	Clean and design overall coefficients of heat transfer, Btu/(hr)(ft ²) (°F)
V	Velocity, fps
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at tube wall-temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³
ϕ	$(\mu/\mu_w)^{0.14}$

Subscripts (except as noted above)

s	Shell side
t	Tube side

CHAPTER 9

GASES

Introduction. The calculation for the heating or cooling of a gas differs in only minor respects from the calculation for liquid-liquid systems. The relationships between gas film coefficients and allowable pressure drops are critically dependent upon the operating pressure of the system whereas for incompressible fluids the operating pressure is not important. The values of film coefficients for gases are generally lower than those obtained for liquids at equal mass velocities, the difference being inherent in the properties of the gases.

Properties of Gases. The properties of gases are compared with those of liquids to emphasize the major differences between them. The viscosities of gases range from about 0.015 to 0.025 centipoise, or about one-tenth to one-fifth the values obtained for the least viscous liquids. Gas viscosities increase with temperature in contrast with liquids and Reynolds numbers are correspondingly large even when mass velocities are small. The thermal conductivities of gases, with the exception of hydrogen, are about one-fifth the values usually obtained for organic liquids and about one-fifteenth of the values for water and aqueous solutions. The specific heats of organic gases and vapors are only slightly lower than those of organic liquids. With the exception of hydrogen the specific heats of inorganic gases and light hydrocarbon vapors range from 0.2 to 0.5 Btu/(lb)(°F). Although the specific heat, viscosity, and thermal conductivity of a gas increase with temperature, the Prandtl number $c\mu/k$ shows little dependence upon temperature except near the critical. The value of $c\mu/k$ calculated at any single temperature serves sufficiently well for the solution of problems involving the same gas at another temperature within reasonable proximity. The values of $c\mu/k$ are given in Table 9.1 for the common gases.

While most of the viscosity, specific-heat, and conductivity data on gases are tabulated for atmospheric pressure, corrections to other pressures may be made by established methods. Viscosities may be corrected by means of the correlation of Comings and Egly¹ given in Fig. 13b or by employing the method of Othmer and Josefowitz.² Specific heats may be

¹ Comings, E. W., and R. S. Egly, *Ind. Eng. Chem.*, **32**, 714-718 (1940).

² Othmer, D. F., and S. Josefowitz, *Ind. Eng. Chem.*, **38**, 111-116 (1946).

corrected by the method of Watson and Smith.¹ These corrections will not be significant, however, unless the pressure of the gas is great. Except at high vacuums the conductivities of gases are not affected by the pressure. Calculation of the density or specific volume of a gas by the perfect gas law is suitable for moderate pressures but may be in error at high pressures. If actual compression data are available, their use is preferable at high pressures, or the perfect gas law may be replaced by a

TABLE 9.1. PRANDTL NUMBERS FOR GASES AT 1 ATM AND 212°F

Gas	$\frac{c\mu}{k}$
Air.....	0.74
Ammonia.....	0.78
Carbon dioxide.....	0.80
Carbon monoxide.....	0.74
Ethylene.....	0.83
Hydrogen.....	0.74
Hydrogen sulfide.....	0.77
Methane.....	0.79
Nitrogen.....	0.74
Oxygen.....	0.74
Steam.....	0.78
Sulfur dioxide.....	0.80

more acceptable equation of state such as that of Van der Waals or Beattie-Bridgman.

Pressure Drop. Equations (7.44) and (7.45) and friction factors obtained from Figs. 29 and 26 can be used for the calculation of the pressure drop on the shell or tube sides of gas heaters or coolers when the average value of the inlet and outlet specific gravities of the gas *relative to water* is used. It is apparent in the case of any gas that the specific gravity varies considerably with its *operating pressure*. The specific gravity of air in an exchanger operated at 150 psia is very nearly ten times the specific gravity when operated at atmospheric pressure, and for a given mass velocity the pressure drop will only be one-tenth as great. Moving in the other direction, air at 7½ psia has a density one-half that of the atmosphere, and the pressure drop for a given mass velocity becomes greater as the operating pressure decreases, an unfavorable consideration in vacuum processes. When a gas is operated at high pressure, however, a relatively large mass velocity can be used without obtaining a pressure drop of an impractical order. When a gas is operated under vacuum, a pressure drop of 0.5 psi may represent a very large portion of the entire head available to move the gas through the exchanger.

¹ Watson, K. M., and R. L. Smith, *Natl. Petroleum News*, July, 1936.

Film Coefficients. Film coefficients can be evaluated with reasonable accuracy through the use of the usual shell-and-tube equations or Figs. 28 and 24. No correction need be made for the viscosity ratio ϕ unless the temperature range is exceedingly great. As mentioned above, the low viscosities of gases result in large Reynolds numbers even where very small mass velocities are used. Large values of j_H are the result, but the correspondingly low thermal conductivity gives film coefficients below those obtained for liquids at equal mass velocities or at equal values of j_H . The commonest applications of gas cooling under pressure are found in the aftercooling and intercooling of gases which have undergone adiabatic or polytropic compression in single- and multistage compressors. Gas-to-liquid heat transfer is also used for the recovery of waste heat from near-atmospheric combustion gases such as *economizers*, but these usually employ a modification known as *extended surfaces*, which will be treated in Chap. 16. When gases are heated, steam is the usual heating medium.

Many of the data reported in the literature for turbulent-flow heat transfer make use of the following transformation:

$$\frac{hD}{k} \left(\frac{c\mu}{k} \right)^{-1/4} \frac{\mu}{DG} \frac{c}{\bar{c}} = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{3/4} \quad (9.1)$$

A new factor j_h is then defined as

$$j_h = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{3/4} \quad (9.2)$$

and is plotted as a function of Re . All the convection equations given heretofore can be plotted as j_h vs. Re in place of j_H vs. Re by simply dividing the values of j_H as given in Figs. 28 and 24 by their respective values of Re . Alternate equations are obtained thereby.

There is considerable merit to the use of j_h in preference to j_H when correlating gases. Using j_H , $c\mu/k$ is a constant, but k in Eq. (6.15a) and (6.15b) must be obtained at the bulk temperature to obtain h , and conductivity data are sparse for gases. Using j_h , $c\mu/k$ is a constant, but only c is required in Eq. (9.1) to obtain h , and this will have been required for the heat load as well.

When gases enter an adiabatic compressor, their isotherms follow the equation $pv^\gamma = \text{constant}$, where p is the absolute pressure of the gas, v its specific volume, and γ the ratio of the specific heats of the gas at constant pressure to constant volume. Applying the perfect gas law, the variation of absolute pressure with *absolute temperature* becomes

$$\left(\frac{T_2}{T_1} \right)_{\text{abs}} = \left(\frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} \quad (9.3)$$

Gases to be cooled or heated at moderate pressures are usually placed in the shell of shell-and-tube equipment to localize corrosion resulting from the cooling water or steam condensate. At higher pressures, however, it is customary to place the gas in the tubes where the pressure is effective only upon the tubes.

Example 9.1. Calculation of an Ammonia Compressor Aftercooler. Dry ammonia gas at 83 psia and at a rate of 9872 lb/hr is discharged from a compressor at 245°F and is to be fed to a reactor at 95°F using cooling water from 85 to 95°F. A pressure drop of 2.0 psi is allowable on the gas and 10.0 psi on the water.

Available for the service is a 23 $\frac{1}{4}$ in. ID 1-2 exchanger having 364 $\frac{3}{4}$ in., 16 BWG tubes 8'0" long and laid out on $1\frac{5}{16}$ -in. triangular pitch. The bundle is arranged for eight passes, and baffles are spaced 12 in. apart.

What will the dirt factor and pressure drops be?

Solution:

Exchanger:

Shell side	Tube side
ID = 23 $\frac{1}{4}$	Number and length = 364, 8'0"
Baffle space = 12 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, $1\frac{5}{16}$ -in. tri.
Passes = 1	Passes = 8

(1) Heat balance:

$$\text{Ammonia gas, } Q = 9872 \times 0.53(245 - 95) = 785,000 \text{ Btu/hr} \quad (\text{Fig. 3})$$

$$\text{Water, } Q = 78,500 \times 1(95 - 85) = 785,000 \text{ Btu/hr}$$

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
245	Higher Temp	95	150
95	Lower Temp	85	10
150	Differences	10	140

$$\text{LMTD} = 51.8^\circ\text{F} \quad (5.14)$$

$$R = \frac{150}{10} = 15 \quad S = \frac{10}{245 - 85} = 0.0625$$

$$F_T = 0.837 \quad (\text{Fig. 18})$$

$$\Delta t = 0.837 \times 51.8 = 43.4^\circ\text{F} \quad (7.42)$$

(3) T_c and t_c : The viscosities will vary too little to require correction. Water will flow in the tubes to prevent corrosion of the shell.

Hot fluid: shell side, ammonia at 83 psia (4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)] $= 23.25 \times 0.1875 \times 12/144$ $\times 0.937$ $= 0.388 \text{ ft}^2$	Cold fluid: tube side, water (4) $a'_t = 0.302 \text{ in.}^2$ [Table 10] $a_t = N_t a'_t / 144n$ $= 364 \times 0.302 / 144 \times 8 = 0.0954 \text{ ft}^2$
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Hot fluid: shell side, ammonia at 83 psia

$$(5') G_s = W/a_s \quad [\text{Eq. (7.2)}]$$

$$= 9872/0.388$$

$$= 25,400 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6') \text{ At } T_s = 170^\circ\text{F},$$

$$\mu = 0.012 \times 2.42 = 0.029 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 15]

$$D_s = 0.55/12 = 0.0458 \text{ ft}$$

$$Re_s = D_s G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.0458 \times 25,400 / 0.029 = 40,200$$

$$(7') j_H = 118 \quad [\text{Fig. 28}]$$

$$(8') \text{ At } T_s = 170^\circ\text{F},$$

$$k = 0.017 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft}) \quad [\text{Table 5}]$$

$$(c\mu/k)^{1/3} = (0.53 \times 0.029 / 0.017)^{1/3} = 0.97$$

$$(9') h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k} \right)^{1/3} \times 1 \quad [\text{Eq. (6.15b)}]$$

$$= 118 \times 0.017 \times 0.97 / 0.0458$$

$$= 42.3 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

(10') (11') (12') Viscosity correction is unnecessary.

Cold fluid: tube side, water

$$(5) G_t = w/a_t$$

$$= 78,500/0.0954$$

$$= 823,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$V = G_t / 3600\rho = 823,000 / 3600 \times 62.5$$

$$= 3.65 \text{ fps}$$

$$(6) \text{ At } t_s = 90^\circ\text{F},$$

$$\mu = 0.82 \times 2.42 = 1.99 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$D = 0.62/12 = 0.0517 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = DG_t / \mu \quad (Re_t \text{ is for pressure drop only})$$

$$= 0.0517 \times 823,000 / 1.99 = 21,400$$

$$(9) h_i = 900 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad [\text{Fig. 25}]$$

$$(10) h_{io} = h_i \times \text{ID}/\text{OD} \quad [\text{Eq. (6.5)}]$$

$$= 900 \times 0.62/0.75 = 744$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{744 \times 42.3}{744 + 42.3} = 40.1 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Design overall coefficient U_D :

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft}$$

(Table 10)

$$\text{Total surface, } A = 364 \times 8'0'' \times 0.1963 = 572 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{785,000}{572 \times 43.4} = 31.7 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{40.1 - 31.7}{40.1 \times 31.7} = 0.0070 \text{ (hr)}(\text{ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Summary

42.3	h outside	744
U_C	40.1	
U_D	31.7	
R_d Calculated 0.0070		
R_d Required	x	

Pressure Drop

(1') For $Re_s = 40,200$, $f = 0.00162 \text{ ft}^2/\text{in.}^2$ [Fig. 29]	(1) For $Re_t = 21,400$, $f = 0.000225 \text{ ft}^2/\text{in.}^2$ [Fig. 26]
(2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)]	(2) $\Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)]
$= 12 \times 8/12 = 8$	$= \frac{0.000225 \times 823,000^2 \times 8 \times 8}{5.22 \times 10^{10} \times 0.0517 \times 1.0}$
$s = \frac{\rho_{\text{gas}}}{\rho_{\text{water}}}$	$= 3.6 \text{ psi}$
$pv = \frac{w'}{(\text{MW})} \times 1545 T_{\text{abs}}$	
$\rho_{\text{gas}} = \frac{w'}{v} = \frac{p(\text{MW})}{1545 T_{\text{abs}}}$ $= \frac{(83 \times 144)(17.1)}{1545(460 + 170)} = 0.209 \text{ lb/ft}^3$	
$s = \frac{0.209}{62.5} = 0.00335$	
$D_s = 23.25/12 = 1.94 \text{ ft}$	
(3') $\Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s \phi_s}$ $= \frac{0.00162 \times 25,400^2 \times 1.94 \times 8}{5.22 \times 10^{10} \times 0.0458 \times 0.00335} \times 1.0$ $= 2.0 \text{ psi}$	(3) $G_t = 823,000$, $V^2/2g' = 0.090$ [Fig. 27] $\Delta P_r = (4n/s)(V^2/2g')$ [Eq. (7.46)]
	$= \frac{4 \times 8}{1.0} \times 0.090 = 2.9 \text{ psi}$
	(4) $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)] $= 3.6 + 2.9 = 6.5 \text{ psi}$
Allowable $\Delta P_s = 2.0 \text{ psi}$	Allowable $\Delta P_T = 10.0 \text{ psi}$

The ability to meet the allowable pressure drop hinges closely upon the density of the gas. If the gas had been air at the same pressure, the density and pressure drop would have been $0.209 \times \frac{29}{17} = 0.357 \text{ lb/ft}^3$ and 1.2 psi. Similarly an exchanger can be used for gases at vacuum pressure only when a very small mass velocity is employed. The latter results in very low transfer rates for vacuum services, values of U_D being as low as 2 to 10 Btu/(hr)(ft²)(°F).

Air Compressor Intercoolers. In the compression of air for utility purposes it is common to subject atmospheric air to four or more stages of compression. The allowable pressure drop in the intercoolers following the initial stages of compression is extremely critical. Assuming that the compressors operate with compression ratios of roughly $2\frac{1}{4}:1$ or $2\frac{1}{2}:1$, a pressure drop of 1 psi in the first-stage intercooler represents a reduction in the total pressure delivered after the fourth stage of $1 \times 2.5 \times 2.5 \times 2.5 = 13.1 \text{ psi}$ and nearly 80 psi after the sixth stage. In addition, the presence of moisture in the inlet air makes it impossible to compute the heat load as a simple sensible-heat change. Suppose saturated air is taken into the compressor at 95°F (during a summer shower), compressed, and then cooled back to 95°F between each stage of compression. The air and water vapor both occupy the same total volume. The compression of a saturated gas raises the dew point above

its initial dew point. Cooling the compressed gas back to its original temperature requires that it be cooled below its dew point. This can occur only if water condenses out of the compressed gas during cooling. The reasoning follows: The original water vapor in the saturated atmospheric air was all that could exist in the original volume of air at 95°F. Its total weight can be obtained from the specific volume of steam, in cubic feet per pound, for 95°F as found in Table 7. After compression and cooling back to 95°F the total volume of air is reduced but the specific volume of steam at 95°F is unchanged. The same specific volume can be maintained in the reduced gas volume only if some of the water condenses out of the gas.

Example 9.2. Calculation of the Heat Load for an Air Intercooler. 4670 cfm of air saturated at 95°F enters a four-stage adiabatic compressor, having a compression ratio of 2.33:1, at atmospheric pressure. (a) How much heat must be removed in the first-stage intercooler? (b) How much heat must be removed in the second-stage intercooler?

Solution:

(a) Inlet 4670 cfm:

Saturation partial pressure of water at 95°F = 0.8153 psi (Table 7)

Saturation specific volume of water at 95°F = 404.3 ft³/lb (Table 7)

The air and water both occupy the same volume at their respective partial pressures.

Lb water/hr entering = $4670 \times 60 / 404.3 = 692$ lb

First stage:

After 2.33 compression ratio

$$p_2 = 14.7 \times 2.33 = 34.2 \text{ psi}$$

$$\left(\frac{T_2}{T_1}\right)_{\text{abs}} = \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma} \quad (\gamma = 1.40 \text{ for air}^*)$$

$$\frac{T_{2\text{abs}}}{460 + 95} = (2.33)^{(1.4-1)/1.4}$$

$$T_{2\text{abs}} = 705^\circ\text{R or } 245^\circ\text{F}$$

Intercooler:

Final gas volume = $4670 \times 60 \times 14.7 / 34.2 = 120,000$ ft³/hr

Water remaining in air = $120,000 / 404.3 = 297$ lb/hr

Condensation in intercooler = $692 - 297 = 395$ lb/hr

Specific volume of atmospheric air = $(359/29) (555/492) 14.7 / (14.7 - 0.8153)$
 $= 14.8$ ft³/lb

Air in inlet gas = $4670 \times 60 / 14.8 = 18,900$ lb/hr

Heat load (245 to 95°F):

Sensible heat:

$Q_{\text{air}} = 18,900 \times 0.25(245 - 95) = 708,000$ Btu/hr

$Q_{\text{water}} = 692 \times 0.45(245 - 95) = 46,700$ Btu/hr

Latent heat:

$Q_{\text{water}} = 395 \times 1040.1 = 411,000$ Btu/hr

Total 1,165,700 Btu/hr

* The correction of γ for the presence of water vapor is usually omitted.

If condensation had not been accounted for, an error of 33 per cent would have resulted. It should also be noted that over half of the water condenses in the first-stage intercooler.

(b) *Second stage:* $p_3 = 34.2 \times 2.33 = 79.8$ psi.

Final gas volume = $4670 \times 60 \times 14.7/79.8 = 51,500$ ft³/hr

Pounds of water remaining in air = $51,500/404.3 = 127.5$ lb/hr

Condensation in intercooler = $297 - 127.5 = 169.5$ lb/hr

Heat load (245° to 95°F):

Sensible heat:

$$Q_{\text{air}} = 18,900 \times 0.25(245 - 95) = 708,000 \text{ Btu/hr}$$

$$Q_{\text{water}} = 297 \times 0.44(245 - 95) = 19,600 \text{ Btu/hr}$$

Latent heat:

$$Q_{\text{water}} = 169.5 \times 1040.1 = \frac{170,700}{898,300} \text{ Btu/hr}$$

Example 9.3. Calculation of the Dew Point after Compression. The dew point and saturation temperature of the saturated inlet air are the same. After the first-stage compression the dew point is raised. What is the dew point when air saturated at 95°F and 14.7 psi is compressed to 34.2 psi?

Solution:

At inlet:

$$\text{Mols air} = 18,900/29 = 652$$

$$\text{Mols water} = 692/18 = \frac{38.4}{690.4}$$

After compression:

$$\text{Partial pressure of water vapor} = (38.4/690.4)34.2 = 1.90 \text{ psi}$$

From Table 7 equivalent to 1.90 psi, dew point = 124°F

In other words, the gas and water vapor are cooled sensibly from 245 to 124°F in the first stage intercooler and the water vapor will start to condense at 124°F.

The Calculation of Coolers for Wet Gases. The calculation of the heat load and film coefficients in the intercoolers of adiabatic compression systems starting with initially dry gases offers no particular difficulty. The heat load is the sensible-heat requirement to cool the gas back between stages. The film coefficient is that of the dry gas.

Coolers which are required to cool wet gases present a number of additional problems. If the wet gas is to be cooled below its dew point, two zones will appear: (1) from the inlet temperature to the dew point in which both the gas and the vapor are cooled sensibly and (2) from the dew point to the outlet temperature, in which the gas and vapor are cooled and part of the vapor condenses. The first zone can be calculated rather simply as a dry gas, but the calculation of the second zone is extremely lengthy. A relatively accurate calculation will be demon-

strated as an example of condensation in Chap. 13, where it will be seen that both the condensation and gas film coefficients are closely related. The film coefficient for the mixture varies considerably from the dew point to the outlet temperature as the concentration of the condensable vapor diminishes. It is also seen that, in any wet gas cooling service if the tube-wall temperature is below the dew point of the gas, even though the gas is not cooled below its dew point, the tube wall will be wet with condensate. As droplets of condensate fall from the tube, they will reflash into the gas and some fraction may drain from the cooler if the temperature should fall below the dew point. However, the film of liquid on the tube actually introduces a resistance film through which the heat must be transferred. If the condensable vapor is water, the resistance can be omitted because of the high conductivity of the film. If it is a vapor whose condensate is a viscous fluid, it may be necessary to calculate the mean resistance of the film from methods given in Chap. 13 based on the properties of the condensate. It is also well to consider that gases which are not particularly corrosive when containing a small concentration of water vapor may be corrosive when dissolved in condensate water at the cold tube wall. Desuperheaters, which are simply gas coolers, frequently operate with part of the surface moist although no actual condensate drains from the system.

The performance of commercial intercoolers for permanent gases saturated at atmospheric pressure with water at 100°F or less can be predicted rapidly by empirical rules. These rules are as follows: (1) Calculate the entire heat load of sensible cooling and condensation as if transferred at the dry gas rate, and (2) use the value of $\Delta t = F_T \times \text{LMTD}$ obtained from the inlet and outlet temperatures of the gas to and from the cooler and the temperatures of the water. These rules are actually the combination of a safe and an unsafe generalization which tend to offset each other. The combined film coefficient for condensation and gas cooling below the dew point is greater than that given by (1). The true temperature difference is less than that calculated by (2), since the log mean for the portion of the heat load delivered from the dew point to the outlet is less than that calculated by the rule.

PROBLEMS

9.1. 3500 cfm of dry nitrogen at 17 psig and 280°F is cooled to 100°F by water with an inlet temperature of 85°F. Available for the service is a 31 in. ID 1-2 exchanger having 600 $\frac{3}{4}$ in. OD, 16 BWG tubes 12'0" long arranged for eight passes on 1-in. triangular pitch. The baffle spacing is 24 in. center to center.

Pressure drops of 2.0 psi for the gas and 10.0 for the water should not be exceeded, and a minimum dirt factor of 0.01 should be provided. Will the cooler work?

9.2. 17,500 lb/hr of oxygen at atmospheric pressure is cooled from 300 to 100°F by water from 85 to 100°. Available for the service is a 31 in. ID 1-2 exchanger containing 600 $\frac{3}{4}$ in. OD, 16 BWG tubes 12'0" long arranged for eight passes on 1-in. triangular pitch. Baffles are spaced 24 in. apart.

What are the dirt factor and the pressure drops?

9.3. 5000 cfm of saturated air at 100°F enters the first stage of a compressor having a 2.45:1 compression ratio. The air is at atmospheric pressure. (a) How much heat must be removed after each of the four stages, assuming a 2.0 psi pressure drop in each intercooler. (b) Available for the first-stage intercooler is a 29 in. ID exchanger having 508 tubes $\frac{3}{4}$ OD in., 14 BWG by 12'0" long arranged for eight passes on 1-in. triangular pitch. The baffle spacing is 24 in. Using cooling water with an 85°F inlet, what are the pressure drops and dirt factor?

9.4. For the second-stage intercooler of Example 9.2 in the text the following 1-2 exchanger is available: 21 $\frac{1}{4}$ in. ID containing 294 $\frac{3}{4}$ in. OD, 14 BWG tubes 12'0" long arranged for eight passes on $1\frac{5}{16}$ -in. triangular pitch. Baffles are spaced 20 in. apart. What are the dirt factor and the pressure drops?

NOMENCLATURE FOR CHAPTER 9

A	Heat-transfer surface, ft ²
a	Flow area, ft ²
a''	External surface per linear foot, ft
B	Baffle spacing, in.
C'	Clearance between tubes, in.
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of tubes, ft
D_e	Equivalent diameter, ft
F_T	Temperature difference factor, $\Delta t = F_T \times \text{LMTD}$, dimensionless
f	Friction factor, ft ² /in. ²
G	Mass velocity, lb/(hr)(ft ²)
g'	Acceleration of gravity, ft/sec ²
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluids, respectively, Btu/(hr)(ft ²)(°F)
h_{io}	Value of h_i when referred to the tube OD, Btu/(hr)(ft ²)(°F)
j_H	Factor for heat transfer $(hD/k)(c\mu/k)^{1/2}$, dimensionless
j_h	Factor for heat transfer $(h/cG)(c\mu/k)^{3/4}$, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Tube length, ft
LMTD	Log mean temperature difference, °F
N	Number of baffles
N_t	Number of tubes
n	Number of tube passes
P_T	Tube pitch, in.
ΔP	Pressure drop, psi
$\Delta P_T, \Delta P_t, \Delta P_r$	Total, tube side and return drop, psi
p	Pressure, psia
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_d	Combined dirt factor, (hr)(ft ²)(°F/Btu)
Re	Reynolds number, dimensionless
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
s	Specific gravity, dimensionless
T_{abs}	Absolute temperature, °R
T_a	Average temperature of hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
T_1, T_2	Inlet and outlet temperature of hot fluid, °F
t_a	Average temperature of cold fluid, °F